How Data Visualisation Works

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Preface

This is a book about data visualisation.

Like many books, it scratches an author's itch.

This particular itch began a long time ago, when first introduced to the works of Edward Tufte. Books like *The Visual Display of Quantitative Information* and *Envisioning Information*[1] inspired an interest in how to produce effective graphics, but they were also frustrating.

Reading a Tufte book to get better at data visualisation is a bit like watching Roger Federer play to get better at tennis. It is possible to glean fragments of insight into the master's skill and technique, but it is difficult to come away with a clear set of rules for how to succeed on our own.

At the same time, encountering the work of William S. Cleveland presented a different problem.[2] Here were hard experimental results with clear implications for effective data visualisations: use length not area; use bars not pies. But they only told us about a small subset of the types of data visualisations that we could create and they only told us about a small subset of the types of questions that we could answer with those data visualisations.

The next decade or so were spent distracted by building tools for *creating* data visualisations[3] before returning again to the question of *designing* data visualisations. In those intervening decades, a lot has happened.

Data visualisation is an area of research for psychologists, for computer scientists, for geographers, and even for statisticians. The literature is wide and deep and complex.

This has significantly reduced the size of the second problem. We know a lot more about a wider range of data visualisation types and data visualisation tasks. Unfortunately, this has introduced a new problem. There are comprehensive and detailed overviews of current knowledge,[4] but it is now difficult to consume and make sense of so much knowledge.

At the same time, there are more accessible and practical guides for creating effective data visualisations.[5] These eschew much of the gory detail in favour of simple, applicable advice. The problem here is that, without the underlying explanations, the guidelines can become just a long laundry list to commit to memory. Rather than just being told that bars should always start at zero, it is more satisfying to know *why* bars should always start at zero.

For some of us there is a need for a middle path. We would like to understand why we are making certain data visualisation choices, but we have brains that can only hold so many ideas at once. For us there is utility in a framework that contains a small number of core concepts, that still covers a wide range of data visualisations, but connects them all with a relatively simple explanation.

This book attempts to provide such a framework. It is a synthesis of a large number of research articles and books and it is a simplification of a large amount of knowledge. It is an attempt to prune the complexity so that we can see the forest through the trees, while at the same time leaving enough foliage that we can still recognise a tree when we see one.

I hope that it helps.

Scope

There is a lot of useful information and understanding about how data visualisations work, but it is distributed throughout and embedded within a very broad, deep, and detailed data visualisation literature. This book aims to produce a simple, coherent explanation that is accessible to a less expert audience.[6]

The focus of this document is **static** data visualisations for **presentation**. The assumption is that our data values have an interesting property and we want to produce a display that makes it easy and efficient for viewers to perceive that property. Some of the information that we cover will also apply to interactive graphics and to exploratory graphics, but the specific requirements of those types of data visualisation will not be addressed.[7]

There is also an emphasis on **statistical graphics** in this book, which is characterised by the use of abstract geometric symbols, such as circles and rectangles, to represent data values, as opposed to scientific graphics, which is focused more on realistic representations of physical objects, or cartography, which is focused on the geographic context of data values.

This document is also focused on what **vision science** has to tell us about effective data visualisation. We will consider factors that influence how rapidly and how accurately information can be extracted from a data visualisation. This will include findings from a broad range of research, encompassing Statistics, Computer Science, Psychology, and Cartography.

However, this document does not cover all fields of research that impact on data visualisation. We will not, for example, address the ethical use of data visualisation, beyond a basic assumption that we want to faithfully represent the properties of the data.[8]

This document is also aimed at a numerate audience, so we assume an appreciation of the importance of collecting the right data in the right way for the questions that we hope to answer.[9]

It is also important to acknowledge that vision science cannot yet tell us everything about how data visualisation works. However, what is known will help us to have some understanding of why some data visualisations are so effective and others are less effective.

As well as being a synthesis of a large range of data visualisation research, this document is also a simplification of that research. Some details are smoothed over and some details are just left out. The goal is to provide a framework that is easy to understand by non-experts and easy to apply to a broad range of data visualisations. References to the relevant literature are provided in footnotes so that the interested, or sceptical, reader can dig deeper.

Software

This book is about the choices that go into *designing* a data visualisation. It does not cover how to use software to implement those design choices and *create* a data visualisation.

All data visualisations in this book were created with R.[10] The code for all data visualisations is available in the github repository for this book.

How to Read this Book

Although the goal is to provide a relatively simple conceptual framework for thinking about data visualisation, this is not a short book. The Executive Summary provides the simplest possible statement of the framework and it may be useful to resurface and revisit this summary if you become lost in the depths of one of the chapters.

A slightly longer summary of the framework is provided in the Review. The Review also provides some guidance on how the framework can be applied to the creation of a new data visualisation and the criticism of an existing one.

Another major issue with a subject this complex and a subject that is studied from so many different angles is *terminology*. There is no consistency across all of the different sources of information from which this book was drawn, so all that we can do is attempt to be consistent with ourselves. Some of the most important terms that will be reused relentlessly throughout this book are highlighted in the Executive Summary. The meaning of these terms is hopefully approximately what you already think, but they will be made clearer as the book progresses.

All section and figure cross-references are hyperlinks, so clicking them will navigate to the relevant section or figure. Even more helpfully, hovering the mouse over a hyperlink will show a preview of the relevant section or figure. This is particularly effective for figures because it obviates the need to reproduce the same image multiple times; just hover over the figure link to see the image that the text is referring to.

Acknowledgements

This book is based upon a large number of research articles and books, but in order to make the text appear more accessible and friendly, there are no formal citations in the main text. There are, however, a large number of footnotes, which provide the links between what is being described in the main text and the underlying research, and those footnotes contain proper citations. The real work is not being ignored, it is just being hidden from view.

This book has been generated using the Quarto publishing system. If any parts of the book look nice to you, it is probably thanks to Quarto.

I am also contractually obliged to thank my wife, Julia. I am also geunuinely indebted to Julia for the brutally honest feedback that she provided on early drafts of the text. If any parts of the book make sense to you, it is probably thanks to Julia.

Notes

[1] *The Visual Display of Quantitative Information* (Tufte 1983) and *Envisioning Information* (Tufte 1990) are just two of a series of influential and visually attractive books that also includes *Visual Explanations* (Tufte 1997) and *Beautiful Evidence* (Tufte 2006).

[2] There were several studies by Cleveland, some in collaboration with Robert McGill, which were summarised and extended in two books: *The Elements of Graphing Data* (William S. Cleveland 1985b) and *Visualizing Data* (William S. Cleveland 1993).

[3] Some of that work is described in Murrell (2019) and some more of it is described here.

[4] *Information Visualization: Perception for Design* (Ware 2021) reviews a very wide range of research that has relevance to data visualisation in one way or another and *Visualization Analysis and Design* (Munzner 2014) provides a comprehensive and theoretical coverage of the complete process of creating effective data visualisations.

[5] For example, *Creating More Effective Graphs* (Robbins 2005), *Data Visualization: A Practical Introduction* (Healy 2018), and *Fundamentals of Data Visualization: A Primer on Making Informative and Compelling Figures* (Wilke 2019).

[6] Besides scratching the author's itch, this book addresses an identified need. For example, Sönning (2022) deconstructs how a line plot works in great detail, but there is a market for something that is more accessible to and applicable by a less-expert audience.

"Going forward, we must do a better job of translating the academic research into practice, making it easier for academics and nonacademics alike to create useful, well-designed graphics." (Vanderplas, Cook, and Hofmann 2020)

The aim is to provide an overview that is similar in scope to more expert frameworks like (Kosslyn 1989), but more consumable by a less expert audience. The goal of this document is to provide a description that is sufficiently straightforward for the non-expert to consume, hopefully without making too many experts roll in their graves or send them to an early one. In most cases, it will be easy to see for ourselves that a visual effect is real through a simple diagram or data visualisation, but references to the relevant theories and experimental results in the literature will be provided in footnotes like this one.

[7] Some examples of books that include a discussion of interactive graphics are Cook and Swayne (2007), Theus and Urbanek (2008), and Unwin, Theus, and Hofmann (2006).

[8] Alberto Cairo's writing often includes the ethical dimensions of data visualisation, e.g., Cairo (2014) and Cairo (2016b).

The *Data by Design* project is a striking example of work with a focus on ethical data visualisation (Emory Digital Humanities Lab 2021).

[9] We are assuming that the data we have is sufficient for the question(s) of interest and contains no *substantive* problems, in the sense of Healy (2018).

Examples of books that provide a more comprehensive treatment of the entire data visualisation process are Cairo (2019) and Munzner (2014).

An example of a much shorter, but still broader view of the data visualisation process is Bertini (2025).

[10] Many of the diagrams in this book were also created with R (R Core Team 2025), but some of diagrams were created using graphviz (Ellson et al. 2002).

License

Executive Summary

- Creating a data visualisation involves an **encoding** of **data values** as **visual features**.

  For example, numeric **counts** are **encoded** as the **heights of bars** to create a bar plot.

- Reading a data visualisation involves a **decoding** of **visual features** to obtain **data values**.

  For example, the **heights** of bars in a bar plot are **decoded** to obtain **counts**.

- The **effectiveness** of a data visualisation depends on the effectiveness of the **decoding**.

  For example, a bar plot is effective for comparing **counts** because the visual system is effective at comparing the **heights** of bars.

- More **complex encodings** produce more **complex visual features**.

  For example, a scatter plot **encodes two numeric variables** as the x- and y-locations of points, which produces the **position in space** of points.

- More **complex visual features** lead to more **complex decodings**.

  For example, the **position in space** of points in a scatter plot is **decoded** as the **correlation** between two numeric variables.

- An **effective data visualisation encodes** data values to visual features that facilitate an effective **decoding** from visual features back to information about the data.

1 Introduction

Visualization is essentially a mapping process from computer representations to perceptual representations, choosing encoding techniques to maximize human understanding and communication.[11]

A data visualisation presents data values in a visual format. In an effective data visualisation, this allows properties of the data to be perceived very rapidly and without conscious effort. As an example, consider the data shown in Table 1.1, which shows the crime rate for youth offenders in New Zealand, for the ages 14 to 16, for each year from 2011 to 2021. With the purely text-based, tabular presentation of the data in Table 1.1, it requires considerable time and effort to determine any trends in the crime rates.

Table 1.1: A table of the number of distinct youth offenders, per 10,000 of popuplation, aged 14 to 16 from 2011 to 2021. It is difficult to perceive trends in the crime rates from this purely text-based, tabular presentation of the data.

age	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021
14	583	524	446	378	318	305	280	281	270	244	219
15	712	619	528	443	369	339	315	304	272	262	246
16	810	723	615	482	426	365	328	334	314	302	257

By contrast, with a simple line plot data visualisation like Figure 1.1 we can perceive trends like the decrease in crime rates over time easily and immediately. It is also easy to perceive more detailed properties like the fact that the trend in crime rates for 16-year-olds is not completely monotonic and the fact that the crime rate for 15-year-olds (the blue line) is always greater than the crime rate for 14-year-olds (the red line) and less than the crime rate for 16-year-olds (the green line). The right choice of visual display can result in a very effective communication of information.

Figure 1.2 shows a different data visualisation of the same data, this time a heatmap. As in Figure 1.1, we are representing data values using shapes, rectangles this time, rather than the pure text in Table 1.1. However, with Figure 1.2, while very basic trends are apparent, like an overall decrease in crime rate, perceiving detailed changes in the crime rates is very much harder compared to Figure 1.1. This shows that choosing the right visual display is not just a matter of using shapes instead of text.

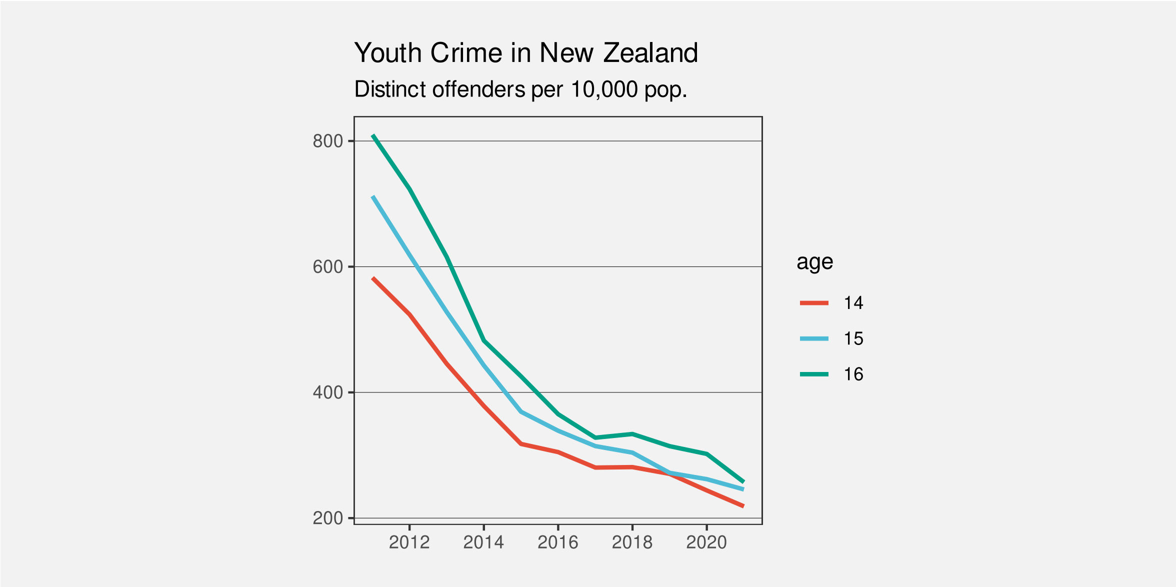


Figure 1.1: A line plot of the crime rate amongst youth offenders aged 14 to 16 from 2011 to 2021. Detecting trends in the crime rates is very fast and easy with this data visualisation.

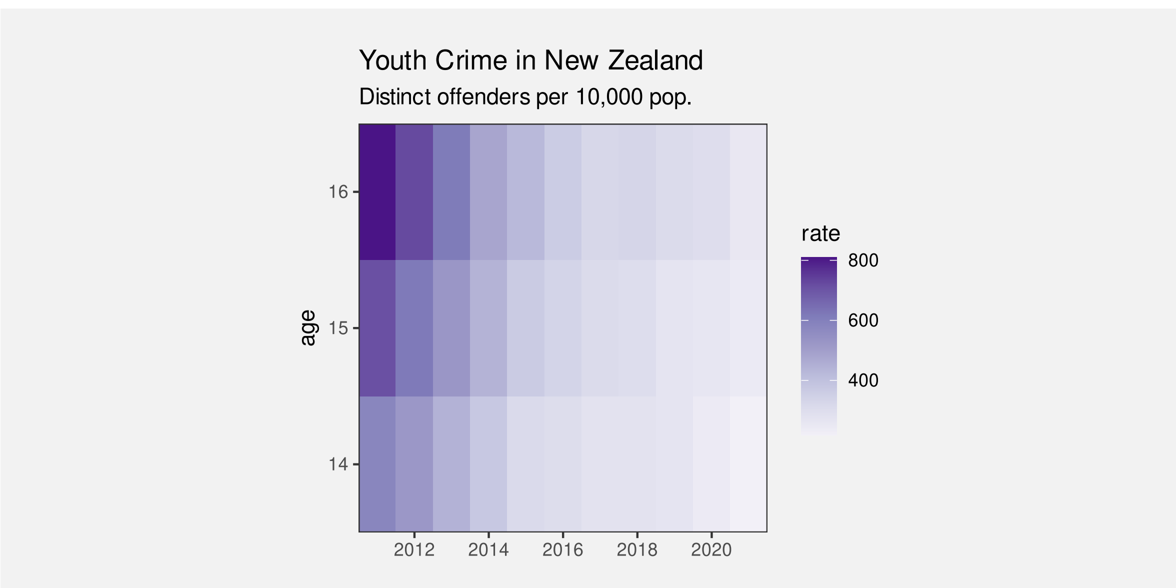


Figure 1.2: A heatmap of the crime rate amongst youth offenders aged 14 to 16 from 2011 to 2021. Some basic trends are visible, but details are harder to perceive compared to Figure 1.1.

Although Figure 1.1 is very effective for perceiving trends in crime rates for each age group, it may be less effective if we are interested in a different question. For example, we might be interested in whether the *total* crime rate is completely monotonic. Does the *sum* of the red, blue, and green lines always decrease year on year? This shows that choosing the right data visualisation depends very much on what information we want to communicate.

Suppose that we are interested in what the *exact* crime rate was for 16-year-olds in 2011. Neither Figure 1.1 nor Figure 1.2 is the best option for answering this question. The best option in this case is actually Table 1.1. Sometimes text rather than shapes is the most effective visual display.

1.1 Understanding why

This book is about how to choose the right data visualisation. As we have already seen, this depends heavily on what information we want the data visualisation to communicate.

One way to help people to choose a data visualisation is to list the types of information that we want to communicate and, for each situation, provide a data visualisation to use. For example, if we have time series data and wish to observe trends over time then we should choose a line plot like Figure 1.1. If we want to communicate individual data values precisely, then we should choose a table like Table 1.1. There are many examples of this sort of "directory" of visualisations[12], which is one reason why this book takes a difference approach.

Rather than providing a directory of visualisations, we will be looking deeper at the building blocks of data visualisation and we will seek an understanding of why different building blocks work better than others at communicating different sorts of information. For example, why do the lines in Figure 1.1 communicate trends more effectively than the rectangles in Figure 1.2?

This understanding of building blocks will provide us with a way to *reason* about the effectiveness of a data visualisation. We will be able to justify the choice of a particular data visualisation. You may have heard that statisticians hate pie charts.[13] But do you know why? Do pie charts have any redeeming features? Conversely, bar plots are extremely popular, but why are they so good? Knowing more about why different plots work for different situations will help us to make better decisions about when to use what sort of data visualisation.

1.2 Encoding

What is a data visualisation? For the purposes of this book, we define a data visualisation as a conversion of information into a visual representation. We will refer to this process as **encoding** information.

The most obvious example of an encoding is the conversion of **data values** into **data symbols** (Figure 1.3). For example, in Figure 1.1, all of the data values for 14-year-olds, the number of offenders for each year, have been encoded to an orange line. The data values for 15-year-olds have been encoded to a blue line and the data values for 16-year-olds have been encoded to a green line. In other words, each row of Table 1.1 is encoded to a separate coloured line.

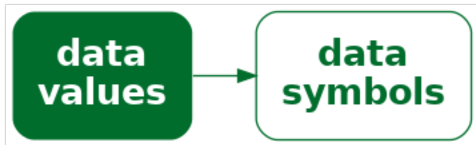


Figure 1.3: A data visualisation can be described by the encodings that it employs to represent data values as data symbols.

The heatmap of the same data (Figure 1.2) provides a different visual representation because it employs a different encoding. In this case, the data symbol is a rectangle, with one data symbol for each combination of age and year, and with the colour of the rectangle reflecting the number of offenders of that age for that year. In other words, each cell of Table 1.1 is encoded to its own coloured rectangle.

Deciding on which data visualisation to use can be thought of in terms of deciding how to encode data values.[14]

1.3 Decoding

What is a data visualisation for? As we saw in the very first section of this chapter, the value of a data visualisation lies in our ability to rapidly and effortlessly extract information. In other words, a data visualisation allows us to convert from a visual representation back to information. We will refer to this process as **decoding** information.

A simple example of a decoding is the extraction of a single data value. For example, we can decode the number of 14-year-old offenders in 2011 from Table 1.1. More complex decodings involve extraction summaries of multiple data values. For example, we can decode the overall decline in youth crime from the lines in Figure 1.1.

The purpose of encoding data values to a particular visual representation is to enable the viewer to extract properties of the data very rapidly and very easily.

In order to produce an effective data visualisation, we need to **encode** data values to data symbols in a way that facilitates **decoding** data values from the data symbols (Figure 1.4).

The journey that we take throughout this book will involve exploring a variety of different ways that information can be **encoded** to a visual representation. We will encounter many

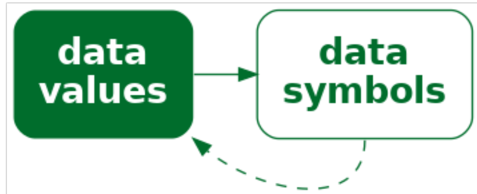


Figure 1.4: The effectiveness of a data visualisation will depend on how well our visual system can **decode** from the data symbols to the properties of the data.

different data visualisations, but we will analyse them in terms of the fundamental ways that they encode information.

The effectiveness of a data visualisation will depend on how well information can be **decoded** from the visual representation. By identifying the different encodings that we can use, and understanding what information can be decoded from different encodings, we will come to understand why some data visualisations are more effective than others for different tasks.

1.4 Summary

Data visualisations can be very effective for communicating information.

However, a data visualisation that is effective for communicating one type of information may be ineffective for communicating another type of information.

The goal of this book is to explain why some data visualisations are more effective than others at communicating different types of information—how data visualisation works.

We will focus on how information can be **encoded** to create a visual representation. We will characterise a data visualisation in terms of the encodings that it uses to convert data values into data symbols.

The effectiveness of an encoding will depend on how well we can **decode** the information that we want from a visual representation. We will judge a data visualisation in terms of how well data values can be recovered from the data symbols.

Notes

[11] This quote is from an early web resource on data visualisation that was produced by the ACM SIGGRAPH Education Committee (Owen et al. 1999).

[12] For example, see Part I of Wilke (2019), particularly Chapter 5.

[13] In fact, pie charts have received a lot of hate from all quarters for over a century:

"In a sense, it might be construed as an insult to a man's intelligence to show him a pie chart" Karsten (1925, p98).

"the only worse design than a pie chart is several of them … pie charts should never be used." Tufte (1983, p178).

For some counter-arguments, see Section 7.2.

[14] The idea of characterising a data visualisation as a set of *encodings* from data values to visual representations aligns with the idea of *mappings* from Wickham (2010).

2 The Visual System

In this book, we are going to explore a range of ways that information can be **encoded** to create different data visualisations. For example, we can encode data values as different-coloured lines to create a line plot (Figure 1.1) and we can encode data values as different-coloured rectangles to create a heatmap (Figure 1.2).

The choice of encoding depends on how well information can be **decoded* from the data visualisation. For example, if we want to communicate trends then encoding data values as lines will be more effective than encoding data values as rectangles because it is easier to decode trends from lines.

Understanding why some encodings are more effective than others will require an understanding of how the human visual system works. In order to understand the effectiveness of an encoding from information to a visual representation, we need to know how well information can be decoded from the visual representation.

In this chapter, we focus on the **decoding** of information from a visual representation. In this chapter we will establish a very simple model of the visual system, which will help us to understand the effectiveness of the different encodings that we consider throughout the book.[15]

2.1 The eye

In order for us to decode a data visualisation, we first have to see it. Light must emit from a computer screen or reflect off a page and enter the eye. Light passes through the pupil, is focused by the lens, and is projected onto the retina at the back of the eye. The retina contains around 100 million light-sensitive nerve cells and signals from those are combined into about 1 million nerve fibres in the optic nerve, which passes signals on to the brain (see Figure 2.1). A very large amount of visual information is gathered by the eye and passed on to the brain.

> One reason why data visualisation is effective is because our visual system gathers an enormous amount of information.

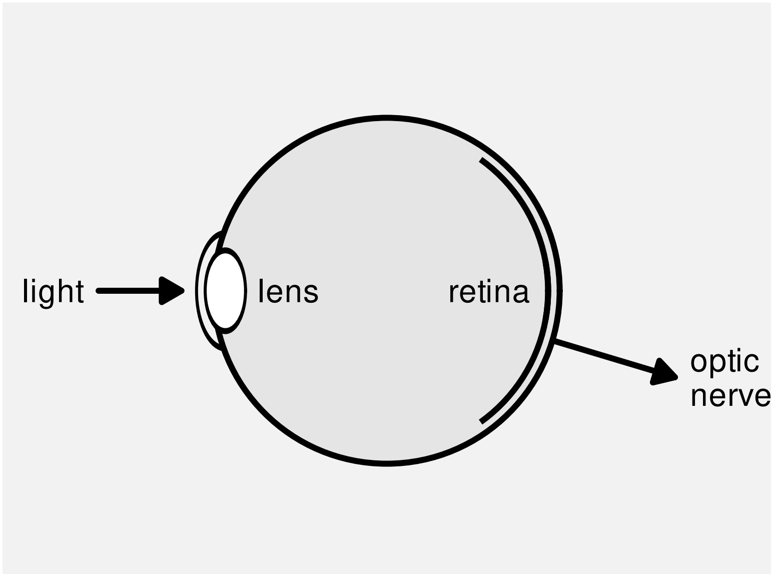


Figure 2.1: A very simple diagram of the human eye. Light enters via the pupil and is focused by the lens onto retina. Light-sensitive neurons in the retina pass signals via the optic nerve to the brain.

2.2 Visual processing

Processing of the signals from the optic nerve occurs in several areas of the brain.[16] Very basic **visual features** are extracted first, such as **position**, **length**, **angle**, **colour**, and **pattern**.[17] This low-level information is passed on and built upon to produce higher-level visual elements, such as regions and **shapes**. Information is then also integrated from prior knowledge to assist with the identification of **objects**—things like trees, animals, and faces.[18]

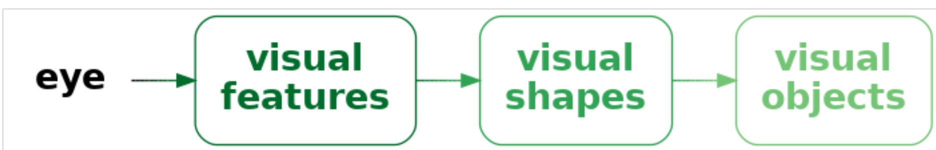


Figure 2.2: A very simple model of visual processing. Basic visual features like position, angle, and colour are identified first, then collections of those features are combined into visual shapes, and, finally, some shapes are recognised as familiar visual objects.

Lower-level processing occurs subconsciously and more rapidly than higher-level processing, which can require conscious effort. Furthermore, lower-level processing occurs in parallel and can involve many instances of visual features at once, whereas higher-level processing will only focus on a small number of items or a single item at once. On the other hand, lower-level visual features are rapidly discarded as our attention shifts, while higher-level visual objects are more persistent and may even become part of long-term memory (Figure 2.3).

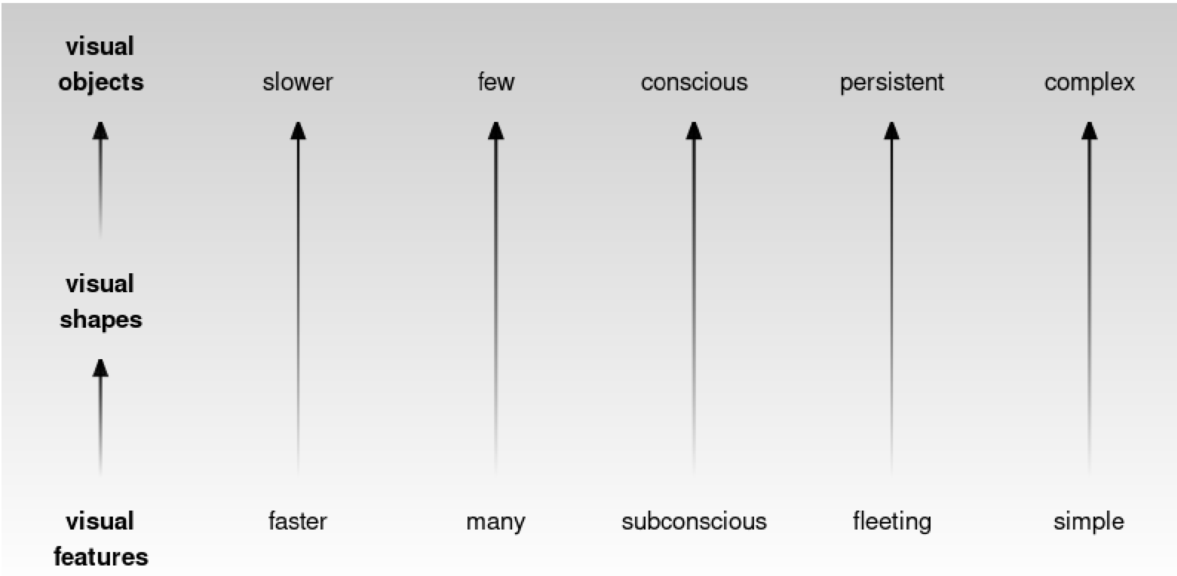


Figure 2.3: Early visual processing identifies large numbers of simple visual features very rapidly and without conscious effort. Subsequent processing takes longer, involves fewer, more complex visual shapes and visual objects, and may involve conscious mental effort, but the result can represent more complex information that is more memorable.

This means that, if we **encode** data values as simple **visual features**, like length, angle, and colour, then it will be possible to rapidly **decode** large amounts of simple information. In addition, a smaller amount of more complex information will be decoded from higher-level processing of those low-level visual features.

For example, Figure 2.4 shows an image that is made up of many small items: tiny triangles with different **positions**, **lengths**, **angles**, and **colours**. The angle of the triangles represents wind direction and the length of the triangles represents wind strength. Blue triangles are over the sea and green triangles are over land.[19] We can effortlessly decode the basic **visual features**—angle, size, and colour—of individual triangles.

The visual system can also very quickly process the visual features of many triangles at once. There is no need to consciously focus on every individual triangle in the image and, at a higher level of processing, darker and lighter **visual shapes** are identified as regions of different colours. For example, a band of lighter wind forms a line from the top-left corner down and to the right at a roughly 45-degree angle. At a higher level of processing again, at least for the denizens of Oceania, the green regions are recognised as a very familiar **visual object**: the outline of New Zealand.

> Encoding data values as simple visual features is effective at conveying simple information because the visual system decodes visual features rapidly and subconsciously.
>
> The visual system also processes many visual features at once, producing more complex visual shapes or visual objects, which allow more complex information to be decoded.

2.3 Visual focus

Figure 2.5 shows a slightly more detailed view of the structure of the eye (compared to Figure 2.1). The retina contains two types of light-sensitive cells: **cones** and **rods**. The cones are sensitive to colour and are concentrated very densely at the centre of the retina, in an area called the **fovea**. Rods only detect the amount of light (monochrome vision) and are spread more sparsely towards the periphery of the retina.

This arrangement means that, despite the experience of a complete, detailed field of view, we only get detailed and colourful information from a small foveal region (less than 5 degrees of visual angle). The information from our peripheral vision is less colourful and lower-resolution.[20]

Furthermore, when we view an image, we do not simply stare at the centre and see everything. Instead, we perform a series of **fixations**, which are brief periods of focus on one area of the image, with **saccades** in between, which are rapid changes in the location of our focus. Figure 2.6 demonstrates that where we look within an image makes a difference.[21]

This means that, in order to decode detailed information from a data visualisation, we will need to switch focus between different areas of detail. For example, Figure 2.7 shows eye-tracking data from one person viewing an image for 10 seconds.[22] The original image is shown on the left and eye-tracking data is shown on the right, with hollow black circles showing the position of the coloured circles in the original image. In the eye-tracking image, the fixations are shown as purple circles and the saccades are shown as lines between the circles. For example, we can see that the viewer has fixated multiple times upon the large, colourful circles in the original image and has also fixated on the text at the top and bottom of the image.

This is effectively an argument for keeping things simple. If we can only decode information from an image in a piecemeal fashion, then the fewer locations that we are required to focus on the better.

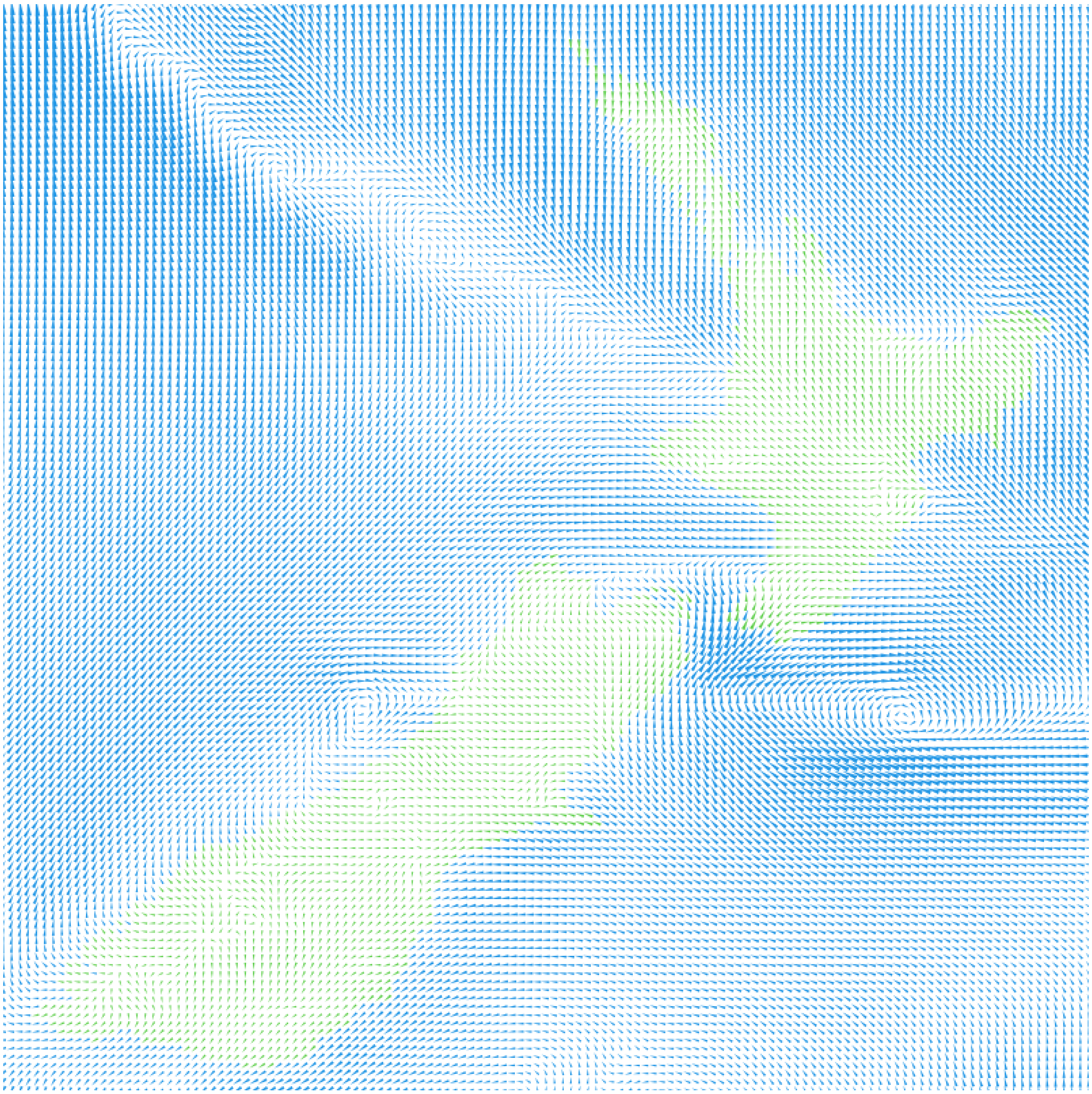


Figure 2.4: A visualisation of wind strength and wind direction that is made up of lots of small lines of different lengths and different angles and different colours.

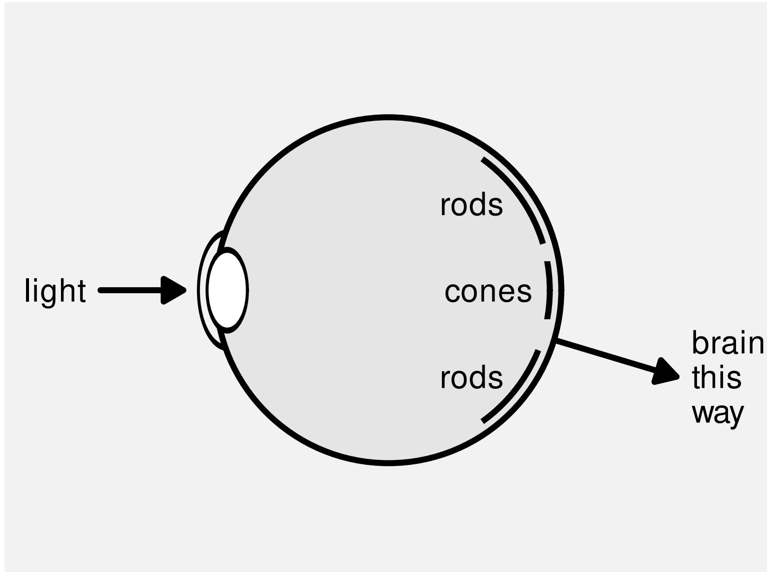


Figure 2.5: A slightly less simple diagram of the human eye. There are two kinds of light-sensitive cells in the retina. Cones are colour-sensitive and focused very densely at the centre of the retina. Rods are only light-sensitive and are spread less densely towards the periphery of the retina.

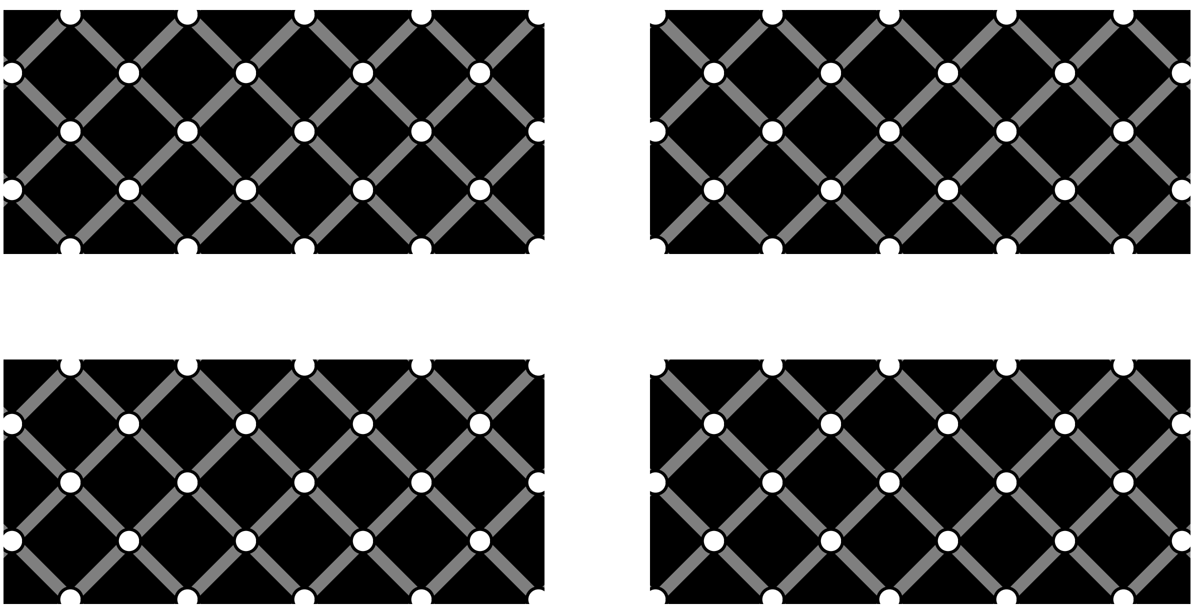
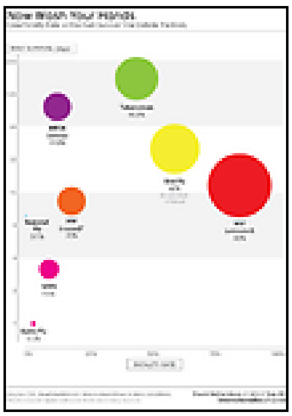
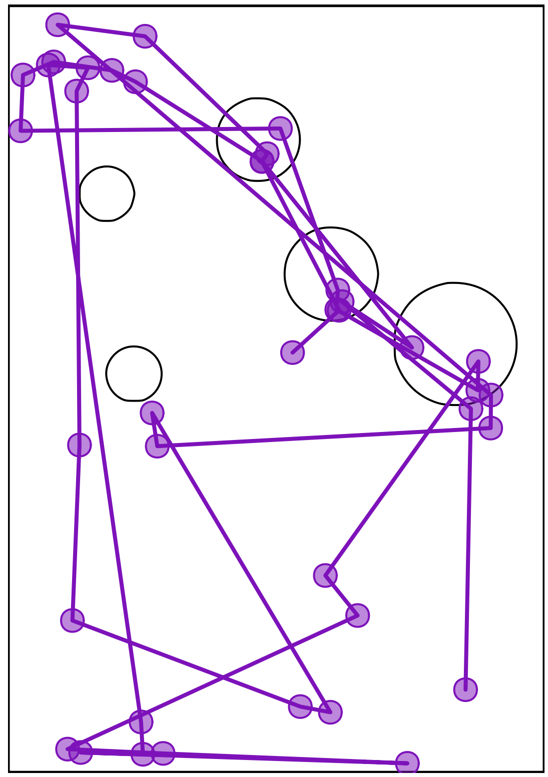


Figure 2.6: The extinction illusion. Look at one corner of the image and then switch focus to a different corner of the image. At a certain viewing distance, the white dots should disappear when we are not focused on them. We may also see black dots within the white dots in our peripheral vision.



(a) A thumbnail of the original image.



(b) Fixations and saccades from eye-tracking data.

Figure 2.7: Eye tracking data (on the right) from the MASSVIS project that shows fixations (circles) and saccades (lines between circles) for one person viewing an image for 10 seconds. A thumbnail of the original image is shown on the left.

> A data visualisation will be slower to decode and require more effort to decode if it requires the viewer to constantly switch focus.

2.4 Visual memory

When we switch focus between different areas of an image, we need to remember information from a previous area of focus. How well can we hold visual information in memory? The most important feature of visual memory is that it is very limited. We can only hold in memory a small amount of information at once.[23]

Figures 2.8 and 2.9 provide a simple demonstration of the limitations of visual memory.

It is easier and faster to decode information from visual elements if they lie within the scope of a single visual focus. If we have to switch focus, it not only takes longer to decode information, but we may hit limits on our ability to remember visual elements from previous areas of focus.[24]

The limits on visual memory reflect well-known limits on working memory and those limits can also be worked around via familiar mechanisms such as chunking.[25] For example, we may only be able to hold in memory a small number of basic visual features such as the colours of circles, but we can effectively hold many more basic visual features if we memorise more complex visual elements that are composed of many basic visual features. In other words, we can remember several **visual shapes** or **visual objects** even if we cannot remember all of the individual **visual features** within those visual shapes or visual objects (Section 2.2). For example, we can hold in memory multiple country flags even though we cannot necessarily hold in memory all of the invidual colours on those flags.

Once again, the obvious conclusion is to keep things simple, so that there is less to remember during decoding, though this is tempered by the fact that more complex visual shapes may allow more to be remembered, especially if they are recognisable visual objects.

> A data visualisation will require more effort to decode if the viewer is required to switch focus and remember visual information across multiple areas of visual focus.

2.5 Visual differences

The ability to gather a large amount of basic features from an image very quickly (Section 2.2) is supplemented by rapid visual processing that combines and summarises the information. These subconscious processes extract some properties from an image without any mental effort.

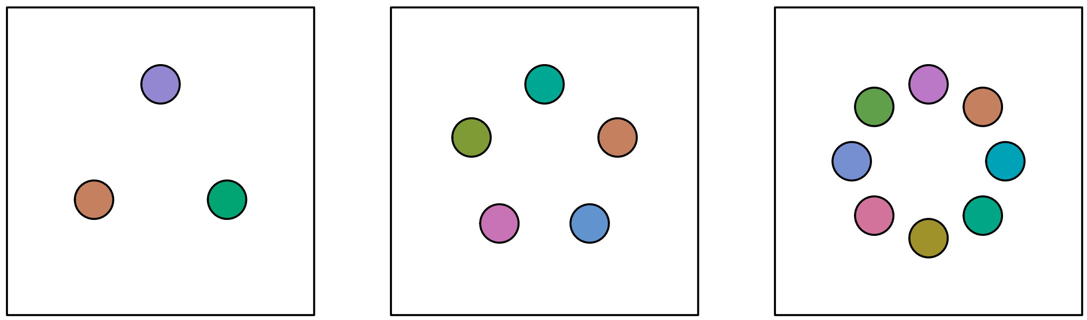


Figure 2.8: A memory task. Focus on the circles in one of the three squares then turn the page and try to tell how many of the circles have changed colour in the corresponding square in 2.9. The task should be obviously harder when there are more circles to remember, which reflects the limits on visual memory.

For example, in Figure 2.10a we can detect the one teal dot amongst the 24 orange dots without effort. Figure 2.10b shows that this task remains effortless even if the number of dots is much larger.[26]

This ability to identify an item that is different from all other items within an image also applies to other basic features like size, as shown in Figure 2.11.

If an item differs on more than one visual feature, the effect is even stronger (Figure 2.12). This means that the visual system is very sensitive to items that are **different** in terms of basic **visual features**.

On the other hand, if the distractor items are similar on one visual feature, but different on another, and if the arrangement of the items is less structured, then finding a particular item of interest becomes much harder. For example, if we look for the single larger teal dot in Figure 2.13, it is much harder to find than it is in Figure 2.12. The visual system's sensitivity to **different** items is much greater when all other items are the **same**.[27]

This means that encoding data values as data symbols that differ only in terms of basic visual features will be effective for decoding unusual data values.

> A data visualisation will be effective for decoding unusual data values if the data are encoded using differences in basic visual features, especially if everything else remains the same.

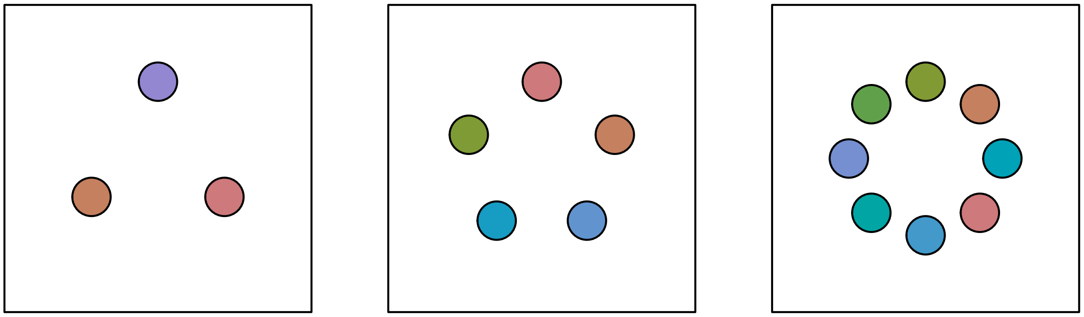


Figure 2.9: A memory task. Look at the circles in one of the squares in 2.8 on the previous page and then come back here and try to tell how many circles have changed colour in the corresponding square. The task should be obviously harder when there are more circles to remember, which reflects the limits on visual memory.

2.6 Visual attention

Where we focus within an image is not random. The figures from Section 2.5 show that our attention is automatically drawn to the **larger**, **brighter**, and more **colourful** items within an image. This happens due to very early visual processing and without conscious thought.[28]

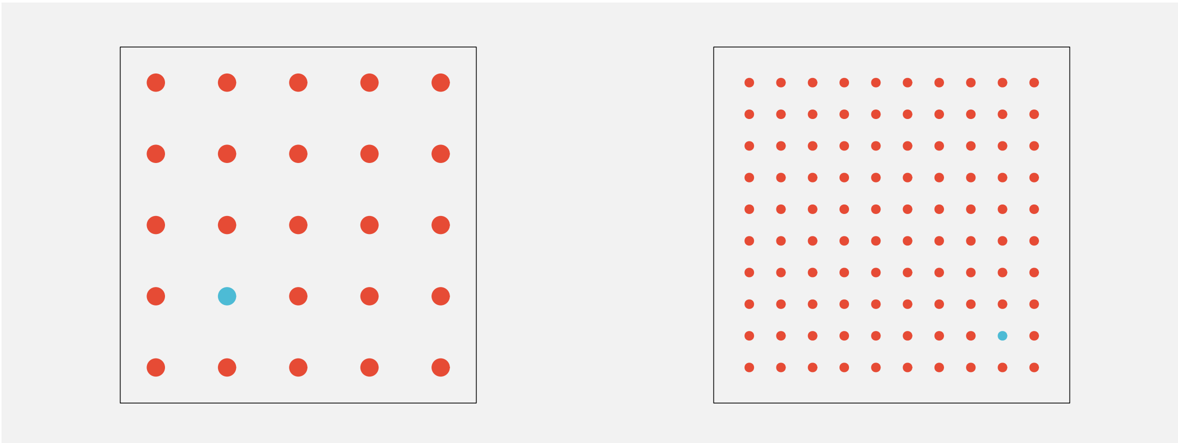
> A data visualisation will be effective for decoding important data values if it uses basic visual features like colour to draw the viewer's attention.

Where we focus within an image, and what information we extract, is also affected by conscious goals—what are we looking for?[29] For example, when you were looking for the larger teal dot in Figure 2.13, did you notice that there was also a single small orange dot? Now that your attention has been drawn to that task, it will be easier to find the smaller orange dot.[30]

> A data visualisation will be effective for decoding important data values if it includes instructions that direct the reader's attention.

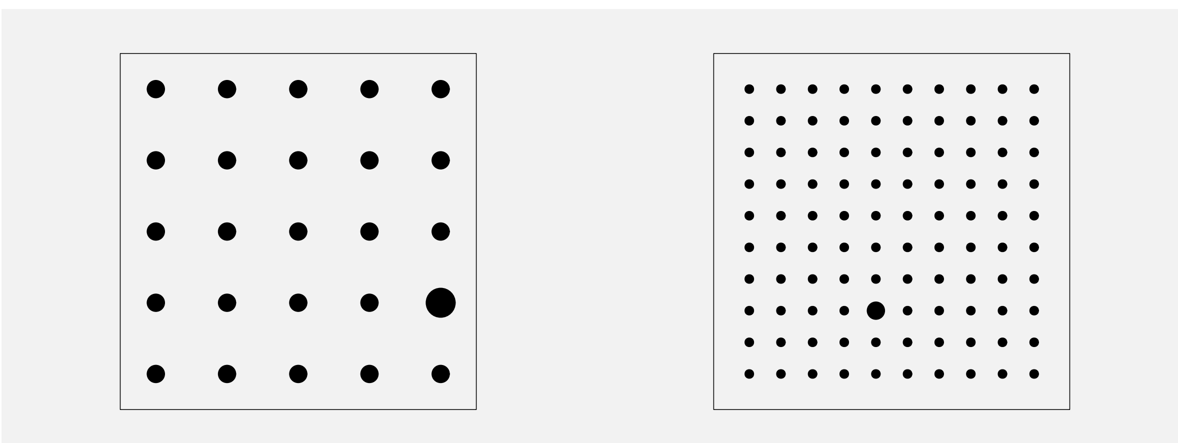
2.7 Visual similarities

In Figure 2.13, we can see several groups of items: orange dots, teal dots, large dots, and small dots. These separate groups of items are identified based on their **similarity**—each group of items have the same size and the same colour. The flip side of being able to easily identify items that are different (Section 2.5) is that we can easily identify items that are the same.



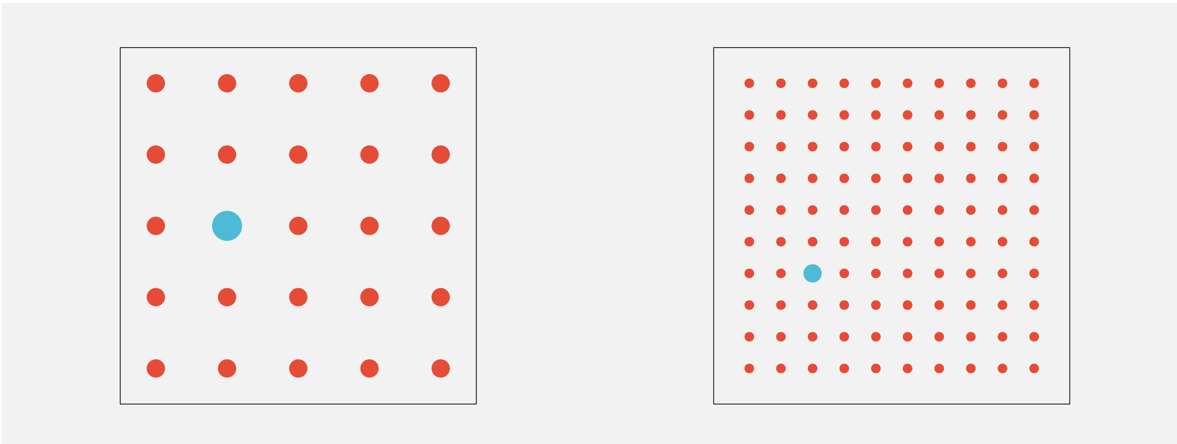
(a) One teal dot amongst 24 orange dots. (b) One teal dot amongst 99 orange dots.

Figure 2.10: Amongst a collection of dots, we effortlessly perceive one dot that has a different colour.



(a) One larger dot amongst 24 smaller dots. (b) One larger dot amongst 99 smaller dots.

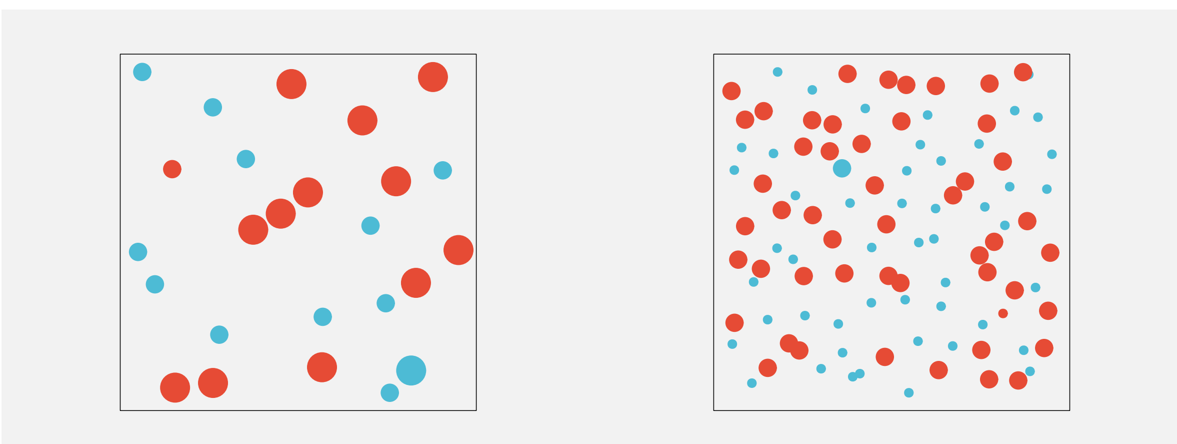
Figure 2.11: Amongst a collection of dots, we effortlessly perceive one dot that differs only by size.



(a) One larger teal dot amongst 24 smaller orange dots.

(b) One larger teal dot amongst 99 smaller orange dots.

Figure 2.12: Amongst a collection of dots, we effortlessly perceive one dot that differs by both colour and size.



(a) One larger teal dot amongst 24 dots that differ by colour and/or size.

(b) One larger teal dot amongst 99 dots that differ by colour and/or size.

Figure 2.13: We have to work much harder to identify a unique item when there is more going on.

The grouping of similar items within an image also happens automatically and without conscious effort. Figure 2.14 shows pairs of dot matrices that are very similar, but in one matrix of each pair we see rows and in the other matrix we see columns. For example, in the middle pair, items that are the same colour produce groups that are either rows or columns.

There are other visual properties that mean we automatically perceive items as groups: items that are close together (items that have similar **positions**) are seen as a group so we see either rows of black dots or columns of black dots on the left of Figure 2.14; items that are connected by lines are also seen as a group so we see either connected rows or connected columns on the right of Figure 2.14.[31]

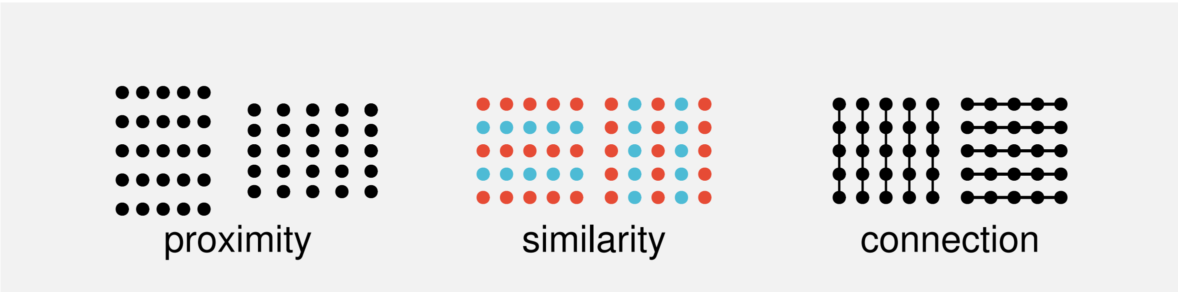


Figure 2.14: In each matrix of dots, we clearly see either rows or columns. In the pair of matrices on the left, making the dots slightly closer together horizontally produces rows, and making the dots slightly closer vertically produces columns. In the middle pair of matrices, using the same colour on each row produces rows, and using the same colour on each column produces columns. In the pair of matrices on the right, drawing vertical lines produces columns and drawing horizontal lines produces rows.

These grouping effects are very powerful and can even be used to override each other in some combinations. Figure 2.15 shows that an enclosing border can be stronger than a connecting line, connecting lines can be stronger than proximity, and proximity can be stronger than similarity.

Another type of automatic grouping is demonstrated in Figure 2.16. On the left is a collection of dots. We naturally perceive a horizontal line and a circle from these dots, as shown in the centre of Figure 2.16. We automatically complete lines, particularly smooth curves, even when the curve is broken, as in the central set of dots. We do not tend to complete lines with sharp corners, as indicated on the right of Figure 2.16.[32]

All of this means that if we encode data values from the same group using the same position or colour, or if we connect them with lines, or if we enclose them within a border, then the visual system will automatically decode the group information.

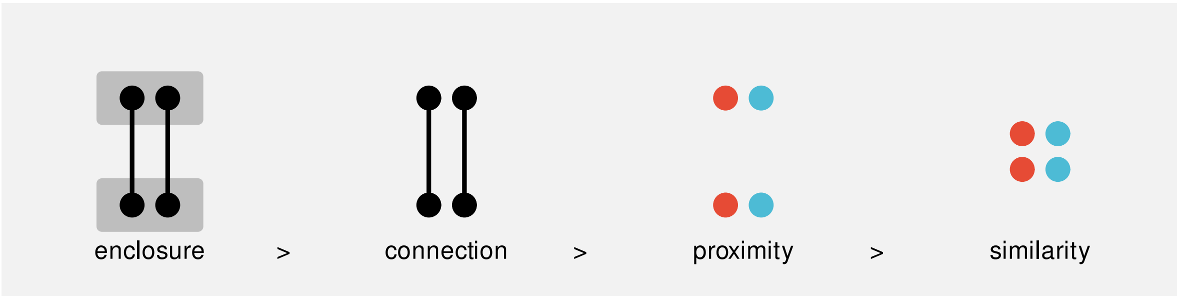


Figure 2.15: For each set of four dots, we automatically perceive two pairs of dots. On the left, the enclosing grey rectangles produce an upper pair and a lower pair, despite the fact that the left and right pair of dots are connected by lines. Second from left, the connecting lines produce a left pair and a right pair, despite the fact that the top pair and the bottom pair are closer to each other. Second from right, the positions of the dots (their proximity) produces an upper pair and a lower pair, despite the fact that the left and right pairs are coloured the same. On the right, the same colours produce a left pair and a right pair.

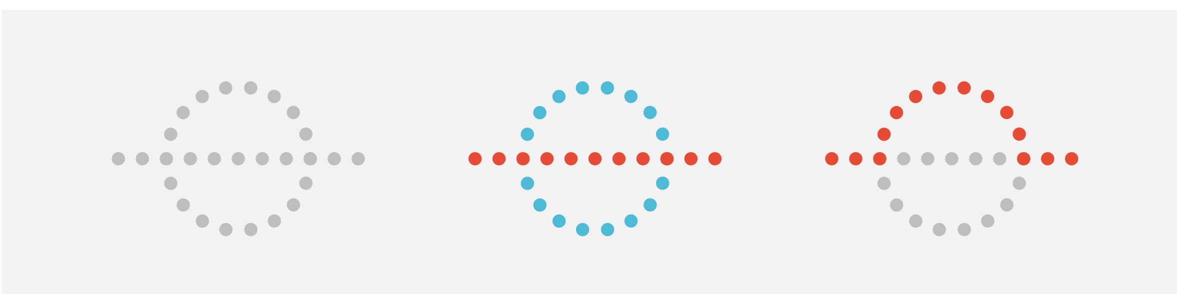


Figure 2.16: We automatically complete curves.

> One way that a data visualisation can be effective is by relying on on the automatic decoding of visual groups (proximity, similarity, etc).

2.8 Visual simplicity

Figure 2.17 shows an arrangement of dots, similar to Figure 2.16. The collection of dots on the left has an ambiguous interpretation. Is this a single shape, or two overlapping shapes (as indicated in the centre of Figure 2.17), or some complex combination of multiple shapes (as indicated on the right of Figure 2.17)?

Our visual system has a preference for the simple interpretation. Given an ambiguous arrangement of visual elements, we will tend to perceive whatever is most orderly, regular, and coherent.[33]

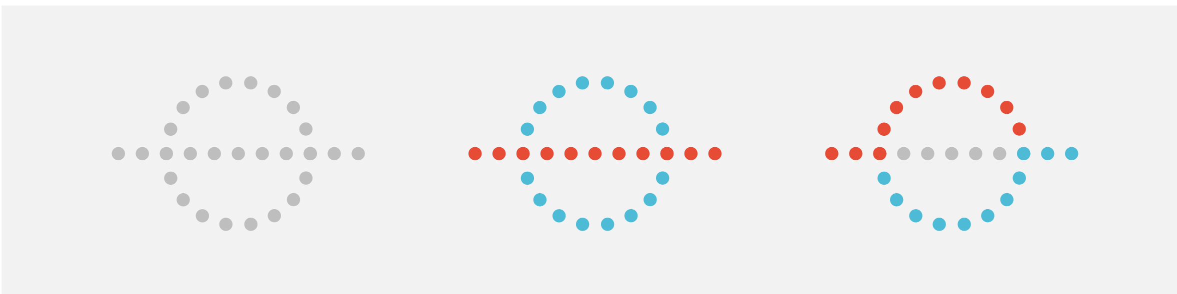


Figure 2.17: We automatically perceive simple shapes over more complex interpretations.

Figure 2.18 shows that we also perceive groups when items have a symmetric arrangement.[34]

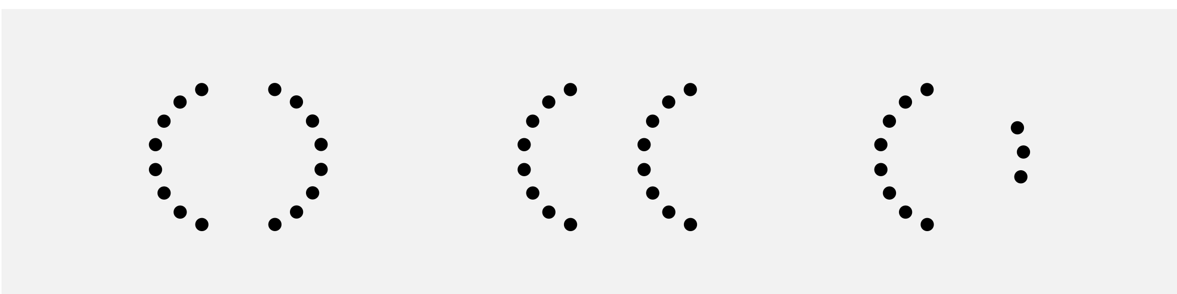


Figure 2.18: We automatically perceive groups from symmetric arrangements. The pair of symmetric curves on the left naturally go together, but the pairs of curves in the middle and to the right do not.

This is another argument for simplicity in a data visualisation, but also for regularity and orderliness. These results suggest that we should design a data visualisation so that it leads the viewer where they already want to go rather than trying to force them to decode a more complex interpretation.

> A data visualisation may be easier to decode if it is orderly and lends itself to a simple interpretation.

2.9 Visual tasks

Identifying whether visual elements are the same or different (Section 2.5) is an example of a very simple visual task. Figure 2.19 (on the left) shows another simple visual task: judging the ratio of two different lengths. These tasks are rapid and require very little effort.

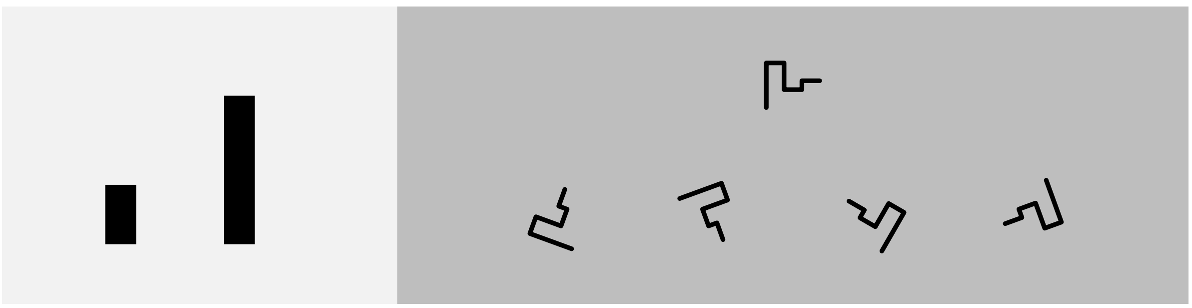


Figure 2.19: On the left is a simple visual task, to estimate the relative length of the two lines. On the right is a more complicated visual task, to decide which of the four bottom shapes is *not* a rotated version of the top shape.

However, on the right of Figure 2.19 is an example of a task that is much more difficult. In this task, we have to identify which of the four shapes in the bottom row is *not* a rotated version of the shape at the top. This task in requires much more deliberate effort and time to perform.[35]

The point here is that we should create data visualisations that can be decoded using simple visual tasks. That is how we can fulfil the promise of data visualisation. It is far less useful to create a data visualisation that forces us to expend significant mental effort in order to decode useful information.

> A data visualisation will be effective if it relies on automatic and rapid visual processing rather than requiring conscious mental effort.

2.10 Dangers of the visual system

The value of data visualisation comes from the visual system's ability to rapidly gather and process a large amount of visual information. However, while data visualisations usually consist of artificial and simple images presented on a two-dimensional page or screen, the visual system has evolved for the unstructured, messy, and three-dimensional natural environment. This means that the visual system will sometimes generate a more sophisticated, but erroneous interpretation of an image from a very simple arrangement of basic shapes. In other words, we are vulnerable to visual illusions.

For example, Figure 2.20 shows the Muller-Lyer Illusion and the Ebbinghaus Illusion. In the former, two horizontal lines of the same length appear to have different lengths and, in the latter, two orange circles of the same size appear to have different sizes.[36]

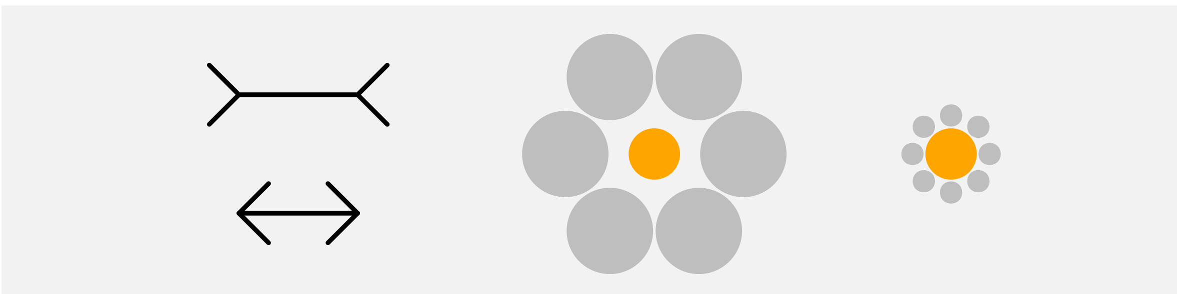


Figure 2.20: The two horizonal lines in the Muller-Lyer illusion on the left have exactly the same physical length, but the top line appears longer. The two inner circles in the Ebbinghaus illusion on the right have exactly the same physical radius, but they appear to be different sizes.

Figure 2.21 shows another illusions involving simple lines. Every line segment in this image is the same physical length, but we perceive some segments as longer than others.[37]

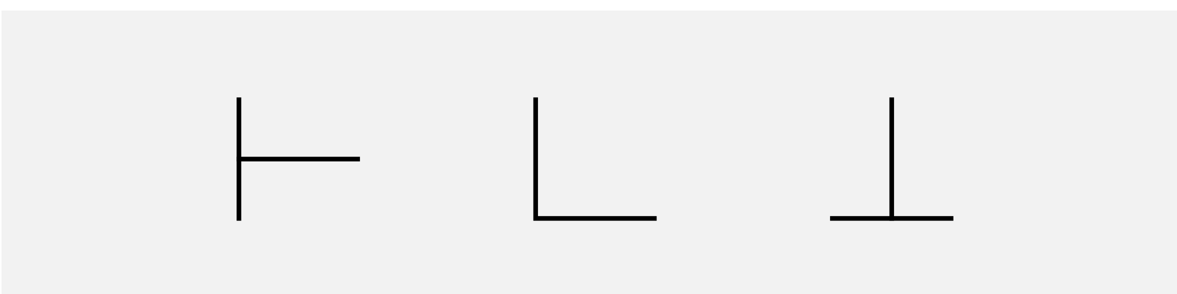


Figure 2.21: All of the horizontal and vertical line segments in this image are exactly the same physical length, but our visual system perceives some segments as longer than others.

Figure 2.22 shows two images that are composed of three simple parallelograms. The three parallelograms are arranged separately on the left, which means that we decode them as three simple shapes. However, for the arrangement of the parallelograms on the right, the natural decoding is a 3-dimensional box, even though the image is clearly 2-dimensional.

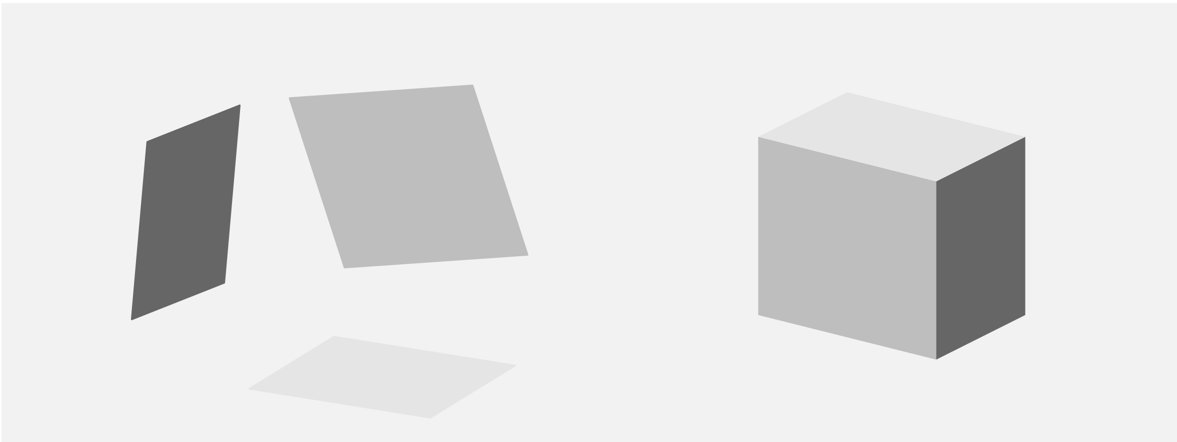


Figure 2.22: The image on the right is made up from the three simple parallelograms shown on the left. Because of their arrangement and their shading, we are tempted to decode the three parallelograms on the right as a 3-dimensional object.

These examples demonstrate that the visual system can produce incorrect or unintended results for even very simple visual tasks based on very simple visual elements. If we encode data values in such a way that we produce a visual illusion then the decoding will be wrong and will not lead back to the orginal data values.

> Visual illusions can destroy the effectiveness of a data visualisation because they lead to an incorrect decoding of information.

2.11 Summary

There are features of the human visual system that mean that we can decode some information extremely rapidly and without effort:

- A very large amount of basic information is gathered at once about simple visual features like positions, lengths, and colours.
- Large, bright, colourful items automatically attract attention.
- We automatically identify groups of items within an image based on similarity of basic visual features like position and colour, plus connecting lines and enclosing borders.

On the other hand, there are limitations of the visual system that suggest encodings that we should avoid:

- Detailed information is only available at the centre of the visual field.
- Visual memory is extremely limited.

These features suggest that encoding data values as basic visual features and generating simple, orderly data visualisations will lead to rapid and effortless decoding of information.

Notes

[15] Simple descriptions of the human visual system can be found in many articles and books on data visualisation, for example, Cairo (2012), Franconeri et al. (2021a), and Sönning (2023). See Ware (2021) for a more in-depth, but still very accessible description. Wikipedia also has some very helpful content (Wikipedia contributors 2025).

[16] A rough estimate is that more than 50% of the brain is involved in visual processing (Van Essen, Anderson, and Felleman 1992).

[17] Low-level visual processing involves the creation of *feature maps* that identify colours, edges, and orientations, along with the spatial arrangement of those basic features within the field of view (Ware 2021, 146).

Patterns, or *textures*, are perceived at a slightly higher level, based on low-level perception of spatial frequencies (fine detail versus coarse detail).

[18] A slightly more detailed overview is provided in Ware (2021; 2021, 20).

[19] Figure 2.4 is based on wind data from the National Oceanic and Atmospheric Administration's Operational Model Archive and Distribution System (NOMADS). The data were accessed on September 14 2024 and processed using the {rNOMADS} package (Bowman and Lees 2015), with raw data interpolated to a finer grid.

[20] This is not to say that peripheral vision is useless. Peripheral vision contributes significantly to our field of view (Stewart, Valsecchi, and Schütz 2020; Rosenholtz 2016), and is as capable as foveal vision at detecting motion (Mckee and Nakayama 1984). However, our ability to perceive fine details is limited to a relatively small area of focus.

[21] This image is a demonstration of the extinction illusion, which is related to the better-known Hermann grid illusion (Ninio and Stevens 2000).

This illusion is not explained simply by differences in foveal versus peripheral perception—there is currently no complete explanation for the effect—but it serves to show that we are getting different information from different regions of the field of view.

[22] The plot was originally from The Economist, but this image was taken from the resources provided on the MASSVIS project web site. The eye tracking data were taken from the same web site. Cite MASSVIS article. Only a thumbnail version of the original image is shown in Figure 2.7 in order to comply with terms of use of the MASSVIS data set.

[23] Some limitations of visual memory are briefly mentioned in Ware (2021) (pages 397-405 and pages 423-425).

Schurgin, Wixted, and Brady (2020) mentions a limit of only “three or four items”!

The limitations of visual memory are also more complex than just a number of visual items (Marois and Ivanoff 2005):

“The first limitation concerns the time it takes to consciously identify and consolidate a visual stimulus in visual short-term memory (VSTM), as revealed by the attentional blink paradigm. This process can take more than half a second before it is free to identify a second stimulus. A second, severely limited capacity is the restricted number of stimuli that can be held in VSTM, as exemplified by the change detection paradigm. Finally, a third bottleneck arises when one must choose an appropriate response to each stimulus. Selecting an appropriate response for one stimulus delays by several hundred milliseconds the ability to select a response for a second stimulus (the ‘psychological refractory period’)”

[24] Tufte refers to this as drawing “within the eyespan” (Tufte 1990, 67)

“In understanding human visual perception, an important component consists of what people can perceive at a glance. If that glance provides the observer with sufficient task-relevant information, this affords efficient processing. If not, one must move one’s eyes and integrate information across glances and over time, which is necessarily slower and limited by both working memory and the ability to integrate that information.” (Rosenholtz 2023).

[25] Miller’s famous paper “The Magical Number Seven, Plus or Minus Two: Some Limits on Our Capacity for Processing Information” (Miller 1956) talks about limits on working memory, but also how to “escape” those limits, for example, by chunking information.

[26] The ability to detect features without conscious effort is called *pre-attentive* processing or *pop out* (Treisman 1985).

[27] For example, from Wolfe and Horowitz (2017): “Salience of a target increases with difference from the distractors (target–distractor heterogeneity) and with the homogeneity of the distractors (distractor–distractor homogeneity) along basic feature dimensions.”

[28] Saliency maps (e.g., Itti, Koch, and Niebur 1998; Matzen et al. 2018), which attempt to predict where the viewer will focus attention, are based on the idea that attention is drawn to basic visual features like colour, angle, and size.

[29] Having a conscious target to search for not only directs our visual focus (Section 2.3), but also filters the processing of visual input to exclude visual features that are not of interest (Knudsen 2020). In other words, to some extent, we see what we are looking for!

[30] Wolfe and Horowitz (2017) describe “Bottom-up, stimulus-driven guidance” versus “Top-down, user-driven guidance”.

[31] These visual effects are known as the Gestalt Grouping Principles (Wagemans et al. (2012)). The Proximity Principle says that items that are close to each other form a group. The Similarity Principle says that items that have the same colour or size form a group. The Connectedness Principle says that items that are connected to each other form a group. The Common Region Principle says that items within the same bounded region form a group. Todorovic (2008) provides a very accessible description of these principles and several others.

[32] This is the Gestalt Continuity Principle (Todorovic 2008): Items that lie along a smooth curve form a group.

[33] This is the Gestalt Prägnanz Principle (Todorovic 2008): The tendency to seek simplifying structure and to find order within the chaos of an image.

[34] This is the Gestalt Symmetry Principle (Todorovic 2008): Items that exhibit symmetry will form a group (in preference to items that do not exhibit symmetry).

[35] “the comprehension of visual displays tends to be easiest when simple pattern identification processes are substituted for complex cognitive processes” (Carpenter and Shah 1998, 98)

[36] The original publication of the Muller-Lyer illusion (in German) is Müller-Lyer (1889), though in that, the stimulus described is a single line with three sets of arrows, as shown below.

The Ebbinghaus Illusion is also known as the Titchener Illusion after its English-language populariser (Titchener 1916).

Various evolutionary explanations of these illusions have been proposed, e.g., (Gregory 1963) and (Howe and Purves 2005).

[37] An early in-depth exploration of the Vertical-Horizontal Illusion was provided by Künnapas (1955).

3 Visual Features

In Chapter 1, we described a data visualisation as an **encoding** from data values to data symbols (Figure 1.3). In Chapter 2, we established some basic properties of the visual system, which will help to explain how well the **encoding** of data values to data symbols can be reversed to **decode** from data symbols back to data values (Figure 1.4). We also identified three levels of visual processing: **visual features**, **visual shapes**, and **visual objects** (Figure 2.2).

There are many possible ways to encode data values to data symbols. For example, in Section 1.2, we saw that exactly the same data values could be encoded in different ways to produce a line plot (Figure 1.1) and a heatmap (Figure 1.2). We also saw that different data visualisations made some decodings easier and others more difficult. For example, it is easier to decode trends from the line plot than the heatmap.

The remainder of this book explores different sorts of encodings and why they they lead to more or less effective decodings.

3.1 Bar plots

Figure 3.1 shows a bar plot of the total number of offenders aged 14 to 16 in New Zealand in 2021, for different ethnic groups. The data values are shown in Table 3.1.

Table 3.1: A table of the total number of offenders aged 14 to 16 in 2021 for different ethnic groups.

	group	count
54	Unknown	1589
52	Pasifika	328
51	Māori	2869
53	European/Other	1833

A bar plot is a good example of a very simple sort of **encoding** where each data value is encoded as a basic **visual feature** of a data symbol (Figure 3.2). For example, every count value in Table 3.1 is encoded as the length of a bar in Figure 3.1.

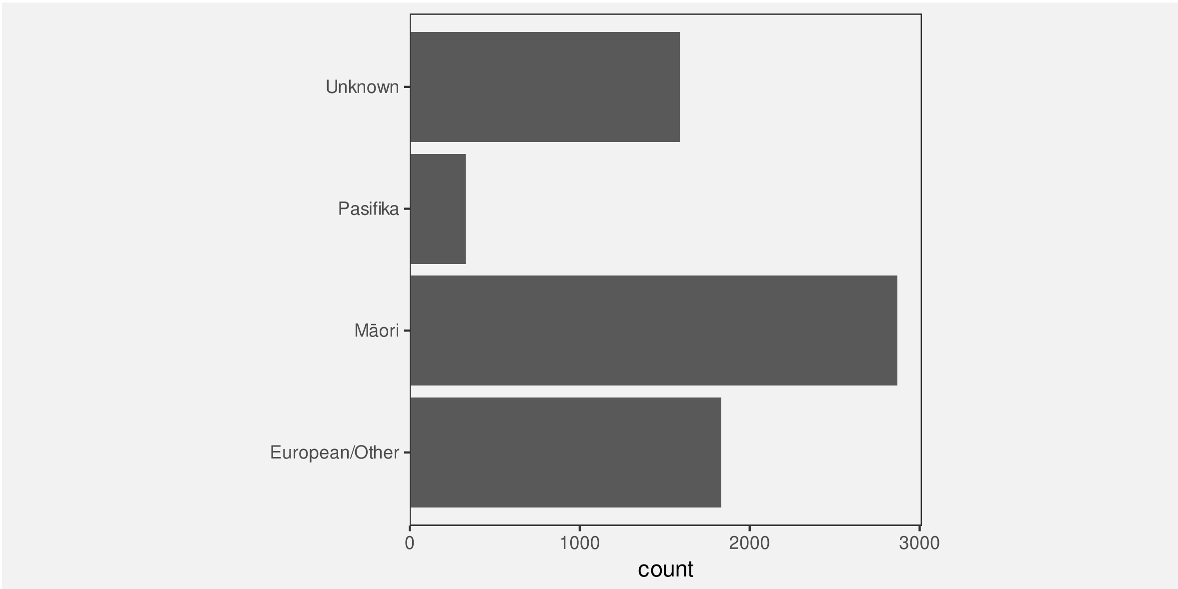


Figure 3.1: A bar plot of the total number of offenders for different ethnic groups in 2021. The data symbols in this plot are the horizontal bars.

In addition, in a bar plot, each data value is encoded to its own data symbol (Figure 3.2). For example, every count value in Table 3.1 is encoded as the length of a *separate* bar in Figure 3.1.[38]

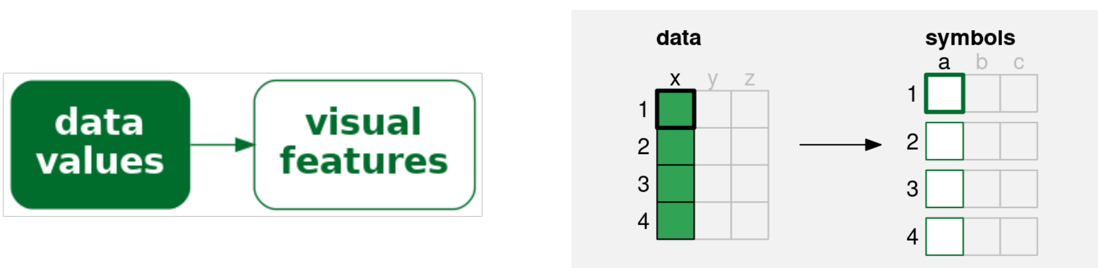


Figure 3.2

A simple data visualisation encodes each data value as a **visual feature** of a data symbol. Each data value (`x`) is encoded to one visual feature (`a`) of a single data symbol. For example, if there are four data values, we end up with four data symbols, each of which encodes a single data value.

Data visualisations that make use of this sort of encoding are useful for **decoding** and comparing individual data values. For example, we can decode a count value from the length of each bar in Figure 3.1 and we can compare two counts by decoding the lengths of two bars.

This allows us to answer simple questions about individual data values. For example, what is the largest total number of offenders and is the total larger for Māori or European/Other offenders?

3.2 Visual features

The data symbols in Figure 3.1 are a set of bars or, put more simply, rectangles. We have seen that each data value from Table 3.1 is encoded to one of these rectangles. The number of offenders for each ethnic group is encoded to a rectangle.

However, we can specify more precisely how individual data values are being encoded as a **visual feature** of a data symbol. In this case, the total number of offenders for each ethnic group is encoded as the **length** of a rectangle (Figure 3.3). Each data value is encoded to the length of one rectangle.[39]

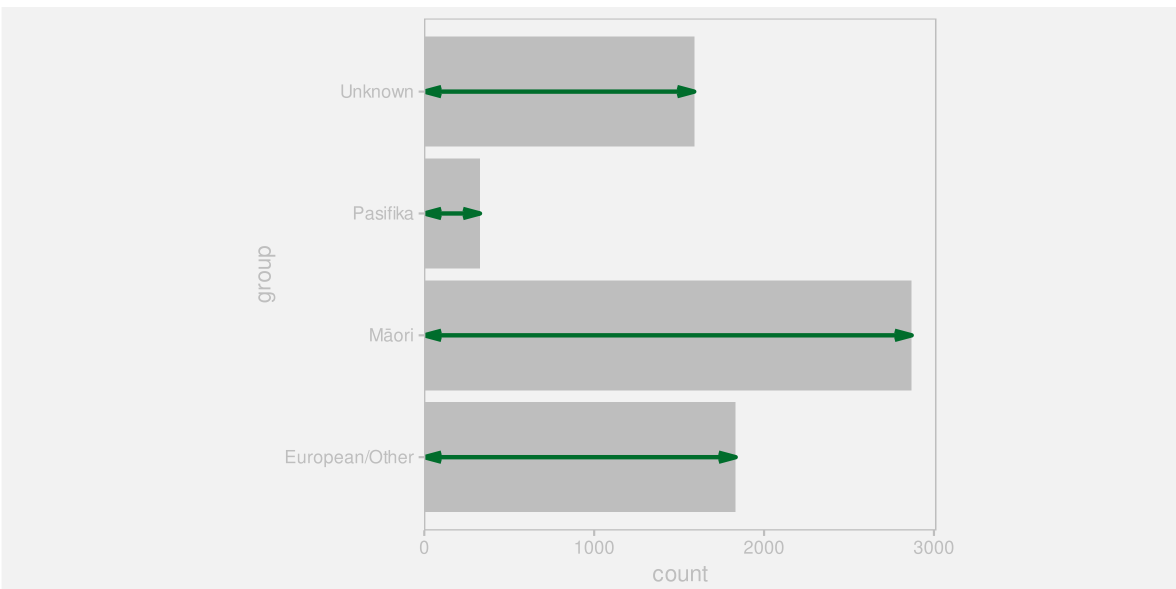


Figure 3.3: A bar plot of number of crimes for different ethnic groups in 2021. The data values, the total number of crimes, have been encoded to the **lengths** of the bars (the green lines).

There is another encoding happening in Figure 3.1. Each ethnic group data value in Table 3.1 is also encoded to a rectangle in Figure 3.1. Each ethnic group is encoded as the vertical **position** of one rectangle (Figure 3.4).

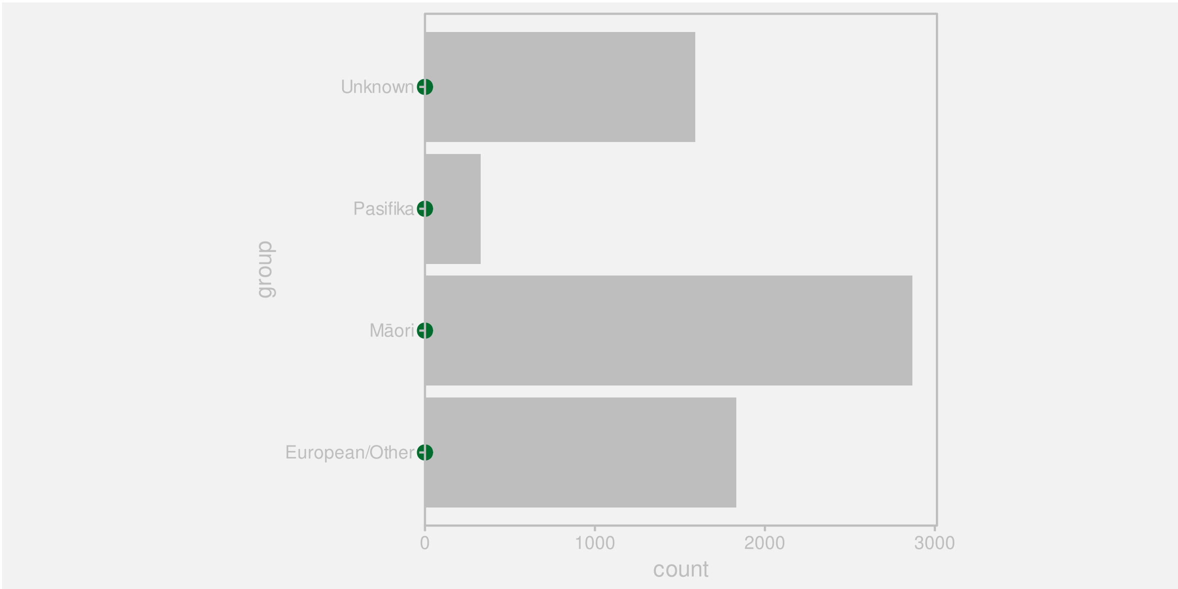


Figure 3.4: A bar plot of number of crimes for different ethnic groups. The data values, the ethnic groups, have been encoded to the **positions** of the bars (the green dots).

Length and **position** are examples of basic **visual features** (Section 2.2). One way to make an effective data visualisation is to encode data values as the basic visual features of a data symbol. This is effective because basic visual features are identified rapidly and without conscious effort by our visual system.

> One reason why a bar plot is an effective data visualisation is because it involves an encoding from data values to the basic visual features of data symbols—the positions and lengths of bars—and basic visual features are decoded rapidly and subconsciously by our visual system.

Figure 3.5 shows some other examples of basic visual features that are commonly used to encode data values in a data visualisation.[40] We will explore each of these in more detail in the following sections.

3.3 Quantitative features

As we discussed in Chapter 2, the real purpose of encoding data values as data symbols is to make use of the visual system to **decode** back to data values from the data symbols. In this chapter, we are considering encodings from data values to basic visual features, so we need to consider how well we are able to **decode** from **visual features** to **data values** (Figure 3.6).

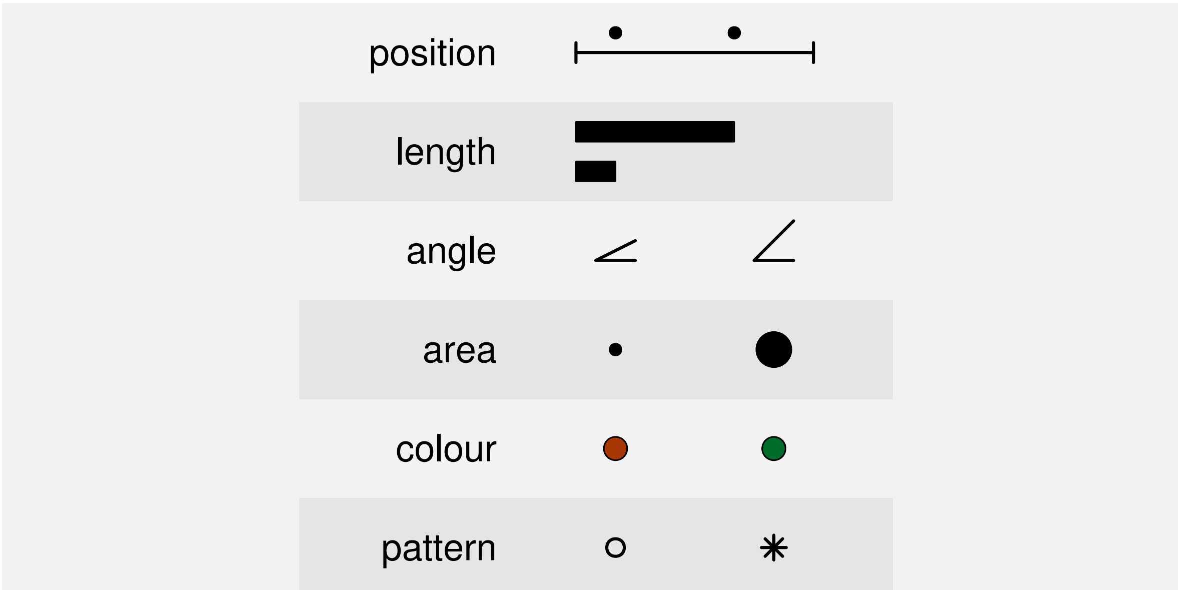


Figure 3.5: Some examples of basic visual features are **position**, **length**, **angle**, **area**, **colour**, and **pattern**.

Figure 3.6: When we encode data values as visual features, we want to encode to a visual feature that allows us to decode the data values from the visual feature.

One issue to consider is whether a visual feature is able to represent all of the information in the data values.[41] Some types of data contain more information than others.

For example, **quantitative** data consist of numeric values, which naturally possess not only an ordering, but information about the size of differences between values. For example, the year 2020 comes after the year 2010 and there is a gap of 10 years in between. This is also referred to as **interval** data. **Ratio** data is quantitative data that also contains a "zero" reference value, so there is also information about absolute size and ratios of differences. For example, 200 offenders is twice as many as 100 offenders.

If we have quantitative data, we would ideally want to encode the data to a visual feature from which we can decode numeric (interval or ratio) values.

Figure 3.7 shows that **position**, **length**, **angle**, and **area** are visual features that allow **quantitative** data values to be decoded. For example, a bar that is twice as long as another bar comfortably represents a data value that is twice as much as another data value. All but **position** also possess a natural representation of zero.

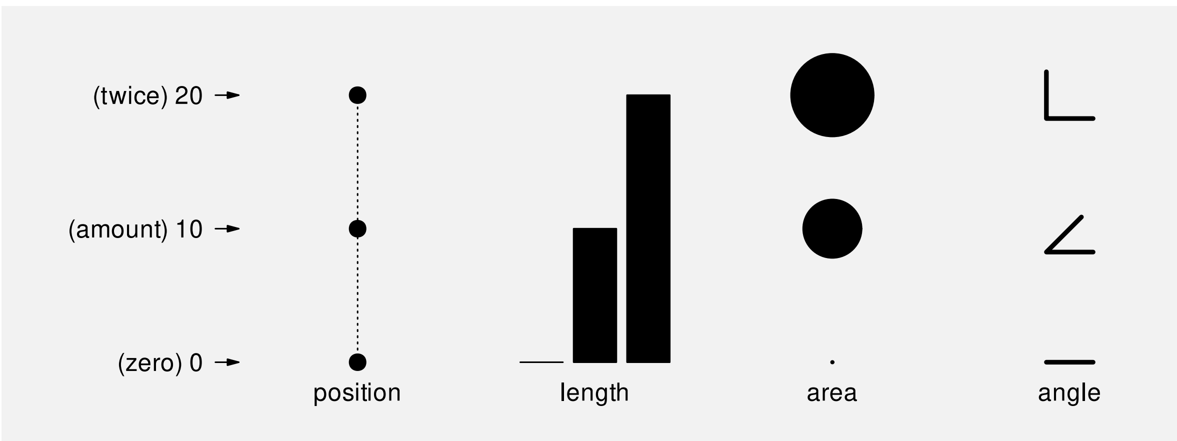


Figure 3.7: The visual features position, length, area, and angle are all appropriate visual features for visualising quantitative data values.

By contrast, Figure 3.8 shows that **colour** and **pattern** do *not* allow decoding of representing quantitative values. For example, the colour blue is not twice the colour green and a circle is not 20 more than a triangle. We can tell that there are three *different* values, but in effect, these visual representations *lose* information. It is not possible to decode the data values from the visual represenations.

If we encode quantitative values as colour or pattern, we *lose* information because we cannot decode quantitative values from those visual features.

The bar plot in Figure 3.1 visualises the total number of offenders, which are quantitative data values. These quantitative values are encoded as the **lengths** of bars, which is a visual feature

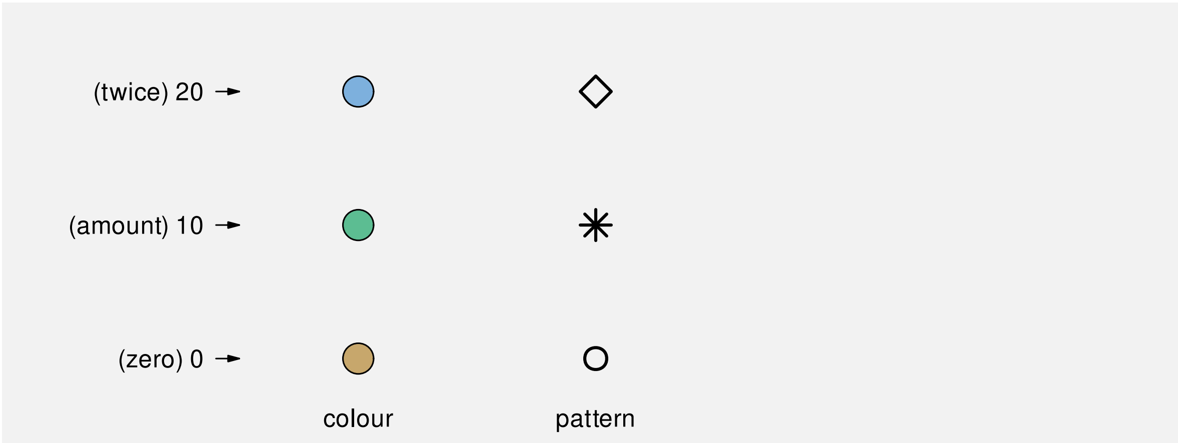


Figure 3.8: The visual features **colour** and **pattern** are not capable of visualising quantitative data values.

that allows us to decode quantitative values.

> One reason why a bar plot is effective for visualising **quantitative data values** is because the quantitative data values, the counts, are encoded to a **quantitative visual feature**, the lengths of the bars.

3.4 Qualitative features

Qualitative data consist of categorical values rather than numeric values. **Nominal** data consists of just group labels, e.g., ethnic groups, and the only information we have is that the categories are different from each other. **Ordinal** data consists of categories that are ordered, but there is no information about the size of differences. For example, a "high" crime level is more severe than a "medium" crime level, which is more severe than a "low" crime level, but we cannot measure the size of the differences.

If we have **qualitative** data values, then we are primarily concerned with being able to decode whether data values are the same or different. Figure 3.9 shows that **position**, **colour**, and **pattern** allow us to decode qualitative data values. For example, blue is clearly different from green, which is clearly different from brown. In Section 2.7 we saw that we also easily decode groups when visual items have the same visual features. For example, we can easily identify all blue items as a separate group from all green items.

By contrast, Figure 3.10 shows that **length**, **area**, and **angle** are not appropriate for decoding qualitative data values. The problem here is not a loss of information, but an *addition* of information. While we can identify whether lengths or areas are different from each other,

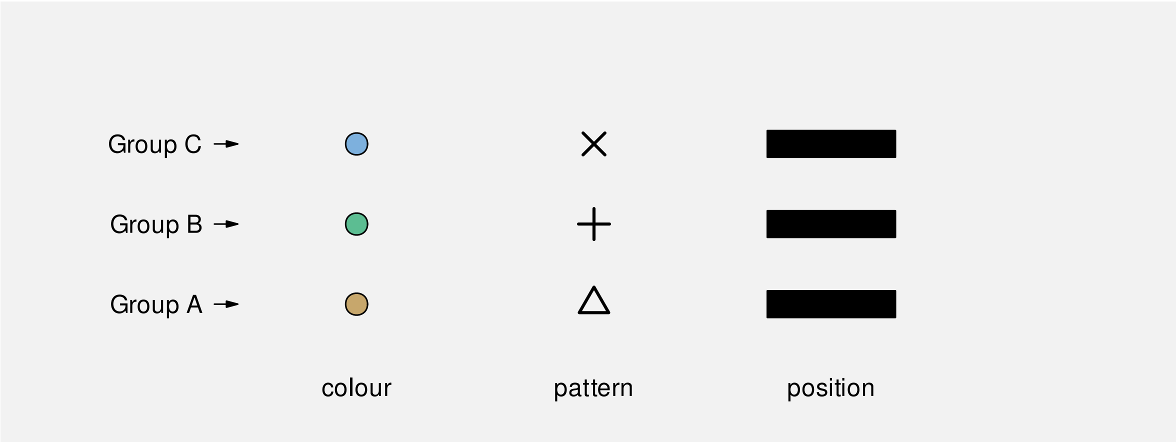


Figure 3.9: The visual features **colour**, **pattern**, and **position** are capable of representing different groups.

these visual features also allow us to decode an *order* from the visual features, and even the size or ratio of the difference between visual features. This is not an effective decoding because it *adds* information that is not present in the data values.

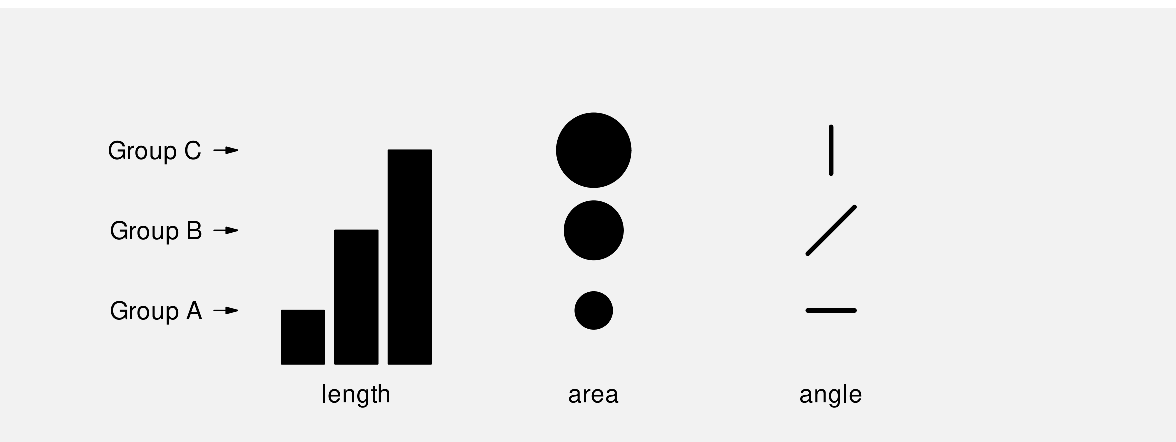


Figure 3.10: The visual features **length**, **area**, and **angle** are not appropriate for representing qualitative data values because they imply an ordering of data values that have no inherent order.

The bar plot in Figure 3.1 encodes the ethnic groups, which are qualitative (nominal) data values, as the **position** of bars, which is a qualitative visual feature.

> One reason why a bar plot is effective for visualising **qualitative data values** is because the qualitative data values, the groups, are encoded to a **qualitative visual feature**, the positions of the bars.

3.5 Accuracy

Another issue that impacts on how well we can **decode** the data values from a visual feature (Figure 3.6) is the **accuracy** with which we can perceive quantitative values from visual features. For example, how well can we perceive the size of the difference between the length of two bars?[42]

It is easy to demonstrate that judgements of this sort are harder for some visual features than others. For example, Figure 3.11 presents a set of three bars, which differ by **length**, a set of three circles, which differ by **area**, and a set of three stars, which differ by **pattern** (number of spikes). Decoding the relative sizes of the bars is easier than decoding the relative sizes of the circles and decoding numeric values from the stars is extremely difficult.

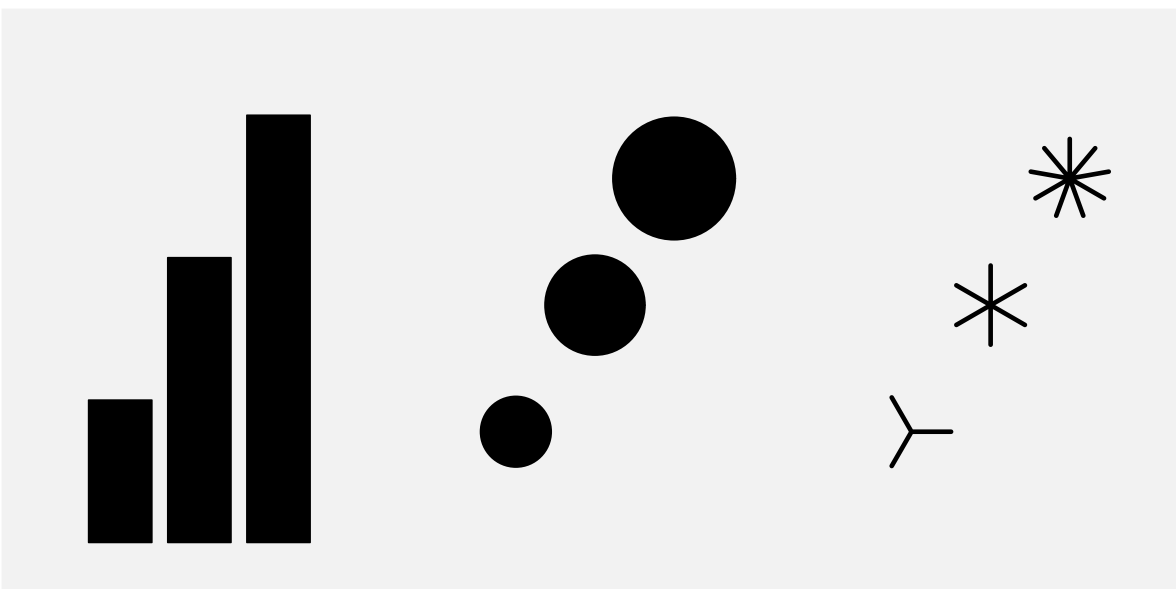


Figure 3.11: A demonstration of differences in accuracy between basic visual features. For each set of symbols, how much "larger" are the "larger" symbols?

This issue has been studied experimentally and Figure 3.12 shows the established ordering of the accuracy of basic visual features, with more accurate visual features at the top.[43] A data visualisation that encodes data values to visual features that are near the top of this ranking will be more effective than one that encodes data values to visual features that are near the bottom.

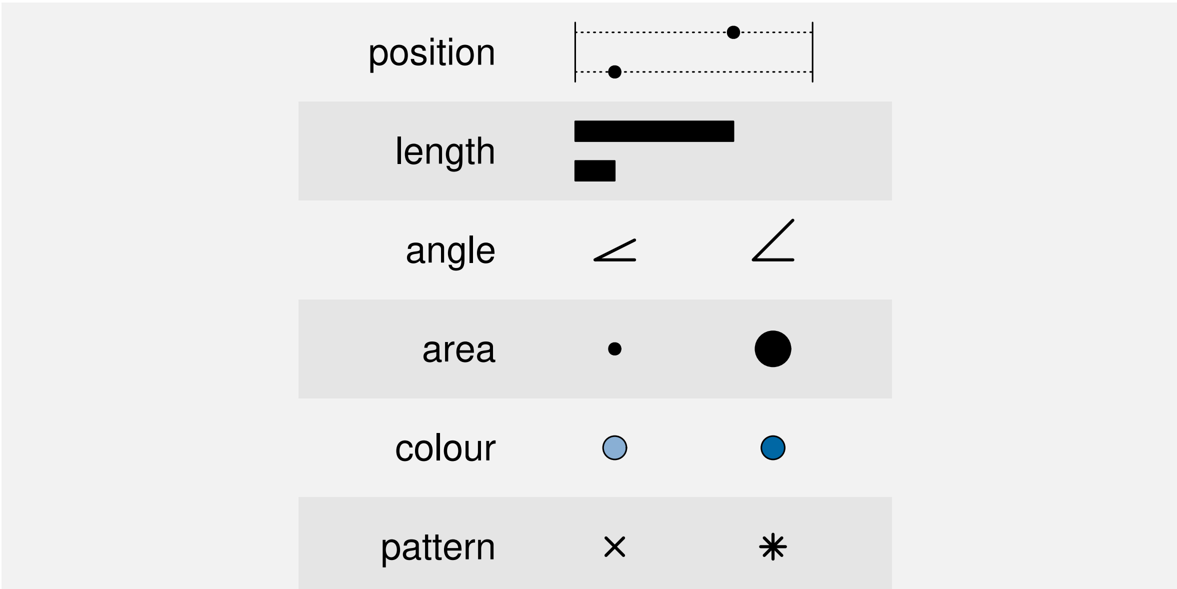


Figure 3.12: An ordering of basic visual features in terms of their accuracy, with more accurate at the top.

The bar plot in Figure 3.1 encodes the total number of offenders, which are quantitative values, as the **lengths** of bars, which is an **accurate** visual feature. This means that viewers can make accurate comparisons between the lengths of the bars.

> One reason why a bar plot is effective for visualising **quantitative data values** is because the quantitative data values, counts or proportions, are encoded as a visual feature that can be decoded very **accurately**, the lengths of the bars.

3.6 Capacity

When we are visualising **qualitative** data values, we are no longer concerned with accurately perceiving the size of differences; all we are concerned with is perceiving whether two values are different or the same. To be slightly more precise, we are concerned with being able to *identify* different values, for example, which colour, out of a set of known colours, is this one?[44]

The issue in this scenario is the **capacity** of a visual feature: how many different categories can be encoded effectively? In particular, how many different categories can be encoded with the viewer able to reliably decode each separate category?

Figure 3.13 shows that **colour** has a limited capacity. A small number of different colours are very easy to differentiate, but the differences become harder to perceive with a greater number of colours.[45] This difficulty only increases with more irregular arrangements.

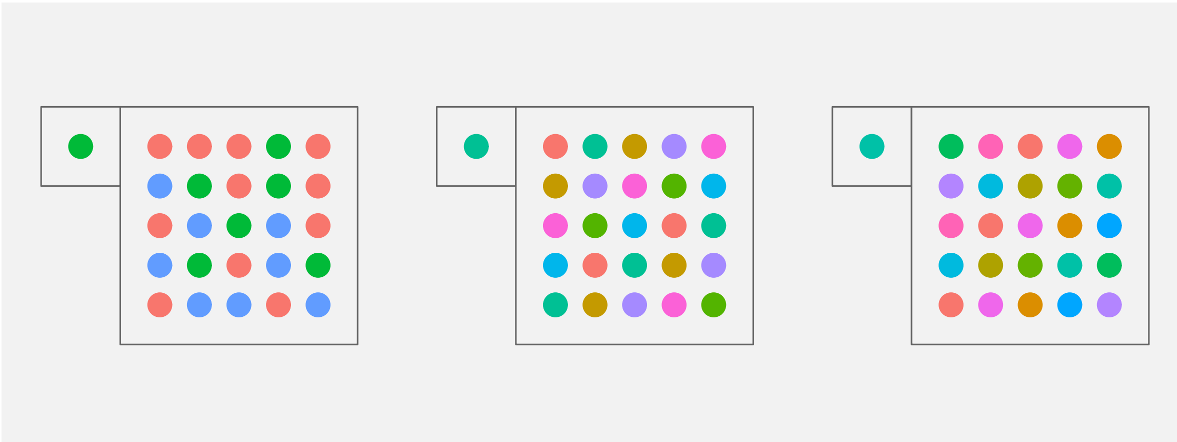


Figure 3.13: Encoding qualitative data values to colour works well for a small number of colours, but less well for a large number of colours. It is quite easy to identify which green the single dot corresponds to amongst three colours on the left, but it is much harder with the seven colours in the middle, harder still for the eleven colours on the right.

Pattern also has a very limited capacity (Figure 3.14), and again this gets worse with less regular arrangements.[46]

By contrast, Figure 3.15 shows that **position** has a greater capacity than both colour and pattern. We can easily identify the location of the single dot even as the number of alternative locations becomes large.

The bar plot in Figure 3.1 encodes the ethnic group as the vertical **position** of bars, which makes it easy to decode the different ethnic groups.

> One reason why a bar plot is effective for visualising **qualitative data values** is because the qualitative data values, the groups, are encoded as a visual feature that has excellent **capacity**, the positions of the bars, so we are able to encode, and decode, a large number of different groups.

3.7 Case study: Dot plots

We have seen that, for several reasons, Figure 3.1 is an effective way to visualise the number of offenders for different ethnic groups. Figure 3.16 shows an alternative representation of these

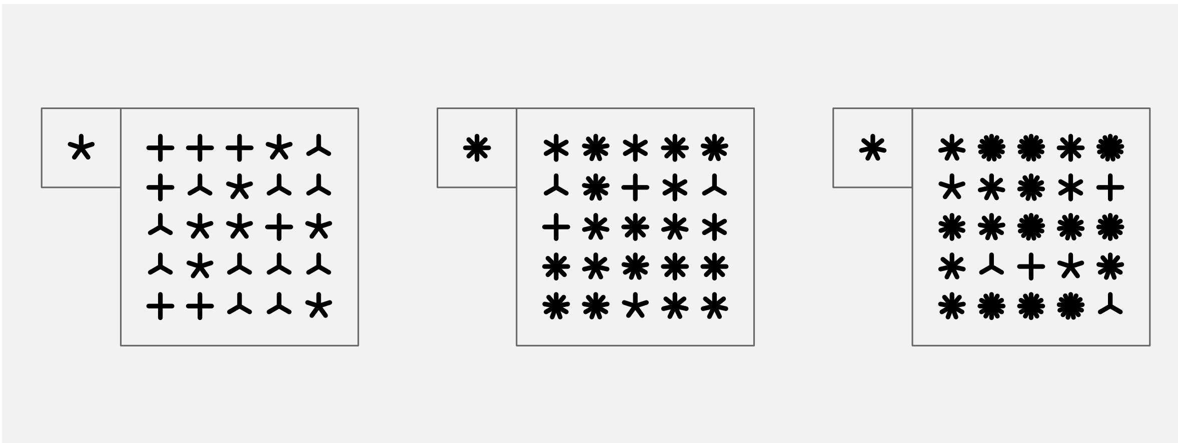


Figure 3.14: Encoding qualitative data values to pattern works well for a small number of patterns, but less well for a large number of patterns.

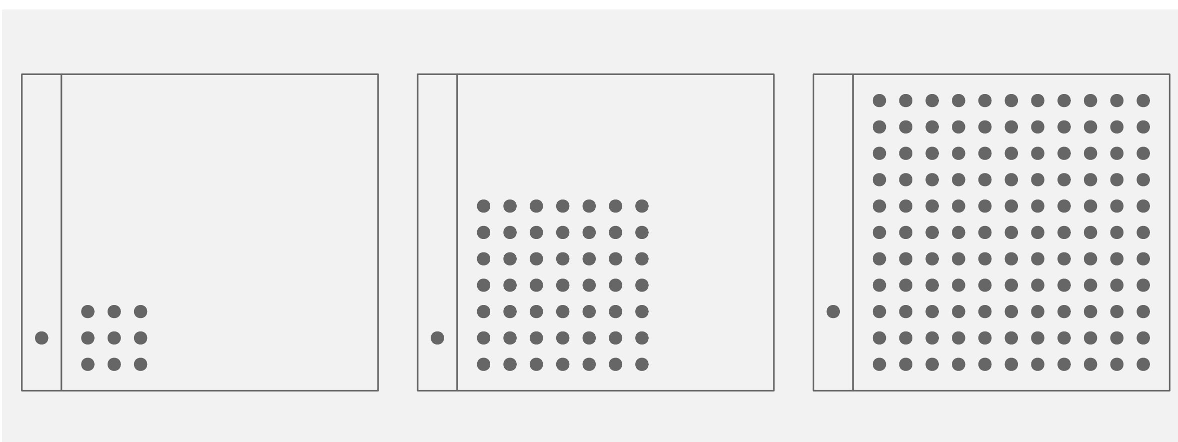


Figure 3.15: Encoding qualitative data values to position works well for both a small number of positions and for a large number of positions.

data in the form of a dot plot.[47]

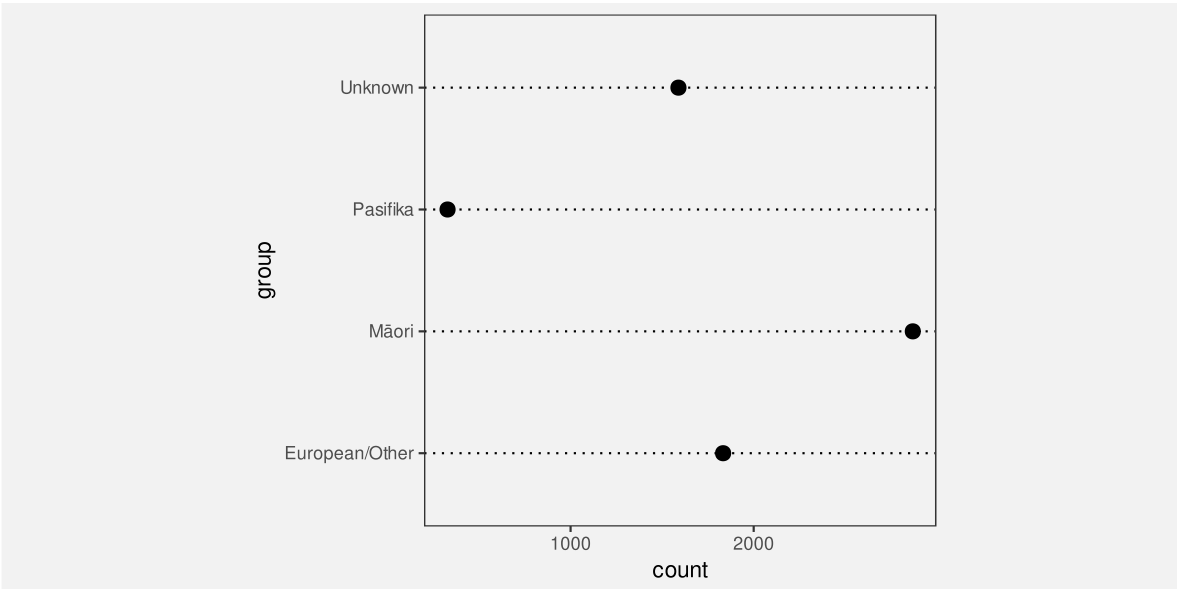


Figure 3.16: A dot plot of the number of crimes for different ethnic groups. The data symbols in this plot are the data points.

Like a bar plot, a dot plot is an effective data visualisation, and for very similar reasons. Dot plots uses a different data symbol—data points instead of bars—but **each data value** is **encoded** as the basic **visual feature** of **one data symbol**. This means that we can **decode** and compare individual data values from the data symbols and answer questions like: what is the difference in the number of offenders between Māori and European/Other?

Furthermore, the quantitative data values (the totals) are encoded as the horizontal **position** of the points, which is a visual feature that is not only capable of representing numeric values, it is also a very accurate visual feature. The qualitative data values (the ethnic groups) are encoded as the vertical **position** of the points, which is a visual feature that is capable of representing qualitive data and is a visual feature with a high capacity.

It could be argued that the dot plot in Figure 3.16 is even *more* effective than the bar plot in Figure 3.1 for decoding the differences between totals because **position** ranks higher than **length** in terms of decoding accuracy (Figure 3.12). On the other hand, if we want to compare totals in terms of their ratios, Figure 3.1 is superior because it includes a representation of zero.

3.8 Case study: Pie charts

Figure 3.17 shows yet another visualisation of the total number of offenders for different ethnic groups, this time in the form of a pie chart. This is a less effective representation of the data than the bar plot in Figure 3.1 or the dot plot in Figure 3.16 and we can use familiar reasoning to see why.

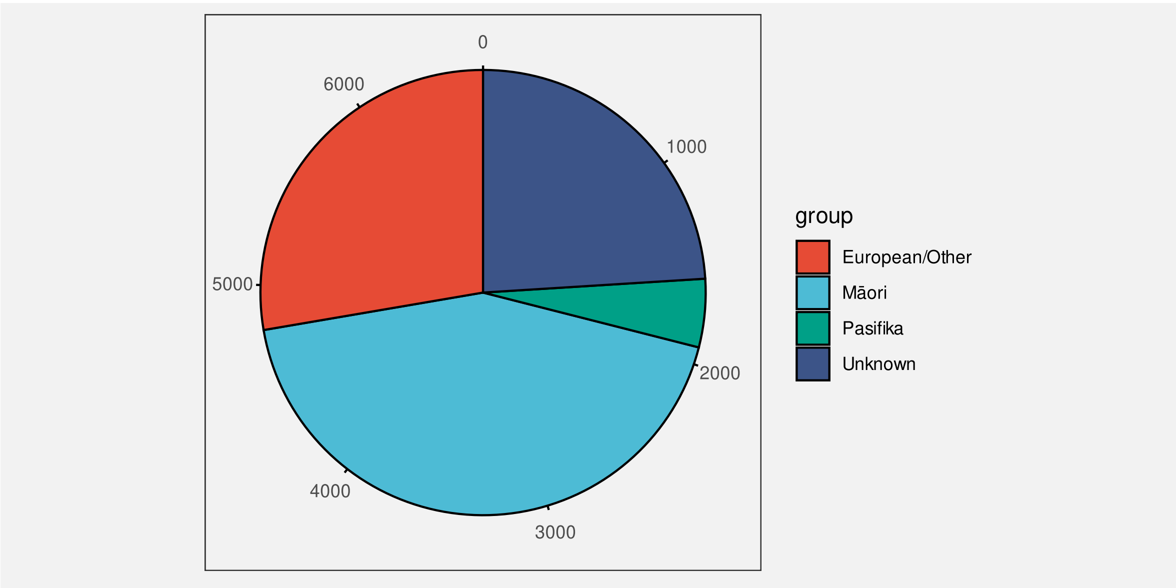


Figure 3.17: A pie chart of the number of crimes for different ethnic groups. The data symbols in this plot are the pie wedges.

The data symbols in Figure 3.17 are wedges or segments of a circle and again we have each data value encoding to a single data symbol. For example, each ethnic group is encoded as a separate wedge of the pie. As with bar plots and dot plots, in a pie chart the data values are encoded as basic visual features of these symbols. However, a pie chart makes use of different visual features: The number of offenders for each ethnic group is encoded as the **angle** and/or **area** of a wedge and each ethnic group is encoded as the **colour** of a wedge.[48] Although **colour** is an appropriate visual feature to represent **qualitative** data, and it has sufficient capacity to differentiate between only four groups, the encoding of quantitative values as **angle** and/or **area** leads to a less accurate decoding of the total number of offenders (Figure 3.12). For example, it is much harder to decode the number of offenders with an unknown ethnicity from the pie chart in Figure 3.17 compared to the bar plot in Figure 3.1 or the dot plot in Figure 3.16.

Although the situation seems bleak for pie charts, we have not yet seen all of the ways that we can assess different encodings of data values to data symbols. We will return to pie charts in Section 7.2 to show that all is not lost.

3.9 Summary

A simple encoding of data values to data symbols involves encoding **each data value to a separate data symbol**. This allows the viewer to **decode** and compare individual data values from the data symbols.

A simple encoding of data values to data symbols also involves **encoding** each data value as a basic **visual feature** of the data symbol, e.g., position, length, area, angle, colour, or pattern.

Position, length, area, and angle are appropriate for encoding quantitative data because we can decode numeric values from these visual features. We can decode position and length more accurately than area and angle.

Position, colour, and pattern are appropriate for encoding qualitative data because we can decode groups from these visual features. We can represent a large number of categories if we use position, but only a few categories if we use colours and patterns.

Notes

[38] This one-to-one encoding between data values and data symbols is called a *bijective* encoding (Ziemkiewicz and Kosara 2009).

[39] Some authors prefer to say that the total number of offenders is encoded to the **position** of the "top" of the rectangle (e.g., William S. Cleveland and McGill 1984), but the difference is not hugely significant because both length and position are excellent visual features in terms of decoding (Section 3.5).

It is also worth noting that there can be a discrepancy between the encoding that the creator of a data visualisation employs and the decoding that the viewer chooses to use (Hill and Imokhai 2025).

[40] In this book, we use *visual feature* to describe a basic visual quality of a data symbol, in place of more established terms like *visual variable* (Bertin 1983), *aesthetic* (Wilkinson 2005), or *visual channel* (Munzner 2014).

This choice of terminology is deliberate so that we can elide the basic visual quality of a data symbol with the early stages of visual processing (Section 2.2). There is not a direct correspondence between these two concepts, but it is a useful simplification.

There are also differences in terminology for individual visual features. For example, we use *angle* where others use *tilt* or *slope* and the underlying visual feature is typically referred to as *orientation*.

The use of *pattern* for what is more commonly referred to as data point *shape* is perhaps the most daring terminological divergence. Data point shape is commonly included as a visual channel, but we would like to avoid confusion with the concept of *visual shape* that we use to represent later stages of visual processing (Section 2.2). The use of *pattern* provides a connection to basic visual features via the early perception of patterns, even though there is more involved in the perception of data point shape (L. Chen 1982; J. Li, Wijk, and Martens 2009).

Much longer lists of basic visual features have been proposed (e.g., M. Chen and Floridi 2013), but we restrict ourselves in this book to a set that has some correspondence with the visual features that are extracted early on in visual processing.

[41] The idea of matching the type of data—nominal, ordinal, interval, or ratio—as well as the categorisation of visual feature types is described in Zhang (1996). Munzner (2014), following Mackinlay (1986), refers to this idea as *expressiveness*.

[42] The pioneering work on the accuracy of different visual features for statistical graphics was carried out by William S. Cleveland and McGill (1984). Their results have been replicated several times in subsequent studies, including Heer and Bostock (2010). The rankings were extended to a larger set of visual features by Mackinlay (1986), although somewhat only by inference from other psychophyscial results, rather than by direct experiment.

Munzner (2014), following Mackinlay (1986), refers to this idea as *effectiveness*.

An argument for ranking the accuracy of visual features can also be made from Steven's Law (S. S. Stevens 1957), which states that the perception of the magnitude of different stimuli follows a power law. For example, the perception of length has a power of approximately one (linear), which suggests that we accurately perceive the magnitude of lengths, while area has a power less than one, which suggests that we underestimate the magnitude of areas.

[43] In William S. Cleveland and McGill (1984), comparisons of bars were treated as **position** judgements rather than **length** judgements, unless the bars were unaligned (see Section 5.2), so this table is a slightly fuzzy representation of their results. However, the general ordering of visual features is basically consistent with William S. Cleveland and McGill (1984).

These rankings are also based on a relative judgement (what proportion of A is B?), which is just one possible decoding task, albeit a common one, particularly for bar plots. The ranking of visual features may differ for other sorts of decoding (McColeman et al. 2022). There is also some evidence that the rankings may differ between individuals even for simple relative judgements (Davis et al. 2024).

[44] The question "Which one is this?" is different from the question "Are these two different?". The former involves "absolute judgements" (Miller 1956) to identify a stimulus from a known set of stimuli. The latter involves relative judgements and just-noticeable differences (JND), which are considerably easier and something we have a much higher sensitivity to. For example, we can differentiate between several million different colours (Pointer and Attridge 1998), but we can only make absolute judgements with up to 10 different colours.

[45] The recommended maximum number of colours varies between sources. However, the consensus is that it is a quite small number.

"Do not use more than 10 colors for coding symbols if reliable identification is required, especially if the symbols are to be used against a variety of backgrounds." (Ware 2021, 126).

Healey (1996) says the effective maximum is 7.

[46] The research evidence for the limited capacity of pattern is suggestive rather than conclusive. The recommendations of experts only extend to very small sets of patterns.

For example, William S. Cleveland (1985b) recommends using the following set of patterns, in this order: `o`, `+`, `<`, `s`, and `w`.

Tremmel (1995) only provides guidance for three different patterns and suggests using separate plots to accommodate a larger number of categories (when relying on pattern).

[47] Dot plots (William S. Cleveland 1985a) are different from scatter plots because they plot one quantitative variable against (at least) one qualitative variable, rather than plotting two quantitative variables against each other.

[48] We are assuming here, as is common, that the viewer will decode from the **angle** of pie wedges. That is not necessarily the case, for example Eells (1926b) found evidence that viewers may decode from the area, arc length, or even chord length of pie wedges. Fortunately, the larger point holds, because most of these alternatives are relatively poor visual features in terms of decoding accuracy.

4 Position

We saw in Chapter 3 that **position** is a very effective visual feature. It is appropriate for encoding quantitative data values (Figure 3.7) and it is appropriate for encoding qualitative data values (Figure 3.9). **Position** is a visual feature with high accuracy (Figure 3.12) and with high capacity (Figure 3.15).[49]

This means that, if we do not encode data values as the **position** of data symbols, it can be more difficult to decode data values from the data symbols.[50] For example, Figure 4.1 shows another data visualisation of the total number of offenders in New Zealand for different ethnic groups (Table 3.1). We have previously seen a bar plot (Figure 3.1), a dot plot (Figure 3.16), and a pie chart (Figure 3.17) of these data. In Figure 4.1, the data symbols are circles, the number of offenders is encoded as the **area** of the circles, and the ethnic groups are encoded as the **colour** of the circles. One reason why it is more difficult to decode data values from Figure 4.1, compared to, for example, Figure 3.1, is because none of the data values are encoded using **position**.

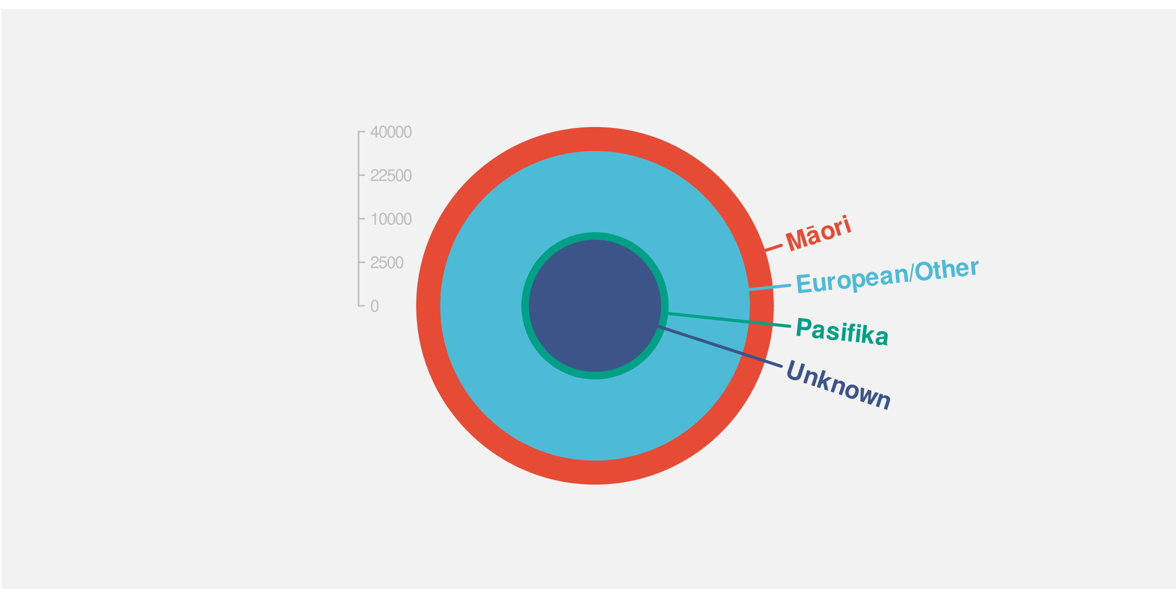


Figure 4.1: A data visualisation that does not make use of position.

However, we have already seen several data visualisations that do not encode *all* data values to position (e.g., Figure 1.1 and Figure 1.2), so there must be more to learn about encoding data values to basic visual features.[51] In this chapter, we look more closely at the encoding of data values to the **position** of data symbols and how well we can decode a data value from the position of a data symbol.

4.1 Perception is relative

In Section 3.5, we saw that **position** was the most accurate visual feature for decoding quantitative values (Figure 3.12). However, that accuracy is for *relative* judgements rather than absolute judgements. In other words, we are good at *comparisons* between positions,[52] which is one of the most important decoding tasks in data visualisation.[53]

For example, the box on the left of Figure 4.2 shows a dot at a vertical positon along a dotted line. We can decode an amount from the position of the dot, but only relative to the ends of the dotted line; the dot is three-quarters of the way from the bottom to the top of the line. We cannot decode an absolute data value from the dot.[54]

The box in the middle of Figure 4.2 shows two dots at vertical positions along two dotted lines. Again, we can only compare the positions of the two dots—the dot on the left is three times as far up its dotted line as the dot on the right.

The box on the right of Figure 4.2 shows two dots again, but this time adds an axis on the right edge of the box. It is now possible to decode an absolute value for each dot (75 and 25), but that is still only possible by making comparisons. We decode the positions of the dots relative to the positions of the tick marks on the axis.

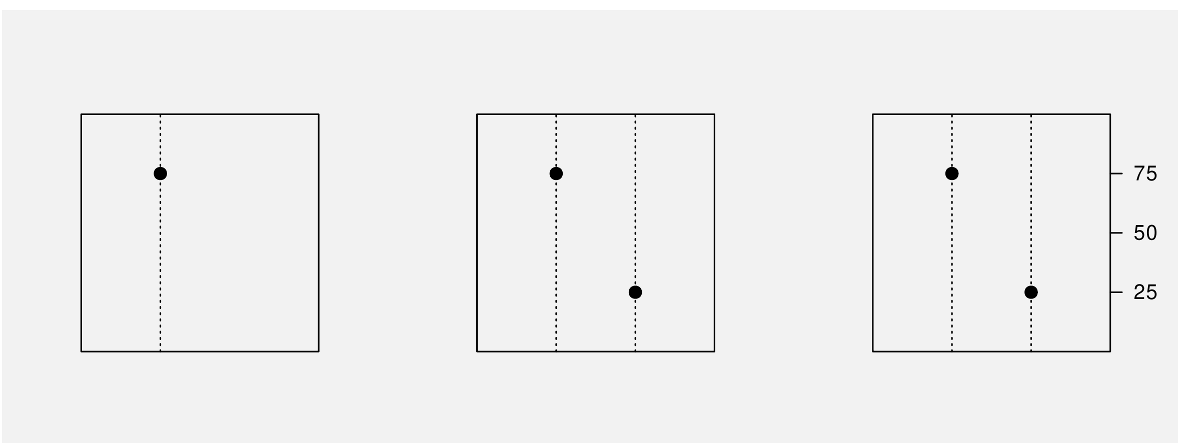


Figure 4.2: Perception of position is relative rather than absolute.

> A data visualisation that encodes a data value as the position of a data symbol is effective because we can decode *relative* positions accurately.

4.2 Unaligned position

We have established that **position** is a very accurate visual feature (Figure 3.12), although that accuracy is for comparisons rather than for decoding individual values. Another limitation on the accuracy of **position** is that it works best when the comparison is between positions with a common baseline. For example, it is easier to compare the two positions, relative to the ends of the dotted lines, on the left of Figure 4.3 than it is to compare the two positions on the right.

Figure 4.3: It is easier to compare two positions on the left because they are **aligned**; they share a common baseline. The two positions on the right are harder to compare because they are **unaligned**.

Figure 4.4 shows yet another data visualisation of the total number of offenders in New Zealand for different ethnic groups (Table 3.1). This data visualisation is similar to Figure 3.16 because it encodes the number of offenders to **position** along a line. However, it is harder to compare the number of offenders between different ethnic groups in Figure 4.4 because the positions are not aligned.

> Decoding data values from positions is most effective when the positions share a common baseline.

4.3 Position is a scarce resource

In Section 3.6 we saw that **position** has a higher capacity for decoding qualitative data values compared to colour or pattern (Figure 3.15). However, position has a limitation that does not affect colour or pattern: if we use the same position for more than one data value, we end up drawing data symbols on top of each other.

For example, in Figure 4.1, we use the same position for all four circles that represent different ethnic groups, so the circles are drawn on top of each other. The total number of youth offenders for each ethnic group is encoded as the **area** of a circle. The largest number of

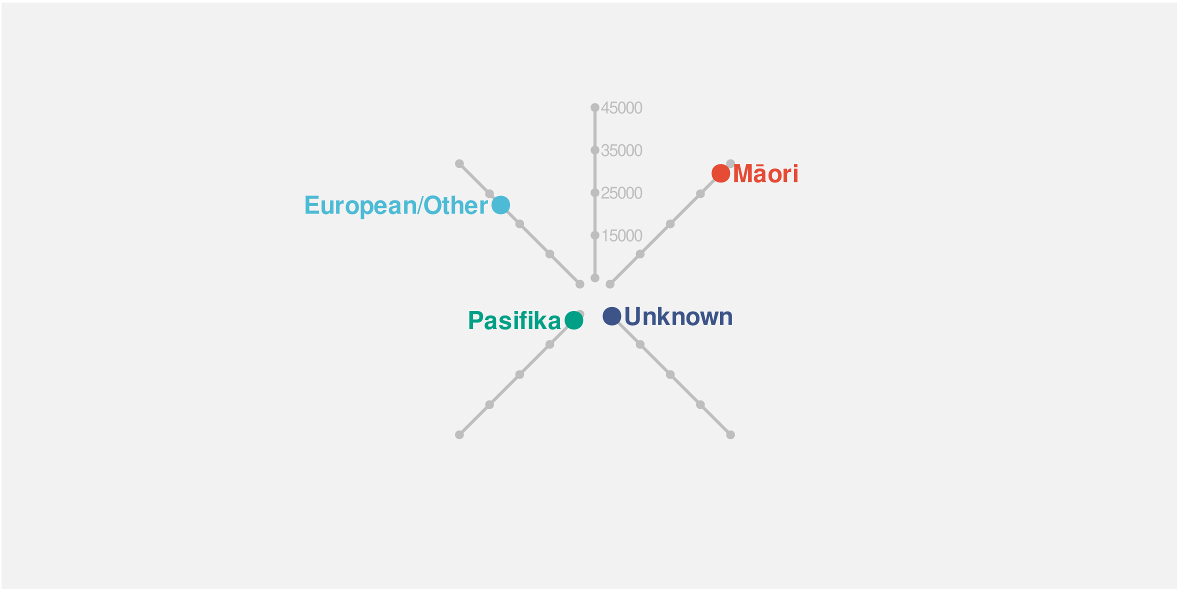


Figure 4.4: A data visualisation that makes use of *unaligned* position.

offenders is for the Māori ethnic group (the orange circle) with European/Other the next largest (the light blue circle). However, because the light blue circle is drawn on top of the orange circle, all that we can actually see is the part of the orange circle that extends beyond the light blue circle.

This represents a serious problem. It is difficult for our visual system to decode from a data symbol to a data value if we cannot see all of the data symbol.[55]

One way to express this limitation is that position is very effective for encoding *different* data values, but it is not very effective for encoding the *same* data value.

> Decoding more than one data value from two data symbols that share the same position is, at best, compromised because we cannot see all of both data symbols.

4.4 Case study: Overplotting

The Rugby World Cup is a competition that is held every four years, with the first event taking place in 1987. We have data on the the number of points that were scored in every World Cup match by Tier One nations (the top 10 rugby-playing nations). Table 4.1 shows a sample of these data.[56]

Table 4.1: A table of the number of points scored by Tier One nations in Rugby World Cup matches, along with the global hemisphere of origin. Each row represents the points scored by one team in one match. There are 294 rows in total, with just the first 6 rows shown here.

team	hemisphere	scored
Argentina	South	8
Argentina	South	18
Argentina	South	13
Argentina	South	28
Argentina	South	10
Argentina	South	7

Figure 4.5 shows a dot plot of the points scored in a match for all teams and all matches. This data visualisation has encoded the number of points scored as the horizontal **position** of data points. In theory, that allows us to decode from data points to a number of points scored. For example, we can see that one team from the southern hemisphere scored just over 100 points in one match.

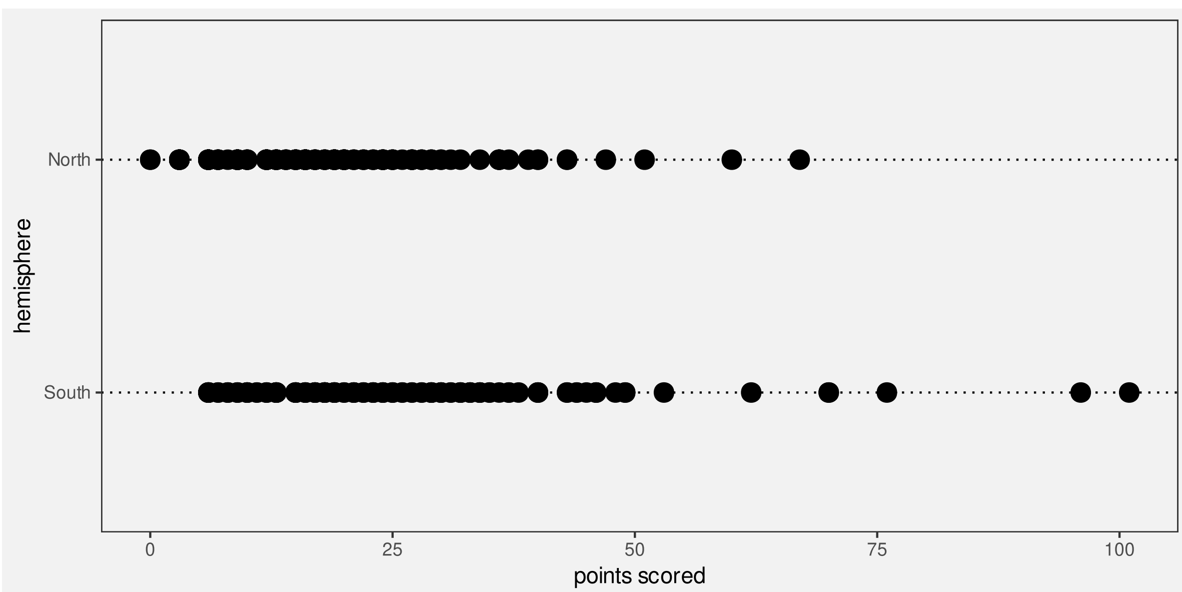


Figure 4.5: The number of points scored in Rugby World Cup games by teams from the northern and southern hemispheres.

However, as Table 4.2 shows, exactly the same number of points were scored by more than one team (from the same hemisphere) and/or in more than one match. This means that there are multiple data points at exactly the same position, which means that there are multiple data points drawn on top of each other, which in effect means that there are data points that we cannot see. For example, we can see that there is a dot at zero points in Figure 4.5 (for teams from the northern hemisphere), but we cannot see that there are actually two dots at that location

Table 4.2: A table of the number of points scored by Tier One nations in Rugby World Cup matches, along with the global hemisphere of origin. Each row represents the points scored by one team in one match. There are 294 rows in total, with just the first 6 rows shown here, ordered by the number of points scored.

team	hemisphere	scored
England	North	0
Scotland	North	0
England	North	3
Ireland	North	3
Ireland	North	3
Italy	North	3

Although it is still possible to decode that, for example, at least one team from the northern hemisphere scored zero points in at least one match, it is not possible to decode whether that was just one team or just one match. We cannot decode an individual data value from each data symbol.[57]

Figure 4.6 shows the same data as Figure 4.5, but with the data points semi-transparent so that we can see that many of the points overlap. The same number of points was scored by multiple teams in multiple matches. This demonstrates one approach to resolving the problem of overplotting, although it is only partially effective here; We can more easily decode points that have been scored many times, but we cannot easily decode the number of times the same set of points was scored. We will see several other approaches to resolving overplotting in later chapters.

4.5 Cartesian coordinates

In Figure 4.5, we have encoded *two* sets of data values using **position**. The number of points is encoded as the horizontal position of the data points and the hemisphere of origin for each team is encoded as the vertical position of the data points. This *cartesian coordinate system* is effective because the visual system is better at decoding horizontal and vertical positions compared to oblique orientations like in Figure 4.4.[58]

Because the horizontal and vertical positions are perpendicular, Cartesian coordinates also mean that we can make use of **position** twice without the position encodings interfering with

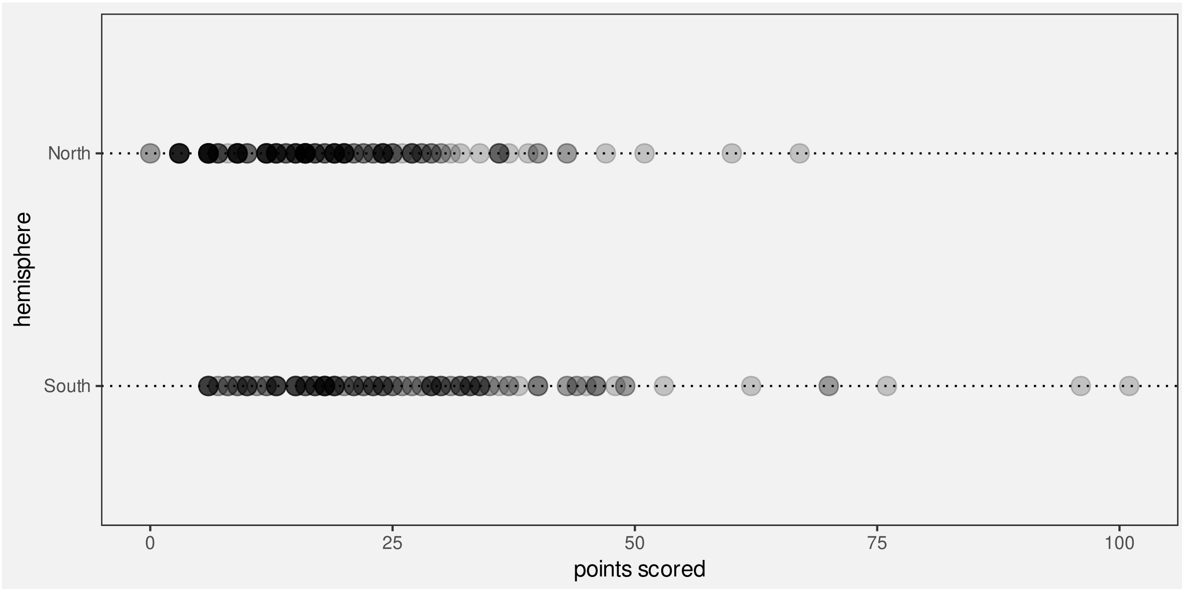


Figure 4.6: The number of points scored in Rugby World Cup games by teams from the northern and southern hemispheres.

each other.[59] For example, in Figure 4.5, we can decode which hemisphere that a data symbol is from separately from decoding how many points were scored. The fact that a team is from the northern hemisphere does not *by itself* imply anything about the number of points scored. Put another way, any difference between the data symbols for northern hemisphere teams and the data symbols for southern hemisphere teams must reflect something about the data.

A good example of a data visualisation that does not employ Cartesian coordinates is a pie chart, like Figure 3.17, though that is also a case where data values are *not* encoded as **position**.

> Cartesian (perpendicular) coordinates are effective because we can encode data values using position *twice*; we can decode data values from horizontal positions and we can separately decode data values from vertical positions.

4.6 Case study: Ternary plots

Table 4.3 shows data on the severity of crimes committed by youth in New Zealand. For each year, we have the proportion of criminal offences that were low, medium, or high severity.[60]

Table 4.3: A table of the proportion of crimes at different levels of severity for each year.

year	Low	Medium	High
2011	0.306	0.624	0.070
2012	0.310	0.616	0.074
2013	0.296	0.631	0.074
2014	0.293	0.638	0.069
2015	0.254	0.656	0.090
2016	0.246	0.655	0.099

Figure 4.7 shows a ternary plot of these data.[61] In this data visualisation, there are three variables (three proportions that sum to 1), and the data values from each variable has been encoded as position. However, the positions are along axes that are not perpendicular. This means that it is not as easy to decode individual data values or compare data values because we are poorer at decoding positions that are not horizontal or vertical. Furthermore, because the three proportions sum to 1, a change in one proportion necessitates a change in at least one of the others. In other words, we cannot easily decode the difference between two data values from the difference in the positions of two data symbols.

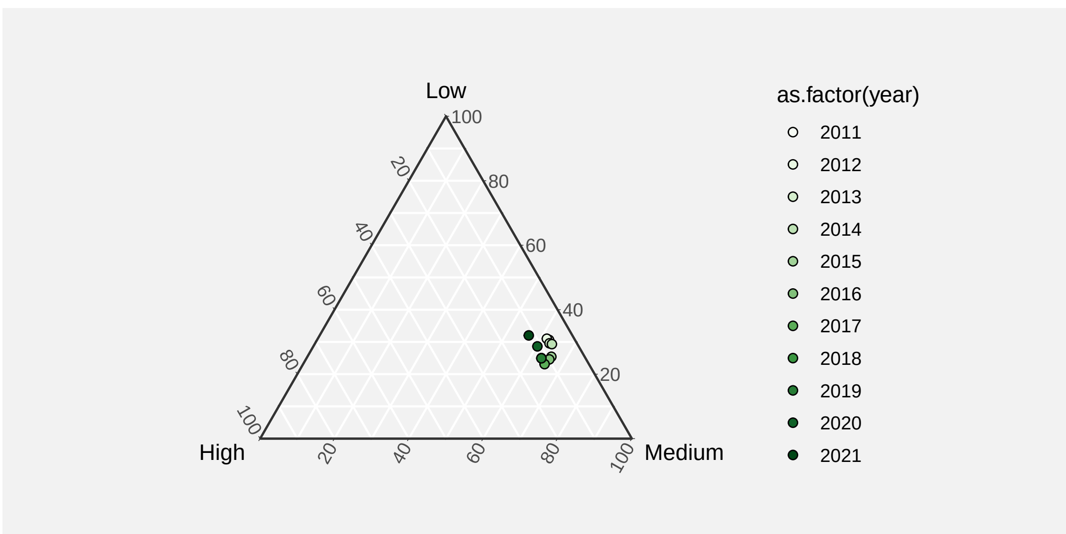


Figure 4.7: A ternary plot of the data in Table 4.3. We can see that for the entire decade around 60% of crimes have been of medium severity, with around 30% of low severity, and only 10% of high severity.

4.7 Non-linear encodings

Table 4.4 shows the total number of youth offenders broken down by type of crime. We have one qualitative variable, crime type, and one quantitative variable, total number of offenders.

Table 4.4: A table of the number of youth offenders in New Zealand in 2021 broken down by type of crime. There are 16 different crime types in total, with only the first 6 shown here.

	type	count
161	Homicides	4
162	Causing injury	1018
163	Sexual offences	190
164	Dangerous acts	653
165	Abductions, threats	273
166	Robbery, extortion	246

Figure 4.8 shows of dot plot of these data, with the total number of offenders encoded as the horizontal position of circles, one per type of crime. This plot is effective for comparing the total number of offenders between types of crime, both in terms of which crimes are more common than others and how much more common some crimes are than others. For example, we can decode that theft is about twice as common as the next most common crime type.

The effectiveness of Figure 4.8 comes not only from the fact that data values are encoded as the **position** of data symbols, but also from the fact that the encoding is *linear*. In Figure 4.8, the horizontal difference between 0 offenders and 5,000 offenders is exactly the same horizontal distance as the difference between 5,000 offenders and 10,000 offenders. The effective decoding of position is dependent on this sort of linear encoding.[62]

When a small number of data values are considerably larger than the majority of data values, as in Figure 4.8, a non-linear scale is sometimes employed in order to see the differences between the smaller values more easily. For example, Figure 4.9 shows the same data and the same encodings, but with a *log scale*. Rather than encoding the number of offenders as the position of the circles, the log (base 10) of the number of offenders is encoded as the position of the circles.[63]

Unfortunately, as a result of the non-linear horizontal scale, we lose the accuracy of the position encoding. We can no longer accurately decode from position to data values because the same horizontal distance means different things in terms of data values at different locations along the x-axis. For example, it is difficult to decode the number of offenders for Miscellaneous crimes; the position of the circle data symbol is visually roughly one quarter of the distance between 10 and 100, but the corresponding data value (18) is *not* one quarter of the distance between 10 and 100. It is even more difficult to decode the size of the difference between

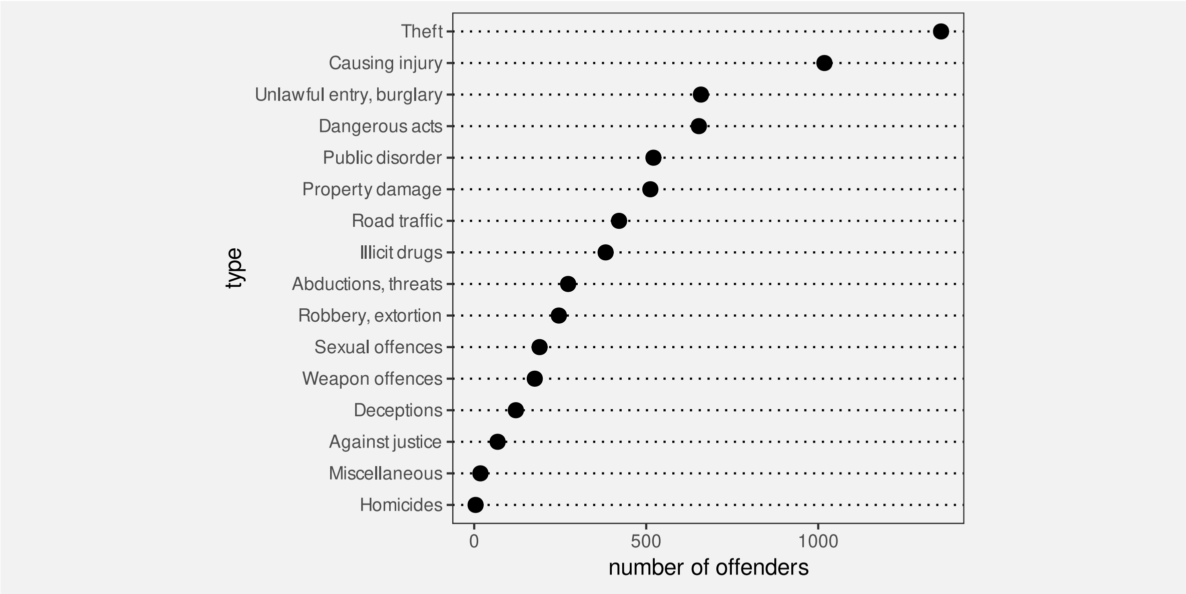


Figure 4.8: A dot plot of the number of offenders in New Zealand in 2021 for different types of crime.

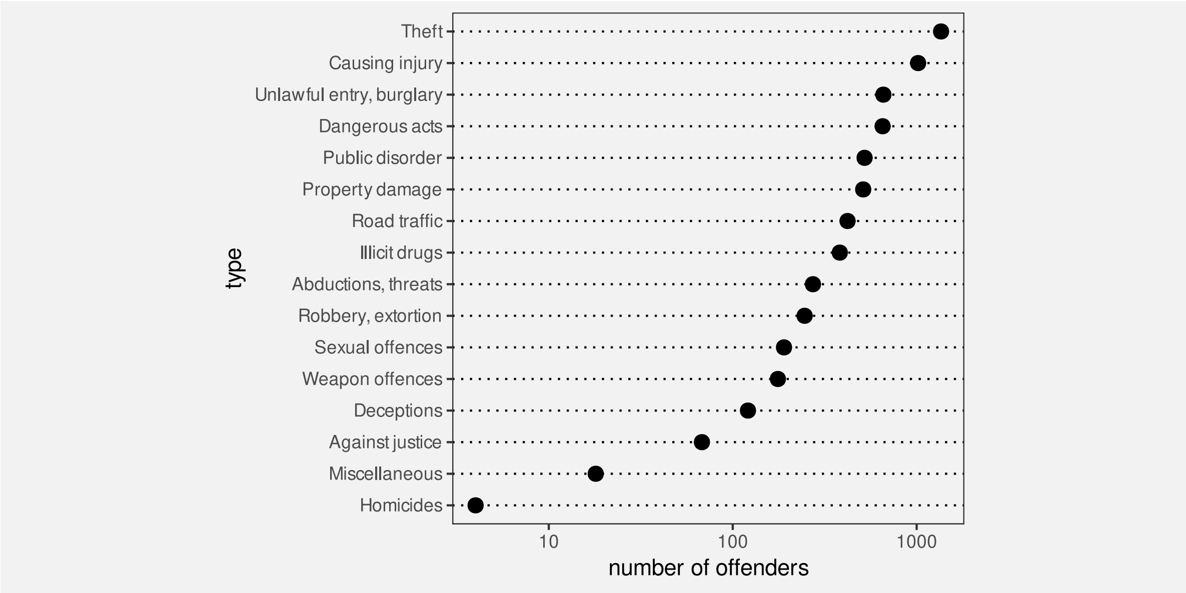


Figure 4.9: A dot plot of the number of crimes for different regions of New Zealand, with a non-linear (log) scale on the x-axis.

the number of offenders for Miscellaneous crimes versus Homicides. Worse still, the difference between Miscellaneous and Homicide is visually similar to the difference between Miscellaneous and crimes Against justice, but the real differences between those data values are very different (14 versus 50).

We can still decode simple ordinal information from Figure 4.9—which type of crimes are more common than others—but we have lost the ability to decode quantitative information from the positions of the data symbols. We have essentially lost information.

Figure 4.10 shows the logged data again, but this time with labels on the x-axis that show logged values rather than the original data values. This shows that, in terms of logged data values, the positions of the data symbols are linear. We can accurately decode from the positions of the circles in Figure 4.10, but only to recover logged data values. If we want to compare logged data values, this is fine, but usually what we want is to compare the original data values—we need to decode to a logged value and then raise 10 to that power. For example, the log of the number of offenders for Causing injury is approximately 3, so the number of offenders is approximately 1000. However, this mental transformation requires much more conscious effort for most people, so we effectively lose the benefit of using a data visualisation, at least for the purpose of decoding quantitative information (Section 2.9).

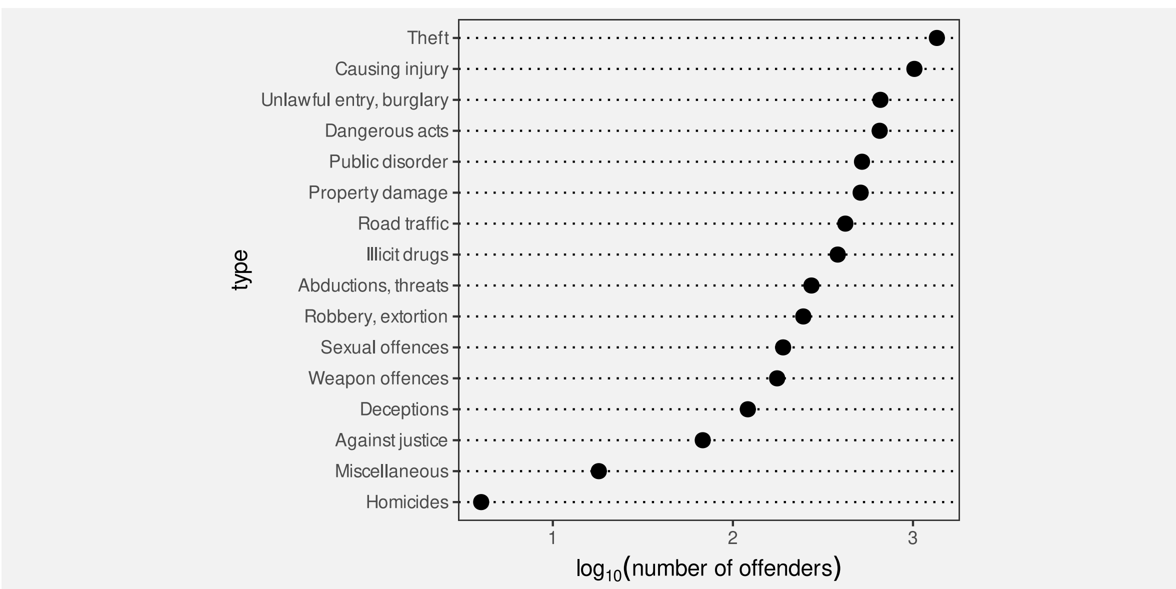


Figure 4.10: A dot plot of the number of crimes for different regions of New Zealand, with a non-linear (log) scale on the x-axis.

Although a non-linear encoding from data values to the positions of data symbols presents problems for decoding quantitative information, there are still situations where non-linear scales can be useful. For example, if we are only interested in ordinal comparisons, then non-linear positions may still be effective. We will see more examples in later sections (Section 6.8, Section 10.9).

> The encoding of quantitative data as the position of data symbols is only effective for decoding quantitative data if the encoding is linear.

4.8 Summary

Encoding data values as the position of data symbols allows for the effective decoding of both quantitative and qualitative information, with the following caveats:

- For quantitative values, what we can accurately decode are *comparisons* between quantitative values, not absolute quantitative values.
- The decoding is most accurate for positions that share a common baseline.
- Encoding identical data values as the positions of data symbols means that the data symbols overlap, which compromises our ability to decode data values from the data symbols.
- We can encode one set of data values as horizontal positions and another set of data values as vertical positions because we can decode horizontal and vertical positions separately.
- Decoding quantitative data values from the positions of data symbols is only accurate if the encoding is linear.

Notes

[49] It can be argued that position is a visual feature that is fundamental to our perception of any image.

Bertin (1967) separates the *planar dimensions* (horizontal and vertical position) from the *retinal variables* (colour, area, angle, etc).

"There is fairly widespread philosophical agreement, which certainly accords with commonsense, that the spatial aspects of all existence are fundamental. Before an awareness of time, there is an awareness of relations in space." (Robinson and Petchenik 1976, 14)

"The concept of spatial relatedness … is a quality without which it is difficult or impossible for the human mind to apprehend anything." (Robinson and Petchenik 1976, 123)

"According to Kubovy (1981), position in space and time have a dominant role in perceptual organization. In relation to visual information displays, the contention for space as an indispensable variable seems indisputable.

For centuries, humans in all cultures have "mapped" the nonvisible to space in order to facilitate understanding." (MacEachren 1995; citing Kubovy 1981)

"We process spatial properties (position and size) separately from object properties (such as shape, color, texture, etc). Furthermore, position in space and time has a dominant role in perceptual organization, as well as in memory." (Meirelles 2013)

[50] Decoding individual data values from individual data symbols is not the only visual task that we need to perform, so it is not *always* true that we should encode data values as **position**. As we explore other sorts of encodings in later chapters, we will see reasons why **position** is not always best.

[51] As Bertini, Correll, and Franconeri (2020) put it: "Why Shouldn't All Charts Be Scatter Plots?"

[52] In the experiments described in William S. Cleveland and McGill (1984), which are the basis of the accuracy ranking in Figure 3.12, "subjects were asked to judge what percent the smaller is of the larger".

[53] Amar, Eagan, and Stasko (2005), in their taxonomy of visual tasks, consider a "low-level comparison as being a fundamental cognitive action" for reading a data visualisation.

"The fundamental task in data analysis is to make smart comparisons - we're always trying to answer the question 'Compared with what?' […] It always comes down to making and showing smart comparisons." (Tufte 2006, 127)

"Almost everything we do with data involves comparison" (Tukey 1990, 329)

"The lesson is that visualization is not good for representing precise absolute numerical values, but rather for displaying patterns of differences or changes over time, to which the eye and brain are extremely sensitive." (Ware 2021, 70)

[54] Even if we have encoded a *proportion* data value to the position of the dot, we would need to know that the bottom of the line represents 0 and the top of the line represents 1 to be able to correctly decode the proportion from the position of the dot.

[55] Wilke (2019) refers to this problem as the *principle of proportional ink*. The amount of ink used to represent a data value should be proportional to the magnitude of the data value.

Another way to decode Figure 4.1 is that there is a thin orange disc outside a light blue disk (which is outside a thin green disk, which is outside a dark blue circle). In this interpretation, we can at least see all of the data symbol, so decoding is possible, but the problem now is that the viewer is using a decoding that is different from the encoding.

The concentric-circles interpretation is perhaps the most parsimonious (Section 2.8), but just the existence of an alternative interpretation is a potential source of confusion and so a cause for concern.

[56] These data were obtained from kaggle (Begbie 2022).

[57] In effect, more than one data value is encoded as the same data symbol, which means that we cannot decode back to individual data values. This is called a *surjective* encoding (Ziemkiewicz and Kosara 2009).

[58] This preference for horizontal and vertical orientations is known as the *oblique effect* (Appelle 1972; B. Li, Peterson, and Freeman 2003)

[59] Put mathematically, horizontal and vertical positions are *orthogonal*, which means that they are *independent*.

[60] This form of data is called *compositional* data; it shows how a total is composed of its different parts.

[61] Ternary plots, also known as barycentric plots, are common in specific areas of science, for example, Earth science where the composition of rocks or soils is studied. (Howarth 1996) provides a history of the development of ternary diagrams.

[62] "The representation of numbers should be directly proportional to the quantities represented." (Tufte 1983)

[63] The log (base 10) of the data values is the power to which we must raise 10 in order to get the data value. For example, if the data value is 100, the logged data value is 2.

$10^2 = 100 \iff log_{10}(100) = 2$

5 Length

Figure 5.1 shows a bar plot of the number of offenders in each police district in New Zealand in 2021 (for ages 14 to 16).

For the reasons outlined in Chapter 3 and Chapter 4), this data visualisation is effective for decoding the number of offenders in different police districts and making comparisons between them.

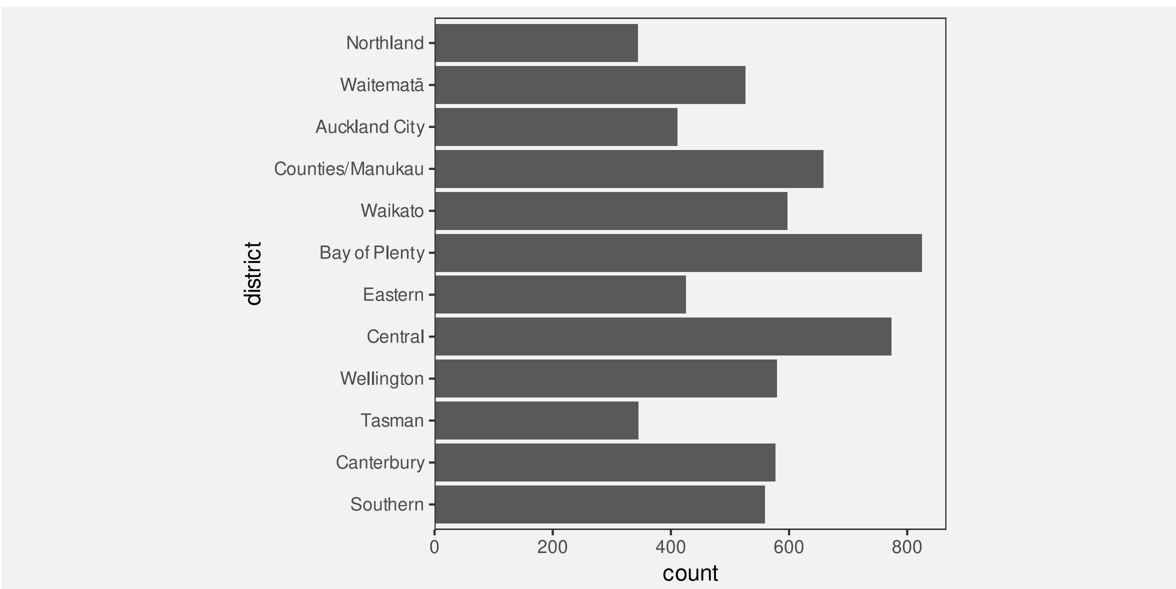


Figure 5.1: A bar plot of the number of offenders for different police districts, with the districts ordered from North to South.

Figure 5.2 shows a bar plot of the same data, but with the bars ordered from longest to shortest. This data visualisation uses the same encodings as Figure 5.1—police districts are encoded as the vertical positions of the bars and the counts are encoded as the lengths of the bars—which means that it is also effective for comparing the number of offenders between police districts.

However, Figure 5.2 is *more* effective than Figure 5.1 for some comparisons. For example, it is easier to compare the Northland district with the Tasman district in Figure 5.2. The fact that two data visualisations with the same encodings can have a different effectiveness suggests that there is more to know about how different visual features work.

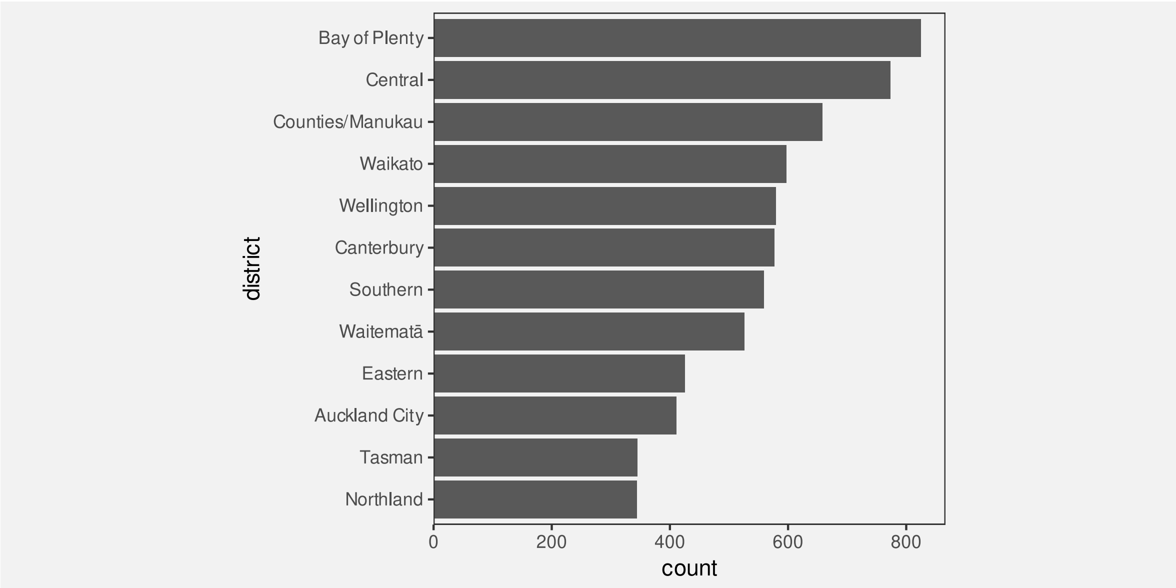


Figure 5.2: A bar plot of the number of offenders for different police districts, with the districts ordered from largest number of offenders to smallest.

This chapter covers some more details about how well we can decode information when we have encoded data values as **length**.

5.1 The distance effect

In Section 3.5, we saw that **position** and **length** were the most accurate visual features (Figure 3.12). However, this accuracy decreases if the two visual features that are being compared are not in close proximity.[64] For example, it is more difficult to compare the lengths of the two black bars in the set of bars at the bottom of Figure 5.3 than it is to compare the lengths of the two black bars at the top of Figure 5.3.

This helps to explain why Figure 5.2 is more effective than Figure 5.1 for some comparisons (e.g., Northland versus Tasman).

> One reason for ordering the bars in a bar plot from longest to shortest is because the ordering places bars that we want to compare closer together.

Figure 5.3: Two bars are easier to compare if they are close to each other (top row). If there is a distance between the bars and/or there are distractors (other bars) in between then comparisons are less accurate (bottom row). On both the top and bottom row, the black bar on the left is slightly higher than the black bar on the right.

5.2 Unaligned length

Figure 5.4 shows that it is harder to accurately compare the **lengths** of two bars if they are **unaligned** and do not share a common baseline (similar to the accuracy of comparing unaligned **positions**; Section 4.2).

Figure 5.5 shows an alternative visualisation of the data on the number of offenders in different ethnic groups. The bar plot of these data that we saw in Figure 3.1 encoded ethnic groups as the **positions** of the bars so that the bars were arranged vertically with a common left-hand edge. In Figure 5.5, the ethnic groups are encoded as different **colours** and the bars are "stacked" horizontally.[65]

Although the number of offenders has been encoded as the same visual feature in both Figure 5.5 and Figure 3.1—in both cases, the number of offenders is encoded as the **lengths** of the bars—comparing those lengths is much harder in Figure 5.5 because the lengths are **unaligned**.

> One reason why stacked bar plots are less effective for some comparisons is because we are forced to compare the unaligned lengths of the bars.

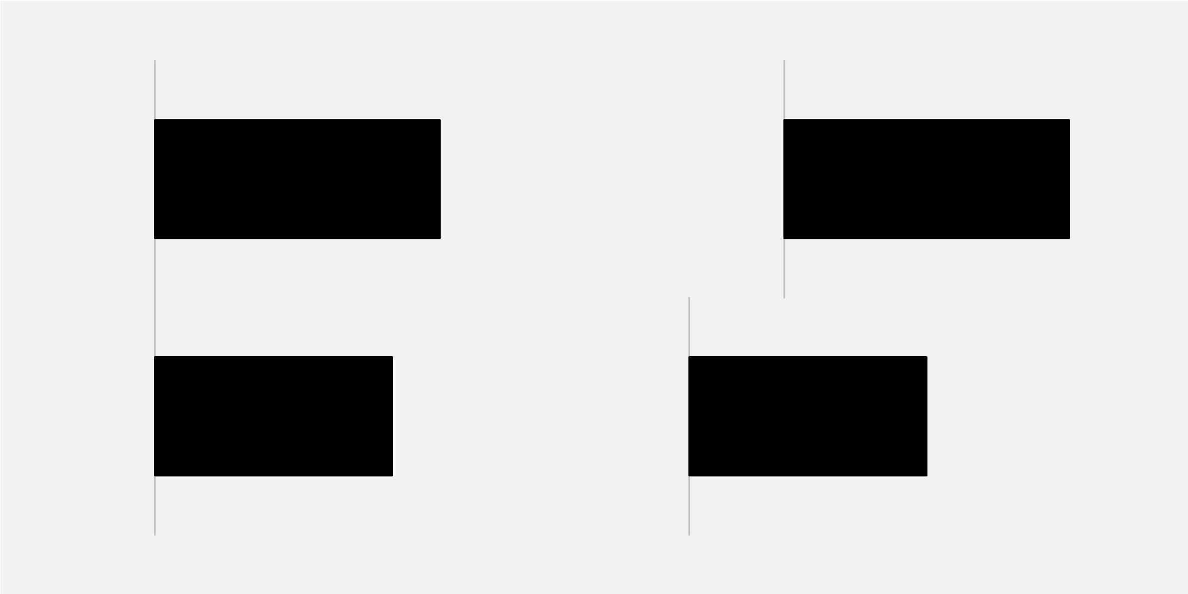


Figure 5.4: It is easier to compare the lengths of the two bars on the left because they are **aligned**; they share a common base. The two bars on the right are harder to compare because they are **unaligned**.

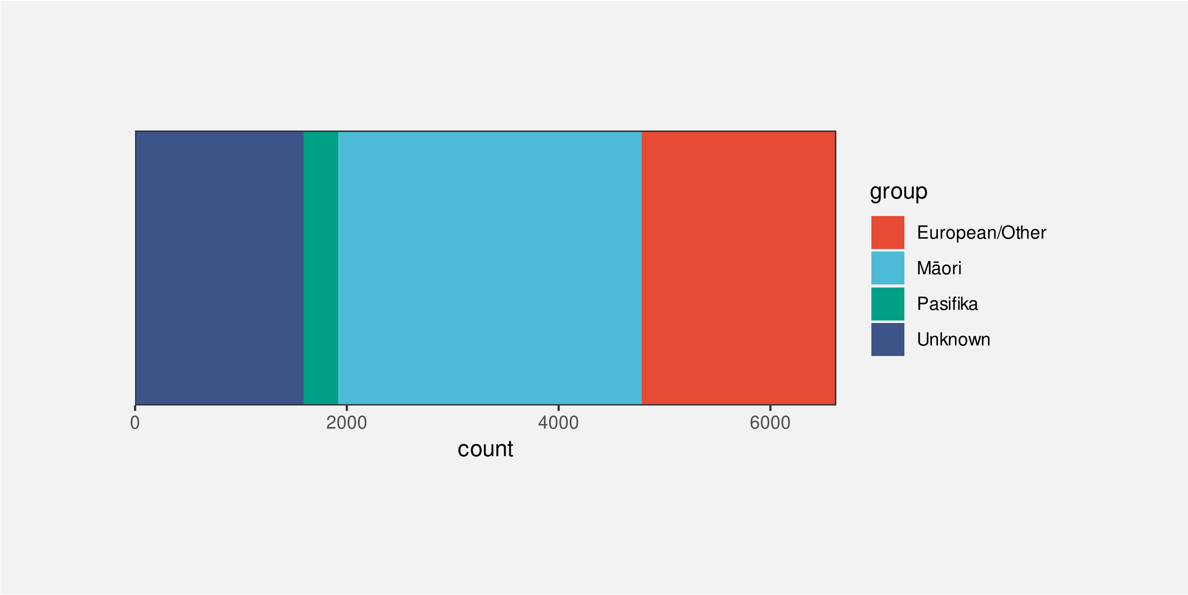


Figure 5.5: A stacked bar plot of the number of offenders in each ethnic group.

5.3 Case study: Stacked versus side-by-side bar plots

Stacking bars, which creates unaligned lengths, is something that arises when we use a bar plot to show a quantitative variable broken down by more than one qualitative variable. For example, Table 5.1 shows data on the number of offenders in different ethnic groups for individual years (from 2011 to 2021). This is a more detailed version of the data from Table 3.1.

Table 5.1: A table of the number of offenders per year aged 14 to 16 for different ethnic groups from 2011 to 2021. There are 44 rows of data, but just the first 6 rows are shown here.

	group	count	year
1	Māori	5957	2011
2	Pasifika	1092	2011
3	European/Other	5775	2011
4	Unknown	194	2011
6	Māori	5458	2012
7	Pasifika	1040	2012

Figure 5.6 shows the number of offenders (quantitative) for each ethnic group (qualitative) and for each year from 2011 to 2021 (treated here as qualitative, or at least discrete). In this data visualisation, the number of offenders is encoded to the lengths of the bars, the years are encoded to the horizontal **position** of the bars, the ethnic groups are encoded to the colours of the bars, and the bars are stacked vertically.

The problem that we face in this situation is that there are four bars to show for each year. If we simply encode the year as the horizontal position of each bar, the bars will overlap and we will have great difficulty decoding data values from the bars (Section 4.3).

Stacking the bars is one solution, but as we have just seen, this makes some comparisons more difficult. For example, we can easily see that the number of offenders in the unknown group increased after 2016, but it is much harder to see what has happened to the European/Other group since 2016.

An alternative visualisation is shown in Figure 5.7. This has placed the four ethnic groups side-by-side for each year. It is now easier to see the changes over time for the European/Other group because all bars are now aligned. Unfortunately, as we saw in Section 5.1, some comparisons are more difficult because the bars are now further apart and have distractors in between.

Yet another variation is shown in Figure 5.8. This time we have placed all years side-by-side for each ethnic group. This provides the clearest view of the changes over time for each ethnic group because the relevant bars are close together and aligned. However, there are other

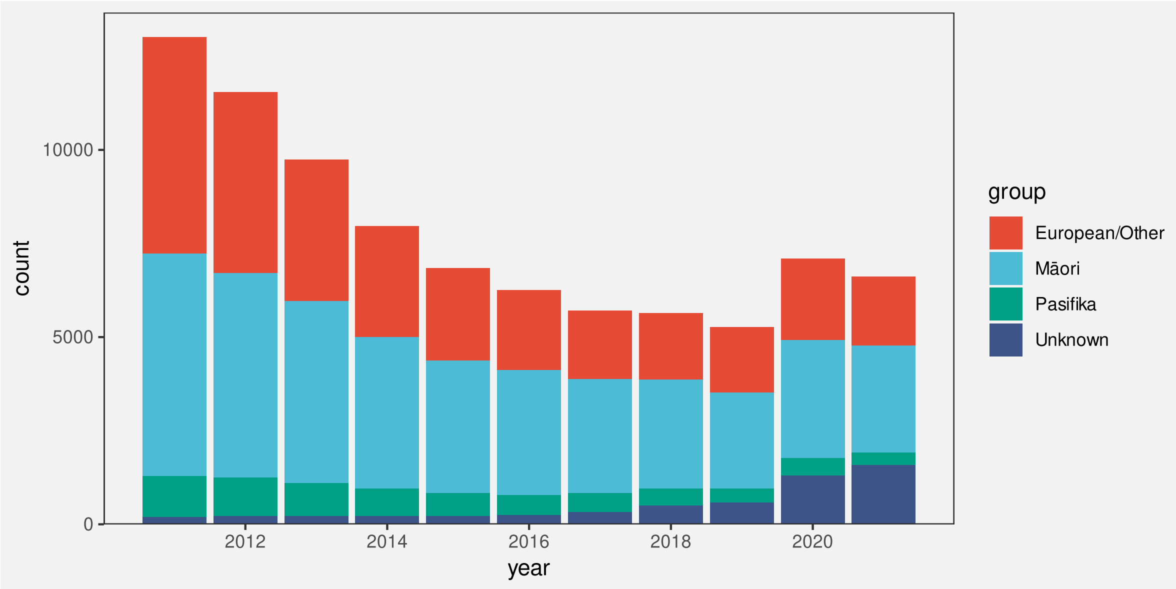


Figure 5.6: A stacked bar plot of the number of offenders in each ethnic group for each year from 2011 to 2021.

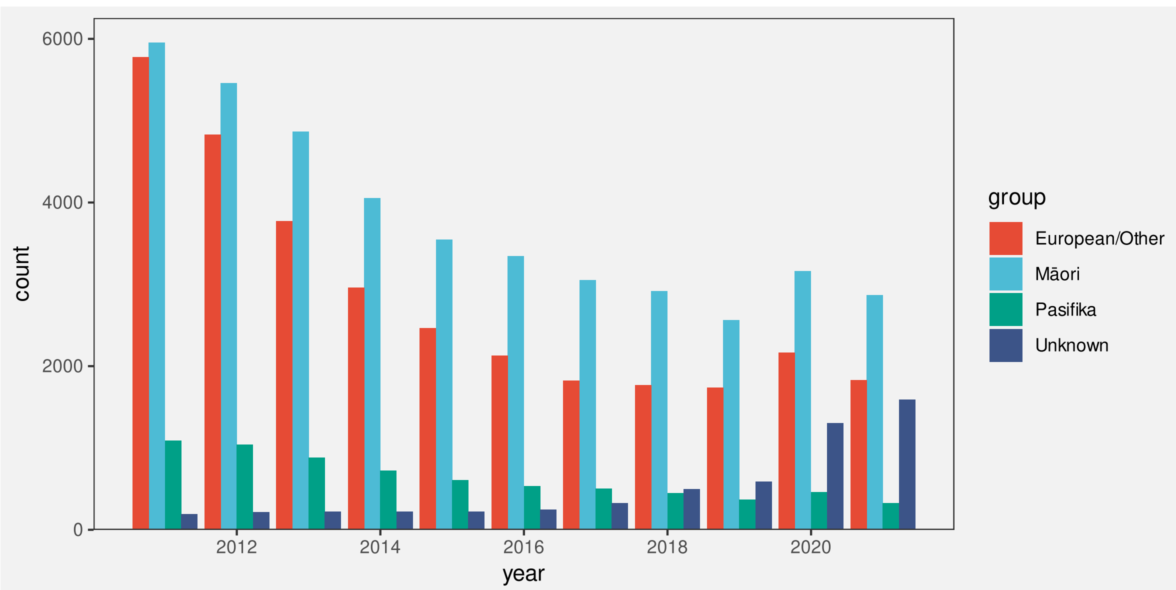


Figure 5.7: A side-by-side bar plot of the number of offenders in each ethnic group for each year from 2011 to 2021.

comparisons that are considerably more difficult because of the distance between the relevant bars, for example, comparing between ethnicities for a specific year.

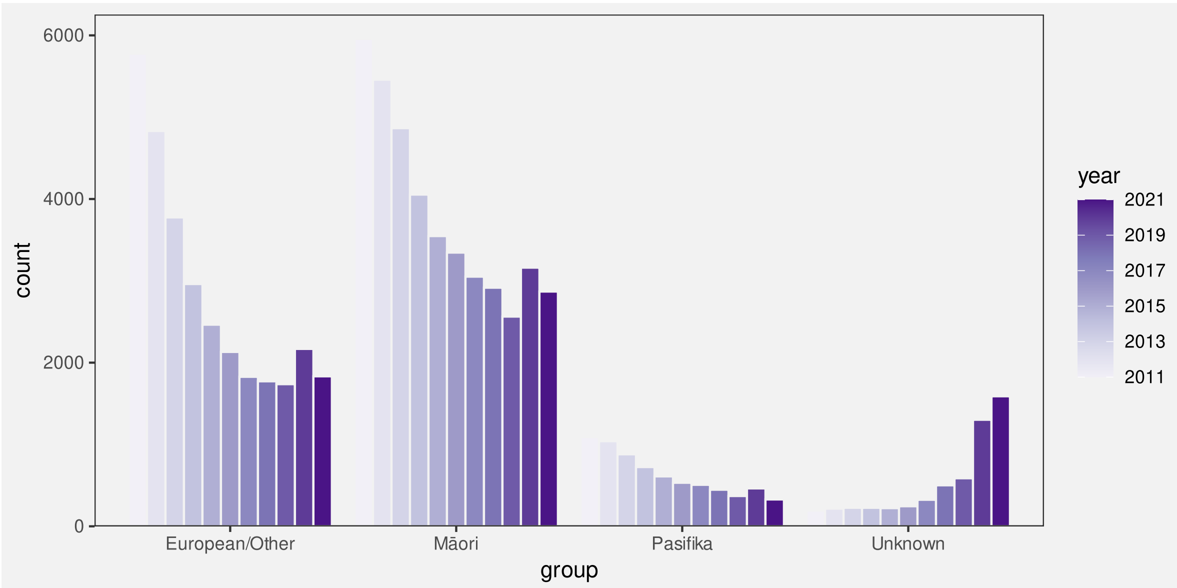


Figure 5.8: A side-by-side bar plot of the number of offenders in each ethnic group for each year from 2011 to 2021.

There is no single best visualisation in general for this situation because different visualisations each have their strengths and weaknesses. If a particular sort of comparison is to be emphasised, then one data visualisation might stand out, but there is also the option of providing more than one data visualisation in order to cover more comparisons.

5.4 Perception is relative

Figure 5.9 shows that the perception of differences between the lengths of bars is relative; it is easier to detect smaller differences when the visual features that we are comparing are smaller.[66] It is easier to see that the pair of bars on the left have different lengths and it is harder to see that the pair of bars on the right have different lengths. All three pairs of bars have different lengths, but the absolute size of the difference is the same, so the size of the difference relative to the bar lengths gets smaller as the bars get longer.

Figure 5.10 shows that we can make it easier to judge the differences by providing reference marks. Each pair of black bars in Figure 5.10 is the same and each pair of bars differs in length by the same small amount. The difference between the bars in each pair is difficult to perceive because the difference is small relative to the lengths of the bars. The reference marks in the middle and right-hand pair of bars provide shorter lengths to compare, so we are able to more

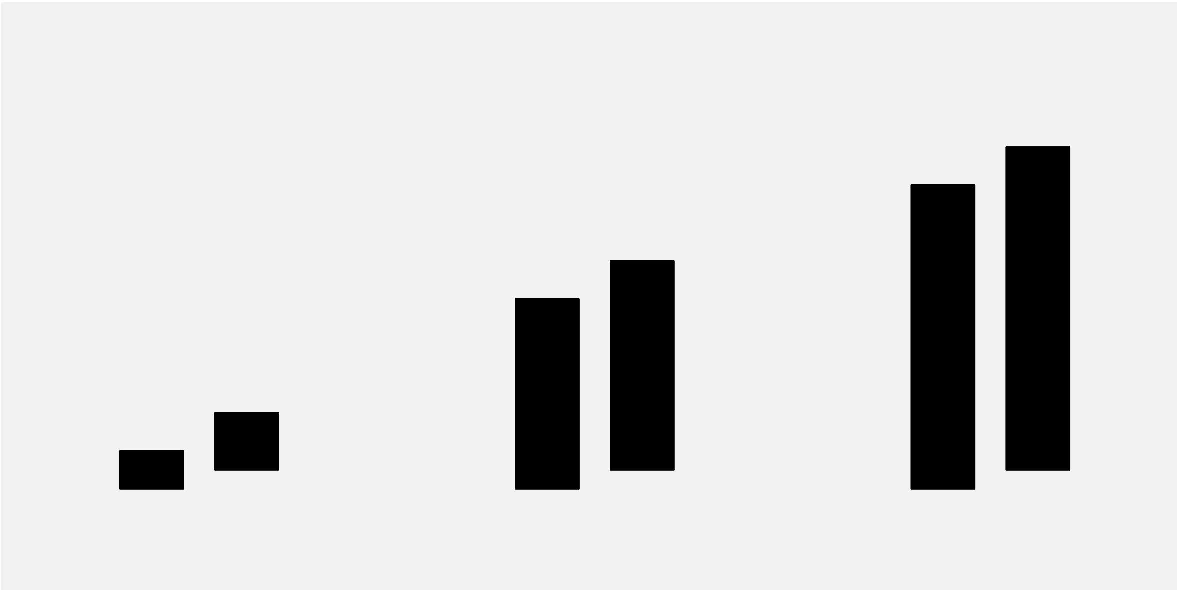


Figure 5.9: A demonstration that perception of differences is relative. The pairs of bars all differ by the same absolute amount, but the difference is more easily perceived when comparing the shorter bars. In the shorter bars, the absolute difference between the bars is a larger difference relative to the length of the bars.

easily perceive the small differences. For example, for the pair of bars on the right, we can compare the lengths of the short white gaps rather than comparing the longer black bars.

Figure 5.11 shows a bar plot of the number of offenders in each police district (Figure 5.1) with some grid lines added. Although it is difficult to compare the bars for Northland and Tasman because they are quite far apart and there are distractors in between (Section 5.1), the addition of the grid lines makes it easier to perform the comparison because we are able to compare the distances from the ends of the bars to the nearest grid line. Those distances are much shorter than the bars so it is easier to perceive that those two bars are very similar.

> One reason why grid lines are effective is because they allow comparisons of smaller lengths or distances, which means that it is easier to perceive smaller changes.

5.5 Summary

Encoding data values as the length of data symbols allows for the effective decoding of quantitative information, with the following caveats:

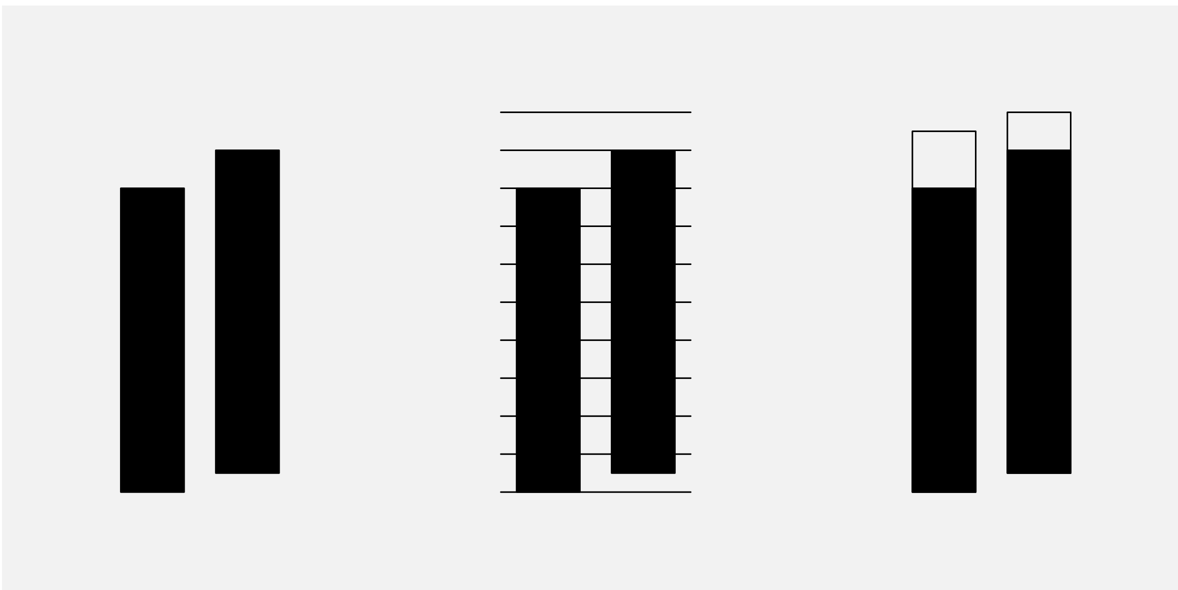


Figure 5.10: A demonstration that reference marks can make it easier to perceive small differences. Each pair of black bars is the same and each pair differs by a small amount. In the middle pair, grid lines make it easier to compare the distances from the ends of the bars to the nearest grid line and, in the right-hand pair, a border added to each bar, where the two borders are the same length, makes it easier to compare the white gaps rather than the bars themselves.

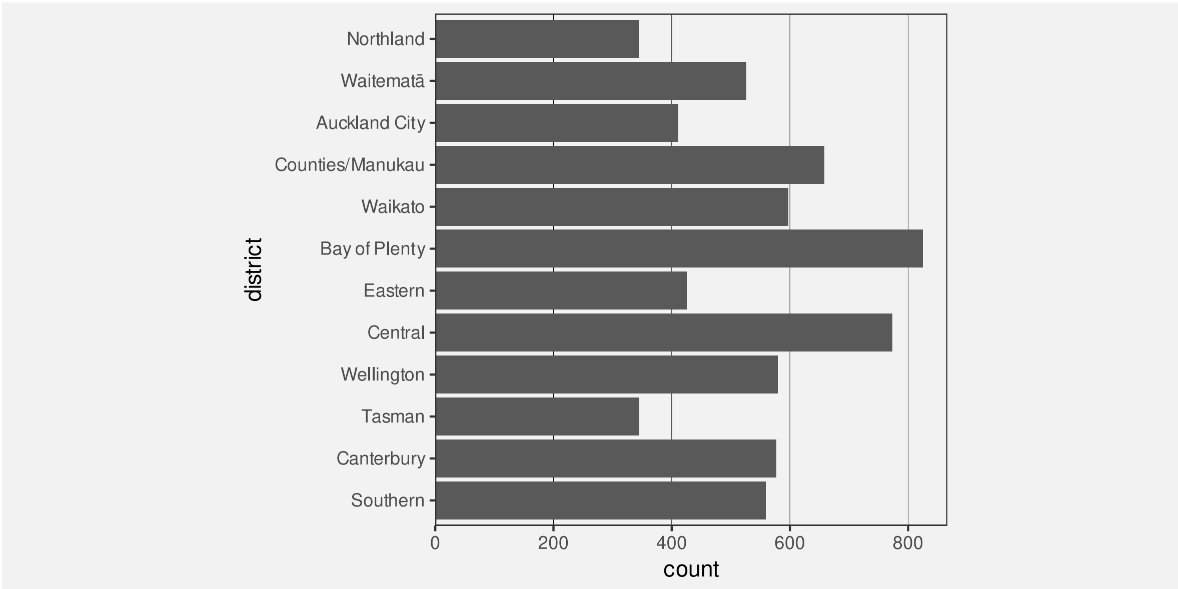


Figure 5.11: A bar plot of the number of offenders for different police districts, with the districts ordered from North to South. This is very similar to Figure 5.1, just with some grid lines added.

- Comparisons between **lengths** are more difficult if the lengths are far apart, especially if there are distractors in between.
- Comparisons between **lengths** are more difficult if the lengths do not have a common baseline.
- Comparisons between **lengths** are easier for shorter lengths.

Notes

[64] The first demonstration of the "distance effect" is attributed to William S. Cleveland and McGill (1984). It has since been replicated several times, for example, by Talbot et al. (2014) and Lu et al. (2021).

[65] In a stacked bar plot, groups are encoded as the positions of the bars and counts are encoded as the lengths of the bars, but the positions are in the same dimension as the lengths (in this case, both position and length are horizontal). In a side-by-side bar plot, the position of the bars is in the opposite dimension to the lengths of the bars.

[66] The idea of perception of differences being relative is a very old result from Psychology known as Weber's Law (Weber 1846; Boring 1942).

6 Colour

Figure 6.1 shows a pie chart of the number of offenders for different levels of crime seriousness in 2021. In this data visualisation, the crime level, which is a **qualitative** variable, is encoded as **colour**, which is appropriate for representing qualitative values (Section 3.4). It is easy to decode the crime levels because it is easy to identify the colour for a particular crime level amongst the set of colours used.

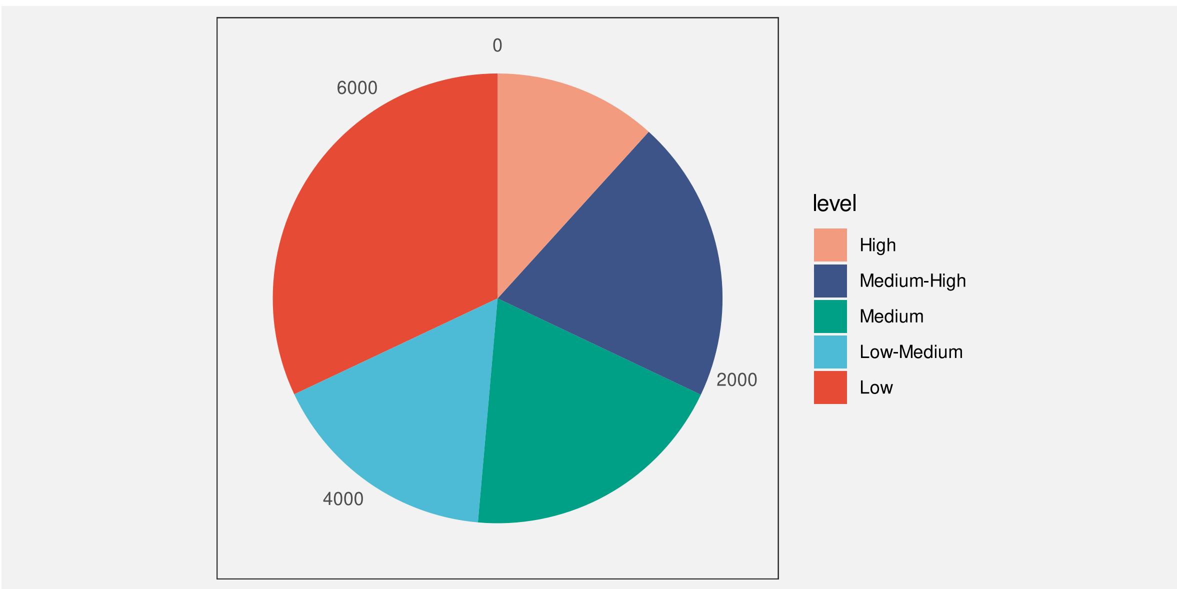


Figure 6.1: A data visualisation of the number of offenders for different levels of crime seriousness.

Figure 6.2 shows another pie chart of the same data, again encoding the **qualitative** level as **colour**. It is again easy to identify each crime level because we can easily identify each separate wedge colour. However, Figure 6.2 has an advantage over Figure 6.1 because we can *also* decode an ordering of the wedges from more serious crimes, which are darker wedges, to less serious crimes, which are lighter wedges.

The fact that there can be two data visualisations, Figure 6.1 and Figure 6.2, with the same encoding—the **qualitative** level of crime is encoded as the **colour** of the wedges—but the two data visualisations have different effectiveness suggests that there is still more to learn about the encoding of data values as colour.

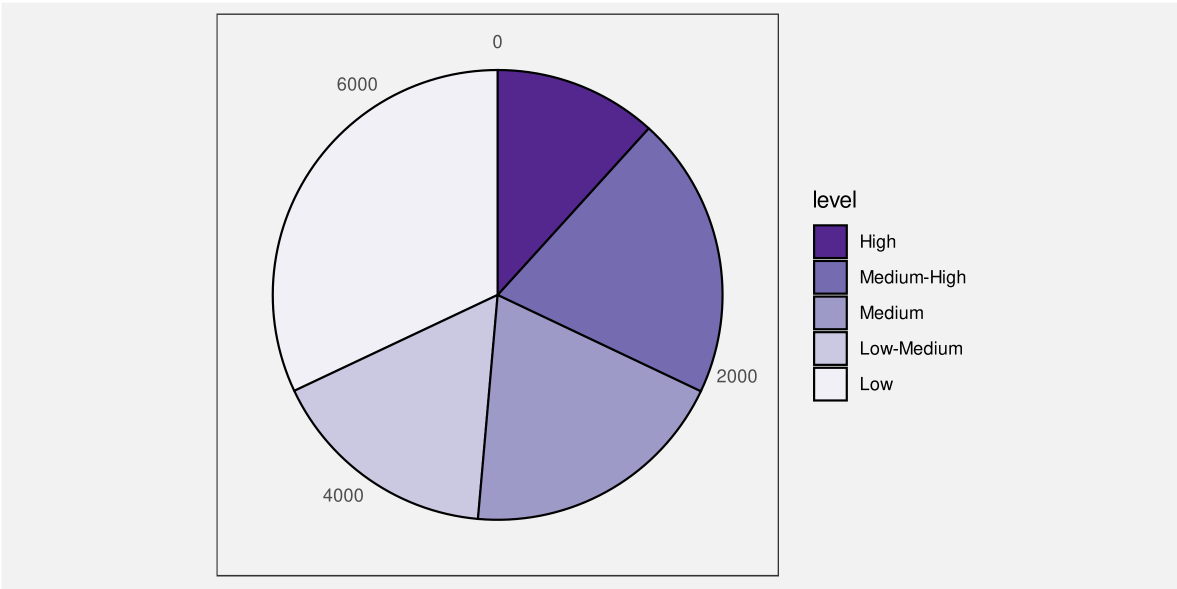


Figure 6.2: A data visualisation of the number of offenders for different levels of crime seriousness.

This chapter covers some more details about how the **colour** visual feature works.

6.1 Hue, chroma, and luminance

We have so far taken a very simplistic approach to **colour**, treating it as if it is a single **qualitative** visual feature (Section 3.4). However, the perception of colour can be divided into three separate visual features: **hue**, **chroma**, and **luminance** (Figure 6.3).[67] Hue describes what we normally refer to as the colour name: reds, oranges, yellows, greens, blues, purples, and so on. Chroma describes the colourfulness, from bright (very colourful) to dull (grey), and luminance describes the lightness, from dark to light. Our visual system is capable of perceiving changes in hue and changes in chroma and changes in luminance.

In Section 3.4, we treated colour as a **nominal** visual feature, but that really strictly only applies to **hue**. For example, in Figure 5.5, we encoded different ethnic groups as different hues. We are very good at decoding nominal data values, or different groups, from different hues, subject to the capacity limits described in Section 3.6.

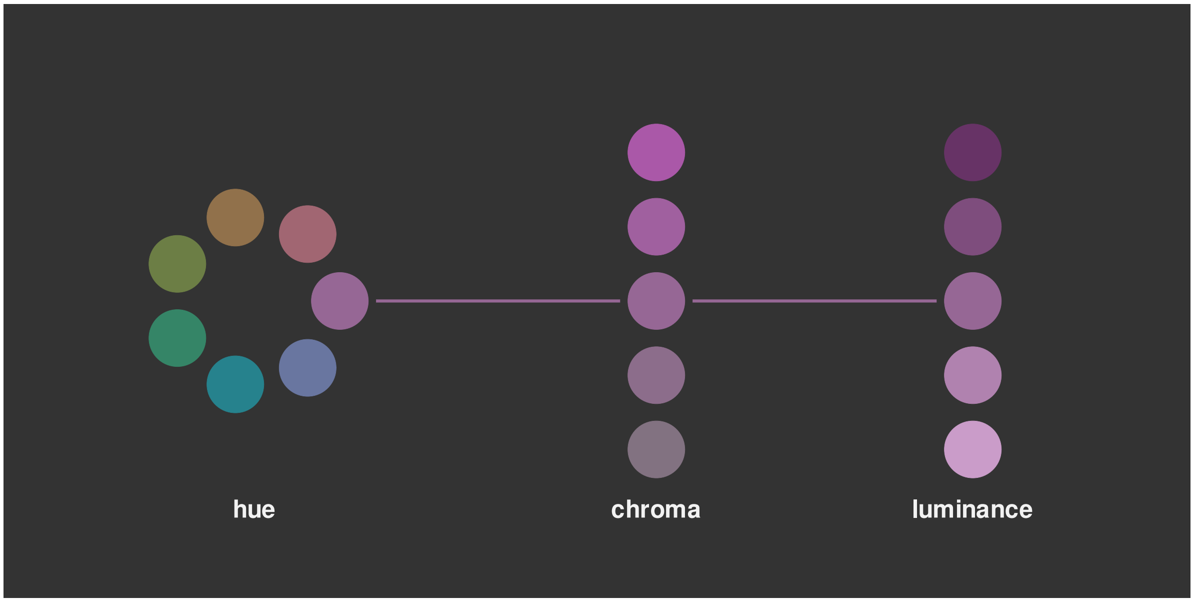


Figure 6.3: The perception of colour can be divided into three components: differences in **hue**, the colours of the rainbow; differences in **chroma**, from bright and colourful to dull and grey; and differences in **luminance**, from light to dark.

6.2 Ordinal visual features

Chroma and **luminance** are both **nominal** visual features in that they can be used to encode different groups, but they are also **ordinal** visual features because we can decode an ordering from a set of colours that differ in terms of either chroma or luminance (or both). A set of colours that transitions from darker to lighter and/or brighter to duller conveys an ordering from larger to smaller.[68]

This helps to explain the advantage of Figure 6.2 over Figure 6.1. In Figure 6.2, the colours differ mostly in terms of luminance and chroma so we are able to perceive changes from larger values to smaller values because the colours transition from darker to lighter.

In this sense, **chroma** and **luminance** are able to convey more information than **hue** because they can represent **nominal** differences *and* **ordinal** differences, whereas hue can only represent nominal differences.[69]

On the other hand, Figure 6.4 shows that chroma and luminance are even more limited than hue in terms of **capacity**. We can only use chroma and/or luminance to represent **nominal** data when there are very few different groups to represent.

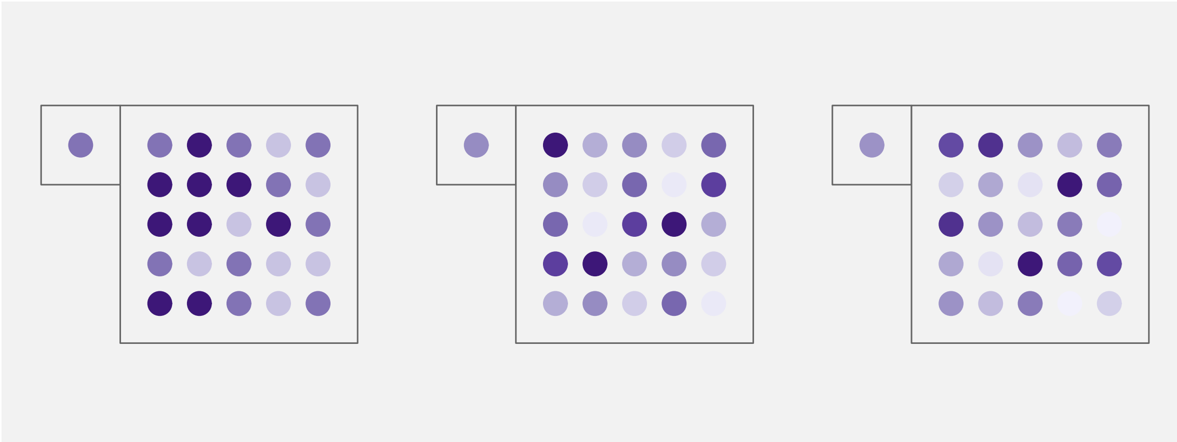


Figure 6.4: Encoding qualitative values to luminance and chroma works well for representing nominal data values, but only for a very small number of values. It is possible to identify which shade of purple the single dot corresponds to amongst the three shades on the left, but it is much harder with the seven shades in the middle, and harder still for the eleven shades on the right. This figure can be compared with Figure 3.13, which demonstrates the capacity of hue for representing nominal data values.

> If we encode data values to the **hue** of data symbols, then we can only decode **nominal** information from the data symbols; do the data symbols represent *different* groups or the *same* group.

> If we encode data values to the **chroma** and/or **luminance** of data symbols, it is also possible to decode **ordinal** information form the data symbols.

6.3 Perception is relative

We saw in Section 5.4 that perception of **lengths** is relative. For example, small differences in length are easier to see when the lengths themselves are short (Figure 5.9).

There is a similar rule for the perception of **colours**. Our perception of luminance and chroma is affected by neighbouring colours.[70] This effect is demonstrated in Figure 6.5: the same colour when surrounded by a darker background appears lighter than it does when surrounded by a lighter background.

This effect also applies to the perception of hue. Figure 6.6 shows a pie chart of the number of offenders in each police district, with the police district encoded to the **colour** of the wedges.

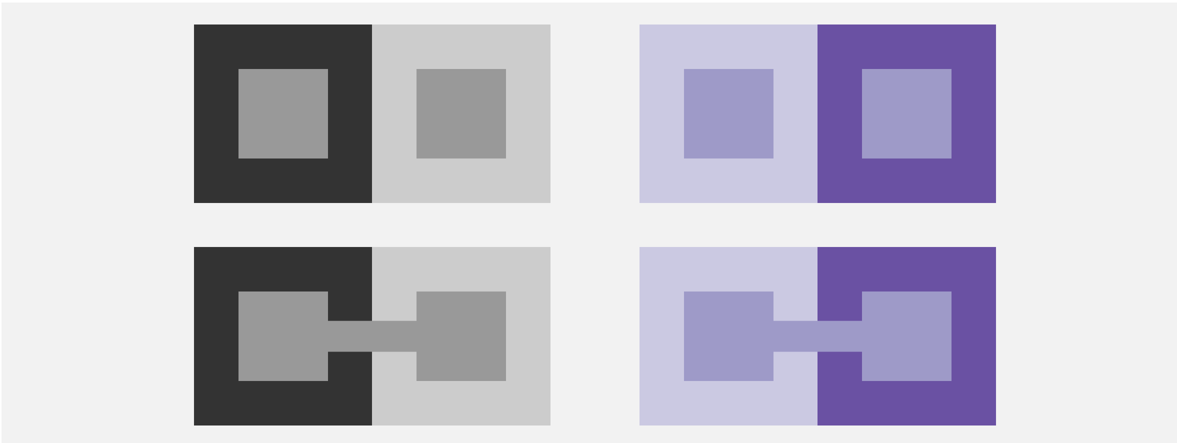


Figure 6.5: The perception of colours is relative. In the top row, the inner rectangles are exactly the same shade of grey on the left and exactly the same shade of purple on the right. The perception of the inner rectangles is affected by the surrounding colour, so the same shade appears darker against a light background and lighter against a dark background. The bottom row draws a line connecting the two inner rectangles to show that they really are the same shade.

There are two issues with the perception of the wedge colours: each wedge is affected by its neighbour so that, for example, some of the blue wedges look more green at the boundary with their more blue neighbour and more blue at the boundary with their more green neighbour; furthermore, the colours in the legend are perceived differently from the corresponding colours in the wedges because the legend colours are surrounded by a light grey, rather than sharing a boundary with a neighbouring colour.

The pie chart in Figure 6.6 is *not* an effective data visualisation because it encodes qualitative data values with many different groups as colour hue. This exceeds the **capacity** of the **hue** visual feature; it is difficult to clearly identify which wedge corresponds to which district (Section 3.6). We have just used it here to demonstrate the relative perception of colours.

Although it does not make the use of hue any more appropriate, for the purposes of demonstration, we can solve the colour perception problems in Figure 6.6 by adding a white border to all of the wedges (Figure 6.7). Now all of the wedges and all of the squares in the legend are perceived relative to a surrounding white border. This means that, not only is each wedge perceived as a solid block of constant colour, but it is easier to match the colours within the wedges with the colours in the legend.

> If we encode data values as the colour of data symbols, we may have difficulty decoding the colours if the same colour is presented with different colours adjacent to it.

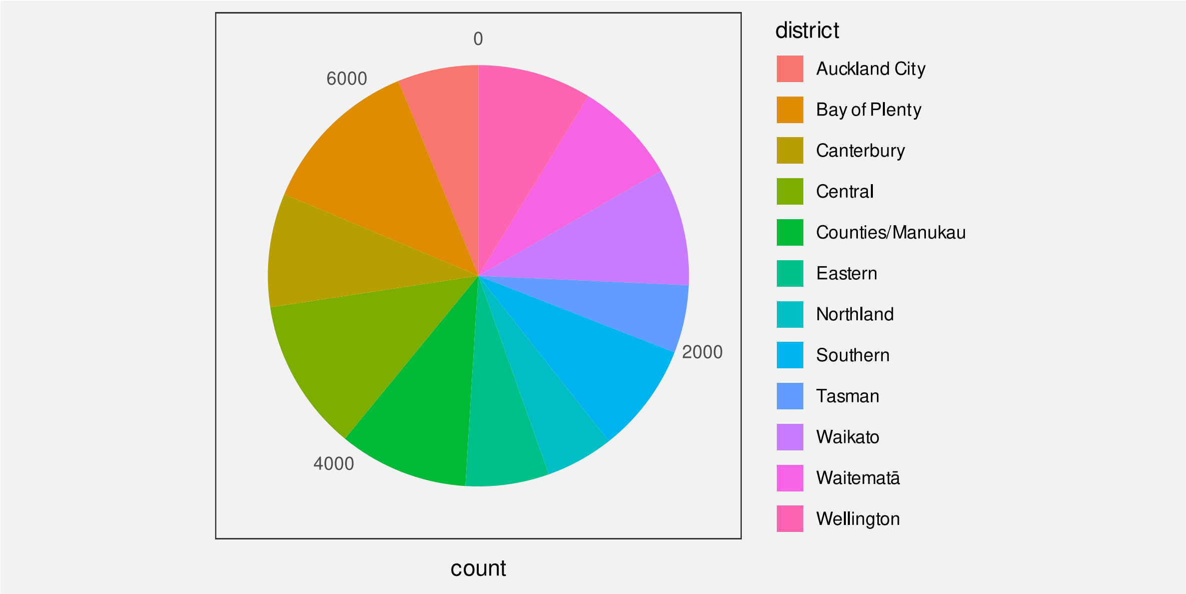


Figure 6.6: This pie chart demonstrates that the perception of hues is relative. The colours within each wedge appear to have a subtle gradient within the wedge because the perception of the wedge colour is affected differently by the neighbouring wedge colours.

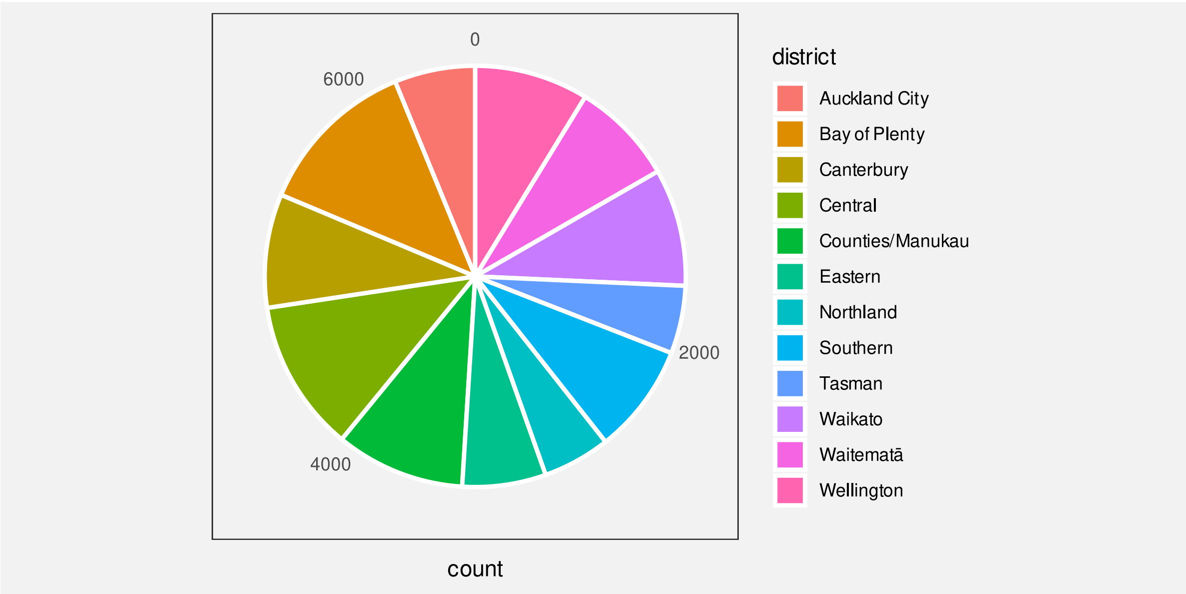


Figure 6.7: The hues in this pie chart are perceived more accurately than those in Figure 6.6 because each wedge is surrounded by a white border.

6.4 Size matters

The perception of colours is also affected by the size of the coloured region. In particular, small regions of colour may blend together with neighbouring colours.[71] For example, on the left of Figure 6.8, the yellow and blue stripes are, in reality, the same colours both in the top set of wide bars and in the bottom set of narrow bars, but the blue bars appear greener in the bottom set. The right side of Figure 6.8 shows that the same colours may appear different when they are used as fill colours for larger regions, like bars, compared to when they are used for thin lines. In this case, the blue line appears darker than the blue bars.

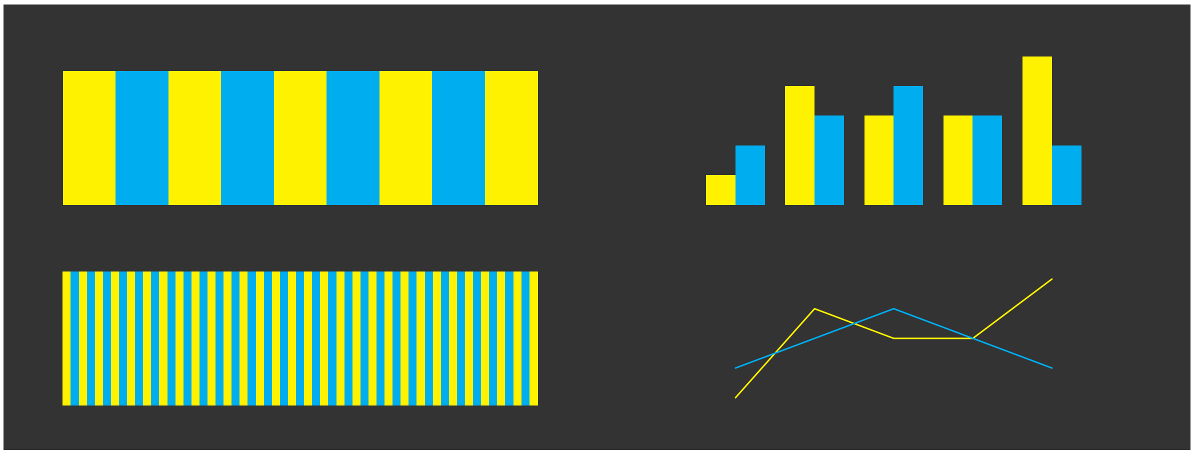


Figure 6.8: The perception of colours varies based on the size of the coloured regions.

> If we encode data values as the colour of data symbols, we may have difficulty decoding the colours if there are large differences in the sizes of data symbols.

6.5 Colours in space

Figure 6.9 shows two-dimensional slices of the set of visible colours arranged in three dimensions.[72] The central column of three coloured regions (surrounded by white circles) show slices of colours with constant luminance. Chroma increases outwards from the centre (indicated by a white dot) and hue changes with angle. In each slice, we can see the familiar rainbow of colours from red, through orange, yellow, green, blue, indigo, and violet, back around to red. The top slice is for high luminance (light) colours and the bottom slice is for low luminance (dark) colours.

The coloured regions to the left and to the right (surrounded by white semicircles) show slices of constant hue. In these slices, Luminance increases from bottom to top and chroma increases away from the flat side of the semicircle. For example, the top-left slice shows the same green hue at different levels of luminance and chroma.

Figure 6.9 provides insight into what sorts of colours are possible and what sorts of differences between colours are possible. For example, from the bottom central slice we can see that it is not possible to produce colours that have a high chroma *and* are very dark. In other words, there is no such thing as a very colourful black. This is also true for colours that are very light.

We can also see that the range of possible chroma values depends on both the hue and the luminance of colours. For example, the top-right slice shows that it is possible to produce a light yellow with either a low chroma or a high chroma, but we can only produce a dark yellow with a low chroma. Equivalently, the range of possible luminance values depends on both the hue and the chroma of colours. For example, we can produce both dark and light low-chroma yellows, but we can only produce light high-chroma yellows.

This means that, if we want to encode data values as the chroma of a data symbol, we can get a wider range of colours—we can get a higher decoding **capacity**—through careful selection of hue and luminance. For example, we can generate a wider range of light greens than dark greens (the top-left slice) and a wider range of reds that are neither too light nor too dark (the centre-right slice).

Figure 6.10 is similar to Figure 6.9 except that, instead of showing the range of colours that it is possible to perceive, it shows the range of colours that it is possible to *produce* using standard technology.[73] The simple message here is that the limits on the range of colours that we can use to encode data values is much smaller than the range of possible colours. On the positive side, the range of colours that we can produce is still very large.

A more important message is that the general points about differences between colours still apply. For example, the range of possible chroma values is dependent on the hue and luminance of colours.

> If we encode data values as the colour of data symbols, the choice of hue, luminance, and chroma affects the capacity of decoding—how many different colours can be effectively decoded.

6.6 Colour harmony

We have seen that **hue** is an appropriate visual feature for encoding **nominal** data values because we can decode differences between hues without any implied ordering between hues. For example, blue is not larger than red.

However, it is not entirely true that there is no sense of *distance* between different hues. The left image in Figure 6.11 shows a colour wheel of hues for a constant luminance and chroma.

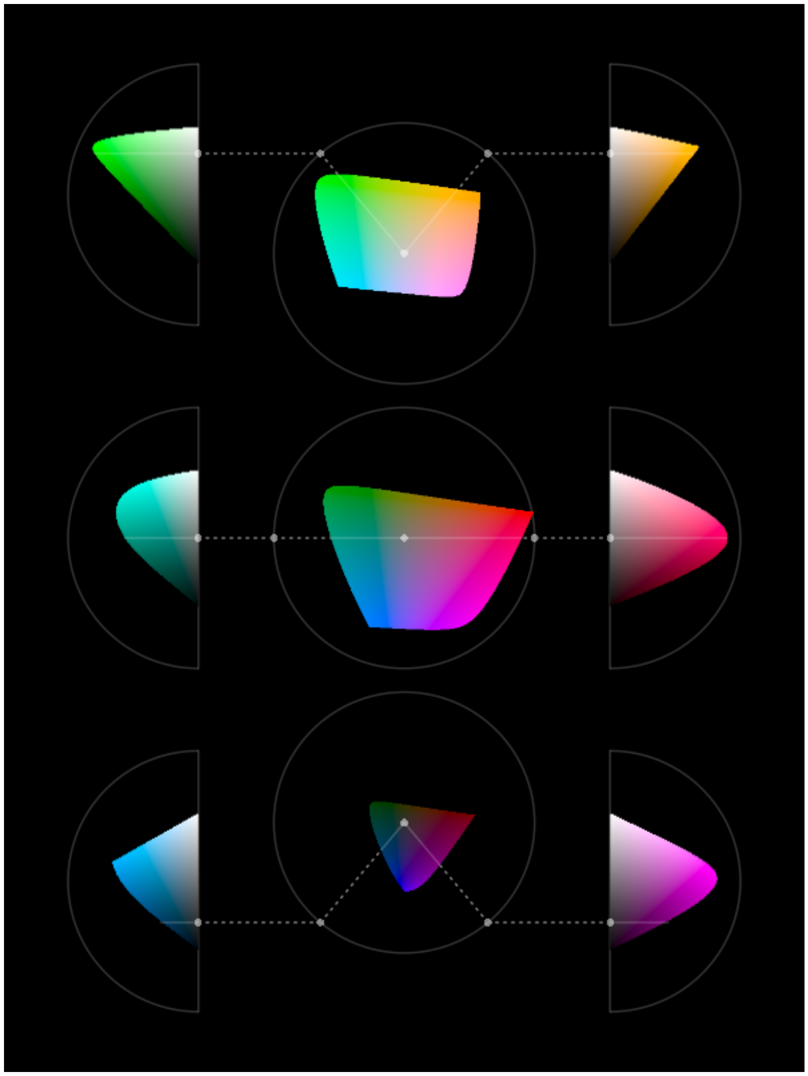


Figure 6.9: Slices through the set of colours that are visible to humans. The central images show colours that differ in terms of chroma and hue, but share a common luminance (from lighter at the top to darker at the bottom). The images to the left and right show colours that differ in terms of chroma and luminance, but share a common hue.

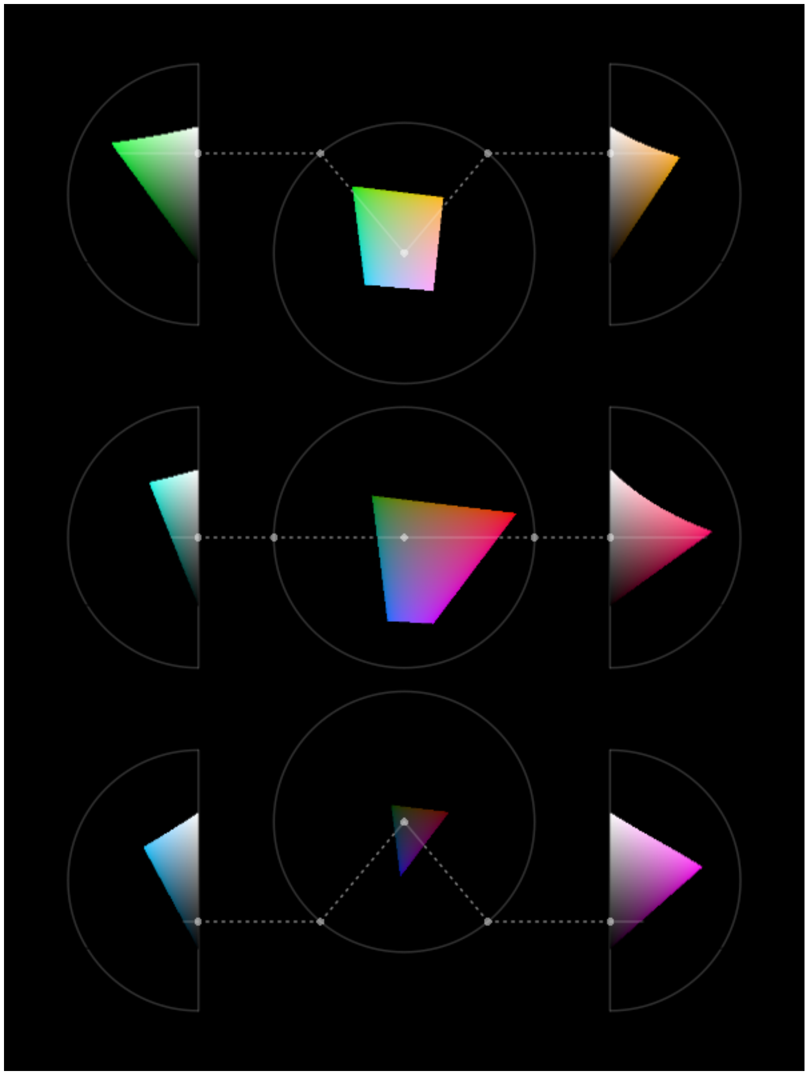


Figure 6.10: Slices through the set of colours that it is possible to produce on a typical computer monitor.

Hues that are close together on this colour wheel are decoded as more similar than hues that are on opposite sides of the wheel. For example, the top two circles on the right of Figure 6.11 have hues that are from the left side of the colour wheel so they appear similar, but the bottom two circles have hues from the left and the right side of the colour wheel so they have a greater contrast.[74]

Figure 6.11: On the left, a colour wheel that shows differet hues for a constant chroma and luminance. At top-right is a pair of hues from the left side of the colour wheel and bottom-right is a pair of hues from opposite sides of the colour wheel. There is greater contrast between the bottom pair of colours and less contrast between the top pair of colours.

> If we encode data values as hue, it is possible to decode some hues as quite similar and other hues as very different.

6.7 Colour vision deficiency

Roughly 10% of the population, primarily men, suffer from a form of colour vision deficiency (CVD). In the most common case, the viewer will have difficulty distinguishing between red and green hues.[75] Figure 6.12 simulates how the colours in Figure 6.6 would appear to a person with severe CVD.

In order to avoid confusion for viewers with CVD, it may be necessary to vary luminance and/or chroma in addition to or instead of hue. For example, Figure 6.13 shows how the

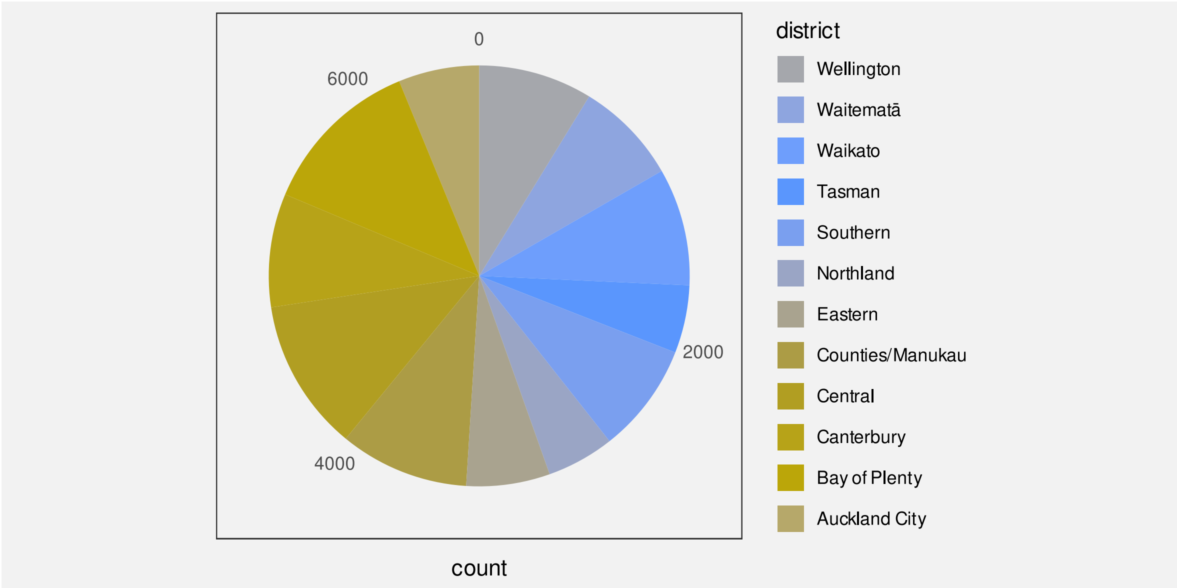


Figure 6.12: A data visualisation of the number of offenders for different police districts. This is how a person with severe deuteranopia might perceive the colours in Figure 6.6.

shades of purple in Figure 6.2 would appear to a viewer with severe deuteranopia. It is not important that a viewer with CVD perceives a different hue in this case because the more important difference is in the luminance and chroma of the colours.

> When we encode data values as the colour of data symbols, some viewers will have difficulty decoding the colours if we do not take into account common forms of colour vision deficiency.

6.8 Non-linear encodings

The top row of Figure 6.14 shows a linear luminance encoding; we have a set of rectangles that change luminance in equal steps. The two rectangles on the left of this row appear very similar, while the two rectangles on the right of this row appear quite different. The darker rectangles appear to change more slowly and the lighter rectangles appear to change more rapidly. In other words, the *decoding* of luminance is *not* linear.

We saw in Section 4.7 that accurate decoding of data values from **position** relies on using a linear encoding. That was because the decoding of position is linear. A linear encoding from data values to position, followed by a linear decoding from position to data values, leads

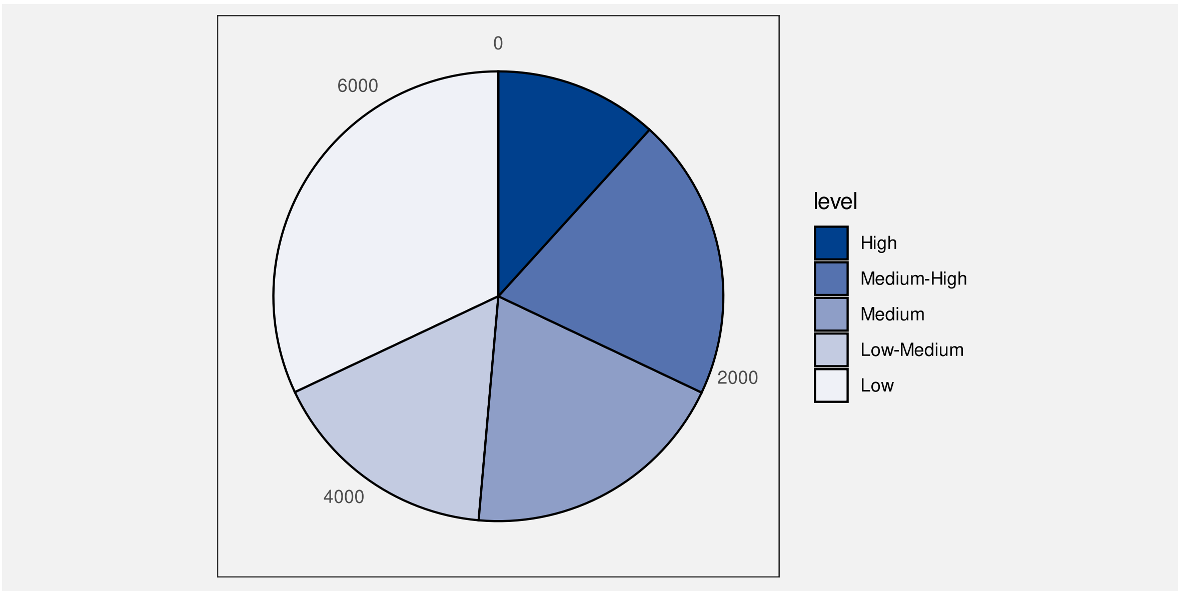


Figure 6.13: A data visualisation of the number of offenders for different levels of crime seriousness. This is how a person with severe deuteranopia might perceive the colours in Figure 6.2.

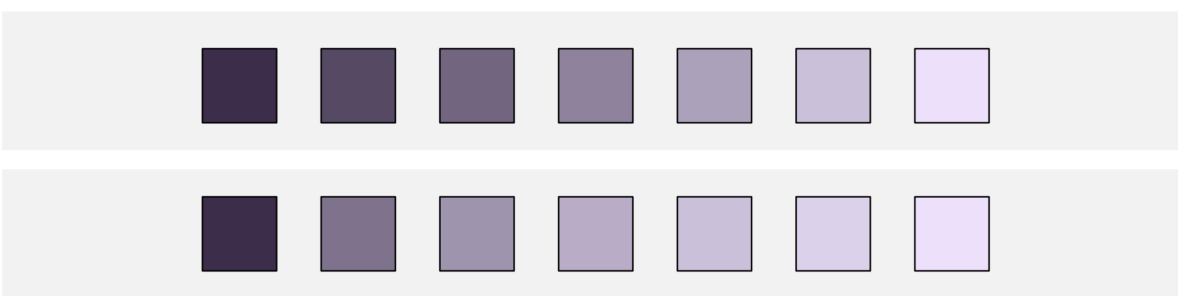


Figure 6.14: The two rows of purple values begin and end at the same level of luminance. The top row encodes luminance linearly, but the bottom row uses a cube-root encoding.

to an accurate encoding. And that is particularly important for encoding quantitative data values.

Part of the reason why it is not appropriate to encode quantitative data values to luminance is that the decoding of luminance is *not* linear. This makes it difficult to accurately decode quantitative data values from the luminance of a data symbol, even if the encoding of the data values to luminance was linear.

However, we can use luminance to encode ordinal data values (Section 6.2). For this purpose, all that we need is to be able to identify that a data symbol is lighter or darker than another data symbol.

> The decoding of luminance is non-linear, which means that luminance is not accurate for decoding quantitative data values. However, luminance is still effective for decoding ordinal data values.

The bottom row of Figure 6.14 shows a luminance encoding that takes larger steps for darker colours and smaller steps for lighter colours. In this row, the purples appear to change more evenly. We can see differences between the darker rectangles as well as between the lighter rectangles.

In other words, a *non-linear* encoding from data values to luminance can be used to generate a larger set of luminance levels that can be differentiated from each other. A non-linear luminance encoding leads to increased **capacity** for decoding data values from luminance.

> The decoding of luminance is non-linear, which means that the capacity of luminance is worse for darker colours than it is for lighter colours. However, a non-linear encoding of data values to luminance can increase the capacity for darker colours.

6.9 Escape capacity

We have previously identified that colour has a limited capacity (Section 3.6 and Section 6.2). In general, we cannot use colour to encode more than around 7 different categories. Section 6.5 also demonstrated that this capacity varies for different combinations of hue, chroma, and luminance.

One way to increase the decoding capacity of a colour encoding is to vary more than one of hue, chroma, and luminance at once. For example, the top row of Figure 6.15 shows a set of colours that only vary in terms of luminance—hue and chroma are held constant (this is the bottom row from Figure 6.14). In the second row of Figure 6.15, in addition to increasing luminance, we have increasing *chroma*, so there are larger differences between colours, but still a clear ordering. Similarly, in the bottom row of Figure 6.14, the *hue* changes slightly as the luminance increases so that the differences are more obvious.

In effect, this is another example of a non-linear encoding. We take a non-linear path through colour space (Section 6.5).

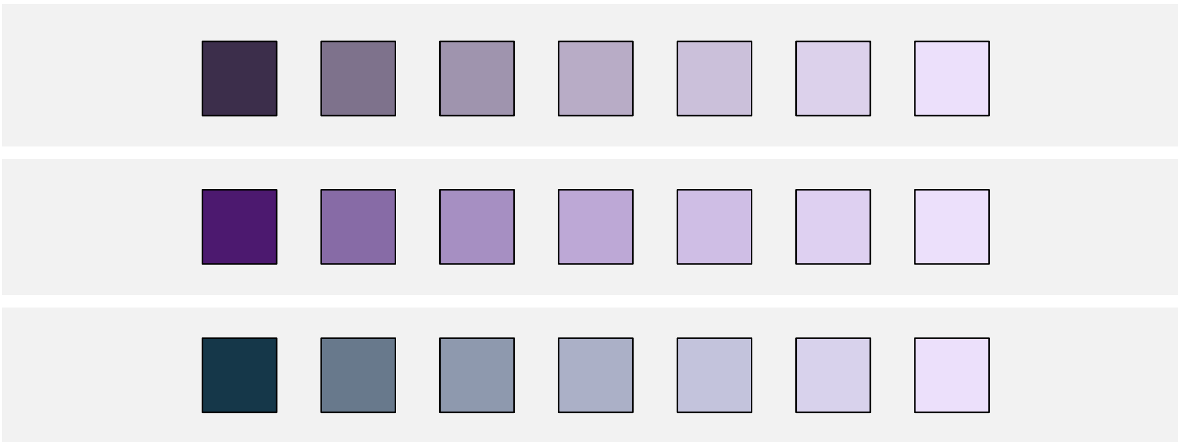


Figure 6.15: The top row holds hue and chroma constant and just varies luminance. It is limited to a very low chroma. The middle row varies chroma as well as luminance, which provides a wider range of colours compared to the top row by allowing a more colourful starting point. The bottom row varies hue and luminance, which provides a wider range of colours compared to the top row by allowing a different-hued starting point.

> When encoding ordinal data values as colour, the capacity can be increased by varying more than one of hue, chroma, and luminance.

Another way to expand the decoding capacity of a colour encoding is to reduce the number of colours that need to be directly compared. For example, if data symbols are arranged in a very regular pattern, like in a bar plot, comparisons between neighbouring data symbols are most important. This means that we can order the colours to maximise neighbour differences rather than maximising all pairwise differences.[76] Figure 6.16 demonstrates this idea with a bar plot of the crime rate in different districts for both 2011 and 2021. This bar plot uses the same set of hues as Figure 6.7, just in a different order so that adjacent bars have very different colours. It is very easy to distringuish between adjacent colours.

A similar idea is to just reuse a smaller set of hues, which means larger differences between adjacent colours, but repeat the set. The regular arrangement of the data symbols means that we can still differentiate between repeated colours because of the bar positions (Figure 6.17). For example, we can differentiate the pink for the Wellington district from the pink for the Eastern district because Wellington is at the end and Eastern is in the middle.

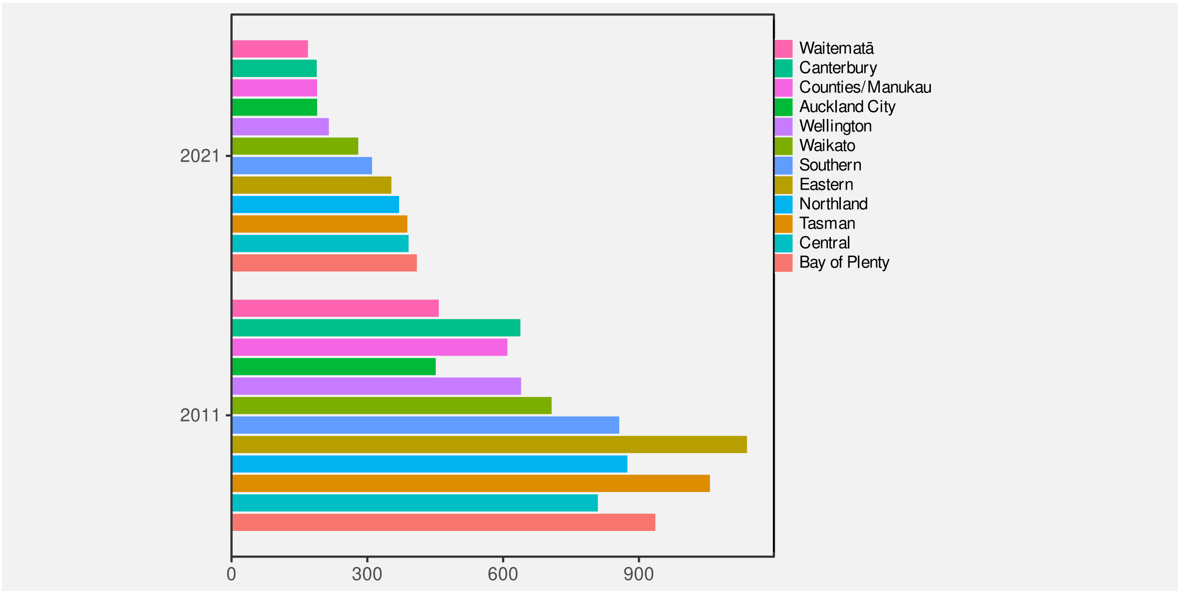


Figure 6.16: The hues in this pie chart are ordered so that adjacent hues are distant from each other.

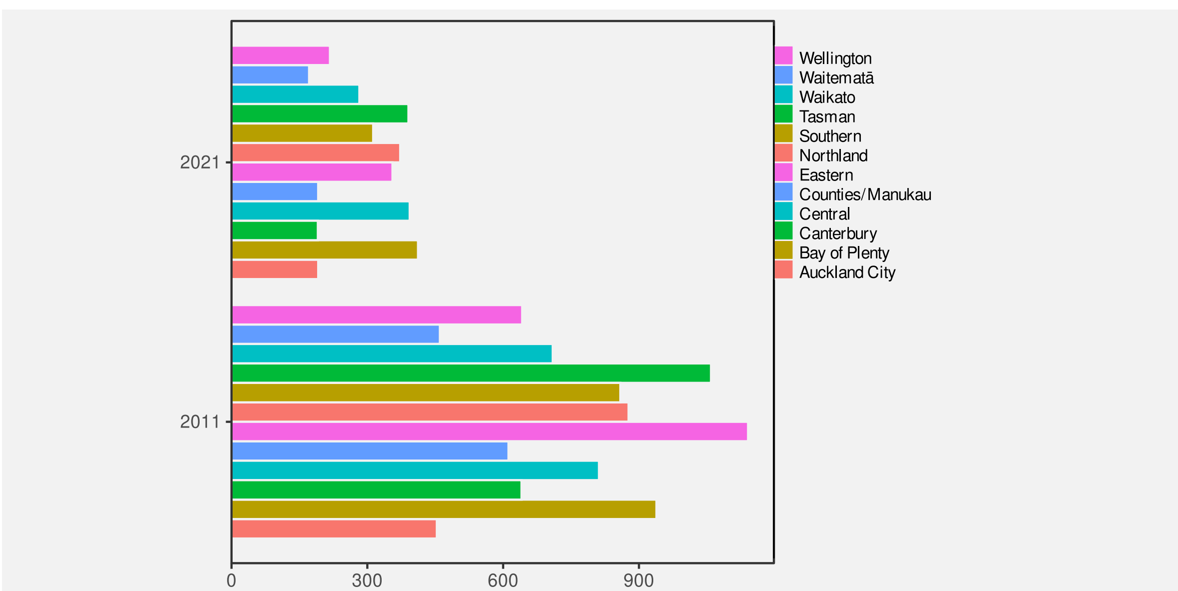


Figure 6.17: The hues in this pie chart are repeated so that adjacent hues are more distant from each other.

> When encoding nominal data values as colour, the capacity can be increased by focusing on adjacent colours rather than all pairs of colours.

6.10 Colour palettes

When we encode nominal or ordinal data values as colours, we have to choose a set of colours, or a **colour palette**. For example, in Figure 6.1 we have to choose a different colour to represent each level of crime.

This choice is difficult because there are many different colours to choose from and often there is no obvious encoding from a data value to a particular colour. For example, what specific hue, chroma, and luminance should we choose to represent a "high crime level"? As we have seen in the preceding sections, there are also many, sometimes competing, criteria involved in selecting effective colours.

As an example, we can consider the problem of selecting a set of 5 different hues to represent 5 different categorical data values, like the palette that is required to represent the difference levels of crime in Figure 6.1.

We know that it is effective to encode qualitative data values as the hue of a data symbol (Section 3.4 and Section 6.1). In order to make the individual hues easy to identify, we can place them evenly around the colour circle. If we do not want the viewer to decode any ordering of the data values from the data symbols, then we should also hold the chroma and luminance of the data symbols constant. The top row of Figure 6.18 shows a palette that follows that simple approach.

The bottom row of Figure 6.18 shows one big problem with this attempt: the hues are very hard to identify for viewers with CVD (Section 6.7).

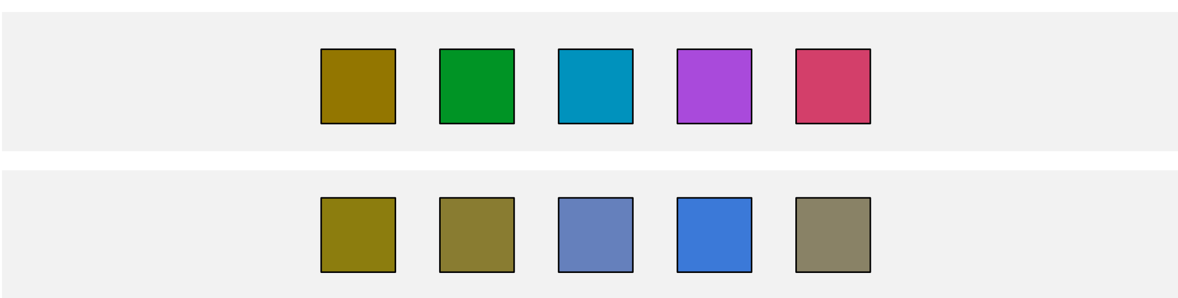


Figure 6.18: The top row shows a colour palette that attempts to vary hue evenly, in order to allow decoding of nominal data values, while holding chroma and luminance constant, in order to avoid decoding of ordinal data values. The bottom row shows how the palette appears to a viewer with red-green CVD.

Figure 6.19 shows another big problem with the palette in the top row of Figure 6.18. An attempt to hold chroma constant for a range of hues will fail on most standard computer displays because they cannot produce high chroma for all hues (Section 6.5).

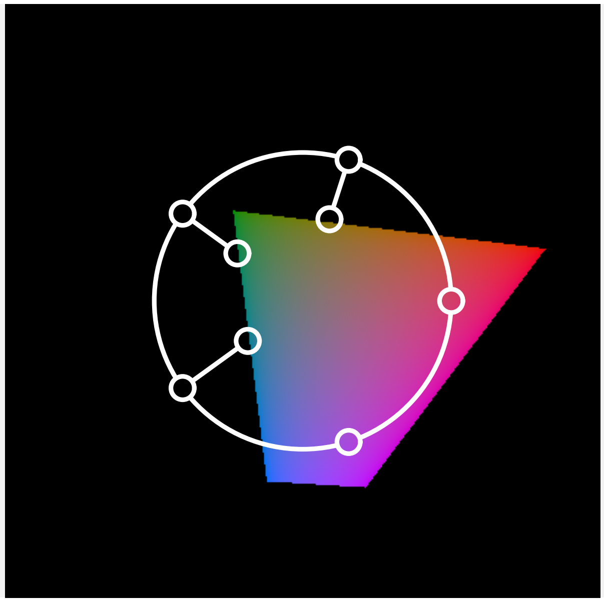


Figure 6.19: A view of the colour palette from Figure 6.18 relative to the range of colours that are possible on a standard computer monitor. The palette attempts to produce the same chroma for different hues, but only a much lower chroma can actually be achieved for some hues. The result is that the brown, green, and blue in the colour palette are less colourful than the purple and red.

Given the difficulty of selecting an effective colour palette, there is a good argument for making use of pre-existing colour palettes that have been designed by experts. For example, the Color-Brewer web site provides tools for choosing colour palettes for nominal and ordinal encodings. Another example is the Okabe-Ito palette, which is specifically designed for encoding nominal data (up to 9 categories) using colours that allow effective decoding by viewers with all types of CVD (Figure 6.20).[77]

> Selecting a colour palette is a complex task so it is a good idea to make use of an existing colour palette that has been designed by an expert.

6.11 Case study: Diverging colour palettes

We have data on the total number of points that were scored and conceded in World Cup matches for Tier One nations at each World Cup. Table 6.1 shows a sample of these data.

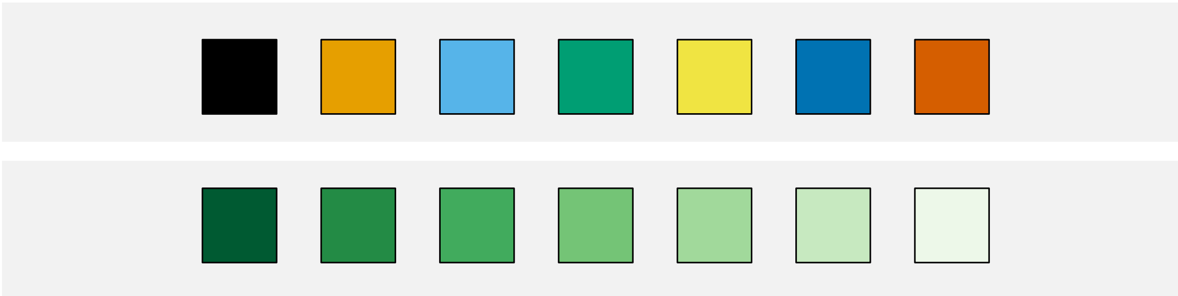


Figure 6.20: The top row shows the first 7 colours from Okabe-Ito. The bottom row is 7 sequential “Greens” from ColorBrewer.

This an aggregated version of the data set that we saw in Table 4.1.

Table 6.1: A table of the total number of points scored and conceded by Tier One nations in Rugby World Cup matches at each World Cup. Each row represents the total points scored by one team in one World Cup. There are 100 rows in total, with just the first 6 rows shown here.

team	year	scored	conceded	diff
Italy	1987	22	95	-73
Italy	1991	27	67	-40
Italy	1995	51	52	-1
Italy	1999	10	168	-158
Italy	2003	22	97	-75
Italy	2007	30	94	-64

Figure 6.21 shows a heatmap of the points differential (points scored for minus points scored against) for Tier One nations at each Rugby World Cup. The data symbols are rectangles, the teams are encoded as the vertical **positions** of the rectangles, and the years are encoded as the horizontal **positions** of the rectangles. These are both effective encodings and we can easily **decode** to teams or years. The points differentials are encoded as the **colour** of the rectangles, but in a slightly complicated way. Positive data values encode to a shade of green, while negative values encode to a shade of brown. In both cases, we have a constant **hue**, with increasing **chroma** and decreasing **luminance** as the data values get further away from zero.[78]

The use of two **hues** creates two different groups (positive and negative values), while the changes in **chroma** and **luminance** convey changes in absolute magnitude (if only for ordinal comparisons).[79] This means that we can easily identify positive versus negative points differentials *and* we can easily decode larger or smaller points differentials (within each of those groups).

> A diverging colour palette is effective because it creates two distinct visual features: a nominal hue to differentiate between two groups and changes in chroma and/or luminance to encode ordinal values within each group.

6.12 Case study: The rainbow

In some fields of research, for example medical imaging and geographic imaging, a common task is to visualise numeric data values on a regular spatial grid. Figure 6.22 shows an example,

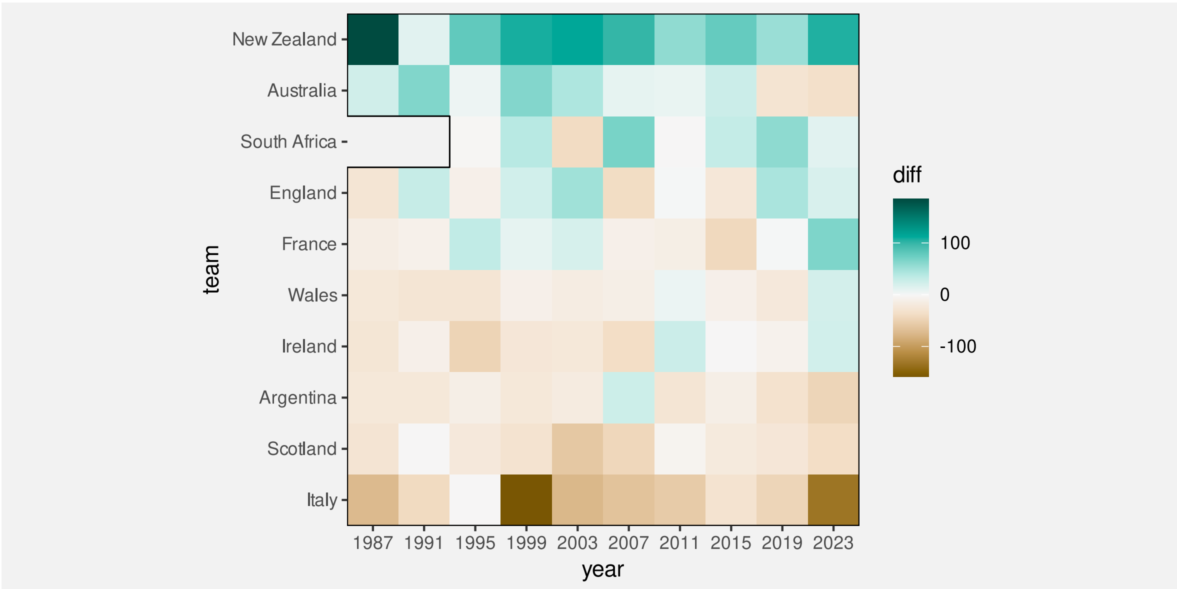


Figure 6.21: A heatmap of the total points differentials for Tier One nations at individual Rugby World Cups. South Africa was excluded from the 1987 and 1991 tournament due to its apartheid policies.

where the grid is a set of longitudes and latitudes in the vicinity of New Zealand and the data values are measures of wind strength.

A common approach is to encode the numeric data values using colour[80] and it is also common to use a rainbow colour palette for that encoding (the image on the right of Figure 6.22). However, there are several problems with this approach.[81]

Figure 6.23 shows the spectrum of colours for rainbow palettes.[82] This shows that the colours within a rainbow palette differ mostly in terms of hue and we have already established that hue is not a good visual feature for encoding quantitative data values (Section 3.4 and Section 6.1).

At the same time, The middle row of Figure 6.23 shows that the rainbow palette also varies in terms of luminance. We know that luminance can be an effective encoding for ordinal data values (Section 6.2), but the luminance changes in a rainbow palette are not monotonic. There are regions of almost constant luminance plus regions where luminance decreases followed by regions where luminance increases.[83] This makes a rainbow palette a poor encoding even for showing the order of numeric data values.

Finally, the changes in luminance and chroma within the rainbow spectrum produce bands of colours (red, yellow, green, cyan, blue, magenta). In other words, the spectrum appears discrete rather than continuous. This can result in a decoding that *adds* structure (discrete levels) to numeric data values that are in reality continuous.[84] For example, Figure 6.22

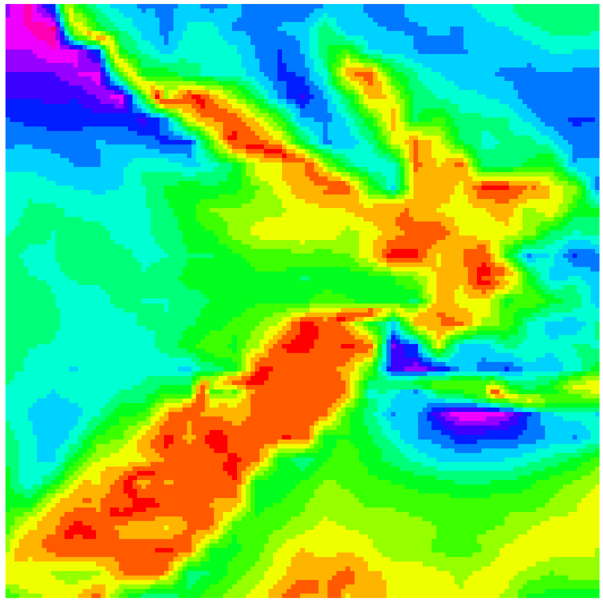


Figure 6.22: A snapshot of wind strength in the vicinity of New Zealand on September 14 2024. This uses the same data as Figure 2.4, but does not show wind direction. The image uses a rainbow colour palette (from red through green and blue to violet; see Figure 6.23).

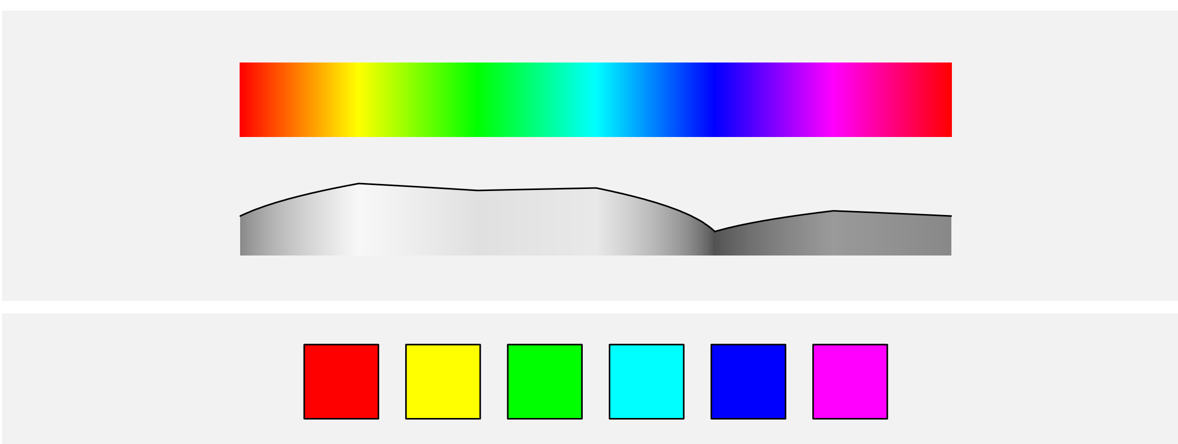


Figure 6.23: The top row shows the spectrum of colours from which a classic *rainbow* colour palette is taken. The middle row shows the changing luminance of the colours in the rainbow spectrum. The luminance of the blue colours is considerably less than the luminance of the yellows and greens and even the reds. The bottom row shows an example rainbow palette of 6 colours taken from the rainbow spectrum.

suggests three main regions–red-yellow, green-cyan, and blue-magenta—but the underlying data values vary more smoothly than that.

6.13 Case study: Viridis

One of the arguments for using a rainbow colour palette is that viewers like that it is colourful. We also saw in Section 6.9 that varying hue in addition to luminance can help to increase the capacity of an ordinal colour scale.[85]

The **viridis** colour palette, shown in Figure 6.24 (on the right), is an attempt to fix the problems of the rainbow palette, while still retaining a significant range of hues. Compared to Figure 6.22, it is much easier to decode regions of stronger winds and regions of weaker winds. This is because the regions of stronger winds are darker than the regions of weaker winds in Figure 6.24, rather than just being different hues in Figure 6.22.

At the same time, the image on the left of Figure 6.24 shows a colour palette that only varies in terms of luminance and we can see that it is easier to see some details in the more colourful viridis map, such as the variations in low wind strength (the yellows) over New Zealand.

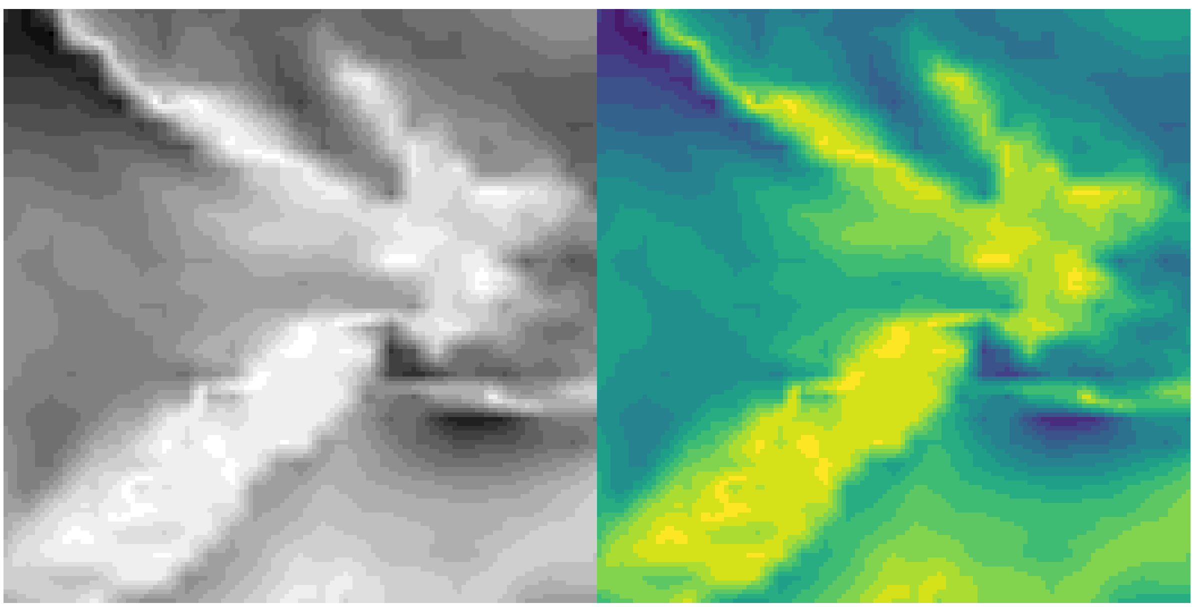


Figure 6.24: A snapshot of wind strength in the vicinity of New Zealand on September 14 2024. This uses the same data as Figure 2.4, but does not show wind direction. The image on the left encodes wind strength as shades of grey (darker is stronger) and the image on the right uses a viridis colour palette (Figure 6.25).

Figure 6.25 shows the spectrum for viridis colour palettes.[86] We can see that there is still a large range of hues, but the change in luminance is now monotonic, which at least allows decoding of ordinal data values. In addition, the range of hues has been selected to be appropriate for viewers with red-green colour blindness (Section 6.7).[87]

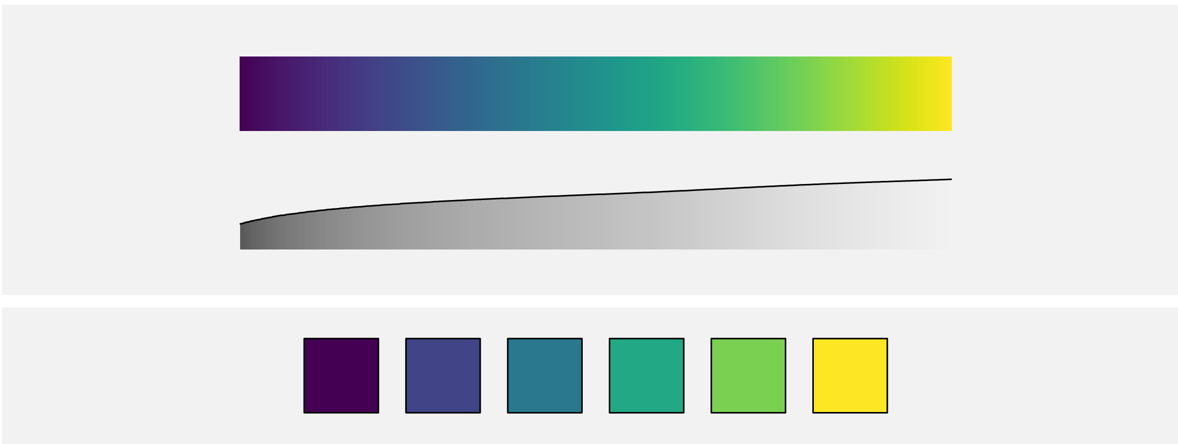


Figure 6.25: The top row shows the spectrum of colours from which a viridis colour palette is taken. The middle row shows the changing luminance of the colours in the viridis spectrum. The luminance change is monotonic. The bottom row shows an example viridis palette of 6 colours taken from the viridis spectrum.

6.14 Summary

Colour is really three visual features: **hue**, **chroma**, and **luminance**.

Hue is excellent for encoding **nominal** data values, though it has a limited capacity.

Chroma and **luminance** can be used to encode **ordinal** data values (as well as nominal data values), but they have even lower capacity.

When we encode data values as colours there are several caveats:

- The decoding of data values from colours is affected by surrounding colours and the size of the data symbol.
- Approximately 10% of viewers are unable to differentiate between red and green hues with similar chroma and luminance.

Selecting which colours should be used to encode data values is difficult to get right and a good solution often involves varying all of hue, chroma, and luminance at once.

Consequently, it is usually a good idea to make use of pre-existing colour palettes that have been carefully designed to avoid most problems.

Notes

[67] The three-dimensional experience of colour arises from the fact that there are three different sorts of cones in the retina (see Section 2.1), each sensitive to a different range of light wavelengths: The L cones respond to long wavelengths (reds); the M cones respond to medium wavelengths (greens); and the S cones respond to short wavelengths (blues). The neural connections that combine the messages from the retina are also divided into three pathways: light-dark, which represents the sum of L, M, and S cones; red-green, which represents the difference between L and M cones; and blue-yellow, which represents the difference between S cones and the sum of L and M cones. For a more detailed description, see Ware (2021).

[68] Schloss, Leggon, and Lessard (2021) and Schloss et al. (2023) are examples of studies that have demonstrated that viewers naturally decode darker colours to larger values.

[69] The stance taken in this book, that chroma and luminance are only appropriate for qualitative (including ordinal) data, is more rigid than in many other places. Chroma and luminance are often included as an appropriate choice for encoding *quantitative* data values as well, albeit a poor one. For example, William S. Cleveland and McGill (1984) includes chroma (as *saturation*) and Munzner (2014) includes both luminance and chroma (as saturation) in their rankings of the accuracy of different visual features.

[70] This relative perception of colours is useful for being able to adapt to a wide range of lighting conditions. See Ware (2021) for a more detailed discussion.

[71] Stone (2012) describes "spreading", which means a blending or averaging of colours over small areas *by the visual system*: "In designing colors for digital visualization systems, one of the most critical factors is the interaction between size and color appearance."

[72] Figure 6.9 shows the extent of the *optimal colour solid*, which is essentially the colours that it is possible to produce by reflecting a light source off a surface that has perfect reflectance. This colour solid was determined assuming specific illumination conditions (D65), and assuming a standard human observer.

The colours are shown in CIE $L^*u^*v^*$ space (*aka* HCL space), which is (roughly) *perceptually uniform* so that perceptual differences between pairs of colours that are the same distance apart within the space are perceived as having the same visual difference.

The colours shown are produced using sRGB specifications, so many of the colours, especially at high chroma, are not realistic.

[73] Figure 6.10 shows the gamut of sRGB colours in CIE $L^*u^*v^*$ space (*aka* HCL space).

[74] In colour theory, hues that are close together on a colour wheel are referred to as *analogous* and hues that are opposite each other are *complementary*.

[75] This is more commonly known as *red-green colour blindness*. Blue-yellow colour blindness is also possible, but much rarer. There are also four distinct categories of red-green colour blindness, though the effect on the colour perceived is similar in all cases. The images used in this book simulate deuteranomaly (the most common form of CVD).

The approximate figure of 10% is from Ware (2021).

[76] This idea is described and demonstrated in Bartolomeo et al. (2025).

[77] The ColorBrewer tools are described in Harrower and Brewer (2003).

Okabe and Ito established the Color Universal Design organization, but the associated web site is no longer available. Okabe and Ito (2008) provides a general discussion of colour-blindness, including the palette used in Figure 6.20. An earlier piece of work that produced a palette of just 4 colours is described in Ichihara et al. (2008).

Zeileis and Murrell (2023) provides an overview of a large number of different palettes, within the context of producing data visualisations in R.

[78] This colour palette is based on the *BrBG* diverging palette from ColorBrewer.

[79] In contrast to a *diverging* colour palette, a colour palette for nominal data is often called a *qualitative* palette and a palette for ordinal or qualitative data is called a *sequential* palette.

[80] The resulting image is often referred to as a *pseudo-colour* image.

[81] The problems with rainbow palettes have been well-known for decades and attempts have been made to stamp out the use of rainbow colour palettes, but they remain popular in some fields of research (Borland and Taylor II 2007; Gołębiowska and Çöltekin 2022).

[82] The rainbow palette is typically a fully-*saturated*, maximum-*value* spectrum of hues—the most colourful and brightest colours that a typical computer monitor can produce.

Saturation and value are essentially less accurate versions of chroma and luminance. For example, a fully-saturated, maximum-value yellow has a higher chroma and luminance than a fully-saturated, maximum-value blue.

The rainbow spectrum is basically a path along the boundary of the colours in Figure 6.10, visiting the most extreme points for different hues. This path clearly is not going to have a simple monotonic change in either luminance or chroma.

[83] A rainbow palette is a colour encoding that is non-linear in an unhelpful way (Section 6.8).

The representation of luminance in Figure 6.23 also shows an example of a 3d illusion (Section 2.10). The curved line combined with the gradients of grey tempt us to see a curved wall of constant height that is receding from us.

[84] Heer and Stone (2012) discuss that viewers tend to group colours by discrete colour names and that those colour groups help to explain how colour palettes are decoded.

[85] Liu and Heer (2018) provide evidence that colour palettes that vary in hue as well as luminance and/or chroma (multi-hue palettes) lead to faster and more accurate decoding than palettes with a constant hue (single-hue palettes). Ware, Stone, and Szafir (2023) make a similar argument.

[86] There is no official publication that explains the development of the viridis colour spectrum, but there is a video on YouTube that is very good (Smith and Walt 2015).

[87] Nuñez (2018) take things a step further with a *cividis* colour palette that provides a CVD-safe version of viridis that also has a wider and *linear* variation in luminance (in Figure 6.25, we can see that the changes in luminance are non-linear and do not span the full range of luminance).

7 Congruent Encodings

In previous chapters, we have seen that, part of the reason why some well-known plots, like bar plots, are effective is because they encode data values to the basic visual features of a data symbol. For example, encoding the number of youth offenders as the **length** of a bar in a bar plot (Figure 5.1). We have also seen that basic visual features like **length** are effective because we can accurately decode data values from those visual features.

The purpose of this chapter is to show that, in order to understand the effectiveness of a data visualisation, we need to consider more than just the accuracy or capacity of our encodings from data values to basic visual features.

7.1 Implicit decodings

Figure 7.1 shows two lines, one longer than the other. We can decode information from these lines without any explicit encoding of data values. For example, we can decode that one line is twice as long as the other without knowing anything about what "twice" represents in terms of data values. In other words, there is an **implicit decoding** from the lengths of the lines to data values (Figure 7.2).[88]

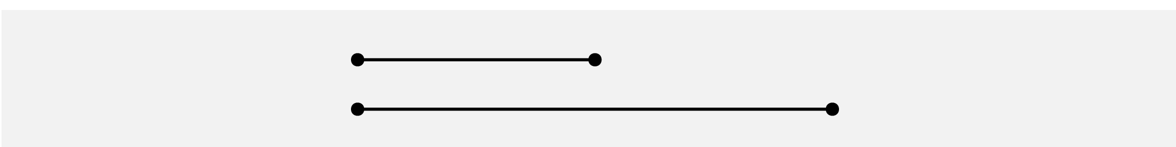


Figure 7.1: We can decode the fact that one line is twice the length of another line without there having to be any explicit encoding of data values to the lengths of the two lines.

When an implicit decoding exists, we can produce a more effective encoding by building upon the implicit decoding. We can explicitly encode data values as visual features in such a way that the decoding is **congruent** with the implicit decoding, in the sense that the implicit decoding leads back to the same data values that we explicitly encoded (Figure 7.3).[89]

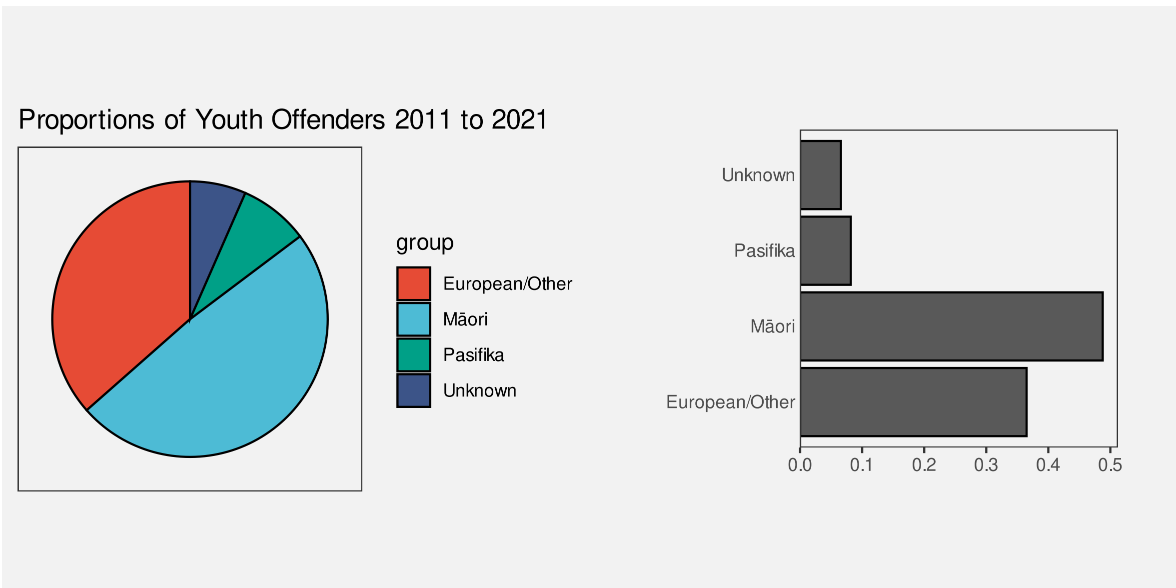
Figure 7.2: Some visual features have implicit encodings, which means that we can decode information from the visual features without having explicitly encoded any data values.

Figure 7.3: If we explicitly encode data values as visual features in a way that is consistent with an implicit decoding, we can more effectively decode data values from visual features.

7.2 Part-to-whole

Figure 7.4 shows a pie chart of the proportions of total youth offenders between 2011 and 2021 in different ethnic groups and a bar plot of the same data. We learned in Section 3.5 that encoding **quantitative** data values, like proportions, as **lengths**, like in the bar plot, is more accurate than encoding the proportions as **angles** or **areas**, like in the pie chart.

However, the superior representation of the fact that Māori offenders make up approximately 50% of the total is the pie chart rather than the bar plot. Although it is easy to determine from the length of the bar and the scale on the x-axis in the bar plot that the bar for Māori is just below 0.5, it is even easier, without any guides at all, to see that the light blue wedge in the pie chart is just under half a circle. It is also easy to see that the red wedge is close to a third of a circle.[90]



(a) A pie chart.

(b) A bar plot.

Figure 7.4: Data visualisations of the proportion of offenders from different ethnic groups.

The reason why the pie chart in Figure 7.4 is more effective than a bar plot for conveying simple proportions like a quarter or a third or a half is because the wedges in a pie chart have an implicit decoding to fractions of a whole. A semi-circle has a strong visual association with *half* of a circle and thirds and quarters are similarly evocative.

> One reason why a pie chart can be effective is because segments of a pie vividly convey simple fractions of a whole.

Figure 7.5 shows that if we place a bar representing a proportion bar within a reference rectangle that corresponds to a proportion of 1, then we can also create a part-to-whole congruence with bars, though the effect is not as strong as it is for pie wedges.

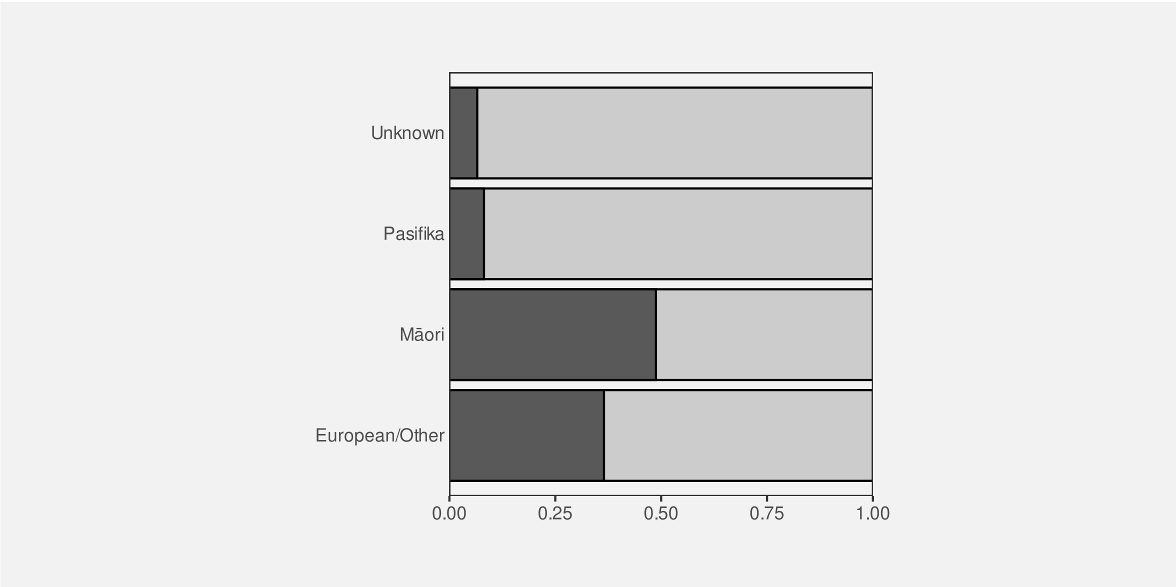


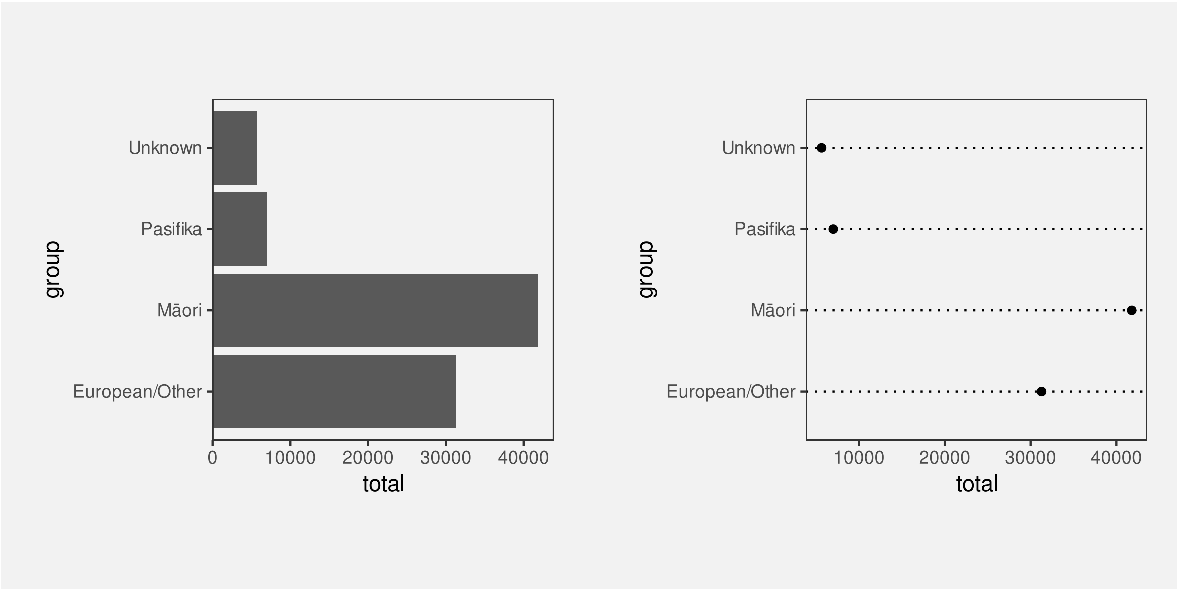
Figure 7.5: A bar plot of the proportion of offenders from different ethnic groups with reference rectangles to enhance the part-to-whole relationship.

7.3 More is more

We saw in Section 3.3 that visual features like **length** and **area** are appropriate for decoding quantitative values. For example, in Figure 7.6, the lengths of the bars are appropriate for decoding the total number of offenders in different ethnic groups.

Part of the reason why **length** and **area** work so well is because a **longer** bar implicitly conveys the idea of *more*.[91] By contrast, the **position** of data points in a dot plot of the same data, as shown in Figure 7.6, do not inherently convey the same sense of *more*. In a bar plot, the data symbols that correspond to larger data values are actually larger data symbols; the presentation of larger data values is more abstract in a dot plot.

Another way that **length** can convey information more naturally than **position** is visualising negative values. For example Figure 7.7 shows the total points differential (points scored minus points conceded) for Tier One nations at Rugby World Cups (see Table 4.1). The *direction* of the bars very clearly indicates the change from positive points differential to negative points differential.



(a) Bars.

(b) Dots.

Figure 7.6: A bar plot and a dot plot of the total number of offenders for different ethnic groups.

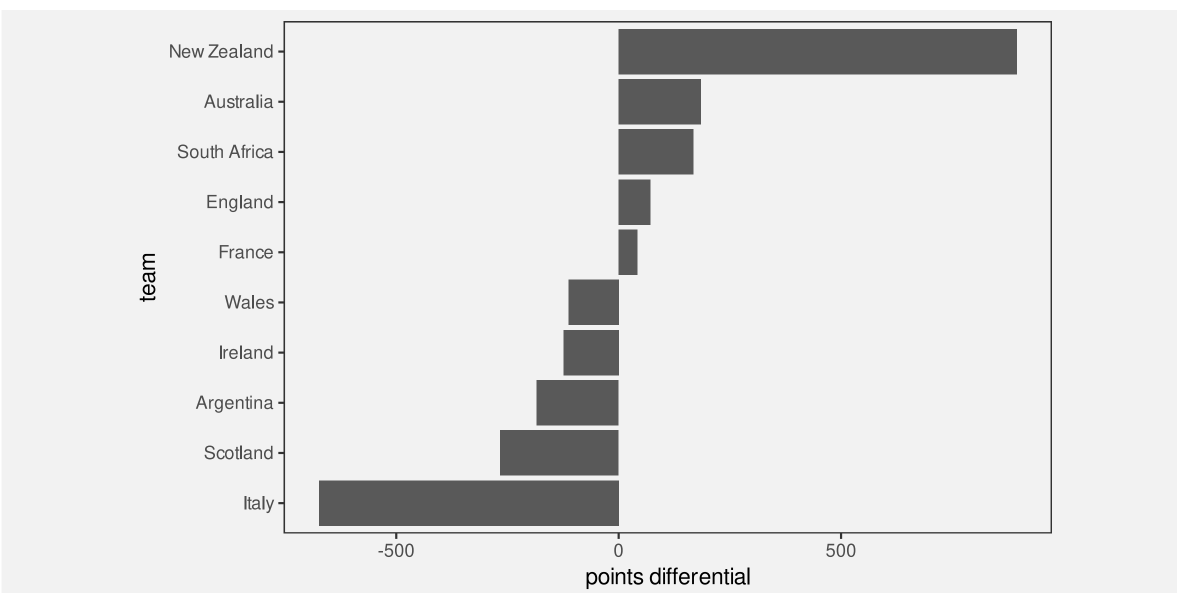


Figure 7.7: A bar plot of the total points differential for Tier One nations at Rugby World Cups.

By comparison, the change from positive to negative points differential is much more abstract in a dot plot of the same data (Figure 7.8).

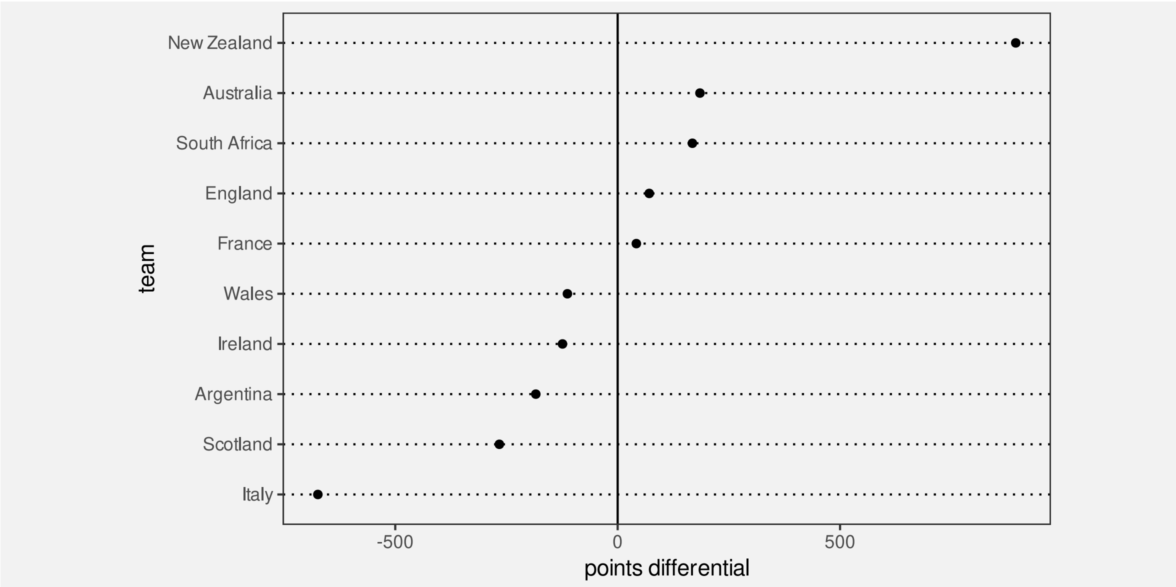


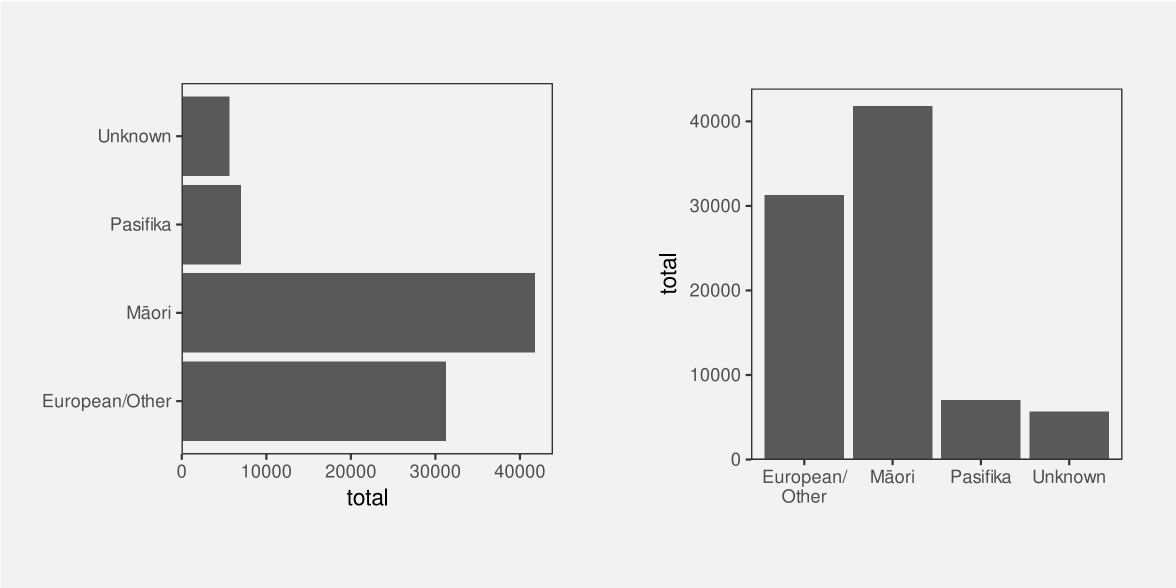
Figure 7.8: A dot plot of the total points differential for Tier One nations at Rugby World Cups.

> One reason why bar plots are effective is because larger data values are represented by larger data symbols.

A more subtle effect can be seen in Figure 7.9, which shows two different bar plots of the total number of offenders in different ethnic groups. In this case, the vertical bars are more natural for conveying count data like this because of the natural correspondence with stacks of items.[92]

7.4 Same is same

In Section 2.5 we saw that the visual system is very sensitive to *differences*, particularly when everything else remains the *same*. This means that encoding *changes* in data values to *changes* in data symbols will be effective because the visual system will notice the differences in the data symbols. More specifically, if a **visual feature** of a data symbol changes then the visual system will notice that change. Furthermore, if a visual feature of a data symbol changes while all other visual features of the data symbol remain the *same*, the visual system will notice the change more easily (Section 2.5). Put another way, visual differences are implicitly decoded



(a) Horizontal bars.

(b) Vertical bars.

Figure 7.9: Data visualisations of the number of offenders from different ethnic groups.

as changes in the data values *and* visual constancy is implicitly decoded as no change in the data values.

For example, in a bar plot like Figure 3.1, the data symbols are bars and only the **lengths** of the bars and the vertical **positions** of the bars are different. The "widths" of the bars and the **colours** of the bars remain the same.

> Another reason why bar plots are effective is because the bars differ in length, but are otherwise the same.

7.5 Dissonant encodings

The fact that **length** and **area** implicitly convey amounts means that care must be taken *not* to confuse that association.[93] For example, Figure 7.10 shows the number of times each team entered the opposition's final third during a match at the 2023 Women's World Cup.[94] The final third of the pitch is shown as darker green regions, broken into five different zones, with a number of entries for each zone and each team represented by a line with an arrow.

In Figure 7.10, longer lines correspond to a larger number of entries. However, the bars start at the halfway line on the pitch rather than at the final third of the pitch. This means that the zero entries by South Africa into the second-from-right zone is encoded as a non-zero-length line. Apart from making comparisons of the line lengths very inaccurate, this creates a confusing visual effect because the viewer is expected to decode a non-zero length to a value of zero.

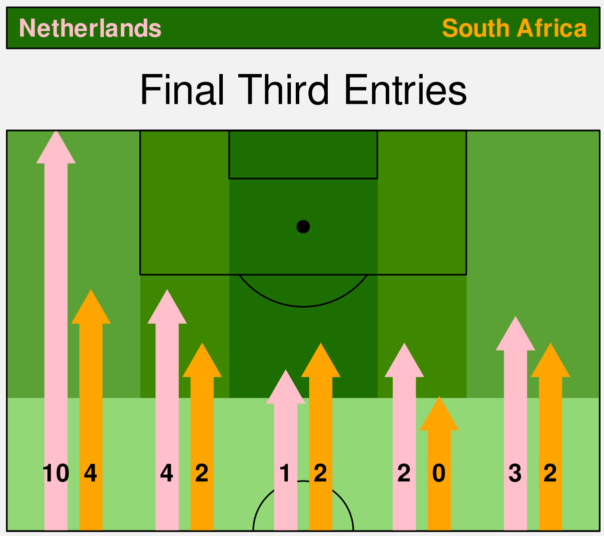


Figure 7.10: The number of times each team entered the opposition's final third in the Women's World Cup match between the Netherlands and South Africa.

For any example of **congruence**, where a visual feature implicitly decodes to a property of the data, there is an opportunity for **dissonance** if the visual feature is used inappropriately. We can improve the effectiveness of a data visualisation by taking advantage of visual congruence and we can reduce the effectiveness of a data visualisation if we create visual dissonance.[95]

If we fail to make our explicit encodings congruent with any implicit encodings—if we create a **dissonant** encoding—we can make it harder and more confusing to decode a data visualisation because there is more than one possible decoding (Figure 7.11).

We saw in Section 7.4 that we want to encode differences in data values to differences in visual features *and* keep everything else the same. Failing to follow that approach will make it harder to perceive changes in the data values. However, a worse error, and another source of visual dissonance arises if we change visual features *without* any underlying change in the data values.[96]

Figure 7.12 shows a data visualisation of the relative number of people aged over 100 for females versus males.[97] There are numerous problems with this data visualisation, but here we will just focus on the horizontal **position** of the cake images. The cakes for females are positioned to the left of the cakes for males, which is acceptable because that difference in

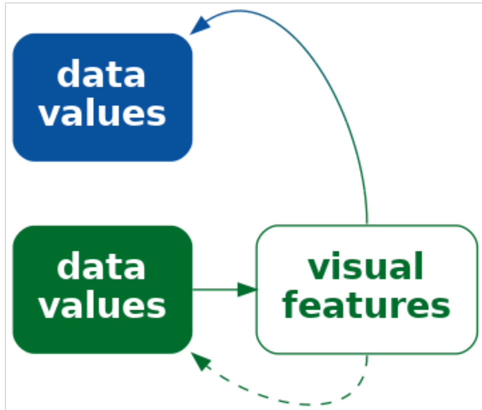


Figure 7.11: If we explicitly encode data values to visual features, but the visual features have an implicit encoding to different data values, we can cause confusion and make it harder to effectively decode data values from visual features.

position reflects a difference in the data—males versus females. However, within each gender, the cakes are also staggered a small amount to the left then to the right. This visual difference in the cake positions does not correspond to any difference in the data values. We are tempted to seek meaning from the the left-to-right staggering of the cakes when no meaning exists.

> A data visualisation is *not* effective if it contains encodings that are dissonant because that can cause distraction and slower and/or incorrect **decoding** of data values from visual features.

7.6 Case study: Time's arrow

Figure 7.13 shows the results of a 2025 poll that asked what was the best thing that the Trump administration had done, comparing results in February to results in September. In terms of encodings, this bar plot makes good use of (vertical) **position** and (horizontal) **length** to encode different policy issues, the difference in dates, and the percent of respondents who voted for each policy. However, when we look at the top two bars, there is a temptation to read the pairs of bars in order from top to bottom.[98] If we do not attend carefully to the legend, we may perceive a *decrease* in the (relative) popularity of the immigration policy and an *increase* in the popularity of DOGE. This is a subtle example of visual dissonance.

Figure 7.14 shows another version of the bar plot, with the bars running vertically. In this plot, the natural reading from left to right leads to the correct interpretation: the (relative)

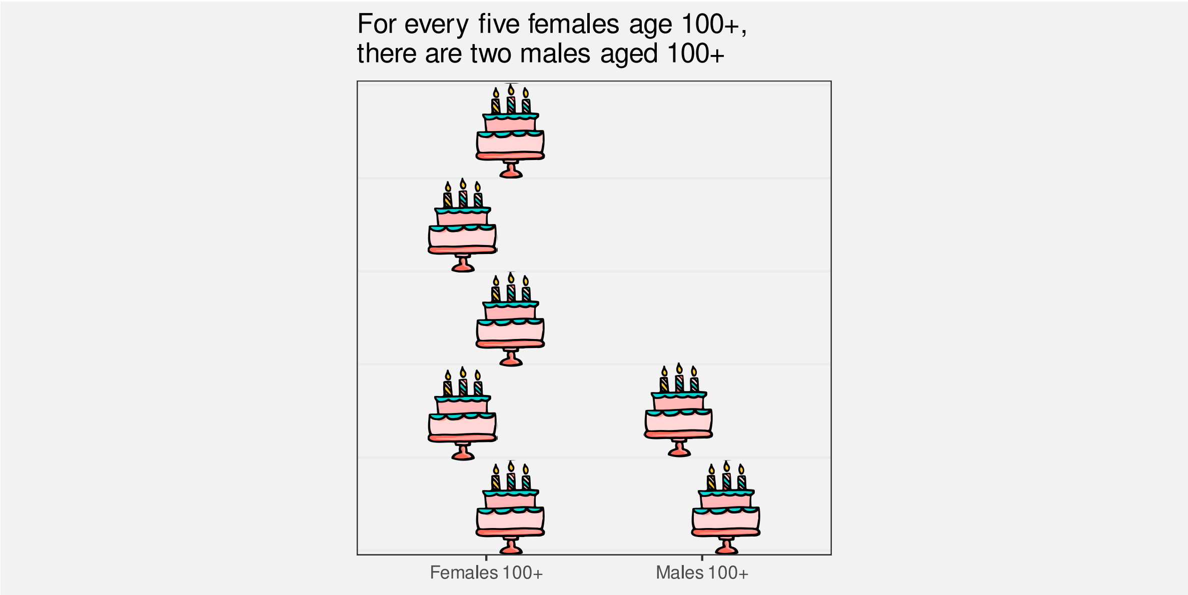


Figure 7.12: A data visualisation of the number of females aged over 100 relative to the number of makes aged over 100.

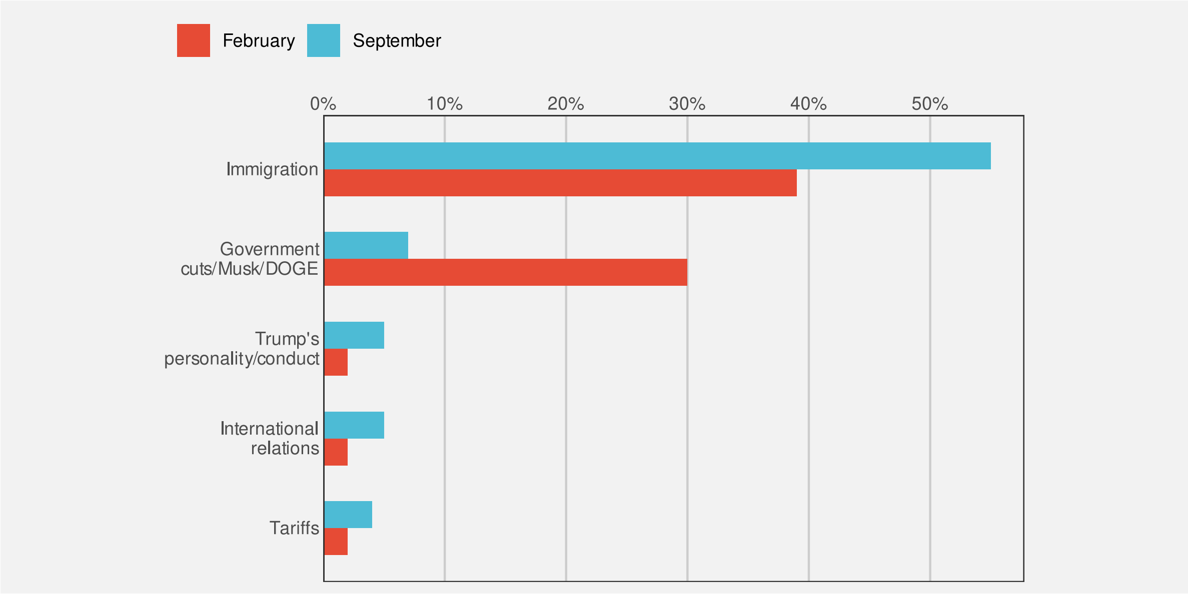


Figure 7.13: Bar plot of most popular Trump policies over time.

popularity of DOGE has declined while the popularity of the immigration policy has increased. This is an example of visual congruence.

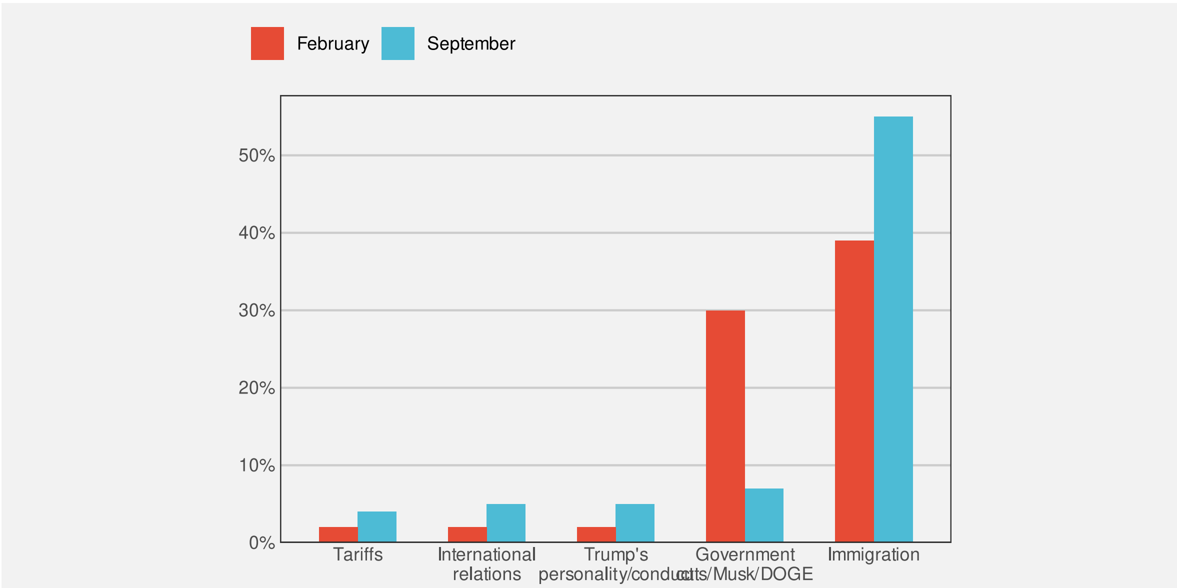


Figure 7.14: Bar plot of most popular Trump policies over time.

7.7 Case study: Jittering

We have data on the the number of points that were scored in every World Cup match by Tier One nations (the top 10 rugby-playing nations). Table 4.1 shows a sample of these data.

Figure 7.15 shows a dot plot of the number of points scored by teams from different hemispheres at Rugby World Cup matches. In this data visualisation, the data symbols are dots, the number of points scored is encoded as the horizontal **position** of the dots, and the hemisphere is encoded as the vertical **position** of the dots. We saw a similar visualisation of these data in Figure 4.5, which demonstrated the problem of overplotting of points that share the same data value.

In Figure 7.15, to ameliorate the overlapping of data points, a random amount of "jitter" has been added the vertical position of the dots. The filled interior of each dot is also semitransparent so that we can see where some points still overlap.

The dot plot in Figure 7.15 is effective for **decoding** to the number of points scored because each data value is encoded as the horizontal position of a single data symbol. Similarly, we can easily decode the hemisphere from each point. The proximity of the data symbols from the same hemisphere means that it is easy to perceive two "rows" of points (Section 2.7).

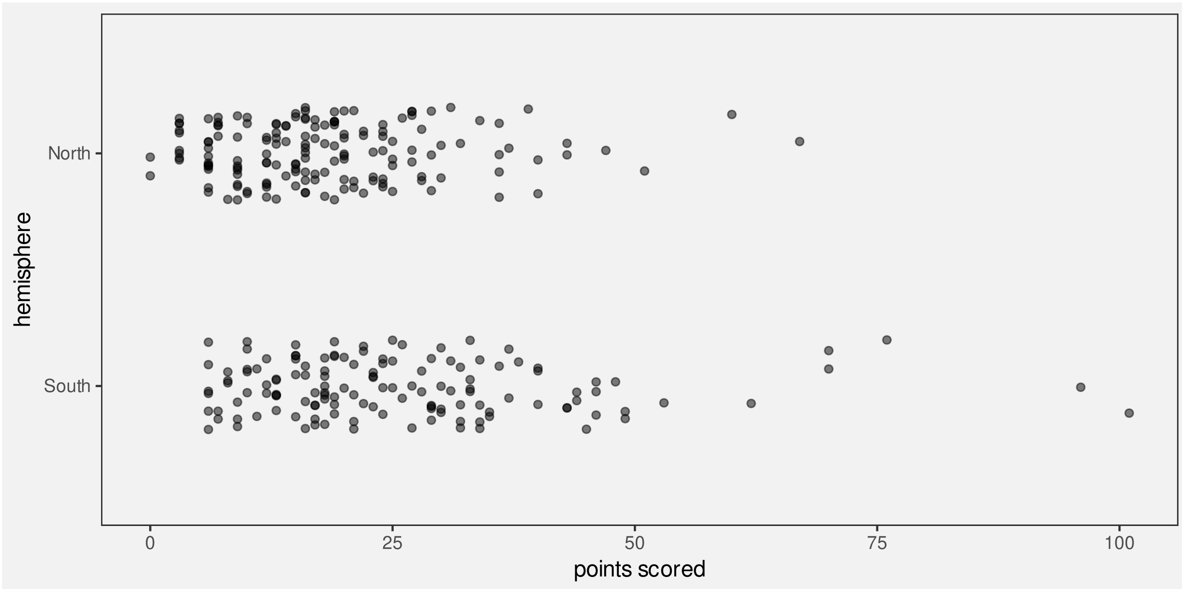


Figure 7.15: The number of points scored in Rugby World Cup games by teams from the northern and southern hemispheres.

On the other hand, the jittering means that there are visual differences between the vertical positions of the dots, within the same hemisphere, that do not correspond to any differences in data values. For example, there are two dots representing matches where a team from the Northern hemisphere scored zero points. The data values for these two dots are identical (zero points scored and northern hemisphere), but they are visually different. The randomness of the jittering may help to dilute any temptation to seek meaning from the artificial visual difference, but the addition of jittering is a potential source of confusion.

> Jittering data symbols is effective because it reduces the amount of overlap amongst data symbols, but jittering may also reduce the effectiveness of a data visualisation because it introduces visual dissonance.

7.8 Summary

The effectiveness of a data visualisation may depend on more than just the accuracy and capacity of visual features.

A **congruent** encoding is one where data values are *explicitly* encoded in a way that is consistent with an *implicit* decoding of the visual feature.

A data visualisation will be more effective if it is visually congruent, for example, data symbols are larger for larger data values or data symbols only change if the data values change.

A **dissonant** encoding is one where data values are *explicitly* encoded in a way that is *inconsistent* with an *implicit* decoding.

A data visualisations will be less effective if it is visually dissonant.

Notes

[88] These implicit encodings are part of the reason why different visual features are appropriate for encoding quantitative verus qualitative data values. The length of a data symbol is appropriate for encoding quantitative data values because we implicitly decode amounts from length.

[89] Tversky, Morrison, and Betrancourt (2002) describe a *congruence principle*: "the structure and content of the external representation should correspond to the desired structure and content of the internal representation," where the "external representation" is a data visualisation and the "internal representation" corresponds to an implicit decoding.

Bertini, Correll, and Franconeri (2020) discuss measures of effectiveness beyond accuracy and capacity and refers to the congruence principle above as well as the idea of *congnitive fit*: "performance on a task will be enhanced when there is a cognitive fit (match) between the information emphasized in the representation type and that required by the task type" (Vessey 1991). The latter was focused on a comparison of plots versus tables (plots for spatial comparisons and tables for more complex cognitive tasks involving symbolic reasoning), but it is analogous to the idea of congruent encodings.

[90] This defence of pie charts has existed for a very long time (Eells 1926a) and continues to receive empirical support in more recent times (Simkin and Hastie 1987).

[91] Kosslyn's Principle of Compatibility states that "A message is easiest to understand if its form is compatible with its meaning", plus explicitly "More Is More" (Kosslyn 2006).

Wilke (2019) has a similar Principle of Proportional Ink, which he attributes to Bergstrom and West (2017). A corollary is that bars in a bar plot should start at zero.

[92] Several authors make an argument for vertical bars being preferred because they correspond naturally to accumulation and growth (e.g., Few 2004).

[93] This is related to visual illusions (Section 2.10). There are decodings that the visual system performs whether we want it to or not, so we want to avoid conflicting with what the visual system does automatically.

[94] This data visualisation is an adaptation of a graphic shown during the live television coverage of the match.

[95] The idea of visual dissonance is just a logical inversion of the idea of visual congruence. This idea is not named in any literature that I could find, though *semantic clashes* (Bertini 2024) perhaps comes closest.

[96] Kosslyn's *Principle of Informative Changes* states that "People expect changes in properties to carry information" (Kosslyn 2006).

[97] This plot is based on a data visualisation from the New Zealand Listener October 14-20 2023.

[98] The idea for this data visualisation, and whether to question the vertical ordering, comes from a PolicyViz Newsletter. The original data source was a Washington Post poll).

8 Encoding Summaries

In Chapter 3 we described a data visualisation as a encoding that **encodes** data values as the visual features of a data symbol (Figure 3.2). For example, a bar plot, like Figure 3.1, encodes the total number of youth offenders to the **lengths** of bars.

We also emphasised the importance of the **decoding** from visual features to data values. For example, the effectiveness of the bar plot in Figure 3.1 relies on our ability to accurately decode the lengths of the bars, particularly relative to each other.

In the simple scenario of a bar plot, the data values are counts (Table 3.1), which are encoded as the lengths of bars, which in turn allows us to decode lengths to retrieve the counts (Figure 3.6). Each **raw data value** (count) is encoded to the length of a bar.

In this chapter, we explore some data visualisations that are effective because they encode **data summaries** instead of raw data values.[99]

8.1 Box plots

Figure 8.1 shows box plots of the number of points scored by Tier One nations at Rugby World Cup matches (Table 4.1), with a separate box plot for teams from different hemispheres.[100] This is an effective data visualisation for comparing the distributions of points scored between hemispheres because we are able to easily see differences between the median values of the countries (the thick vertical lines) and differences between the interquartile ranges of the countries (the horizontal rectangles). For example, we can see that the median points scored by southern hemisphere teams is larger than the median for northern hemisphere teams and the interquartile range of points scored is also wider for southern hemisphere teams.

Box plots are quite different from bar plots and dot plots because the data symbol is a much more complicated shape: there is a vertical line to represent the median, a rectangle to represent the interquartile range (lower quartile to upper quartile), and horizontal lines to represent the extremes of the data values, from the the quartiles to the minimum and maximum values. Points are also included for “outliers” if data values lie further than 1.5 times the interquartile range beyond the quartiles.

Another important difference with a box plot is that it encodes **data summaries**, rather than raw data values, to visual features of the data symbol (Figure 8.2). For example, the median

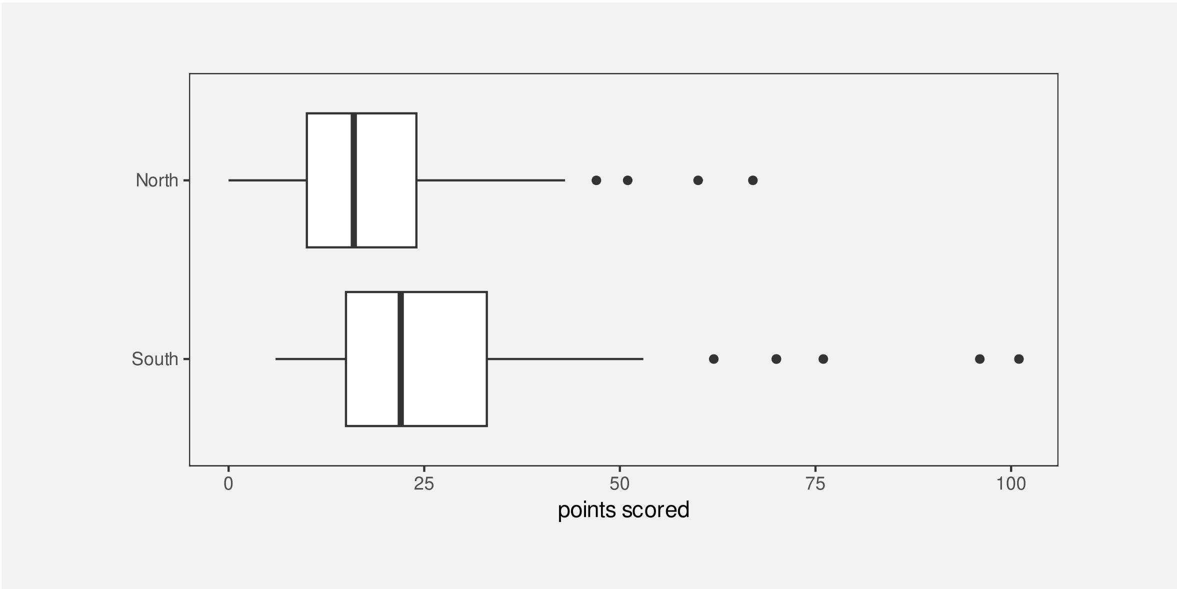


Figure 8.1: Box plots of the number of points scored in all Rugby World Cup games by Tier One nations.

values are encoded to the horizontal **positions** of thick vertical lines and the interquartile ranges are encoded to the **lengths** of horizontal rectangles.

To be explicit, the data values that are being **encoded** as visual features in Figure 8.1 are *not* the raw data values in Table 4.1; the values that are being encoded to visual features in Figure 8.1 are the **data summaries** shown in Table 8.1. The data summaries in Table 8.1 are the values that we can **decode** from the boxplots in Figure 8.1.[101]

Table 8.1: A table of the five-number summaries for the number of points scored by Tier One nations in Rugby World Cup matches, along with the country name. These data summaries are the values that are encoded to visual features to produce the box plots in Figure 8.1.

	minimum	lower quartile	median	upper quartile	maximum
South	6	15	22	33	101
North	0	10	16	24	67

Figure 8.1 is effective for allowing us to compare summaries like the medians between different hemispheres because, although the box plot data symbol is relatively complicated, a box plot still just encodes the data summaries to simple visual features. We know from Chapter 3 that **position** and **length** are effective for **decoding** the data summaries from the visual features (Figure 8.2).

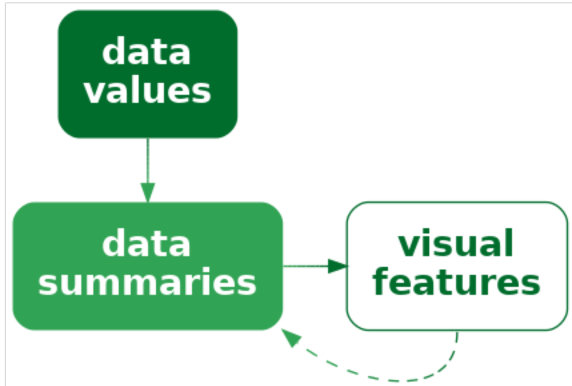


Figure 8.2: A data visualisation may encode data summaries, rather than the raw data, as visual features. This is effective for decoding the data summaries, but it is not effective for decoding the raw data.

Notice how, even though Table 8.1 contains very few values, it is much easier to perceive the differences between the two rows of values using the data visualisation in Figure 8.1 rather than the table of values in Table 8.1. Once again, we can see the power of encoding values to basic visual features. The only differences from previous sections is that we are encoding **data summaries** rather than raw data values and we are encoding to the visual features of a more complicated data symbol.

> A box plot is effective at conveying information about the distribution of data values because it encodes **data summaries** about the distribution, like the median and interquartile range, to basic visual features.

8.2 Visual summaries

Figure 8.3 shows a dot plot of the number of points scored by countries from the different hemispheres in Rugby World Cup matches.[102] This data visualisation uses the same data as Figure 8.1, but it encodes raw data values rather than data summaries and it uses much simpler data symbols (data points). This data visualisation encodes the number of points scored encodes as the horizontal **position** of a data point and it encodes the hemisphere as the vertical **position** of a data point. There is an overplotting problem (Section 4.4), which is solved in this case by offsetting data symbols vertically for repeated data values.

Because this data visualisation encodes raw data values as visual features, it is possible to decode raw data values from the visual features (Figure 3.6). For example, it is possible to see that the lowest score (zero points) for a northern hemisphere team ocurred twice. It is also possible to see that there are scores that have never occurred in any matches: 1, 2, 4, and 5. These features are not visible in the boxplots in Figure 8.1.

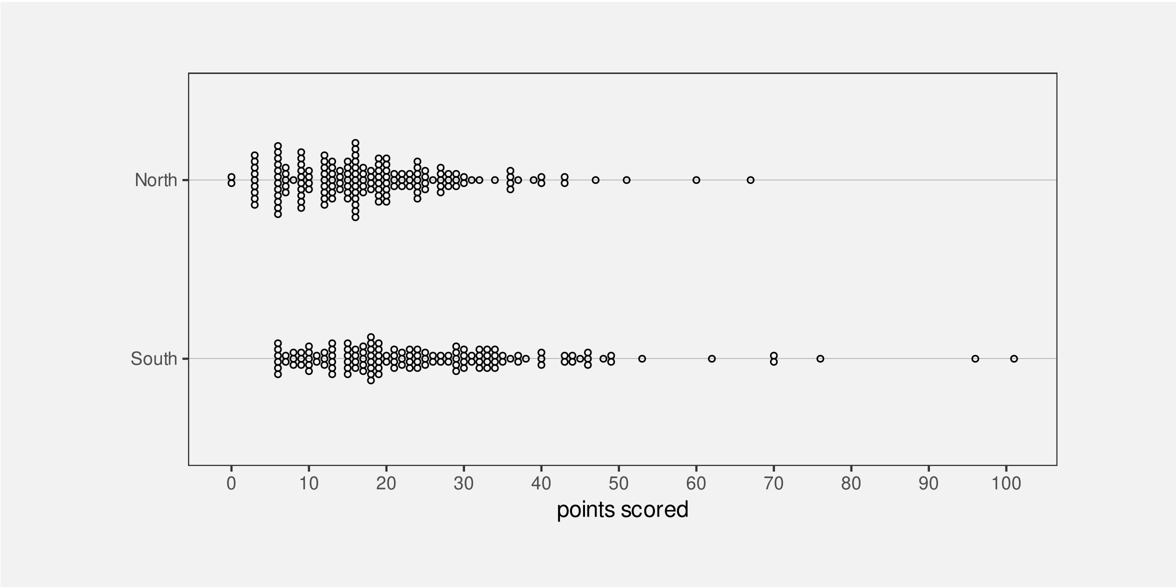


Figure 8.3: Stacked dotplots of the number of points scored by Tier One nations at Rugby World Cup matches.

The fact that we can decode raw data values from a dot plot is not news. That is easy because we have encoded raw data values as accurate visual features (see Section 3.7). However, Figure 8.3 also demonstrates that, in addition to decoding raw data values from visual features, our visual system allows us to decode **data summaries** from visual features.

For example, we can see that the southern hemisphere teams score slightly more points on average than the northern hemisphere teams; the data symbols for southern-hemisphere teams appear shifted to the right of the data symbols for northern-hemisphere teams. In other words, we can perceive **visual summaries**; we can perceive **data summaries** from collections of visual features. For example, we can visually average the **positions** of the data points.[103]

In Figure 8.3, we have encoded raw data values as visual features, which has enabled us to decode raw data values from individual visual features. In addition, **visual summaries** of the visual features enable us to decode **data summaries** from collections of visual features (Figure 8.4).

> A dot plot is effective not just for perceiving raw data values, but also for perceving data summaries like clusters and outliers, and central tendency and spread.

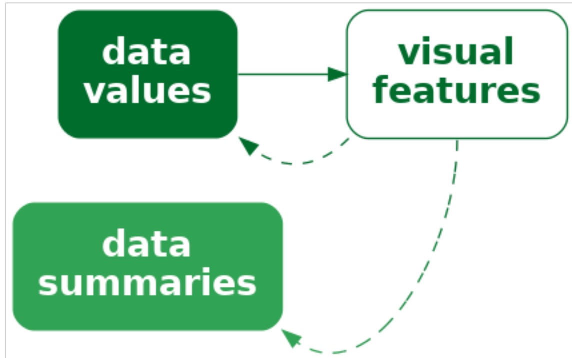


Figure 8.4: A data visualisation that encodes data values as visual features allows us to decode data values, but it may *also* allow us to decode summaries of the data, such as measures of central tendency or variability.

8.3 Case study: Histograms

Figure 8.5 shows a histogram of the number of points scored by Tier One nations at the Rugby World Cups (Table 4.1). A histogram is similar to a box plot in two ways: a histogram is a data visualisation that can be used to convey the distribution of a variable; and a histogram is a data visualisation that encodes **data summaries**, rather than raw values, to visual features (Figure 8.2).

However, a histogram differs from a box plot in two main ways: the data symbol for a histogram is simpler than a box plot, because it consists of just rectangles or bars; and the data summaries for a histogram are more complicated than those for a box plot.

The raw data values for the histogram in Figure 8.5 are the points scored in each game by each country (Table 4.1). To get the data summaries for the histogram, the raw data values are binned; we break the range of the data into equal-sized intervals and count how many data values lie in each interval. The centre of each interval is then encoded as the horizontal **position** of a bar and the count of data values in each interval is encoded as the **length** of a bar.

To be explicit, the values that are being encoded to visual features in Figure 8.5 are not raw data values, but data summaries—intervals and counts—as shown in Table 8.2.[104]

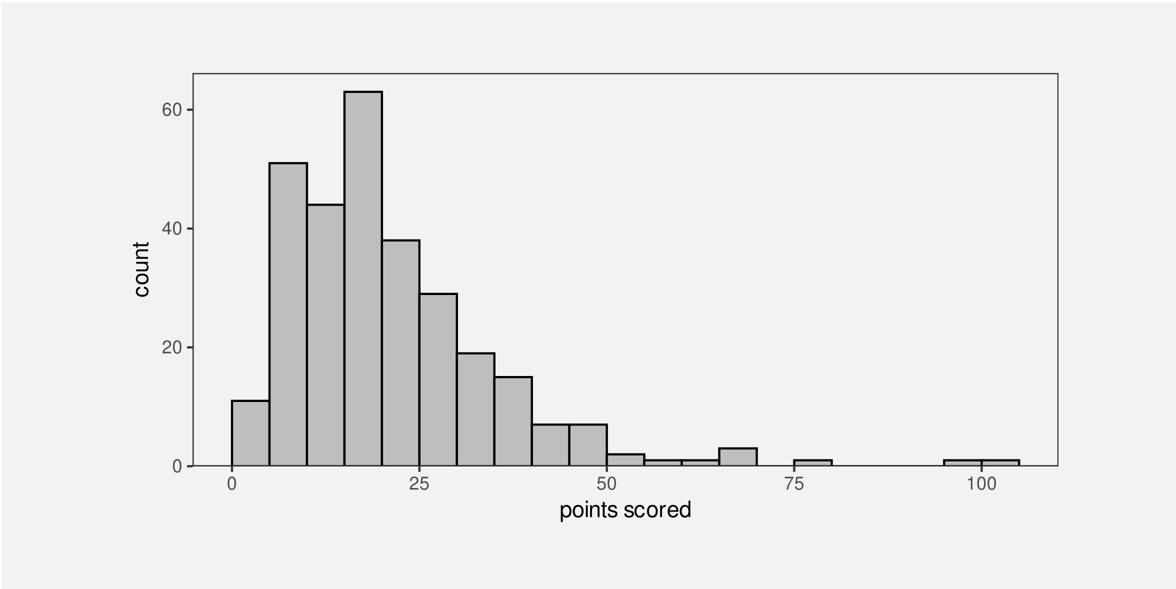


Figure 8.5: A histogram of the points scored per game by Tier One nations at Rugby World Cups.

Table 8.2: A table of intervals that cover the range of the number of points scored by Tier One nations in Rugby World Cup matches, along with the count of the number of points scored that fall within each interval. These data summaries are the values that are encoded to visual features to produce the histogram in Figure 8.1. Not all intervals are shown for reasons of space.

interval	0-5	5-10	10-15	15-20	20-25	25-30	30-35	35-40	40-45	45-50	50-55	55-60	60-65
count	11	51	44	63	38	29	19	15	7	7	2	1	1

As with a box plot, encoding data summaries as accurate visual features, in this case **position** and **length**, means that we can easily decode data summaries from the visual features. For a histogram, this means that we can easily compare the lengths of the bars, which means we can easily compare the counts of data values within each interval. For example, the longest bars represent the intervals that contain a relatively large count of data values, which tells us about the mode of the data (or the number of modes in the data). In Figure 8.5 we can see that the largest number of data values occur around 15 to 20 points and we can see that the number of data values in each interval drops more rapidly above the highest bar than below the highest bar. The latter tells us about the spread and/or skewness of the data.

> A histogram is effective at conveying information about the distribution of data values because it encodes data summaries to basic visual features and those data summaries can be compared to provide information about the distribution of the data.

8.4 Decoding summaries

When we encode a data summary as a visual feature, like the position of the median line of a boxplot (Figure 8.1), we know from previous chapters how effective the decoding is going to be. For example, because the median is encoded as position, we should be able to decode the median very accurately (Section 3.5).

However, if we rely on a visual summary to decode a data summary, for example, comparing the average points scored by teams from different hemispheres based on a collection of data points for each hemisphere (Figure 8.3), we are no longer dealing with just the decoding of a single visual feature. How well can we decode the average from the positions of several data symbols compared to decoding a single data value from a single data symbol? How effective is the decoding from visual features to a data summary (Figure 8.4)?

One issue that arises is that there are many possible visual summaries. For example, we might be interested in the average data value, or the variability of data values, whether there are unusual data values (outliers), or whether there are groups of data values (clusters).[105]

Furthermore, we can produce each sort of visual summary for each possible visual feature. For example, we can decode the average position of data symbols, or the average luminance of data symbols, or the average area of data symbols. We can decode unusual positions, unusual colours, or unusual angles.

Although it is well-established that our visual system is capable of generating visual summaries, less is known about the accuracy of those summaries, partly because the range of possible visual summaries is so large. However, there are a few well-established results. For example, we know that estimating averages from the positions of multiple data symbols is quite accurate, as in Figure 8.3. We also know that estimating averages from the luminance and/or chroma of multiple data symbols can also be effective.[106]

For more qualitative summaries, such as identifying clusters and outliers, we know that performance will be good because these decodings rely on fundamental properties of our visual system that identify unusual items (Section 2.5) and identify groups of items (Section 2.7).

8.5 Dangers of summaries

The benefit of a data visualisation that encodes **data summaries** as visual features is that it allows us to decode data summaries very easily. For example, although we can perceive a general idea of central tendency from the dots in Figure 8.3 (see Section 8.2), it is much easier and more accurate to perceive median values from the thick vertical lines in Figure 8.1.

The flipside is that a data visualisation that encodes **data summaries** as visual features is *not* effective for decoding raw data values. For example, we cannot see from the box plot in Figure 8.1 that the scores 1, 2, 4, and 5 never occurred, while this is clear from Figure 8.3 because the dot plot does encode the raw data values to data symbols.

> Box plots and histograms are *not* effective at conveying raw data values because they do not encode raw data values to data symbols.

Another point to consider with a data visualisation that is based on data summaries is whether the data summaries provide reasonable summaries of the data.[107] For example, Figure 8.6 shows box plots of the number of points *conceded* by each team in Rugby World Cup games. The box plot for southern hemisphere teams suggests a unimodal distribution with a slight right skew.

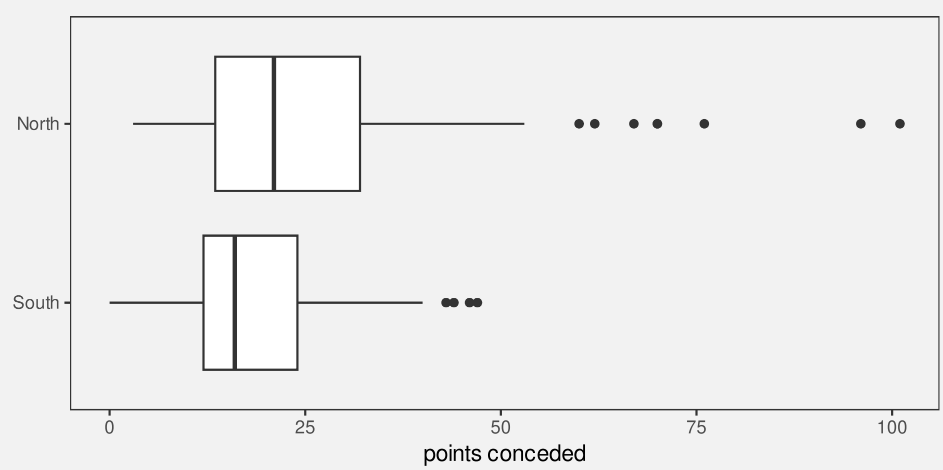


Figure 8.6: Box plots of the number of points conceded in all Rugby World Cup games by Tier One nations.

However, the dot plots in Figure 8.7 tell a different story. This data visualisation shows some evidence that the number of points conceded by southern-hemisphere teams may be bimodal, with some teams only conceding very few points (0, 3, or 6) in a number of games. The data summaries for a box plot assume a unimodal distribution, so a box plot will only ever tell a unimodal story, and that is only reasonable if the data are unimodal.

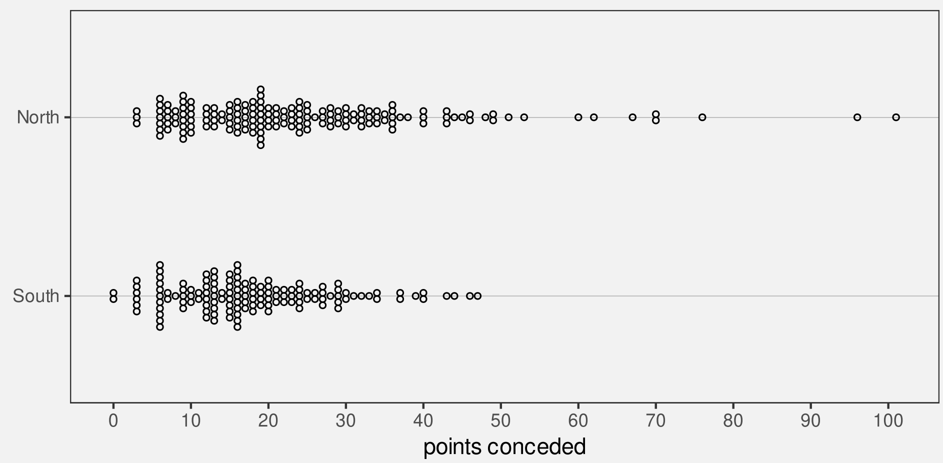


Figure 8.7: Stacked dotplots of the number of points conceded by Tier One nations at Rugby World Cup matches.

> A data visualisation that is based on data summaries can only be effective if the data summaries are sensible summaries of the data.

> A box plot is *not* effective if a five-number summary is not a sensible summary of the data values.

A further complication arises with histograms because we have to *choose* the intervals that are used to calculate the data summaries and that choice can have a large influence on the data summaries. For example, Figure 8.8 shows a histogram of the same data that we used for Figure 8.5, but with a greater number of narrower intervals. This visualisation tells a slightly different story than Figure 8.5. The most common score is now 6 points (rather than between 15 and 20 points) and we see features like gaps at 1, 2, 4, and 5 points and a stronger suggestion of bimodality. There are also point totals that appear strangely low, like 11 and 14.

Unlike a box plot, where the data summaries are essentially fixed calculations,[108] for a histogram we get to choose the data summaries to some extent, which means that we get to choose the story that the data visualisation tells. Figure 8.5 tells a simpler story, but Figure 8.8 tells a story that includes some important details.

When using a histogram for exploring a data set, it is a good idea to try out more than one choice of intervals in order to avoid missing an important story. When using a histogram to present a story about a data set, there is a responsibility to present a story that does not exclude important details.

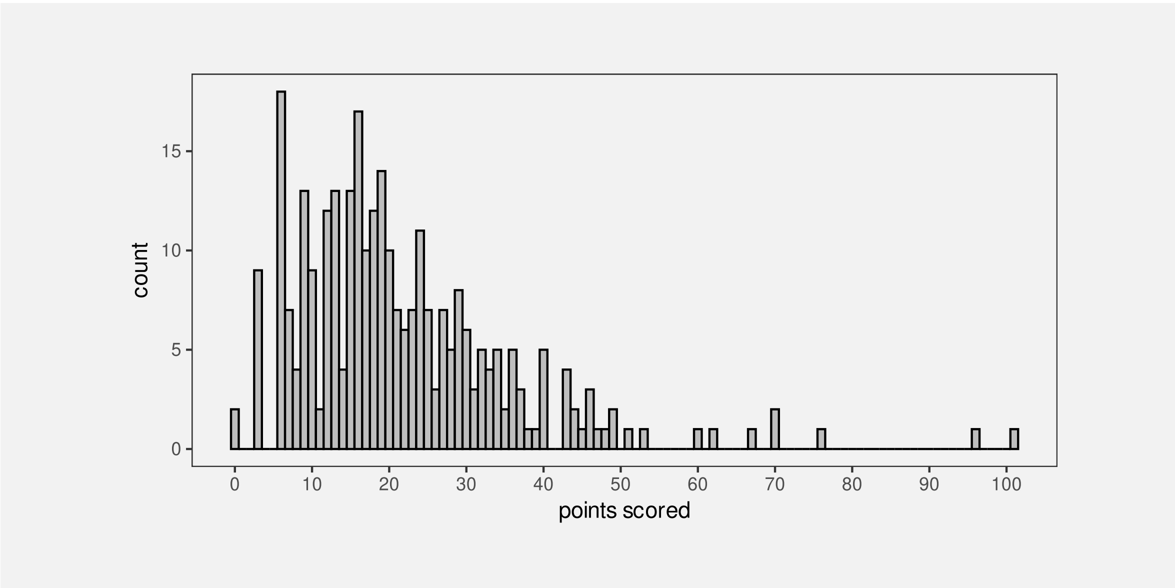


Figure 8.8: A histogram of the points scored per game by Tier One nations at Rugby World Cups. This histogram uses the same data as Figure 8.5, but bins the data using narrower intervals.

8.6 Case study: How charts lie

In this book, we assume that the purpose of a data visualisation is to faithfully and honestly encode data values and the goal is to produce a data visualisation that provides the most effective decoding. Nevertheless, even less honest representations of data can also be thought of in terms of encodings, particularly encodings of data summaries.

For example, Figure 8.9 shows a bar plot of the predicted impact on the top tax rate in the United States from President George W. Bush to President Barack Obama.[109] One problem with this data visualisation is that it does not encode raw data values—the actual predicted top tax rates are 35 and 39.6. Instead, a "data summary" of the tax rate *minus 34* (the values 1 and 5.6) are encoded as the lengths of the two bars. We are unable to decode the raw data from this bar plot—we can only decode the data values that have been encoded—and this leads to an extreme exaggeration of the difference between the two top tax rates.

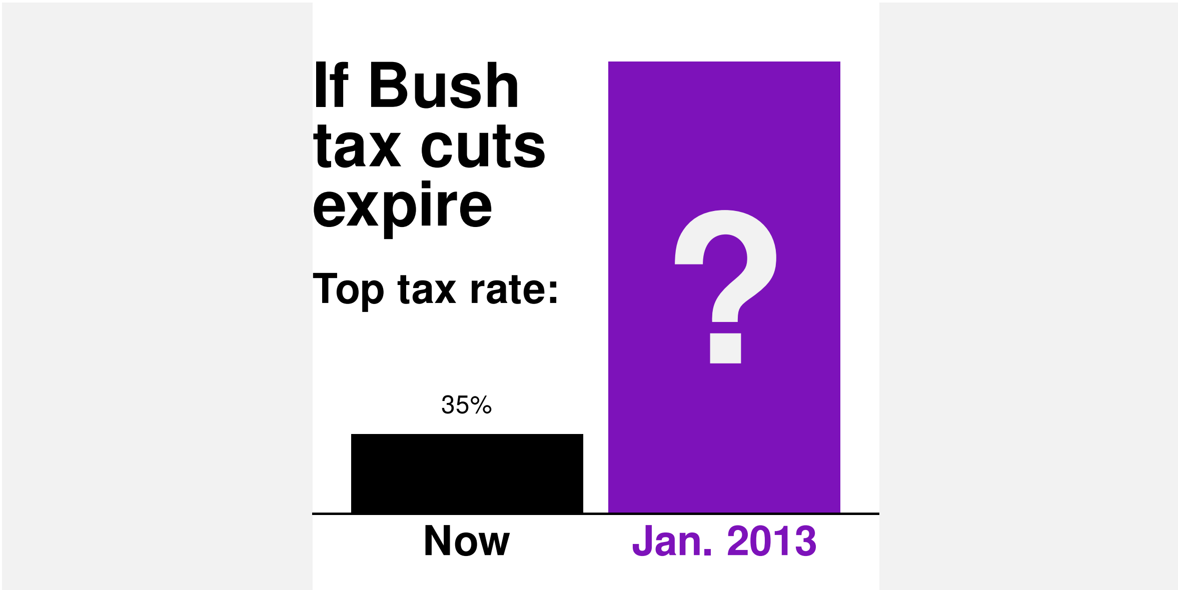


Figure 8.9: A bar plot of the increase in the top tax rate in the United States of America from President George W. Bush to President Barack Obama.

A data visualisation that encodes a data summary that is a distortion of the data or an inappropriate subset of the data is encoding unhelpful information to visual features. It is only then possible to decode unhelpful information from the data visualisation (Figure 8.10).

8.7 Visual downgrades

We have seen that a downside to encoding data summaries, rather than raw data values, is that we lose the ability to decode raw data values from data symbols. However, we pay this price in order to improve our ability to decode data summaries from the data symbols. For example, a boxplot makes it easier to decode the median of a set of data values.

By encoding data summaries, we deliberately lose the ability to decode raw data values in order to gain the ability to decode data summaries.

Another way that we may deliberately lose information is by selecting a visual feature that does not allow complete decoding of the original data values.

For example, Figure 8.11 shows the crime rate for 2021 in each police district of New Zealand. The data values for this map are shown in Table 8.3.

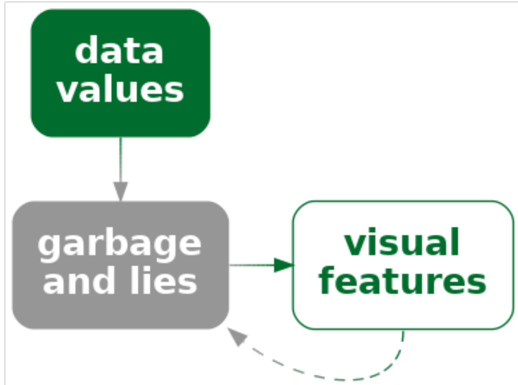
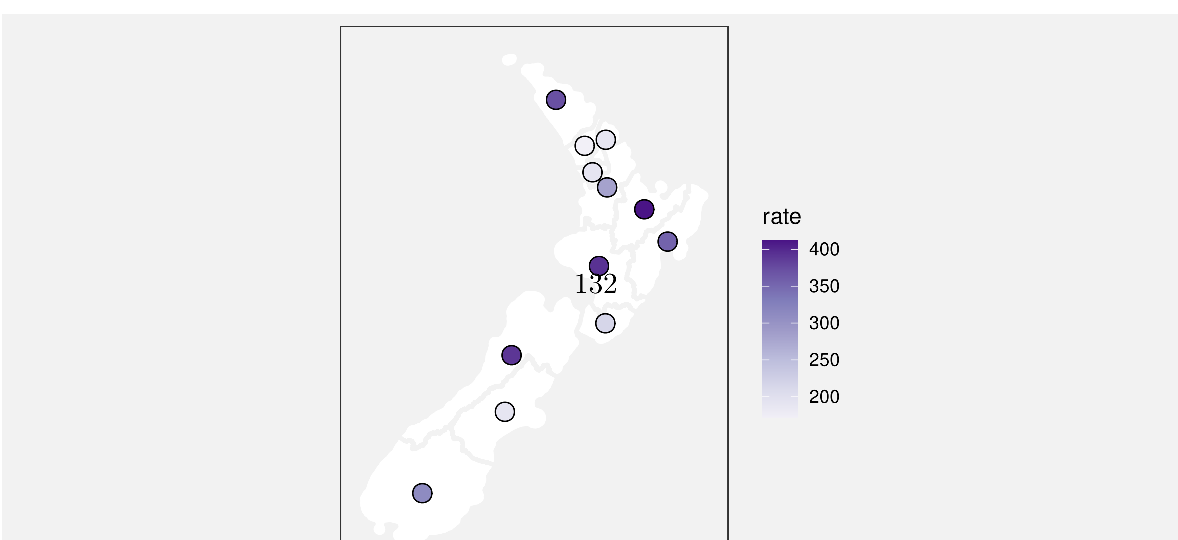


Figure 8.10: A data visualisation may encode an unrepresentative or "cooked" version of the data, rather than the raw data, to visual features. Visual decoding leads back to the lies rather than the real data.

Table 8.3: The average crime rate over the period 2011 to 2021 for each police district of New Zealand.

district	rate
Northland	372.63
Waitematā	171.37
Auckland City	191.53
Counties/Manukau	191.43
Waikato	282.42
Bay of Plenty	411.96
Eastern	355.56
Central	393.59
Wellington	217.47
Tasman	390.83
Canterbury	190.70
Southern	312.89

The data symbols in Figure 8.11 are circles with the geographic location of the (centre of the) police district encoded to the position of the circles and the crime rate encoded to the (fill) colour, particularly the luminance, of the circles. Darker circles represent higher crime rates.



luminance of the circles (Section 3.4).

However, we may accept this loss of information in order to gain the ability to decode the geographic position of the police districts. For example, although we cannot decode exact crime rates, we are able to see that the higher crime rates correspond to more rural areas; the lighter circles appear in the urban regions that include New Zealand's largest cities: Auckland, Wellington, and Christchurch.

In Figure 8.11, we can only decode to **data summaries** of the crime rate because we an only decode the **ranking** of the crime rates, not individual numeric crime rates and not numeric differences between crime rates (Figure 8.12).

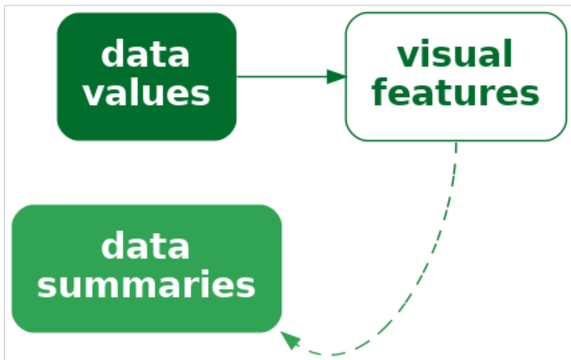


Figure 8.12: A data visualisation may encode data values as a visual feature that does not allow decoding back to the original data values. Instead, it is only possible to decode summaries of the data.

On the other hand, this loss of information is acceptable if the decoding still allows us to make useful comparisons. Put more strongly, the numeric comparisons that we lose access to may not be the most important decodings, so the loss may not be that important.[110]

For comparison, Figure 8.13 shows a simple bar plot of the same crime rates. In this data visualisation, although we can decode and compare the numeric crime rates much more accurately, the rural/urban comparison is much more difficult.

8.8 Case study: Heatmaps

Heatmaps are a common way of representing three-dimensional data where two of the data variables are either qualitative or have been measured on a regular grid and the third variable is **quantitative**. The first two variables naturally encode to horizontal and vertical **position** to produce an array of rectangles, while the third variable encodes to **colour**.

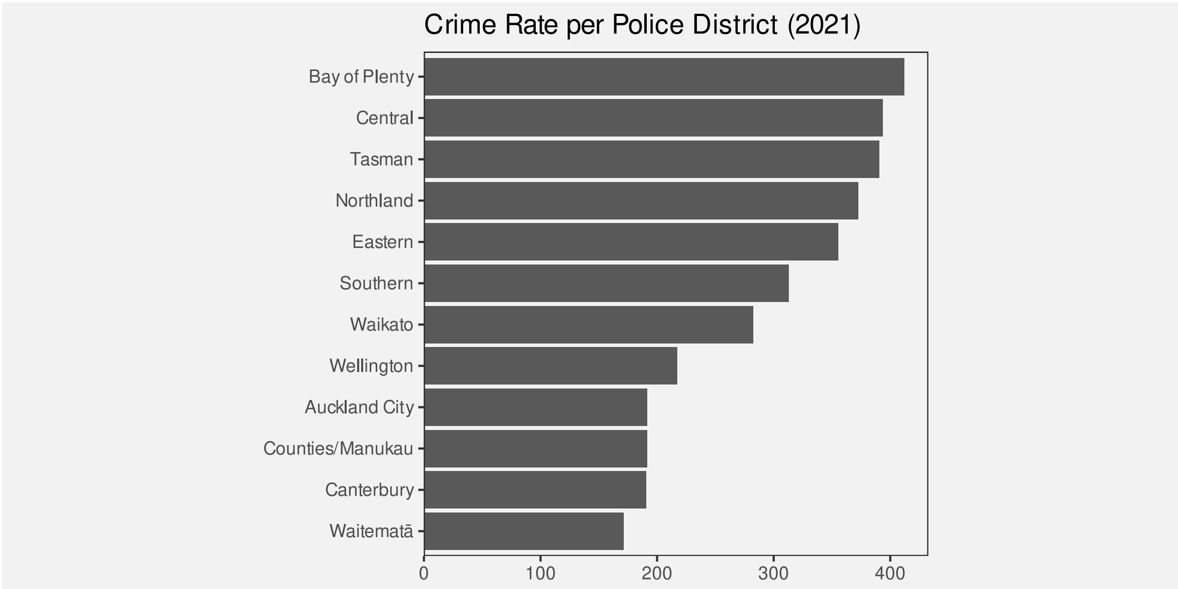


Figure 8.13: A bar plot of the crime rate for 2021 in each police district of New Zealand.

We saw an example of a heatmap in Figure 1.2 and Figure 8.14 shows another example that is a variation on Figure 6.22 and Figure 6.24.

The use of colour to encode quantitative data values is not optimal for accuracy (Section 3.3). However, as we saw in Section 8.7, as long as the colour scheme has monotonic luminance and/or chroma changes, it is still possible to decode ordinal values.

Furthermore, we are able to decode useful visual summaries from the multiple rectangle data symbols on a heatmap (Section 8.2). For example, we can identify regions of darker or lighter colours, not just individual dark or light rectangles.[111]

On the downside, because each rectangle in a heatmap is surrounded by other rectangles of differing colours, it is difficult to accurately decode the colours of individual rectangles (Section 6.3).

In the case of a heatmap, ordinal comparisons are typically what is of most importance. The aim is *not* to be able to decode individual quantitative data values and this moderates the damage that is done by relative perception of the heatmap rectangles.

8.9 Cognitive load

Consider the following sentence about changes in global malnutrition rates:[112]

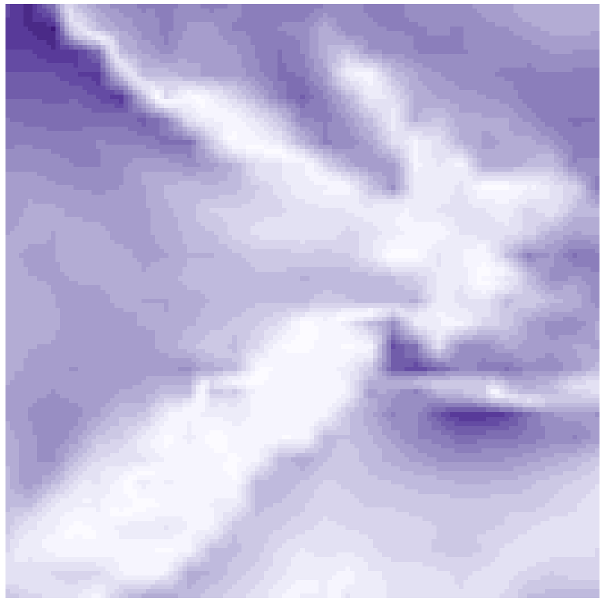


Figure 8.14: A snapshot of wind strength in the vicinity of New Zealand on September 14 2024. This uses the same data as Figure 2.4, but does not show wind direction.

“rising food production reduced the malnutrition rate from 2 in 3 people in 1950 to 1 in 11 by 2019”

In order to comprehend the change in malnutrition rate, we are required to compare “2 in 3” to “1 in 11”. This requires mental effort to calculate the ratios or fractions and then compare them. It is easier to compare the rates if we hold the denominator constant, comparing “2 in 3” to “2 in 22”. It is easier still if we calculate the fractions, comparing 0.667 to 0.091. And it is easier still to compare those fractions visually (Figure 8.15).

The point is that we can reduce the cognitive load that is required of the viewer if we make the mental calculations that are required easier. We can reduce the cognitive load further if we present the results of the calculation directly and even further if we encode the results of the calculation as simple visual features.

In other words, we can calculate **data summaries** or we can force the viewer to perform **visual summaries**.

Conversely, we can add to the cognitive load if we make the visual calculations harder. For example, Figure 8.16 shows a bar plot of the original “2 in 3” versus “1 in 11” data values. We now need to perceive the proportion of blue bars relative to red bars and compare those proportions, which is clearly more work than just comparing the two bars in Figure 8.15.

We ideally want to present data in a way that imposes the least cognitive load on the viewer. In terms of Section 2.9, we want to present the viewer with simple **visual tasks**.[113]

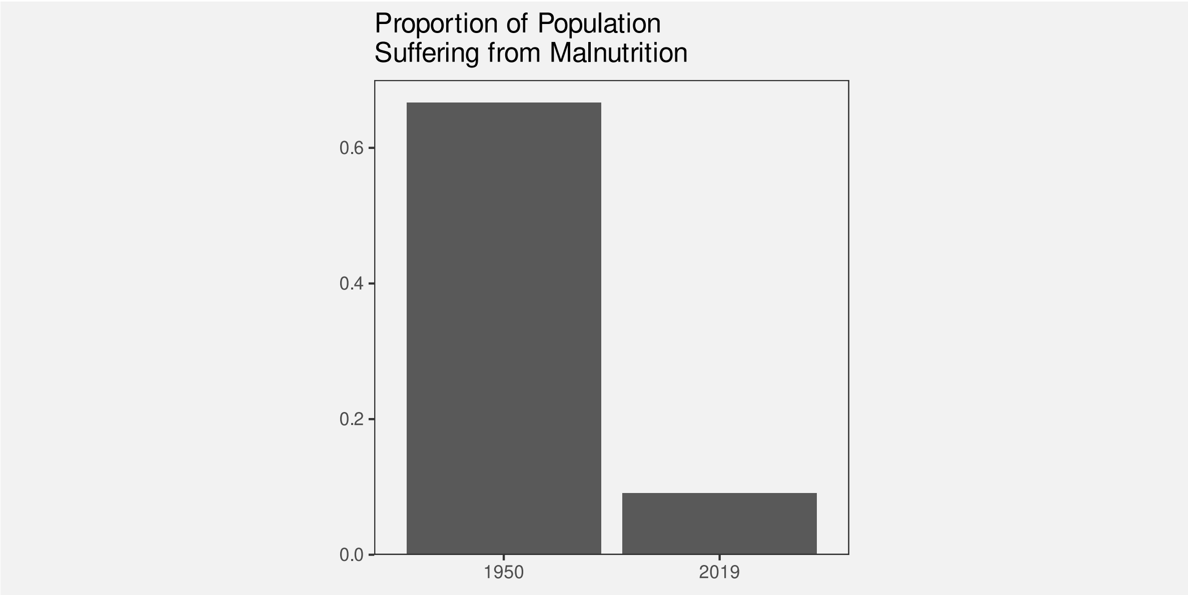


Figure 8.15: A bar plot comparing two fractions.

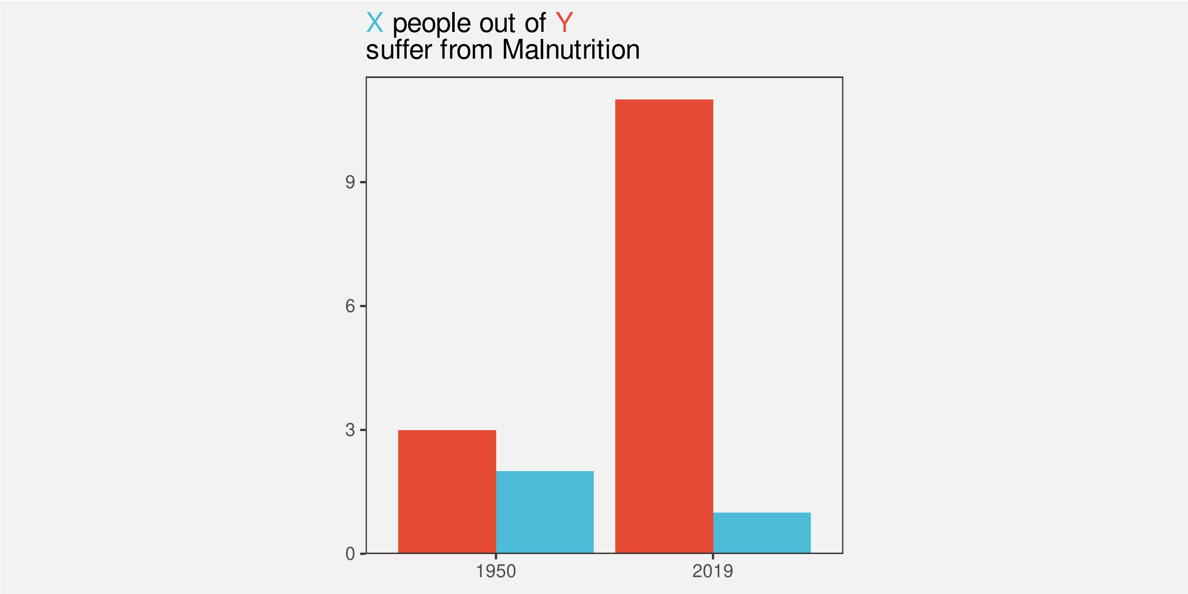


Figure 8.16: A bar plot comparing two fractions.

The best data visualisation will be different for different questions of interest, but if the question of interest involves comparing data summaries, then one way that we can make the viewer's life easier is by calculating summaries of the data and encoding data summaries as visual features, rather than encoding raw data values and asking the viewer to perform visual summaries.

> It is possible in some cases to rely on visual summaries, but we need to be careful not to set a visual task that requires too much mental effort.

8.10 Summary

Data summaries, such as measures of central tendency, measures of variability, and tables of counts, provide information about the distribution of a set of data values.

Some data visualisations, like box plots and histograms, are effective because they encode **data summaries** to visual features. This makes it easy to decode and compare data summaries.

Care must be taken to use data summaries that appropriately summarise the data values.

When we encode raw data values to visual features, like in a dot plot, the encoding to individual visual features allows us to decode and compare raw data values, but, in addition, **visual summaries** of *collections* of visual features allow us to decode and compare data summaries.

A box plot that encodes data summaries to visual features is more effective for perceiving data summaries than a dot plot that relies on visual summaries. However, a dot plot is more effective for perceiving raw data values.

Notes

[99] In this book, we use *data summary* to cover a broad range of transformations of the data. These are transformations that occur prior to the encoding of data values to visual features.

There are several more sophisticated break downs of the pipeline from data values to visual representations. For example, the Information Visualization Reference Model (Card, Mackinlay, and Shneiderman 1999) includes the transformation of data into a standard format (*data transformations*) prior to encoding data values to visual features and also includes the subsequent transformation and scaling of visual features to positions and shapes on the screen or page (*view transformations*).

Wu and Chang (2024) differentiates between *data transformations* and *design-specific transformations*, such as the calculation of summaries for a box plot.

The idea of a *data summary* corresponds well with the idea of a *statistic* in the *Grammar of Graphics* (Wilkinson 2005), although in this book, because we only consider static data visualisations, the connection with the raw data is lost. The *Grammar of Graphics* also includes transformations in the form of *scales* and *coordinates*, which roughly correspond to the *view transformations* of the Information Visualization Reference Model.

[100] John W. Tukey is credited as the inventor of the box plot (Tukey 1977).

[101] Ziemkiewicz and Kosara (2009) refer to this as the *readability* of *aggregations*.

[102] The general idea of arranging overlapping dots so that they are all visible, not just jittering them (Section 7.7), goes back to Wilkinson (1999) and Tukey (1977). The vertically-centred arrangement used here is also known as a *beeswarm plot* (Rosanbalm 2014).

[103] The ability to decode summary statistics from multiple visual objects is known as *ensemble perception*. Cui and Liu (2021) and Whitney and Yamanashi Leib (2018) provide modern overviews.

A related concept is *subitizing*, which is the ability to rapidly identify the number of visual objects, though that only works for very small numbers of objects (Ware 2021).

[104] A bar plot can also be thought of in terms of encoding data summaries rather than raw data. The data values that are encoded as the lengths of bars in a bar plot are tables of counts of raw data values; the raw data values are individual observations of a single qualitative variable, such as the ethnicity of a single youth offender.

See this way, there are raw data values that cannot be decoded from a bar plot. For example, information about an individual offender cannot be decoded from a bar plot because all similar individuals are represented together by a single bar data symbol.

[105] In the visual perception literature, the strict definition of *ensemble* encoding is limited to *statistical moments* (means and variances) (Whitney and Yamanashi Leib 2018), but the data visualisation literature takes a more relaxed approach and includes the identification of clusters and outliers (Szafir et al. 2016). In this book, we embrace the latter, looser terminology.

[106] Gleicher et al. (2013) show that the average position of data points in a scatterplot can be accurately determined and that this is robust to a number of variations in context, such as the number of data points and the presence of distractors.

Albers, Correll, and Gleicher (2014) show that the average luminance of rectangles in a colour field can be determined reasonably well. This study is also a good demonstration of both the multitude of visual summaries that can be studied and the variability of performance across different combinations of visual summaries and visual features.

[107] The classic demonstration of the idea that numerical summaries may not satisfactorily summarise the raw data is Anscombe's quartet (Anscombe 1973). Matejka and Fitzmaurice (2017) provide a more modern update, including Alberto Cairo's *datasaurus* (Cairo 2016a).

[108] It is not true that there is only one way to generate the data summaries for a box plot. McGill, Tukey, and Larsen (1978) describe several early variants and *letter-value plots* are an example of a more modern adaptation (Hofmann, Wickham, and Kafadar 2017).

[109] This is a reproduction of a chart from Alberto Cairo's "How Charts Lie" (p12-13), which itself was a reproduction of a chart shown by Fox News.

[110] Bertini, Correll, and Franconeri (2020) argue that ordinal comparisons are more commonly useful in practice than quantitative comparisons.

"1. Graphics are for the qualitative/descriptive—conceivably the semiquantitative—never for the carefully quantitative."

"2. Graphics are for comparison-comparison of one kind or another - not for access to individual amounts."

Tukey (1993)

[111] Correll et al. (2012) provide evidence for superior ensemble perception of "high" regions within a time series from "colour fields" (heatmaps) compared to traditional time series line plots.

[112] (Smil 2022, 46)

[113] The 10 data visualisation tasks in Amar, Eagan, and Stasko (2005) all correspond to some sort of numerical summary of the data. We can choose to perform the numerical summary and then visualise the result, or we can choose to present the raw data and ask the viewer to perform a visual summary. We should choose whichever will be easier for the viewer.

9 Combining Encodings

Most data visualisations involve more than one data variable. For example, the bar plot in Figure 9.1 involves data on the **gender** of offenders and data on the **count** of offenders for each gender.

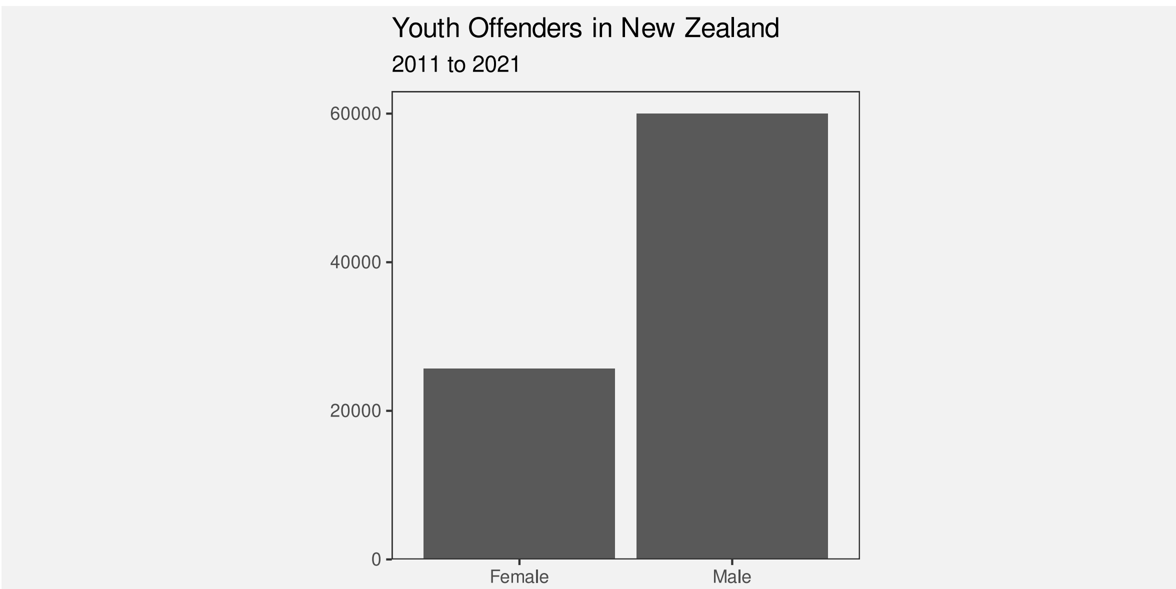


Figure 9.1: A bar plot of the total number of male offenders and the total number of female offenders in New Zealand from 2011 to 2021.

In Figure 9.1, each data value is encoded to a **visual feature** of a data symbol (Figure 3.2). For example, the gender is encoded as the horizontal **position** of the bars and the count of offenders is encoded as the **length** of the bars.

We have so far only looked at encodings from data values to visual features in isolation. For example, we know that the encoding from gender to position is effective because position is appropriate for qualitative data and position has sufficient capacity to differentiate between two genders. We also know that encoding the count of offenders to length is effective because length is appropriate for quantitative data and length provides an accurate decoding.

In this chapter, we begin to consider what happens when encodings are used in combination. For a start, we will just extend our thinking to data visualisations that involve encoding **two** data variables to **two** visual features, like gender and count to position and length. Furthermore, we will restrict ourselves to encodings where each pair of data values are encoded as a single data symbol (Figure 9.2). For example, in Figure 9.1, each combination of gender and count produces a single bar.

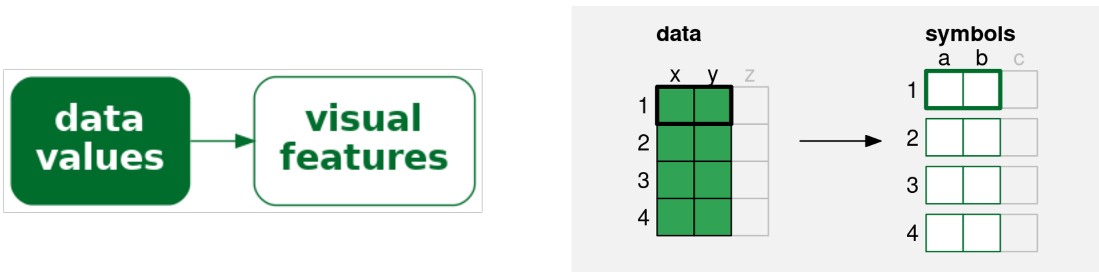


Figure 9.2: A simple data visualisation encodes **two** data values to **two** visual features to create a data symbol. Each pair of data values (`x` and `y`) is encoded to two visual features (`a` and `b`) of a **single** data symbol.

9.1 Independent visual features

A bar plot involves encoding a qualitative variable as bar **position** and a quantitative variable as bar **length** (e.g., Figure 3.1). We know that each visual feature is effective on its own—we can decode from positions to qualitative values and we can decode from lengths to quantitative values—but another reason why the bar plot is effective overall is because the two visual features act independently.[114] Our perception of the horizontal **length** of the bars is not affected by the vertical **position** of the bars.[115]

Figure 9.3 demonstrates that we can also add **colour** to the mix without creating any interactions between visual features. The fact that the bottom two bars are blue and red does not affect our ability to perceive that they have different lengths.

> One reason why a bar plot is effective is because data values are encoded as visual features that can be decoded independently.

9.2 Scatter plots

Table 9.1 shows data gathered on the performance of all twenty teams at the 2023 Rugby World Cup. Most measures are positive, for example, more **points** is better, but there are also a couple of negative measures: more **tackles** may just mean that the team was weak and

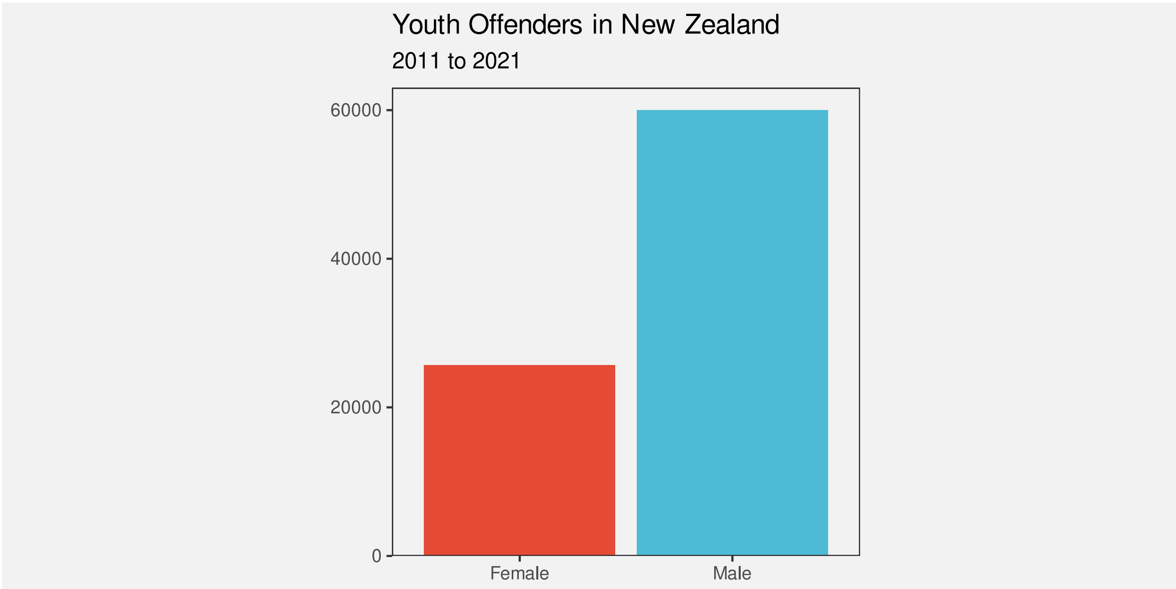


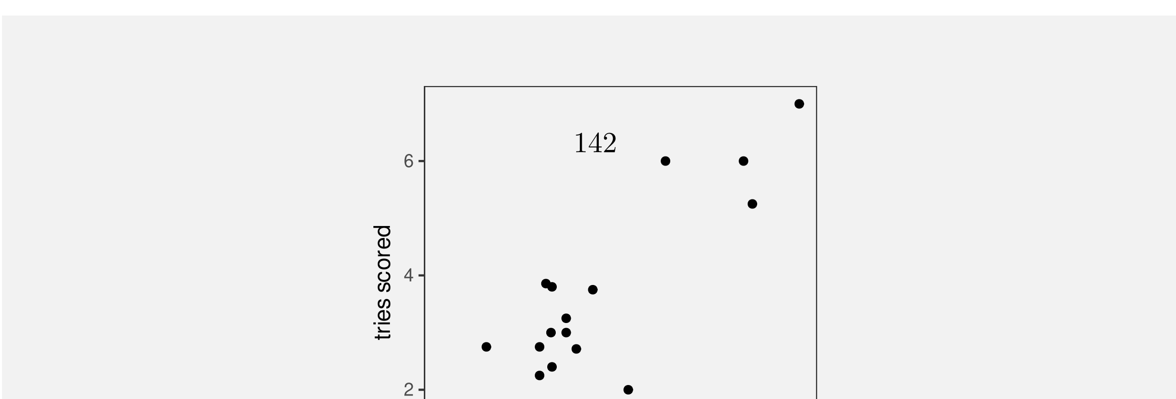
Figure 9.3: A bar plot of the total number of male offenders and the total number of female offenders. The data symbols in this plot come from encoding gender to the vertical positions and colours of the bars and encoding the total number of offenders to the lengths of the bars.

was always being attacked; and more disciplinary **cards** (either **y**ellow or **r**ed) is definitely a bad sign.

Table 9.1: Performance measures for teams at the 2023 Rugby World Cup. Each measure is a per-game average because some teams played more games than others. There are 20 teams in total, with only the first 6 shown here.

country	sphere	ycards	rcards	breaks	tackles	points	converts	offloads	tries	runs
Namibia	South	1.0	0.5	2.5	102.0	9.2	0.5	2.2	0.8	92.0
Romania	North	1.2	0.0	2.8	142.5	8.0	0.8	1.5	1.0	81.0
Chile	South	1.2	0.0	3.8	132.2	6.8	0.5	5.2	1.0	102.2
Samoa	South	1.2	0.2	3.8	109.5	23.0	2.0	9.2	2.8	102.5
Australia	South	0.5	0.0	5.2	108.2	22.5	1.8	7.8	2.8	110.8
Georgia	North	0.5	0.0	5.2	149.8	16.0	1.0	8.0	1.8	114.0

Figure 9.4 shows a scatter plot of the number of "clean breaks" per game versus the number of "tries scored" per game for each team. A clean break means that a team has broken through the opposition defence and a "try scored" means that the team has earned 5 points. A clean break implies an opportunity to score a try, but does not guarantee that a try will be scored.



in the scatter plot in Figure 9.4: the **quantitative** number of clean breaks is encoded as the horizontal **position** of the data points and the **quantitative** number of tries scored is encoded as the vertical **position** of the data points.

We know from Chapter 3 that we should be able to accurately decode the original data values from the positions of the data points. For example, we can easily see that one team scored a little over 7.5 clean breaks per game and we can see that two teams scored about 6 tries per game.

However, there is something else going on in the scatter plot. The two explicit encodings—horizontal and vertical position—**interact** to create an additional **emergent** visual feature: **position in space** (Figure 9.5).[116] For example, we can perceive not only horizontal and vertical distances between data points, but also euclidean distances between data points ("as the crow flies").

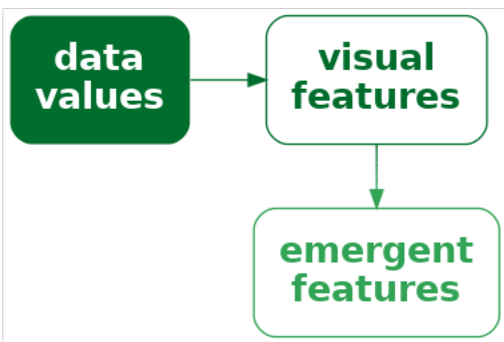


Figure 9.5: If we encode data values as visual features and the visual features interact, the result can be additional visual features. It is possible for the additional visual feature to be dominant relative to the original visual features or even to obscure the original visual features.

> One reason why a scatter plot is effective at conveying relationships between variables is because the encodings of data values to visual features—horizontal position and vertical position—interact to produce an additional visual feature: position in space.

Furthermore, our visual system allows us to decode **visual summaries** (see Section 8.2) from **position in space** (see Figure 9.6).[117] For example, in Figure 9.4 we can easily see a positive **correlation** between the number of clean breaks and the number of tries scored. The points lie roughly along a line from bottom-left to top-right.

We can also easily see **clusters** of data points; the four data points in the top-right corner of the plot appear separate from the other data points because they share a similar position in space (Section 2.7).

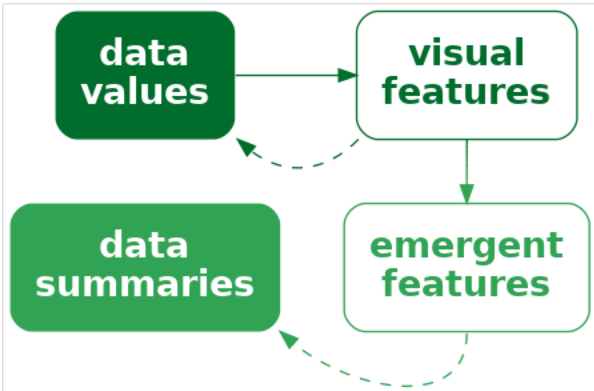


Figure 9.6: If we encode data values as visual features and the visual features interact, the result can be emergent visual features. The original visual features allow us to decode raw data values and the emergent features allow us to decode data summaries.

> One reason why a scatter plot is effective at conveying relationships between variables is because we are able to decode visual summaries from a collection of positions in space.

Although a scatter plot is similar to a bar plot because they both involve two encodings—position and length for a bar plot and horizontal position and vertical position for a scatter plot—there are additional decodings possible from the data symbols in a scatterplot because the additional visual feature of position in space emerges from the combination of horizontal and vertical position.

9.3 Visual features that interact

We have seen that some combinations of visual features, such as **position** and **length** (Figure 9.1) and **position** and **hue** (Figure 9.3) do not interact, while other combinations of features, such as horizontal and vertical **position** (Figure 9.4) do interact. What about other combinations of visual features? Figure 9.7 shows all pairwise combinations of visual features that we have considered (Figure 3.5).[118]

The top row of Figure 9.7 shows that **position** can be combined independently with every other visual feature, except itself (Section 9.2), even when we vary both visual features at once.

For other combinations, we have an array of data symbols, with one visual feature changing down rows and another visual feature changing across columns. If the visual features are independent, we should be able to decode one visual feature from the rows and one visual feature from the columns. For example, for the combination of **hue** and **pattern**, we can clearly see columns of the same hue and rows of the same pattern.

If the visual features interact, we should not be able decode separate visual features. For example, for the combination of **chroma** and **luminance**, all we can see is an array of different shades; there are no clear rows of different chroma or columns of different luminance.

There are also combinations that are not fully independent and have weak interactions between visual features. For example, the decoding of hue, chroma, and luminance is somewhat affected by the area of the data symbol. Stone (2012) provides a clearer demonstration of this effect. Another weak interaction is between hue and luminance; this still allows for diverging colour palettes (Section 6.11).

One clear message from Figure 9.7 is that all visual features interact with themselves. If we try to encode two separate data variables as the same visual feature, it will be difficult to decode the individual data variables. We can also see that combinations of length, angle, area, and pattern produce interations, as do combinations of area, hue, chroma, and luminance. However, combinations of one of hue, chroma, or luminance with any of position, length, angle, area, or pattern are independent.

9.4 Case study: Mosaic plots

Table 9.2 shows a table of counts for offences committed in New Zealand in 2021. For each offence, we know the sex of the offender and what action was taken against the offender.

Table 9.2: Counts of offences committed in New Zealand in 2021 by the sex of the offender and the action taken against the offender.

	Female	Male
Court Action	13950	39526
Non-Court Action	8717	19663
Not Proceeded With	381	933

Figure 9.8 shows a mosaic plot of the data in Table 9.2.[119] This is an effective data visualisation for making several comparisons: the overall proportion of male versus female offenders; within

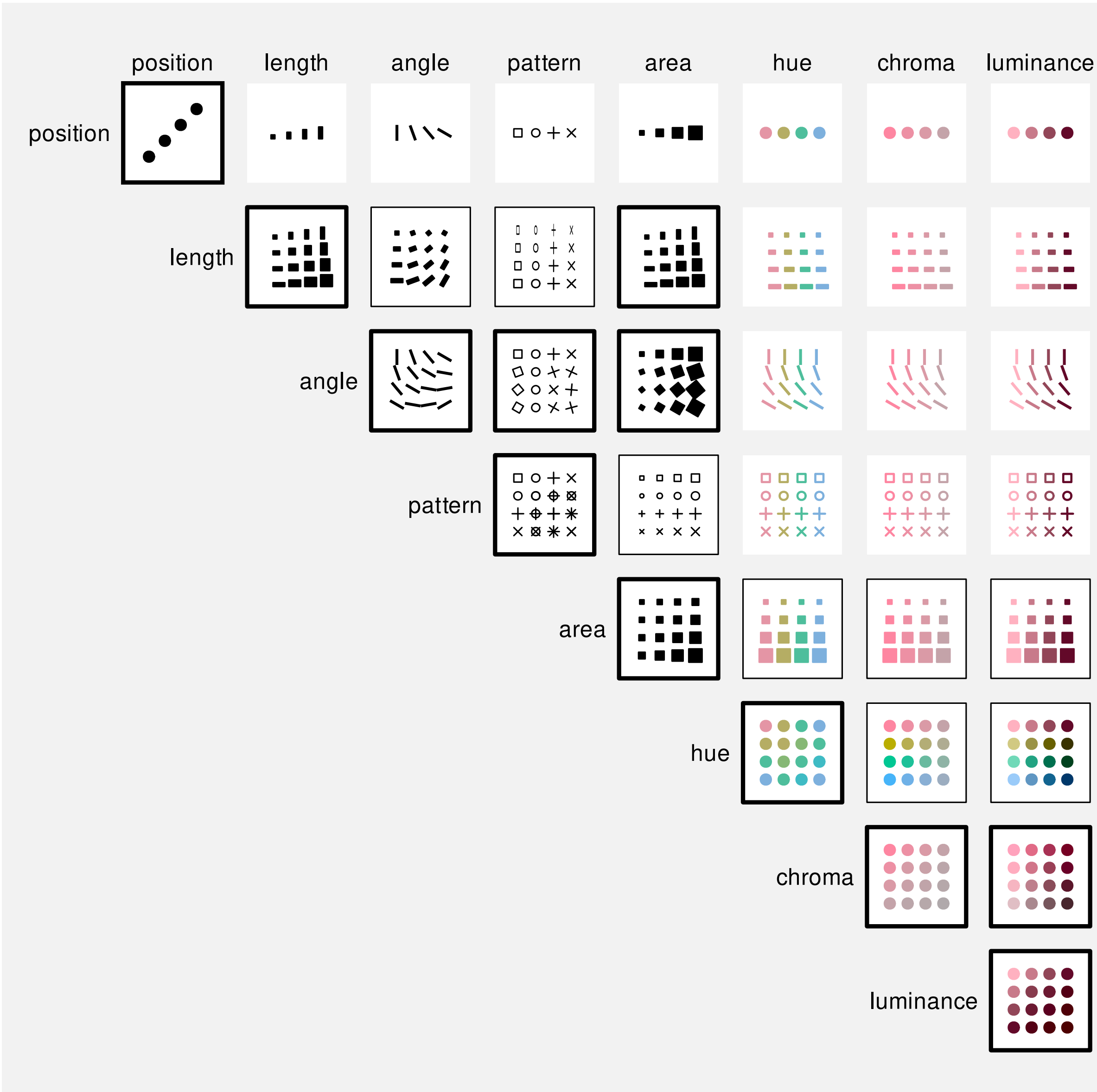


Figure 9.7: All pairwise combinations of basic visual features. Combinations that strongly interact have a thick black border, combinations that are clearly independent have no border, and combinations where there is weak interaction have a thin black border.

each sex, the proportion of different actions against offenders; between sexes, the distribution of different actions; and the overall proportions of combinations of sex and action against offenders.

For example, we can see that there are more male offenders than female offenders, court action is more common than non-court action for both males and female offenders, court action is more common for male offenders than for female offenders, and the most common offences involve male offenders and result in court action.

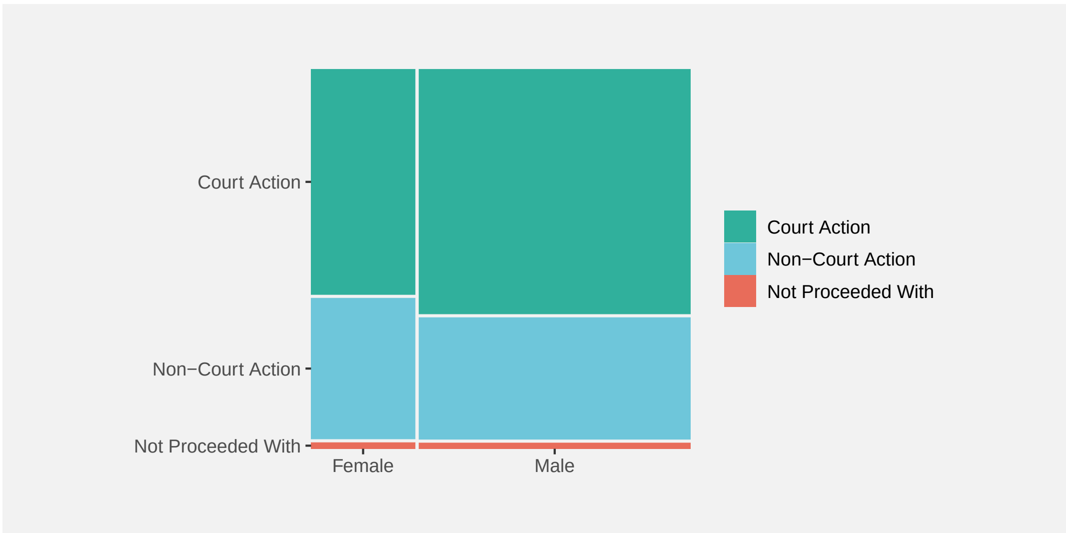


Figure 9.8: A spine plot.

A mosaic plot is another example of a data visualisation that encodes **data summaries** to data symbols rather than raw data. The data symbols are rectangles, the proportions of male versus female offenders are encoded as the widths of the rectangles, and the proportions of different actions taken, within males and females, are encoded as the heights of the rectangles (see Table 9.3).

> One reason why a mosaic plot is effective is because it encodes proportions and conditional proportions, rather than raw counts, to the visual features of rectangles.

A mosaic plot is also an example of a data visualisation that encodes to visual features that **interact**. The explicit encodings involve encoding proportions as the widths and heights of the rectangles and we know from Section 3.3 that this will allow accurate decoding from lengths to proportions.

Table 9.3: The data summaries calculated from Table 9.2 that are encoded as the widths and heights of the rectangles in Figure 9.8.

(a) Widths

	Female	Male
Court Action	0.28	0.72
Non-Court Action	0.28	0.72
Not Proceeded With	0.28	0.72

(b) Heights

	Female	Male
Court Action	0.61	0.66
Non-Court Action	0.38	0.33
Not Proceeded With	0.02	0.02

> One reason why a mosaic plot is effective is because it encodes proportions to lengths.

However, the interaction of these encodings produces an **emergent feature**, which is the **area** of the rectangles (Figure 9.5). Importantly, these areas correspond to additional data summaries, in this case, the overall proportion of offences in each combination of sex and action taken.

> One reason why a mosaic plot is effective is because the encodings to length and height interact to produce area, which represents overall proportions.

One weakness of mosaic plots is that area is not the most accurate visual feature (Section 3.5), so we are not able to decode overall proportions from the rectangle areas as accurately as we are able to decode the proportions that are represented by the separate widths and heights of the rectangles.

Another weakness is that the lengths and heights of the rectangles are unaligned (Section 5.2). This compromises the accuracy of comparisons of widths between categories or the comparisons of heights between categories. This weakness is greater if there are more categories and/or larger differences between categories than we see in Figure 9.8. For example, Figure 9.9 shows a mosaic plot of the proportions of offenders in different age groups and, within age groups, the proportions of actions against the offender at a finer level of detail than in Figure 9.8. If we attempt to compare the proportion of offences that result in "Formal Warnings" in younger age groups versus older age groups, we are hampered by the fact that the pink rectangles do not have a common vertical baseline. As a side note, we are also hampered in that case by the distance and distractors between the rectangles (Section 5.1); the ordering of the variables and the ordering of categories can have an impact on the effectiveness of a mosaic plot just like ordering the bars of a bar plot.

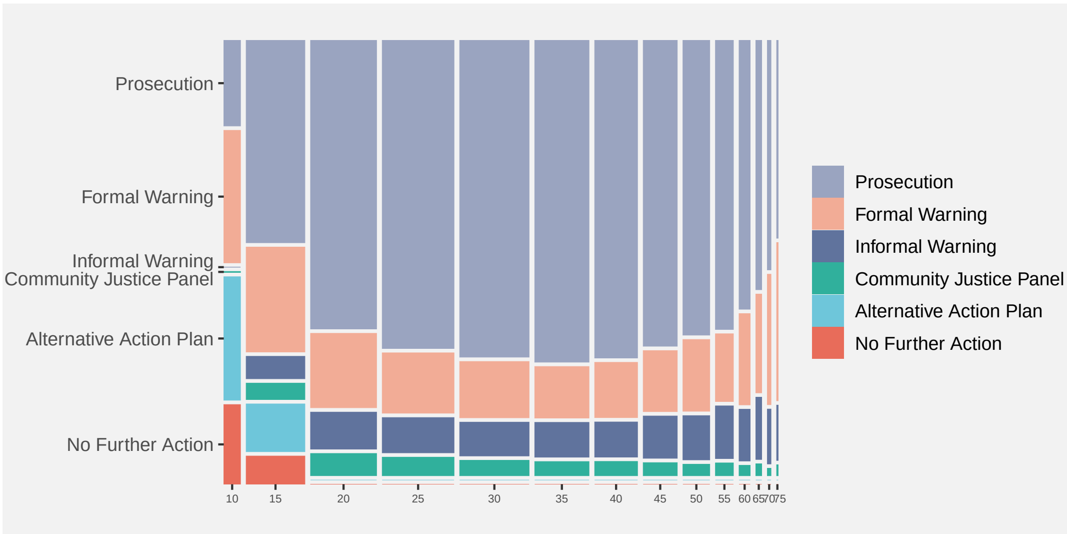


Figure 9.9: A spine plot with problems

These weaknesses are exacerbated for mosaic plots of more than two variables. For example, the mosaic plot in Figure 9.10 shows the proportions of youth versus adult offenders, then, within age levels, the proportions of male versus female offenders, then, within combinations of age levels and sex, the proportions of different actions taken against the offender. Even though there are only very few levels for each variable, our ability to compare heights and widths of rectangles is impaired by inconsistent baselines, distance between rectangles, and distractors.

A final weakness with mosaic plots is that the **shape** of the rectangles changes as well as the **area**. In Section 7.5 we saw that changes in visual features should only reflect changes in the data. Figure 9.11 shows that we can create different rectangles with the same area, but different shapes. There can be rectangles within a mosaic plot with the same area, which represents the same overall probability data value, but with different shapes. The different shapes are a visual signal of differences in the data when in reality no difference exists.

9.5 Case study: Interaction plots

Figure 9.12 shows an interaction plot of the number of offenders in different ethnic groups, comparing 2011 to 2021. The raw data are shown in Table 9.4.

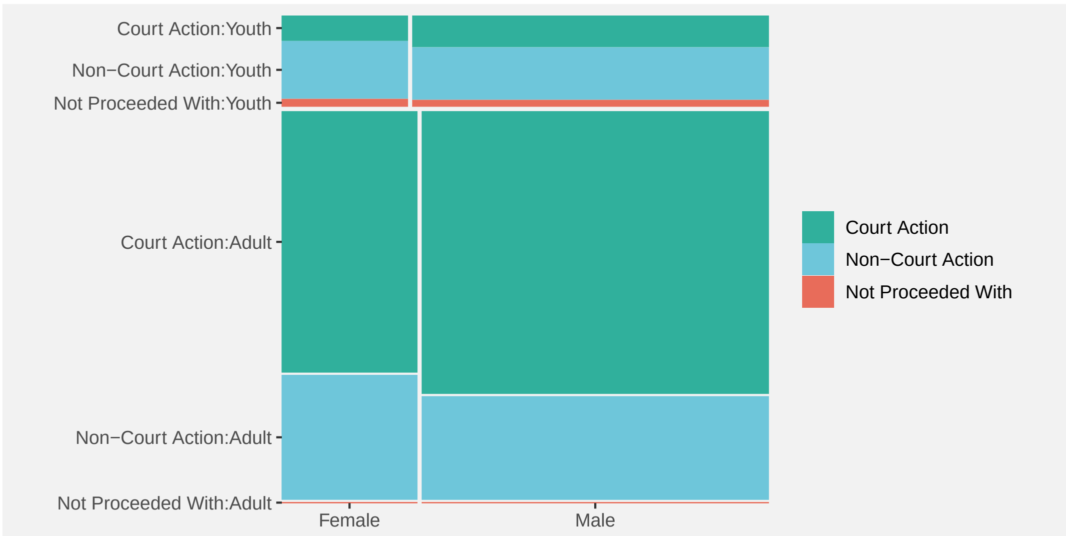


Figure 9.10: A mosaic plot of three variables

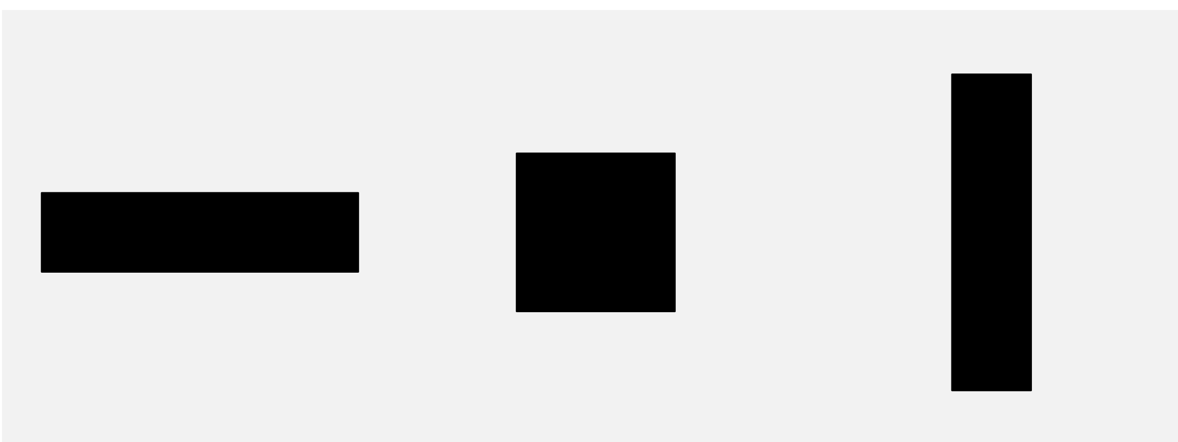


Figure 9.11: Three different rectangles that all have the same area.

Table 9.4: The number of offenders in different ethnic groups in 2011 and in 2021.

group	year	count
Māori	2011	5957
Pasifika	2011	1092
European/Other	2011	5775
Unknown	2011	194
Māori	2021	2869
Pasifika	2021	328
European/Other	2021	1833
Unknown	2021	1589

The purpose of the interaction plot is to show the *change* in the number of offenders between 2011 and 2021 for each of the ethnic groups. If the lines of the interaction plot were parallel we would conclude that the same change has happened to all ethnic groups; the fact that the lines are not parallel indicates that there have been different changes in some ethnic groups compared to other ethnic groups.[120]

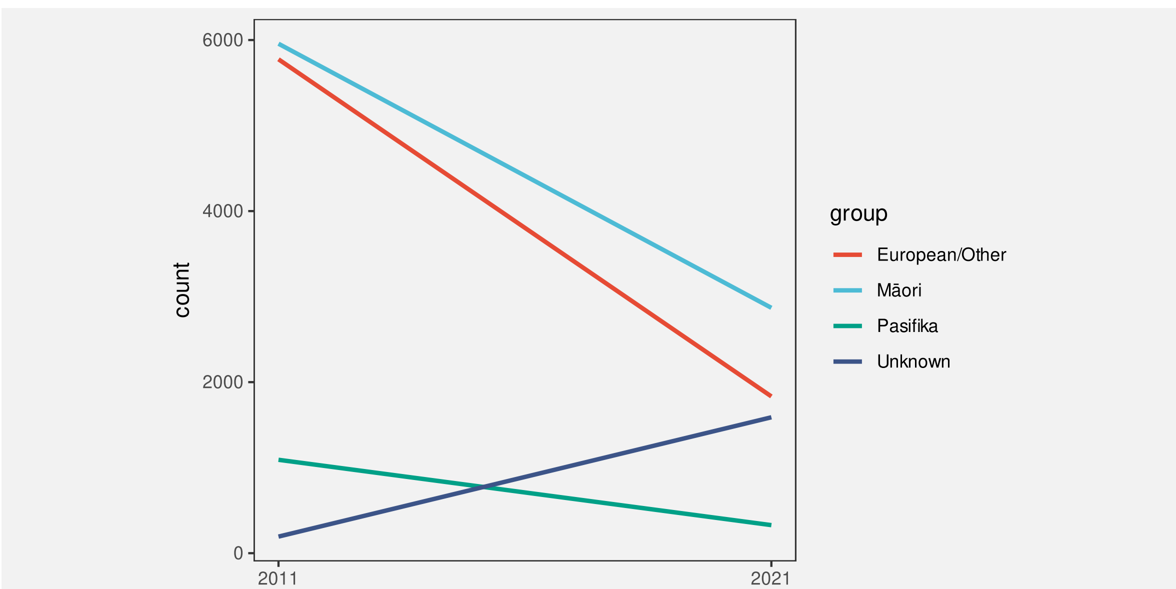


Figure 9.12: An interaction plot showing the change in the number of offenders in different ethnic groups between 2011 and 2021.

The data symbols in Figure 9.12 are straight line segments. The year is encoded as the horizontal start and end **position** of the lines and the number of offenders is encoded as the vertical start and end **position** of the lines. The ethnic group is encoded as the **colour** of the lines. These encodings are effective for decoding the raw data values in Table 9.4 from the end points and the colours of the lines.

In addition, the explicit encodings generate an **emergent feature**: the **angle** of the lines. Furthermore, the angle of the lines allow us to decode a **data summary**: the *change* in the number of offenders between 2011 and 2021 (see Figure 9.6).

> An interaction plot is effective because it allows us to decode changes in the data values from the angles of the lines.

It is more difficult to decode and compare the *change* in the number of offenders for each ethnic group in a bar plot like the one shown in Figure 9.13. Partly this is because we have to decode the lengths of two bars for each ethnic group rather than the slope of a single line.[121]

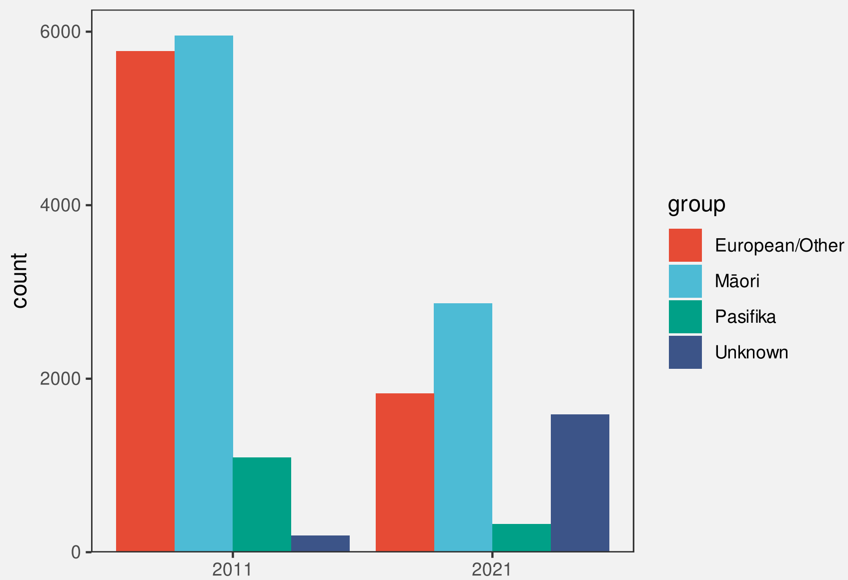


Figure 9.13: A bar plot showing the number of offenders in different ethnic groups in 2011 and in 2021.

9.6 Decoding combinations of visual features

When we combine visual features that are independent, each visual feature can can be decoded separately. In that case, the decoding of the combination can be reduced to the decoding of the individual features. This means that we can determine the effectiveness of a combination of features based on the effectiveness of the individual features. For example, the combination of length and position in a bar plot like Figure 9.1 is effective because length is an effective encoding (for quantitative data) and position is an effective encoding (for qualitative data).

When we combine visual features that interact, the situation is more complicated. There are two parts to consider: the individual features that are being combined and the emergent feature that results from the interaction between the individual features.

One issue is that the individual features may be difficult to decode on their own (Section 9.4). In that case, the effectiveness of the combination is dependent mainly on the emergent feature.

The effectiveness of the emergent feature is straightforward to determine when the emergent feature is one the basic features that we already know about. Examples of this case are combinations of length producing area, as in a mosaic plot (Figure 9.8), and combinations of position producing angle, as in an interaction plot (Figure 9.12). Figure 9.7 shows several other cases of this sort. For example, the combinations of colour features—hue, chroma, and luminance—all just produce different colours.

However, when the emergent feature is not one of the basic features, we need more information to know how that emergent feature will perform. An example of this situation is when a combination of horizontal and vertical positions in a scatter plot produce position in space (Figure 9.4).

Even when the emergent feature is known, another issue is that the emergent feature decodes to a data summary, rather than the raw data values. As we saw in Section 8.4 this creates a problem because there are many possible data summaries to consider and there is less known about how well these decodings perform.

However, in the specific case of decoding a data summary from the positions of multiple data symbols within a scatter plot—decoding correlation from position in space—experiments have shown that the decoding is reasonably accurate. We are quite good at decoding the strength of a linear relationship from a scatter plot.[122]

9.7 Dangers of combining encodings

Figure 9.14 shows a plot of changes in health spending by successive New Zealand governments.[123] In this data visualisation, the average change in health spending over a government's entire term is encoded as the **height** of a bar, with the duration of that government's term encoded as the **width** of the bar. Because height and width are visual features that interact, the result is an **area** for each government's term. However, that area has no sensible interpretation; what does a percent multiplied by a number of years mean? The areas of the bars do not correspond to any property of the data values.

The interaction of visual features can be a boon when the interaction produces an emergent feature that can be decoded to properties of the data (as in a scatter plot, or mosaic plot, or

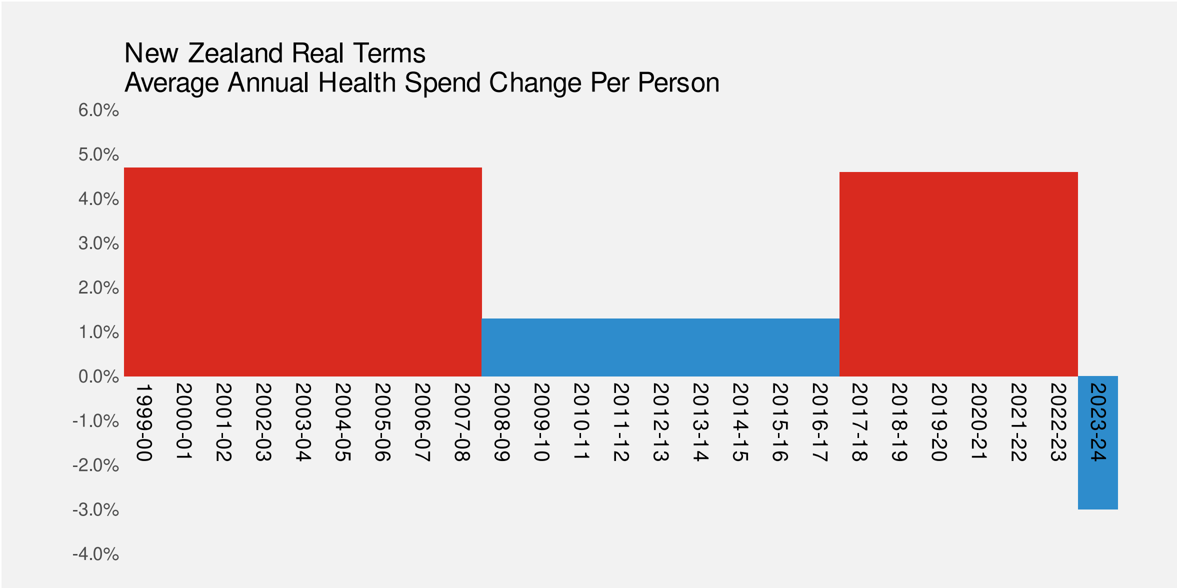


Figure 9.14: A plot showing the differences in health spending for successive New Zealand governments using filled bars.

interaction plot). However, it is possible to create a poor data visualisation if we produce an emergent feature that has no correspondence with data values (Figure 9.15).[124]

> A data visualisation can be misleading if we encode data values using a combination of visual features that have a strong interaction, but the values that we decode from the resulting emergent feature do not correspond to any meaningful summary of the data.

Figure 9.16 shows a different representation of the data. In this case, the average change in health spending is encoded as the (vertical) **position** of a line and the duration is encoded as the **length** of the line. These visual features do not interact, so we no longer have a misleading emergent visual feature.

9.8 Case study: Redundant encodings

Figure 9.17 shows a bar plot of various performance metrics for the football player Lamine Yamal from July 2024.[125] The percentile values show how Yamal ranked against other similar players on each metric. He is a relatively good football player.

Figure 9.17 is a typical bar plot in that it encodes qualitative data values as the (vertical) position of the bars and quantitative data values as the lengths of the bars. However, there

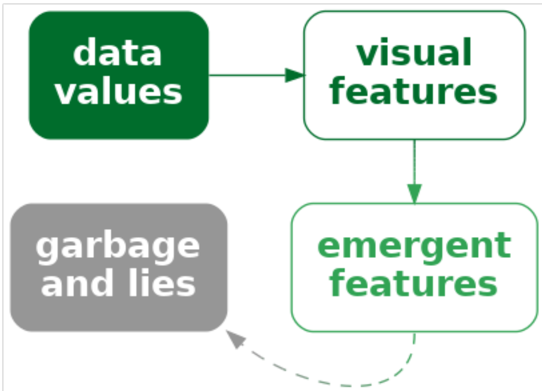


Figure 9.15: If we encode data values as visual features and the visual features interact, the result can be emergent visual features. However, this is only appropriate if the emergent features correspond to some property of the data. Otherwise, the decoding may lead to unhelpful and misleading interpretations.

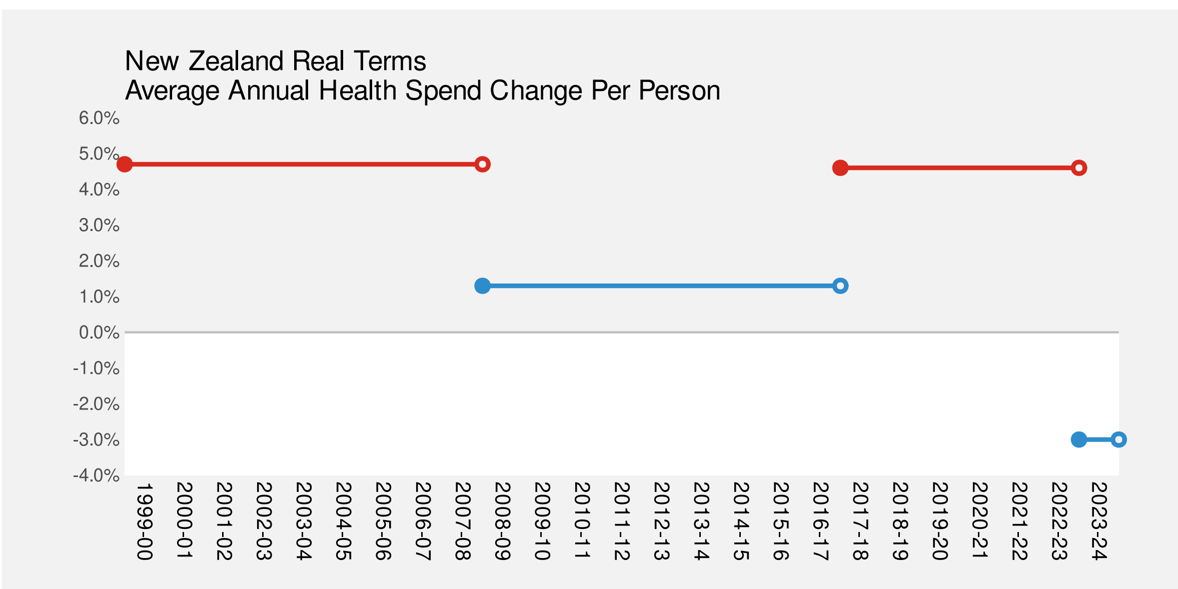
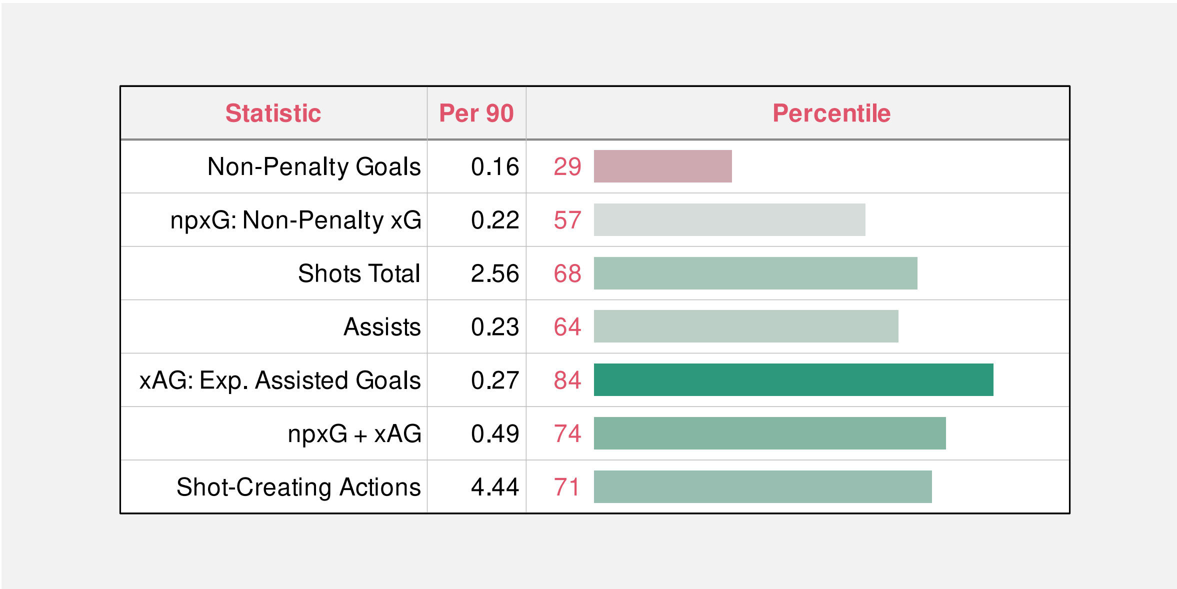


Figure 9.16: A plot showing the differences in health spending for successive New Zealand governments using horizontal lines.

is also a third encoding in Figure 9.17: the percentile values are also encoded as the colour of the bars.[126]



Statistic	Per 90	Percentile
Non-Penalty Goals	0.16	29
npxG: Non-Penalty xG	0.22	57
Shots Total	2.56	68
Assists	0.23	64
xAG: Exp. Assisted Goals	0.27	84
npxG + xAG	0.49	74
Shot-Creating Actions	4.44	71

Figure 9.17: FBREF statistics for Lamine Yamal from July 2024.

The additional encoding of percentile is **redundant** in the sense that we can already decode the perentile from the length of the bars. However, having the additional encoding means that we have two ways to decode the percentile: from the length and from the colour (Figure 9.18).

The effectiveness of a redundant encoding partially depends upon the independence of the two visual features. The features are only redundant if we are able to decode data values from the two features separately, as we can with colour and length.

One useful application of redundant encoding is to accommodate viewers with CVD (Section 6.7). Having a secondary decoding alongside colour allows for the fact that the colour decoding may fail for some viewers. For example, Figure 9.19 shows what Figure 9.17 might look like to a viewer with severe red-green colour blindness. The distinction between red and green bars is gone and that in turn confuses the decoding of the bar luminance (changes in luminance are not monotonic with changes in length). However, thanks to the redundant encoding, the simple comparison between bar lengths remains.

One argument against using a redundant encoding is that we are introducing additional visual complexity. Each individual visual feature may not be as effective as it would be on its own because it is not the *only* visual change (Section 2.5). For example, Figure 9.20 shows a variation on Figure 9.17 with the redundant encoding removed—all of the bars are just the same dark grey. It is arguably easier to decode the lengths of the bars in Figure 9.17 because

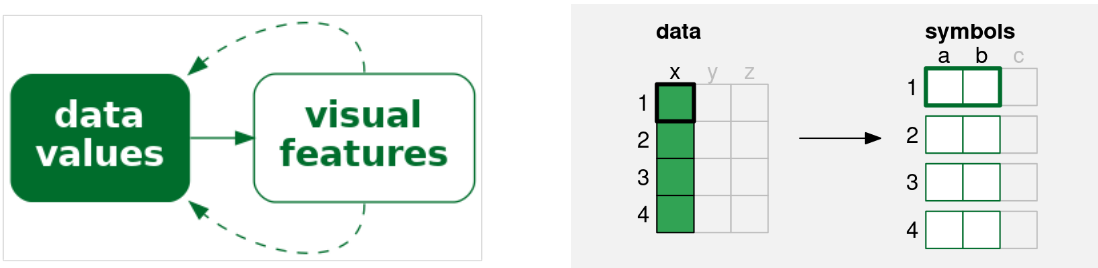
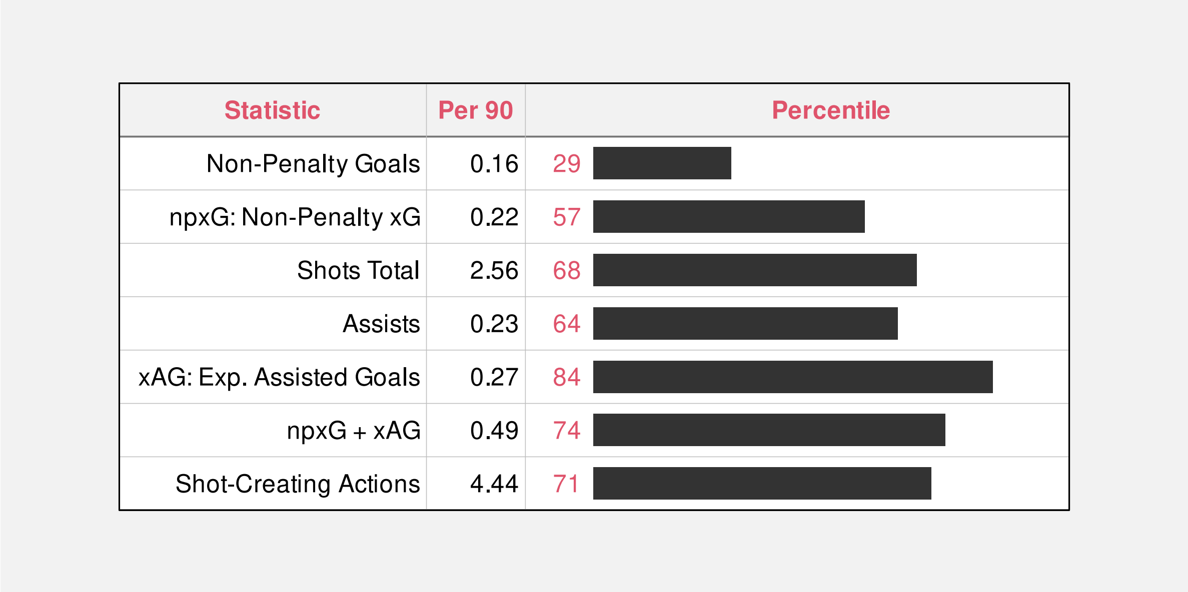


Figure 9.18: A redundant encoding encodes a single data value to multiple visual features. Each data value (`x`) is encoded to not just one visual feature (`a`), but also a second visual feature (`b`), of a single data symbol. If we encode data values using multiple visual features, then multiple decodings are possible.

Statistic	Per 90	Percentile
Non-Penalty Goals	0.16	29
npxG: Non-Penalty xG	0.22	57
Shots Total	2.56	68
Assists	0.23	64
xAG: Exp. Assisted Goals	0.27	84
npxG + xAG	0.49	74
Shot-Creating Actions	4.44	71

Figure 9.19: FBREF statistics for Lamine Yamal from July 2024. This is a version of Figure 9.17 with the colours adjusted to simulate severe red-green colour blindness.

the bars are less complex visual objects. This is even more true when comparing with the confusing changes in luminance of the bars in Figure 9.19.[127]



Statistic	Per 90	Percentile
Non-Penalty Goals	0.16	29
npxG: Non-Penalty xG	0.22	57
Shots Total	2.56	68
Assists	0.23	64
xAG: Exp. Assisted Goals	0.27	84
npxG + xAG	0.49	74
Shot-Creating Actions	4.44	71

Figure 9.20: FBREF statistics for Lamine Yamal from July 2024. This is a version of Figure 9.17 with the redundant colour encoding removed.

9.9 Summary

Almost all data visualisations involve *combinations* of encodings. More than one set of data values are encoded as more than one visual feature of data symbols.

The encodings involved in a bar plot—quantitative data values encoded as lengths and qualitative values encoded as position—are effective because we are able to perceive some combinations of visual features, such as length, position, and colour **independently**, which means that we can effectively decode both position and length from a bar plot.

A scatter plot is effective for perceiving relationships between variables because the encodings of quantitative data values to both horizontal and vertical positions **interact** to produce **position in space** and our visual system is capable of producing useful **visual summaries** from position in space, such as correlation.

Independence between visual features is useful when we want to decode separate data values. Interactions between visual features is useful when we want to produce an emergent feature that allows us to decode data summaries.

Conversely, independence between visual features is of no help if what we want is to decode a data summary from an emergent feature. Furthermore, interaction between visual features is unhelpful, or even misleading, if we cannot decode any meaningful information from the emergent feature that results from the interaction.

Notes

[114] Independent visual features are called *separable*, in contrast to *integral* features that interact. An early use of these terms in the context of data visualisation (or at least *statistical graphics*) is Carswell and Wickens (1988).

A further distinction is made for *configural* visual features, which are combinations that interact, but still allow decoding of the individual visual features (Carswell and Wickens 1990).

These ideas can be traced back to the much less readable Garner (1974).

[115] To be honest, there is an interaction between position and length because lengths that are further apart from each other vertically are harder to compare (Section 5.1). However, compared to visual features that strongly interact, length and position are relatively independent.

[116] The use of *emergent* visual features in this book, to describe the result of *integral* (or *configural*) visual features interacting, corresponds loosely to various usages of *emergent* in the literature.

MacEachren (1995) uses *emergent* in this way, but, for example, Cragin and Pomerantz (2015) and Palmer (1999) use *emergent* more to describe visual features that arise from combinations of separate visual objects (data symbols).

"The arrangement of several dots in a line give rise to emergent properties, such as length, orientation, and curvature, that are different from the properties of the dots that compose it." (Palmer 1999, fig. 2.1.5)

[117] The ability to decode correlations from scatter plots is another example of *ensemble* perception (Szafir et al. 2016; Rensink 2017).

[118] Munzner (2014) provides some examples of combinations of *visual channels* in Figure 5.10. Ware (2021) gives a broader selection of combinations of *display attributes* in Figure 5.24.

[119] This two-dimensional mosaic plot could also be called a spine plot. Mosaic plot is the more general term that includes plots of more than two qualitative variables.

Mosaic plots also provide a simple visual check for independence between factors. For example, in Figure 9.8 we can see that the action taken is reasonably independent of the sex of the offender because the heights of the rectangles are similar for both Male and Female offenders (the conditional proportions are roughly the same).

[120] An interaction plot is typically used in a more experimental setting where the x-axis might be different treatments and the groups might be different patient groups and the y-axis might be a measure of response to treatment. The outcome is then whether the treatment has had the same effect for different patient groups (or not).

[121] Pinker1990GraphComprehension (p. 111) discusses the problem of extracting different sorts of information from different data symbols, in particular bars versus lines.

[122] Examples of studies that demonstrate accuracy for decoding correlation, although with a tendency to underestimate, are Rensink and Baldridge (2010) and Cui, Kini, and Liu (2024).

On the other hand, Braun et al. (2025) claim that we are more likely to fit a line to a scatter plot based on orthogonal distances from line to points, rather than vertical, and hence more likely to over-estimate a slope.

These results also only apply to *linear* correlation, but the visual system is also very capable of decoding *non-linear* correlations between variables. This is one example where the flexibility of a visual summary makes it superior to a numerical summary. There are numerical summaries of non-linear correlation, such as distance correlation, but a visual summary provides much more useful information about a non-linear relationship.

[123] Figure 9.14 is based on Figure 2 from Huskinson (2024).

[124] The idea that combining visual features that interact is more appropriate for decoding data summaries (and vice versa) is similar to the *proximity compatibility principle* (Wickens and Carswell 1995).

Shah and Hoeffner (2002a) use the following nice quote from earlier work (Carswell and Wickens 1987):

"Integrated, object-like displays (e.g., a line graph) are better for integrative tasks, whereas more separable formats (e.g., bar graphs) are better for less integrative or synthetic tasks such as point reading."

[125] Figure 9.17 is based on an image produced by the FBREF web site for football statistics and history.

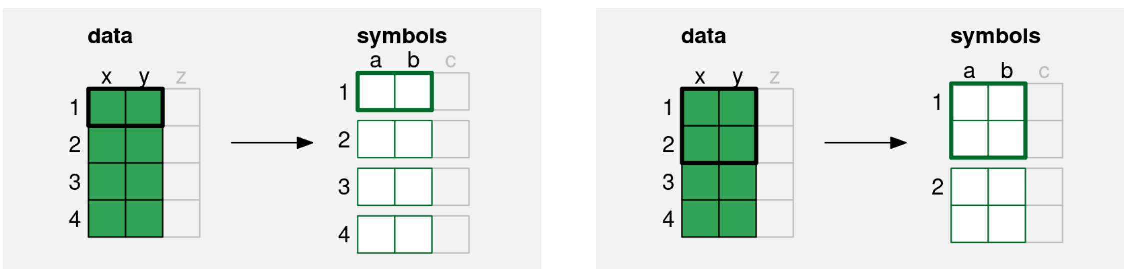
[126] In fact, because the colours are from a diverging colour palette (Section 6.11), there are arguably four encodings: whether the percentile is below 50 or above 50 is encoded as a red or green hue and the amount above or below 50 is encoded as the luminance/chroma of the bar.

[127] Redundant encodings are not universally supported in the literature either. For example, VanderPlas and Hofmann (2017) provide evidence that additional cues improve plot perception and Borkin et al. (2016) report that "redundancy helps effectively communicate the message". However, Franconeri et al. (2021b) are less in favour: "redundant encoding should be avoided in most cases, except when used to make visualizations accessible for viewers with color-vision impairments."

10 Visual Shapes

All of the data visualisations that we have considered so far have encoded a **single data value** from one or more variables to a **single data symbol** (Figure 10.1a). For example, a bar plot like Figure 3.1 encodes the number of offenders for one ethnic group to one bar. A scatter plot like Figure 9.4 encodes the number of clean breaks and the number of tries for one team to one data point.

In this chapter, we consider data visualisations that encode **multiple data values** from one or more variables to a **single data symbol** (Figure 10.1b). In this chapter, we begin to consider more complex data symbols that produce **visual shapes**.[128]



(a) (b)

Figure 10.1: (a) A simple data symbol, like a bar in a bar plot or a data point in a scatter plot involves encoding **one** row of data values to the visual features of **one** data symbol; a single pair of data values produces a single data symbol (this is a reproduction of the diagram in Figure 9.2). (b) A more complex data symbol is produced if we encode **multiple** rows of data values to the visual features of **one** data symbol. Multiple data values from `x` and `y` are encoded to the visual features, `a` and `b`, of a single data symbol. Multiple pairs of values produce a single data symbol.

10.1 Line plots

Figure 10.2 shows a line plot of the number of offenders per year in different ethnic groups from 2011 to 2021 (Table 5.1). This data visualisation is very effective for perceiving changes over time in the number of offenders within each group as well as comparing the changes over time between groups.

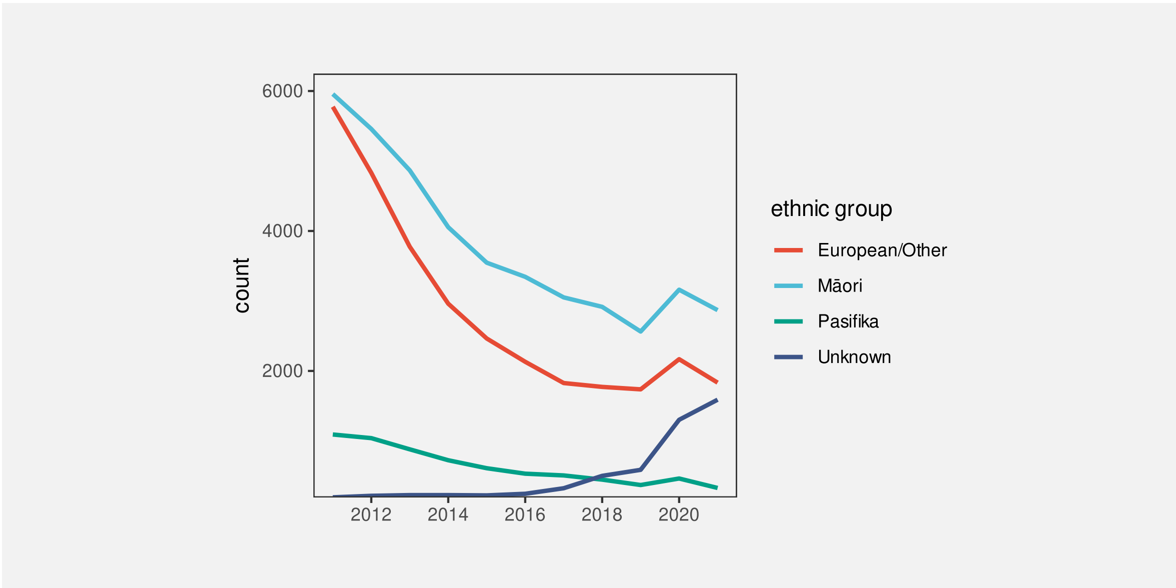


Figure 10.2: A line plot of the number of offenders per year in different ethnic groups from 2011 to 2021.

There are some familiar encodings in this data visualisation: year is encoded to horizontal **position**, the number of offenders is encoded to vertical **position**, and the ethnic group is encoded to **colour**. However, the **data symbols** in this data visualisation, the coloured lines, are different from any previous examples that we have seen. Although there are 44 rows of data, there are only 4 data symbols. Multiple rows of data are used to create each data symbol. Each coloured line in Figure 10.2 represents 11 pairs of year and count values.

This contrasts with visualisations like bar plots that use simpler data symbols. For example, the bar plot of the same data in Figure 5.7 has 44 data symbols (44 bars).

Figure 10.2 is effective at conveying changes over time because the coloured lines create **visual shapes** (Figure 10.3) and those visual shapes convey information about collections of data values. For example, we can perceive slopes and curves and peaks and valleys in each of the coloured lines.[129]

By comparison, in Figure 10.4, which encodes each data value to its own data symbol, it is much harder to decode the slopes, curves, peaks, and valleys. Despite the colours of the data symbols, it is even harder to group the different sets of data symbols together; the connecting lines between the data values in Figure 10.2 are a very strong encoding that the data values belong to the same ethnic group (Section 2.7).

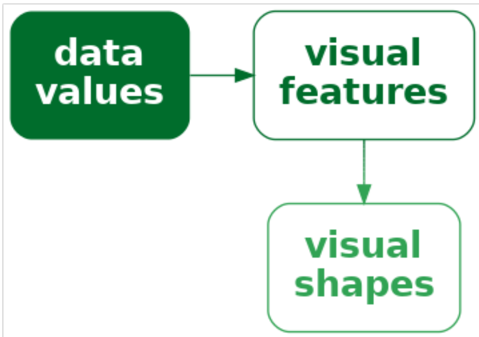


Figure 10.3: If we encode multiple data values to multiple visual features of a single data symbol, like in a line plot, we create a smaller number of more complex data symbols, **visual shapes**, that contain a larger amount of information.

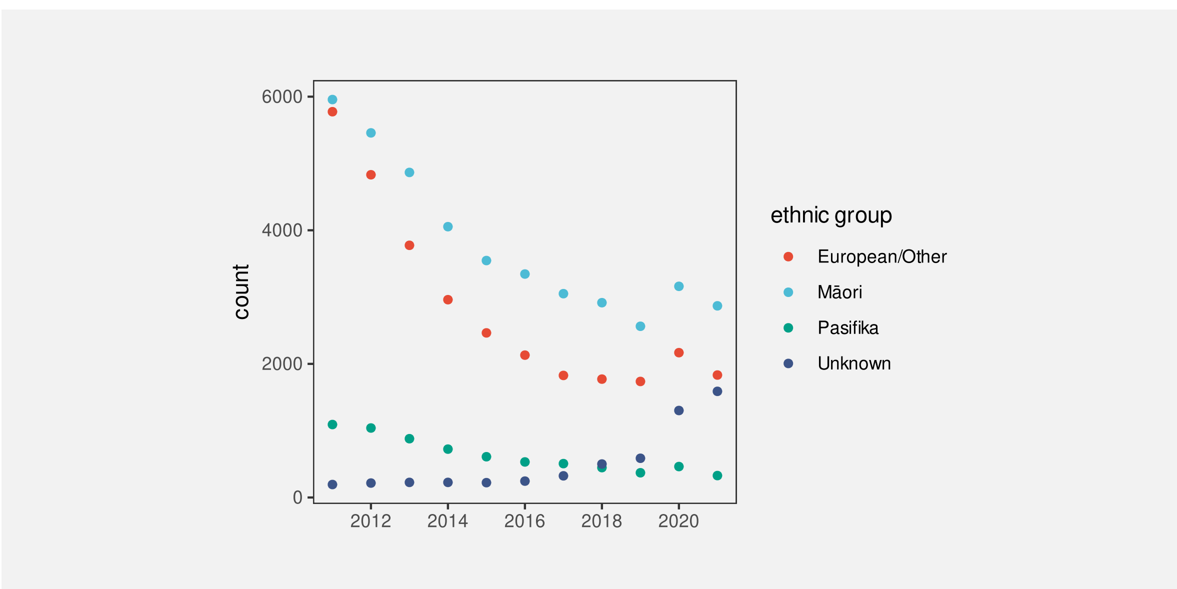


Figure 10.4: A scatter plot of the number of offenders per year in different ethnic groups from 2011 to 2021.

In terms of the simple visual processing model from Section 2.2, these more complicated shapes can convey higher-level information, at the expense of a little more visual processing, and with a limit on how many shapes we can process simultaneously (Figure 2.3). In terms of Chapter 8, the more complex data symbols create visual shapes that allow us to decode **data summaries** (Figure 10.5). For example, in Figure 10.2 we are able to decode data summaries such as the overall trends in the number of offenders, periods of more rapid change versus periods of slower change, and local maxima and minima.[130]

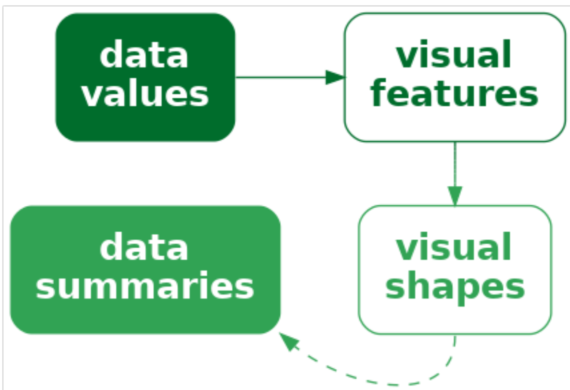


Figure 10.5: If we encode data values to a **visual shape** that contains a large amount of information, then we may be able to decode higher-level **data summaries**.

This ability to decode data summaries, such as peaks and slopes, when we use lines as data symbols is even more apparent if we compare Figure 10.2 with Figure 5.7. When we have bars as data symbols, these data summaries are much less apparent.

In terms of Chapter 7, the use of lines for data symbols is also **congruent** with trends in the data; we naturally associate slopes with increases and decreases in data values.[131] For example, it is very natural to decode a downward trend from the lines in a line plot like Figure 1.1. While it is also possible to decode trends from the coloured rectangles in a heatmap like Figure 1.2, the decoding is less immediately obvious.

> One reason why line plots are effective is because they encode multiple data values into a complex data symbol that creates a visual shape, from which we are able to decode data summaries.

On the other hand, the more complex data symbols in a line plot, where each data symbol contains multiple data values, are less effective at encoding the *individual* raw data values. The coloured lines in Figure 10.2 are not simple shapes from which it is easy to decode basic **visual features**. This means that, although we may be able to decode data summaries, it is not as easy to decode the raw data values (Figure 10.5). For example, compare the ease with which we can determine the number of European/Other offenders in 2013 from Figure 10.2 versus Figure 5.7. Figure 5.7 has simple data symbols that encode a single data value as a basic visual feature of a single data symbol—the length of a bar—so we can easily decode the individual raw data values from the bars.

This deficiency is significant because the main encodings in a line plot are from **quantitative** data values to (horizontal and vertical) **position**. We know from Chapter 3 that these should be very effective encodings that allow us to accurately decode the raw data values. However, in a line plot, we are not encoding a single data value to a basic visual feature of a single, simple shape. Instead, a single data value is being encoded, along with many other values, to a basic visual feature of a complicated shape. In effect, the individual data values lose their individual identity.[132]

> One reason why a line plot is *less* effective for decoding raw data values is because each raw data value is not encoded to its own data symbol.

The creation of visual shapes from multiple data values bears some similarity to the creation of emergent features by combining visual features (Section 9.2). In both cases, we have *explicit* encodings from data values to basic visual features, but we also produce an *implicit* additional visual effect. The difference here is that, with visual shapes, we are collapsing multiple data values into a data symbol, so we run a higher risk of losing the ability to perceive individual data values.[133]

There is also a similarity between visual shapes and visual summaries (Section 8.2). In both cases, we gain the ability to decode data summaries from collections of data values. The difference is that, with visual summaries, we are summarising many simple data symbols that each represent individual data values, whereas with a visual shape we decode the data summary from a single data symbol that obscures the individual data values. In the case of a visual summary, the individual data values can still be decoded, but with a visual shape that is more difficult.

10.2 Case study: Scatter plots with lines

Figure 10.6 shows a data visualisation of the number of offenders per year in different ethnic groups (Table 5.1). This is very similar to Figure 10.2, but we have added a data point at every horizontal and vertical **position** as well as the lines between positions.

Like Figure 10.2, we can decode overall trends from the line data symbols, but we can also decode individual data values more easily because now every individual data value is also encoded to its own data symbol. By employing both simple data symbols that encode individual data values and complex data symbols that encode multiple data values as visual shapes, we get the best of both worlds.

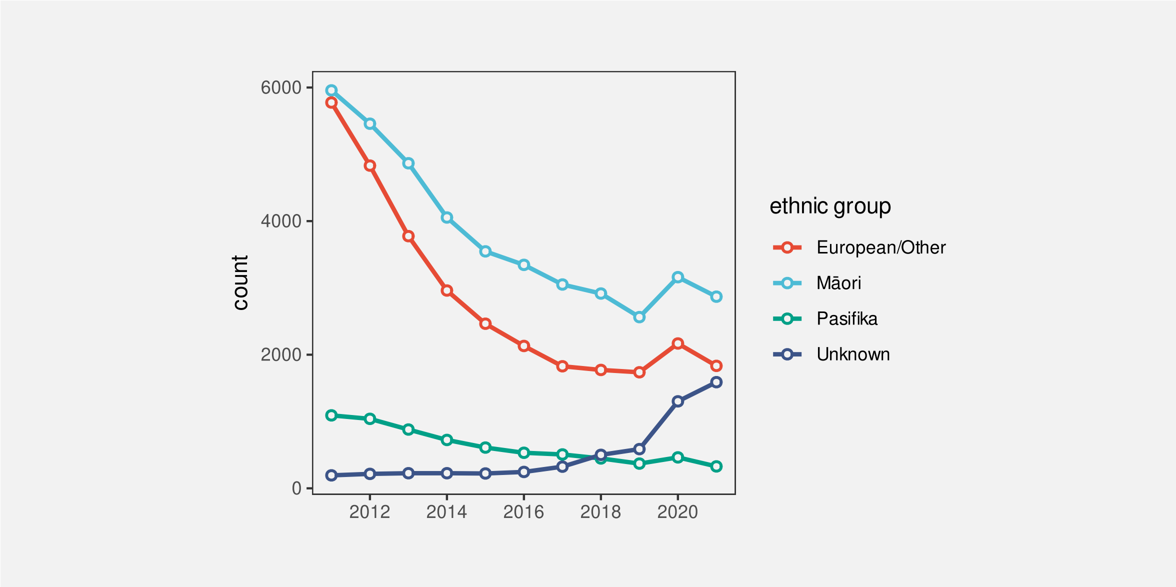


Figure 10.6: A plot of the number of offenders per year in different ethnic groups from 2011 to 2021 with both lines and points.

10.3 Case study: Density plots

Figure 10.7 shows a density plot of the number of points scored by Tier One nations at the Rugby World Cups (Table 4.1). As with a histogram (Figure 8.5), this data visualisation is effective for conveying the distribution of data values. For example, we can decode that there is evidence of a single mode and right skew.

Also like a histogram, a density plot encodes **data summaries**, rather than raw data values, to visual features (Table 10.1). In the case of a histogram, the data summary values are counts within bins (Table 8.2) and each data summary produces a separate bar. In the case of a density plot, the data summary produces density estimates and these are encoded to a single data symbol, a line, to create a **visual shape**. The density estimates are encoded to the horizontal and vertical **positions** of a single line data symbol. The visual shape allows us to decode summaries of the density estimates. For example, whether there is a single maximum density and how the density changes to either side of that peak.

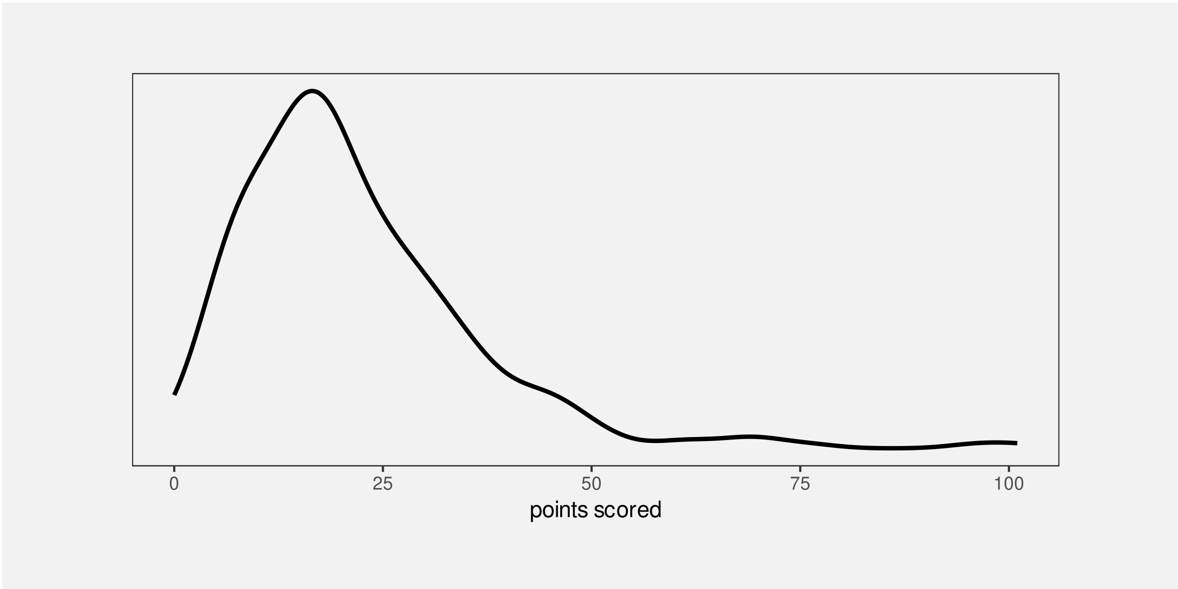


Figure 10.7: A density plot of the points scored per game by Tier One nations at Rugby World Cups.

Table 10.1: A table of density estimates at a sample of values over the range of the number of points scored by Tier One nations in Rugby World Cup matches. These data summaries are the values that are encoded to visual features to produce the density plot in Figure 10.7. There are 512 rows of values, but only the first 6 are shown here.

scored	density
0.00	0.0055
0.20	0.0059
0.40	0.0063
0.59	0.0067
0.79	0.0071
0.99	0.0075

Figure 10.8 shows the path from raw data values to data summaries (density estimates), which are encoded as the visual features (horizontal and vertical position) of a visual shape (a line) in a density plot and the summary information that we can decode from the visual shape. We cannot decode the raw data values from a density plot, nor can we very easily decode the data summaries (the density estimates). The only real purpose of the data symbol in a density plot is to allow us to decode **summaries** of the data summaries, which are visual features of the density estimates (like the number of peaks and where the peaks occur).

> A density plot is effective because it conveys high-level features of density estimates of the raw data.

Another commonality between density plots and histograms is that we get to choose how the data summaries are calculated from the raw data values. In the case of a density plot, we must choose a *kernel* and a *bandwidth*, both of which affect the smoothness of the density estimates. In other words, the choice of kernel and bandwidth affects the shape of the line data symbol and that affects the data summaries that we will decode (peaks, slopes, etc). In a histogram, we may need to explore multiple choices of bin width and in a density plot we may need to explore more than one kernel and bandwidth (Section 8.5).

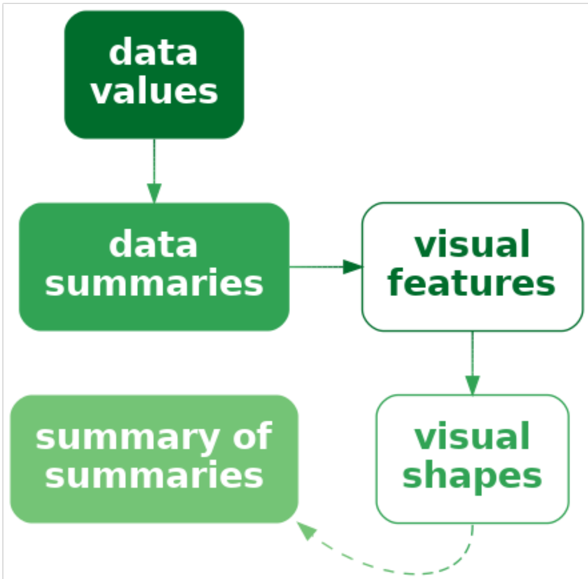


Figure 10.8: A density plot encodes data summaries to visual features to create a visual shape from which we can decode **summaries** of the data summaries.

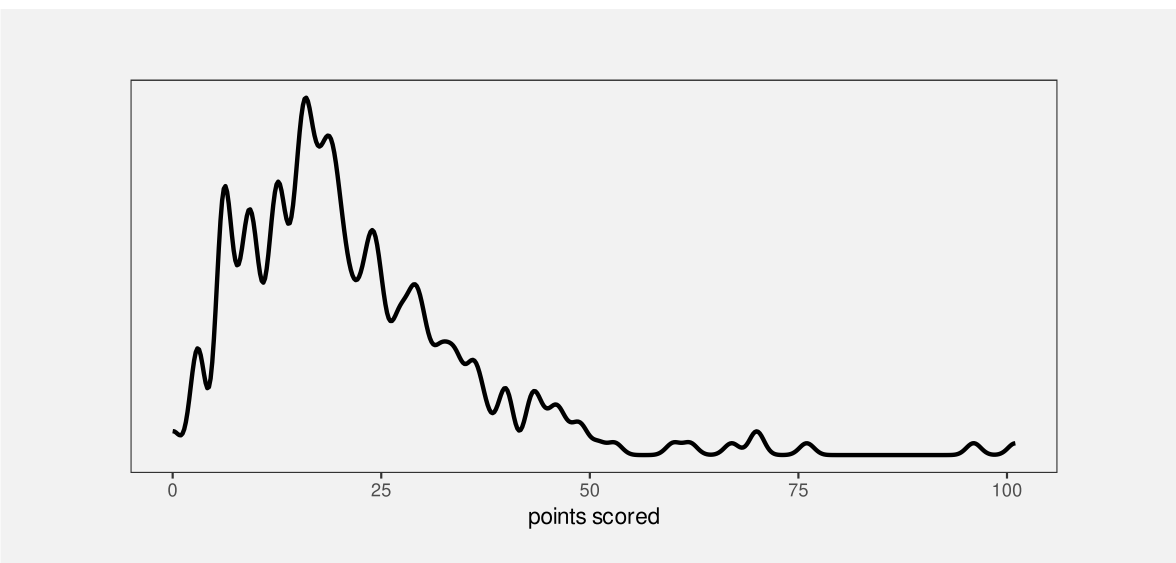


Figure 10.9: A density plot of the points scored per game by Tier One nations at Rugby World Cups. This is a variation of Figure 10.7 using a much narrower bandwidth, so the density curve is much less smooth.

it difficult to perceive the separate histograms and this is not satisfactorily solved by adding semitransparency. By comparison, multiple density curves create very little overlap, and even though the orange curve passes "under" the blue curve, we effortlessly perceive the orange curve as a single, coherent shape (Section 2.7).

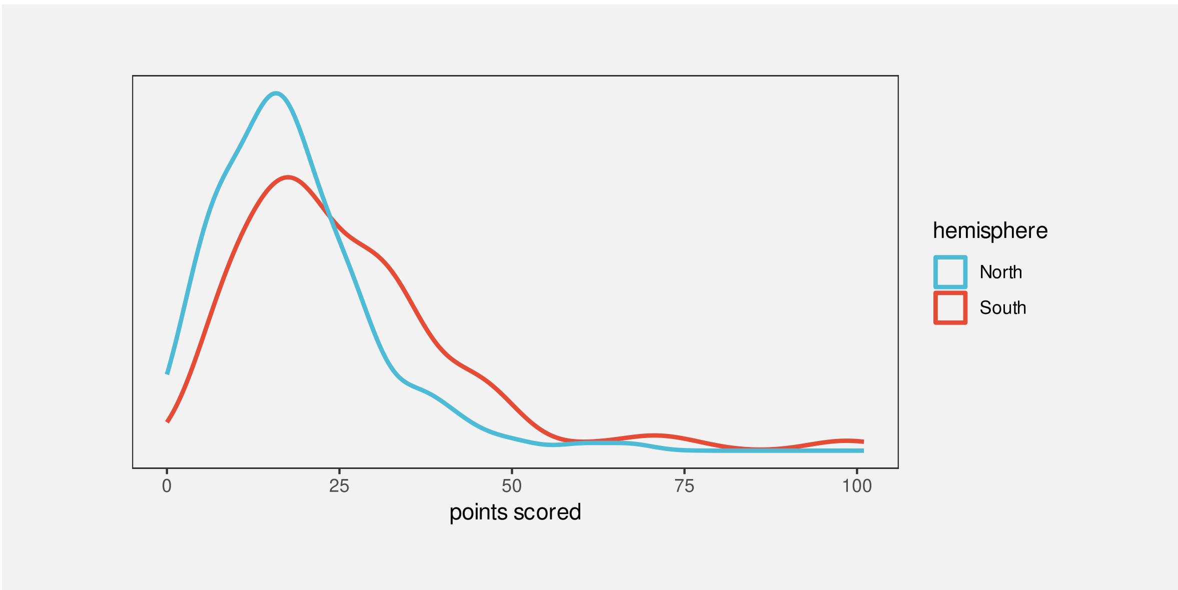


Figure 10.10: Density plots of the points scored per game by Tier One nations at Rugby World Cups, broken down by hemisphere.

10.4 Visual features of visual shapes

We have emphasised that a data symbol like a line in a line plot is the result of encoding multiple data values as the **positions** that the line is drawn between. This results in a **visual shape** that is more effective for decoding **data summaries** rather than decoding the individual data values.

However, the examples that we have seen in Figure 10.2 and Figure 10.10 also involve another encoding: qualitative data values, like ethnic group or hemisphere, are encoded as the **colour** of the lines. This encoding is consistent across the entire data symbol. In effect, there is a simple encoding of qualitative data values to a basic **visual feature** of the visual shape.

This means that it is possible to decode individual data values, such as ethnic group or hemisphere, from a visual shape if the data values have been encoded as a single consistent visual feature of the visual shape (Figure 3.6).

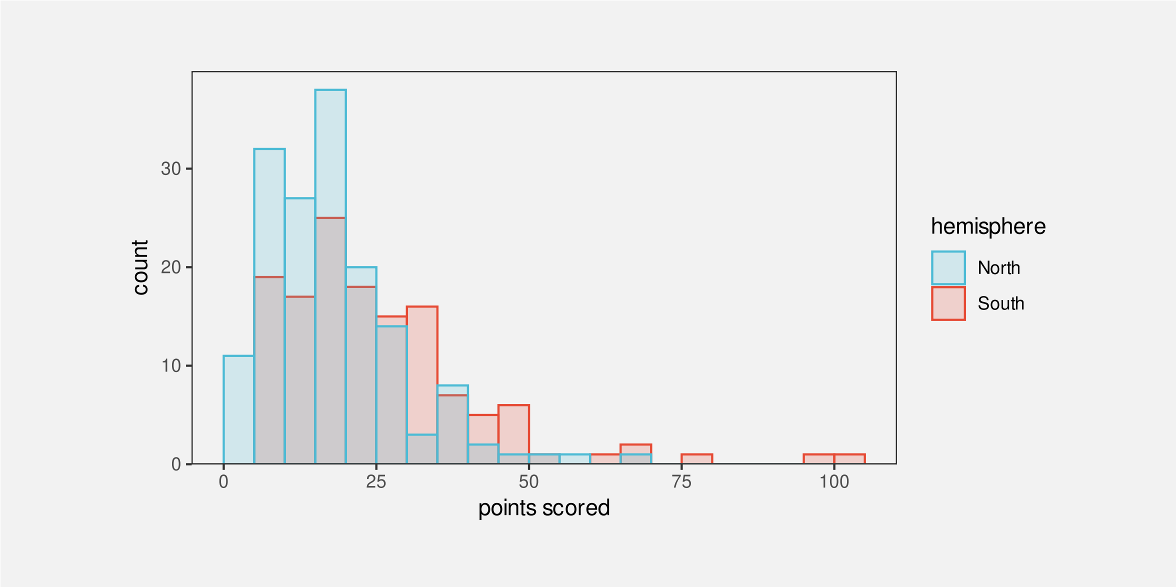


Figure 10.11: Histograms of the points scored per game by Tier One nations at Rugby World Cups, broken down by hemisphere.

10.5 Aspect ratio

Figure 10.12 shows a variation on Figure 10.2 using a different **aspect ratio**—the height of the plot divided by the width of the plot. Figure 10.2 has an aspect ratio of 1 (the plot region is square), but Figure 10.12 has an aspect ratio of only 0.2 (the plot region is much shorter than it is wide).

Figure 10.12 shows that the aspect ratio of the plot has a large effect on the visual shapes of the data symbols, which means that the aspect ratio has a large effect on the data summaries that we decode from the visual shapes. For example, in Figure 10.12, we can still see a general downward trend in three of the lines, with one line increasing. However, the peak or jump in values at 2020 is much less obvious. In other words, we decode different data summaries from Figure 10.12 compared to Figure 10.2 purely because of the difference in aspect ratio.

There is no single rule for what the correct aspect ratio should be,[134] so we may need to experiment with more than one aspect ratio to check what visual shapes are produced and whether it is possible to decode the features of interest.

Figure 10.13 demonstrates that a similar problem exists for decoding data summaries from scatter plots (Section 9.2). These plots encode the number of offloads (a fancy pass under pressure in rugby) as the horizontal position of data points and the number of tries (scoring actions) as the vertical position. The correlation between the two measures can be decoded

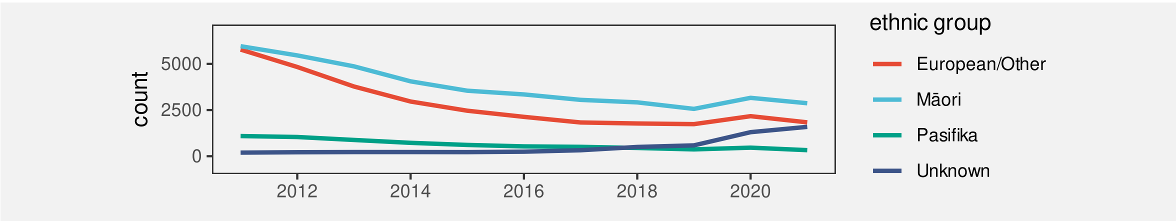


Figure 10.12: A line plot of the number of offenders per year in different ethnic groups from 2011 to 2021.

from the emergent position in space of the data points. The data is exactly the same for all four plots.

The decoding of trends in the points on a scatter plot has similarities to the decoding of trends from visual shapes in a line plot[135] and the shapes that we decode from a collection of data points are also affected by the aspect ratio of a plot. For example, the top-left plot in Figure 10.13 suggests a stronger relationship between offloads and tries than the bottom-left plot simply because the trend appears flatter in the bottom-left plot and that is solely thanks to the lower aspect ratio of the bottom-left plot.

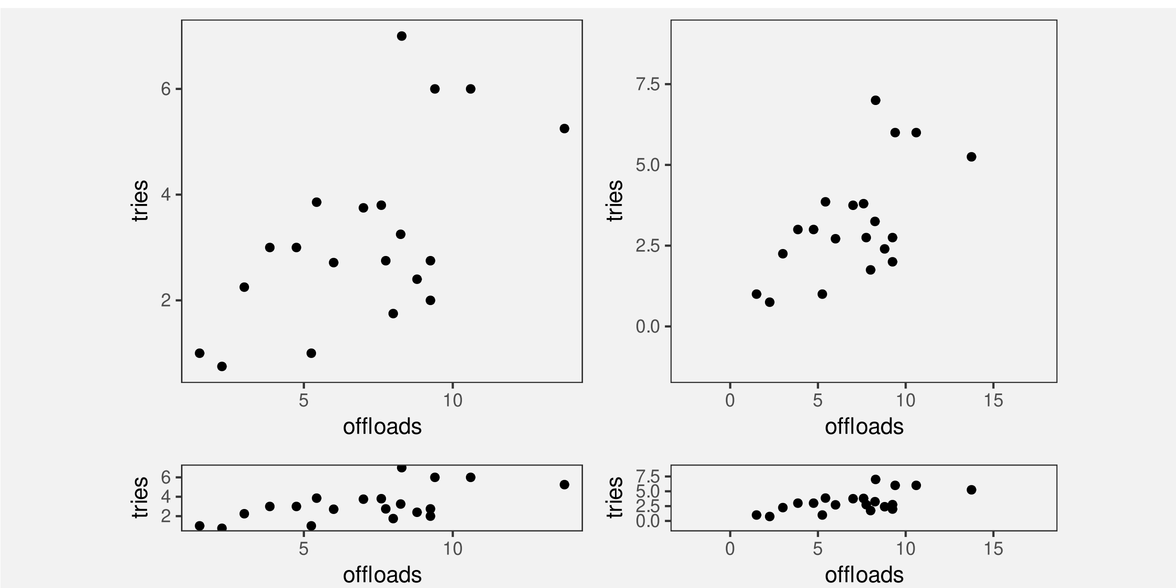


Figure 10.13: A scatter plot of the number of times a team breaks through the opposition defence and the number of tries that a team scores (both are per-game averages) for teams at the 2023 Rugby World Cup.

A related issue is the amount of empty space around the data points in a scatter plot. IN Figure 10.13, the relationship appears stronger in the top-right plot compared to the top-left plot simply because the data points are clustered more densely thanks to the different ranges on the plot axes.[136]

10.6 Decoding visual shapes

As we saw in Section 10.4, if data values are encoded as a single, consistent visual feature of a visual shape, for example the colour of a line, then the original data values can be decoded from the basic visual feature, as described in earlier chapters for simpler data symbols like bars.

For data values that are encoded as different values of a visual feature, such as the positions that a line is drawn between, decoding from a visual shape results in data summaries (Figure 10.5), for example peaks and valleys. The visual system's ability to rapidly and effortlessly identify a wide range of data summaries is part of why data visualisation is so effective. It is difficult to calculate data summaries numerically from the raw data with the same flexibility.[137] On the downside, as we saw in Section 8.4 and Section 9.6, the huge range of possible data summaries means that less is known about how accurately we can decode any particular data summary.

There is experimental support for the intuitive sense that we can perform well at decoding qualitative features from shapes, such as peaks and valleys, as well as decoding ordinal comparisons, such as steeper and shallower.[138] However, as Section 10.5 demonstrated, the decoding of slopes is limited to *physical* slopes, which are dependent on the aspect ratio, rather than decoding absolute rates of change in the data.[139]

10.7 Case study: Box plots

Figure 10.14 shows box plots of the number of points scored by Tier One nations at Rugby World Cup matches. We have discussed box plots previously as examples of data visualisations that encode data summaries to data symbols (Section 8.1). For example, the median points scored is encoded as the horizontal position of a thick vertical line for each team. The only raw data values that are directly encoded are the outliers, which are encoded as the horizontal position of data points for a few teams. A box plot is effective for decoding the median for each team because the median has been encoded as basic visual features of simple data symbol.

Box plots are also examples of encoding more than one data value (or data summary) to a single data symbol. For each team, we have a complex data symbol made up of a rectangle and horizontal and vertical lines, plus possibly some data points. This creates a visual shape that allows us to decode additional data summaries. For example, some box plots, like the one for England, are symmetric, with the median line at the center of the rectangle and with horizontal lines of similar lengths to either side. Other box plots, like the one for Argentina, are asymmetric, with the median towards one end of the rectangle and a longer horizontal line

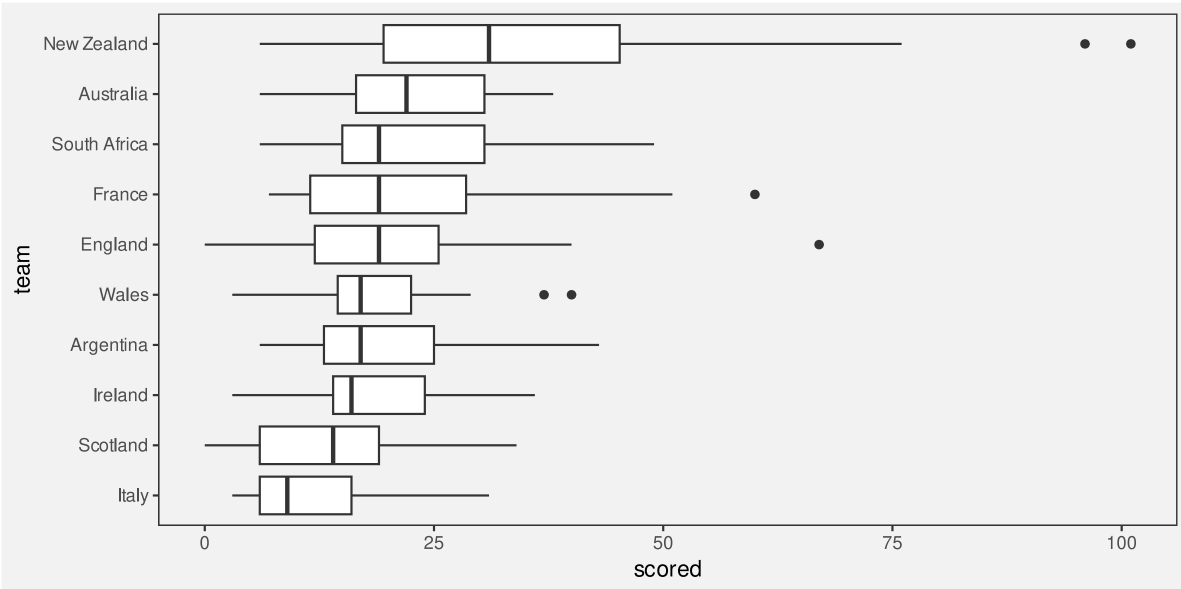


Figure 10.14: Box plots of the number of points scored by Tier One nations at Rugby World Cups.

to one side. In other words, we can decode information about the skewness of the data from the visual shape (Figure 10.5).

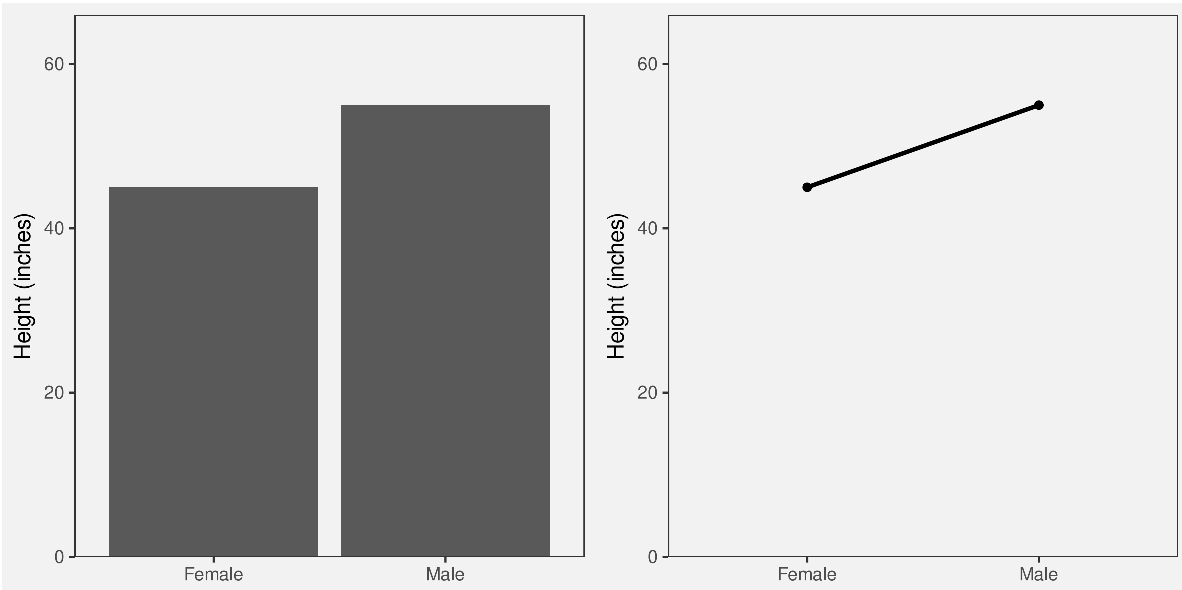
10.8 Dangers of visual shapes

Figure 10.15 shows two data visualisations of average heights for males and females. We have already discussed that it is easier to decode the individual average height values from the bar plot because each average height is encoded as its own visual feature—the height of one bar. We have also discussed that it is easier to decode trends from a line plot (Section 10.1).

However, in Figure 10.15, the temptation to decode a trend from the line plot is in fact misleading. A line implies continuity, in this case even a linear trend. However, the data values on the x-axis are qualitative categories (biological sex), rather than a continuum. This means that decoding a trend from females to males does not actually make sense.[140]

A visual shape can be decoded to a data summary, but that is only useful if the data summary is appropriate and not misleading (Figure 10.16).

Figure 10.17 shows another example of an undesirable shape decoding. The two curves in Figure 10.17 are identical except that one has been shifted vertically. In other words, the vertical distance between the two lines is constant. In a data visualisation, we are typically interested in that vertical distance: what is the difference in y between the two lines?



(a) A bar plot supports decoding individual values. (b) A line plot supports decoding trends.

Figure 10.15: Two data visualisations of average heights of females and males.

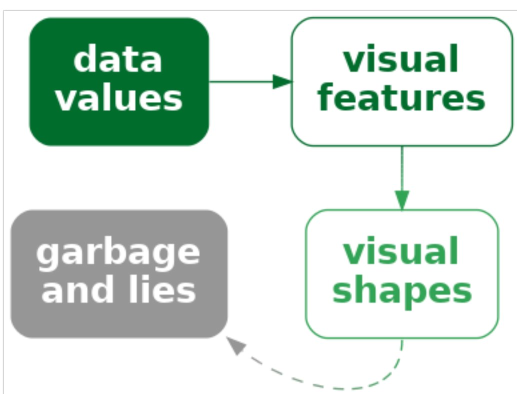


Figure 10.16: Encoding data values to the visual features of a complex shape may not be effective if the complex shape decodes to unhelpful or misleading information.

However, the natural decoding of the distance between the lines focuses on the perpendicular distance, so the two lines appear to be converging.[141] In other words, we decode the shape of the gap between the lines to be a narrowing gap.

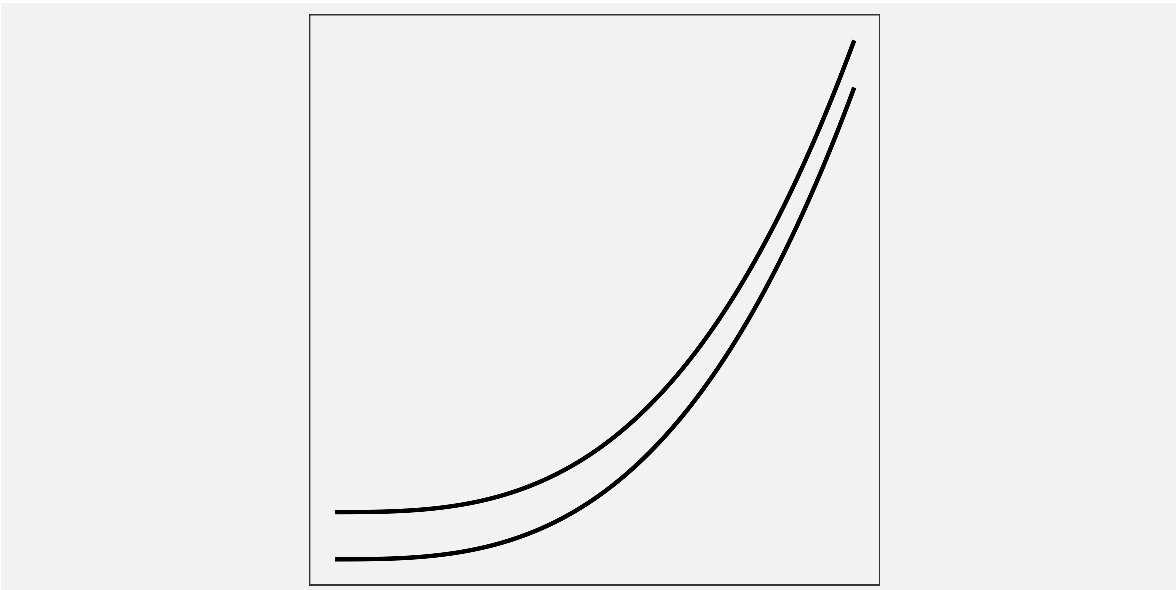


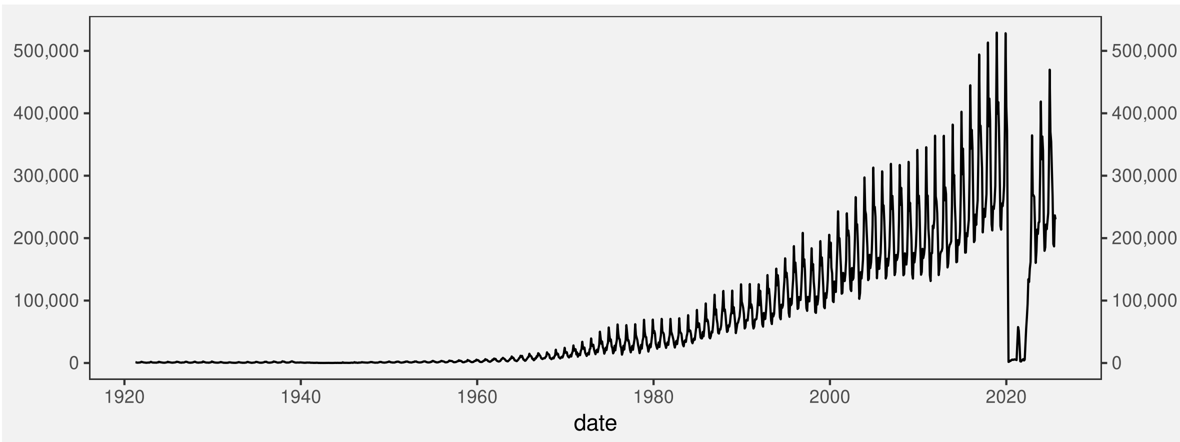
Figure 10.17: The two lines in this plot are just the same curve shifted vertically—the vertical distance between the lines in constant. However, we decode the visual shape between the lines as a decrease in the gap between the lines.

10.9 Non-linear encodings

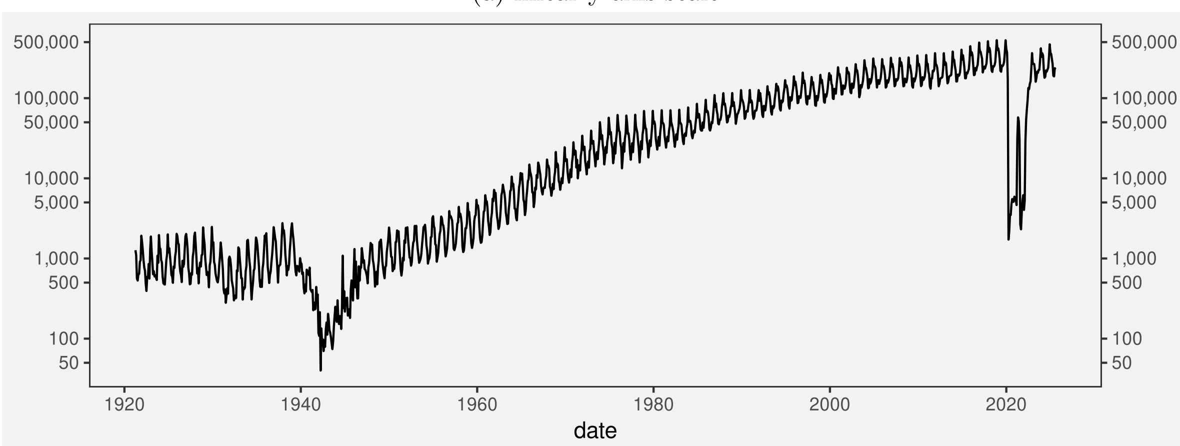
Figure 10.18 shows time series plots of the number of international arrivals to New Zealand over the last century.[142] Figure 10.18a employs a linear scale on the y-axis and, because the amount of international travel has changed so dramatically over time, it is impossible to decode any detail from the first half of the twentieth century.

We saw in Section 4.7 that one approach in this sort of situation is to use a non-linear scale. Figure 10.18b uses this approach by employing a log (base 10) scale on the y-axis. The advantage of the non-linear scale is clear: it is now possible to see details in the data for the first half of the twentieth century. For example, we can now see the effect of the Second World War (1939–1945), in addition to the effect of the COVID pandemic (2020–2021), and we can see that the seasonal patterns are similar over time.

An important point is that *all* of these decodings are from the visual shapes in Figure 10.18b. We are able to decode an overall trend, local dips within the trend, and regular spikes within the overall trend.



(a) linear y-axis scale



(b) log (base 10) y-axis scale

Figure 10.18: Figures reproduced from Zacks and Tversky (1999)

However, as we pointed out in Section 4.7, the decoding of simple visual features is compromised by the use of a non-linear scale. For example, it is very difficult to decode the final value of the time series in Figure 10.18b because the same distance between the tick marks on the y-axis does not represent the same difference in data values (as emphasised by the choice of tick mark labels). As we can see from Figure 10.18a, the final data value in the time series is *not* half-way between 100,000 and 500,000.

It is *possible* to decode quantitative values from Figure 10.18b. For example, the height of the seasonal changes remained fairly constant over time. However, that is an observation about the *logged* data, not the raw data values.[143] The danger of a log scale is that our natural decoding of position and length is now incorrect. For example, (in the absence of Figure 10.18a) we might be tempted to incorrectly decode from Figure 10.18b that the absolute seasonal change in arrivals has remained constant over time. In other words, the natural decoding of simple visual features can be misleading in the presence of a non-linear scale.

10.10 Case study: log-log plots

Figure 10.19 shows a scatter plot of metabolic rate versus body mass for a wide range of lifeforms.[144] Because of the very wide range of body mass values and metabolic rates, with the largest animals being many orders of magnitude larger than the smallest bacteria, almost all of the data is crushed into the bottom-left corner.

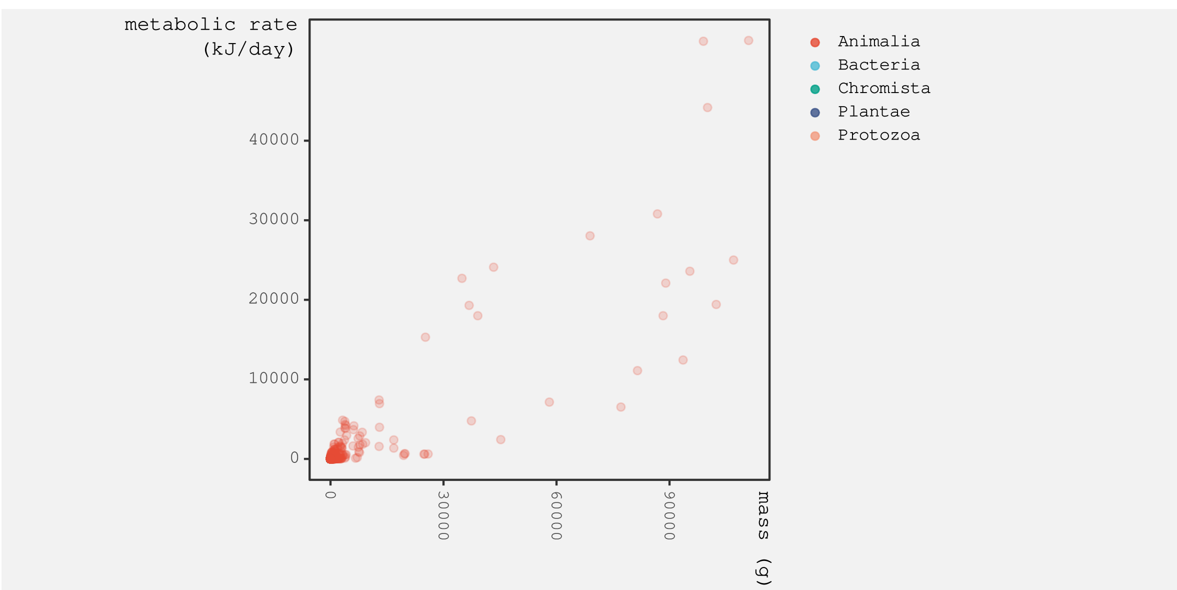


Figure 10.19: Body mass versus metabolic rate for a wide variety of lifeforms.

Figure 10.20 shows the same data, but this time with log (base 10) scales on both axes. While there is still considerable overlap, thanks to the large number of observations, we can now see data for the other Kingdoms beyond Animalia.

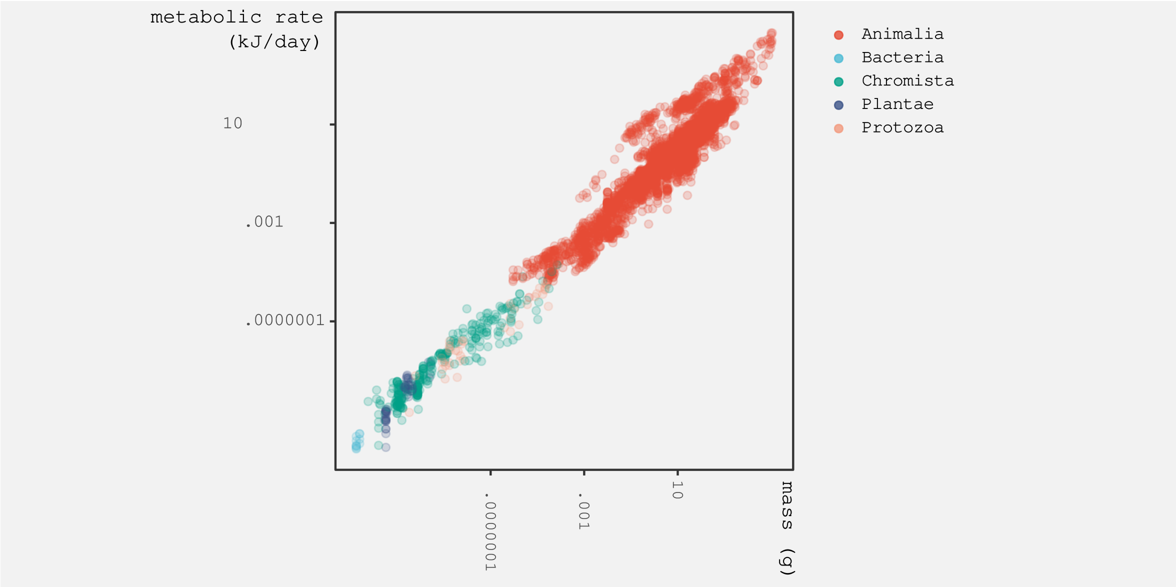


Figure 10.20: Body mass versus metabolic rate for a wide variety of lifeforms. Log (base 10) scales.

What we can also see in Figure 10.20 is a clear **visual shape** amongst the data symbols—a strong linear relationship. It is useful to be able to decode this linear shape from a plot with log scales because it indicates a power relationship between the two variables, in this case metabolic rate and body mass.[145] It is also possible to decode ordinal information, such as the fact that animals are larger and use more energy than bacteria.

However, decoding the basic **position** of data symbols from a plot with log scales is very difficult because equal steps along either axis do *not* decode to equal steps between data values (as indicated by the axis labels).
For example, we cannot easily decode *how much* bigger animals are than bacteria. There is even danger, for the uninitiated, in decoding the linear **visual shape** because it does *not* decode to a linear relationship between the raw data values.

Once again, we have a data visualisation that can be usefully decoded in a particular way, but which also contains encodings that are difficult and potentially misleading.

10.11 Summary

Encoding multiple rows of data values to a single data symbol produces a **visual shape**, like a line on a line plot.

The main benefit of a data symbol that is a visual shape is that data summaries, such as modes, skewness, local maxima and minima, and trends over time, can can be decoded from a visual shape. On the downside, decoding raw data values from a visual shape may be harder, compared to decoding raw data values from a simple data symbol like a bar.

One danger with visual shapes is that they depend on aspect ratio and scale. The same encodings can be made to decode to different data summaries, so we are responsible for selecting an appropriate aspect ratio and scale.

Another danger is that a visual shape may not necessarily convey a useful data summary. We need to only create visual shapes in a purposeful manner and avoid accidentally creating visual shapes that may confuse or mislead.

Notes

[128] In this chapter, we focus on data symbols that represent multiple sets of data values, like lines. It could be argued that there are data symbols that are visual shapes that arise from encoding a single, qualitative data value to a data symbol, for example, different data points—circles versus triangles versus squares—in a scatter plot. However, in this book, that case is referred to as a encoding from qualitative data value to the *pattern*—a basic visual feature—of a data symbol (Section 3.2).

[129] Few (2006, 10) discusses this difference in a comparison between a density plot (mostly see a shape) and a histogram (mostly see bar heights).

[130] "not only is a line a single perceptual unit, and hence there will be less to hold in short-term memory, but we are good at detecting slope differences—which convey the relevant information" (Kosslyn 1989, 211)

[131] According to Zacks and Tversky (1999) people use *trend words* (increasing/decreasing) versus *discrete words* (higher/lower than) *and* are faster to judge absolute values from bars but angle from line graphs.

[132] Ziemkiewicz and Kosara (2009) refer to this as a *non-injective* encoding and point out that it makes it impossible for there to be a decoding, because multiple data values encode to the same data symbol. This is in contrast to a *bijective* encoding, which is one-to-one between data values and data symbols (Section 3.1).

This problem also relates to the issue of drawing data symbols on top of each other (Section 4.3). If two data symbols completely overlap, for example if two data points on a scatterplot overlap, then it is impossible to decode both individual data values.

Figure 4.5 of Bartonicek (2023) has an interesting, if pathological, demonstration of this problem using a simple bar plot. The bar plot has a single bar that represents the sum of counts for two categories. In this demonstration there is also a bar just for one of the counts, so that, with unnecessary mental effort, it is possible to recover the "hidden" data value, but we can imagine an even more pathological demonstration which would encode two data values to a single bar, without encoding one of the data values to its own bar. In that case it would be completely impossible to decode the two data values from the single bar.

[133] Peebles (2008) describes a line plot as *configural*—rather than either *separable* (Section 9.1) or *integral* (Section 9.2)—which means that it is possible to decode individual data values and data summaries, but that configural displays lead to poorer decoding of individual data values. This also links to the *display proximity principal* (Wickens and Carswell 1995).

[134] The *median-absolute-slope criterion* suggests choosing an aspect ratio such that the median absolute slope of the line segments between points is 45° (also known as the *45 degree banking rule*) (William S. Cleveland, McGill, and McGill 1988). This is shown to produce the most accurate decoding of slopes.

[135] William S. Cleveland, McGill, and McGill (1988) draws on the idea of *virtual lines* between the data points, from Marr (1982) and K. A. Stevens (1978), to relate the decoding of trends from lines in a line plot with the decoding of trends from data points in a scatter plot.

[136] William S. Cleveland (1994) states that "our estimation of correlation is strongly affected by the area of the data rectangle relative to the area of the scale-line rectangle" and provides demonstrations (Figure 4.15 and 4.16).

[137] The survey of anomaly-detection techniques by Chandola, Banerjee, and Kumar (2009) gives a sense of how difficult it is to provide a general-purpose numerical technique for detecting features of interest in data values.

[138] Proma et al. (2025) showed that people are good at reproducing (therefore identifying) trends, peaks, and valleys, if less so at reproducing periodicity, noise/variability.

Correll and Heer (2017) showed that decoding of trends from line plots (and scatter plots) is quite accurate, at least relative to ordinary-least-squares models.

[139] William S. Cleveland, McGill, and McGill (1988) points out that the accuracy of decoding slopes is only relevant to comparing the *relative steepness* of different regions of a line. This is not accuracy in terms of decoding the actual rate of change in the data. For that purpose, we should encode the actual rate of change directly.

[140] The data visualisations are based on Figure 2 from Zacks and Tversky (1999). An example of one subject's decoding of the line plot was: "The more male a person is, the taller he/she is"

[141] This decoding of the difference between lines is described in William S. Cleveland and McGill (1985). The recommended solution is to encode the difference between the lines, rather than requiring the viewer to perform the subtraction visually. Gleicher et al. (2011) call that solution *explicit encoding*.

Braun et al. (2025) describe a similar issue with estimating the line of best fit from data points in a scatter plot. Their results showed a bias towards orthogonal distance to the line of best fit rather than vertical distance.

[142] These plots are based on plots posted by Peter Ellis on a StackExchange thread. The data are from Stats NZ and licensed by Stats NZ for reuse under the Creative Commons Attribution 4.0 International licence.

[143] This observation does have a sensible interpretation: the consistent difference in logged values sugests that the size of seasonal changes is consistent *relative to* the absolute size of arrivals. In other words the seasonal change in arrivals *divided by* the absolute number of arrivals is consistent over time. However, that interpretation is not a simple, sub-conscious decoding of visual features; it requires a more sophisticated mental effort and a self-discipline to ignore the more naive decoding.

[144] These data are from Castro (2025).

[145] This power relationship between metabolic rate and body mass is known as *Kleiber's Law* (Kleiber 1947).

11 Multiple Encodings

Figure 11.1 shows a data visualisation of the performance measures for teams in the 2023 Rugby World Cup (Table 9.1). This visualisation is different from anything we have seen previously because it involves **multiple encodings**. The data symbol in this case is a simple data point, but we have encoded data values to *five* different visual features of each data point.

The number of clean breaks is encoded as the horizontal **position** of the data points and the number of tries is encoded as the vertical **position** of the data points. The number of points scored is encoded as the **area** of the data points, the hemisphere is encoded as the **shape** of the data points, and the number of runs made is encoded as the **colour** of the data points.

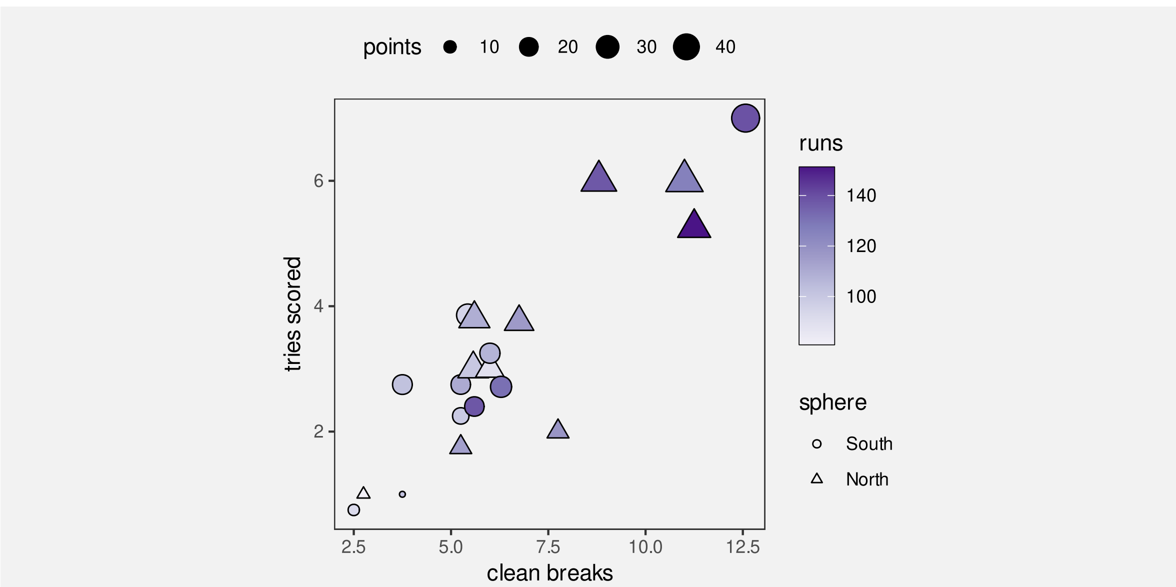


Figure 11.1: A data visualisation of the performance measures for teams in the 2023 Rugby World Cup.

Figure 11.1 is effective in some ways because some of the encodings involve effective visual features. For example, we can easily decode the number of clean breaks and the number of tries from horizontal and vertical positions. We can also decode a correlation between clean breaks and tries from the emergent feature of **position in space** (Section 9.2).

However, other encodings are less effective. For example, it is less easy to accurately decode number of points scored from size and even harder to decode the number of runs made from colour. These are poor encodings for quantitative data values.

In addition, we are hampered by the fact that many different visual features are changing at once, so it is harder to decode changes in any one visual feature (Section 2.5), plus there are interactions between some visual features, which can affect the decoding of individual features (Section 9.3). For example, the comparison of data points in terms of their **size** is made harder by the concurrent changes in **pattern**; it is difficult to tell whether a triangle has the same area as a circle.

One goal of visualising multiple variables like we have in Figure 11.1 is to discover or display multivariate relationships between the variables. For example, we can see that there are four teams in the top-right corner of the plot that are all relatively large and relatively dark in colour, which suggests that there is a group of teams that run more, make more clean breaks, score more tries, and score more points compared to the other teams. However, the decoding of multivariate relationships is not necessarily immediate nor effortless from Figure 11.1.

This chapter looks at some of the different approaches to visualising several variables at once. How do we encode multiple variables so that we can decode multivariate relationships? And can we still decode individual data values from individual variables at the same time?

11.1 Scatter plot matrices

One issue with Figure 11.1 is that it attempts to encode each data variable to a separate visual feature. Once we have used both horizontal and vertical **position**, we have to start using less accurate and less appropriate visual features, like **area** and **colour** for quantitative data values (Section 3.5). If we attempted to encode all of the variables in Table 9.1, we could actually run out of visual features to use (Figure 3.5).

An alternative approach is to encode more of the data variables to the *same* visual feature. In particular, we can encode more variables to **position**.[146]

Figure 11.2 shows a scatter plot matrix of the performance measures for teams in the 2023 Rugby World Cup (Table 9.1). These are the same data values that are encoded in the complicated scatter plot in Figure 11.1. Apart from hemisphere, each different variable is encoded as a horizontal and vertical position and a scatter plot is created for each combination of variables. For example, the number of runs is encoded as the top row and the left column of scatter plots. Within each scatter plot, two variables are encoded as horizontal and vertical position. For example, the second plot in the first row is a scatter plot of the number runs versus the number of clean breaks.

We can decode which variables are being combined because this is just a qualitative decoding from position. Within each scatter plot, the encoding of data values as position allows us to decode and compare individual data values and this provides a superior decoding for most variables compared to Figure 11.1. For example, it is easy to decode from the position of data points in the second scatter plot on the top row that the lowest number of runs made was about 80. This decoding is much more difficult from the colour of the data points in Figure 11.1. We can also identify clusters and outliers within each scatter plot, which is another improvement over Figure 11.1.

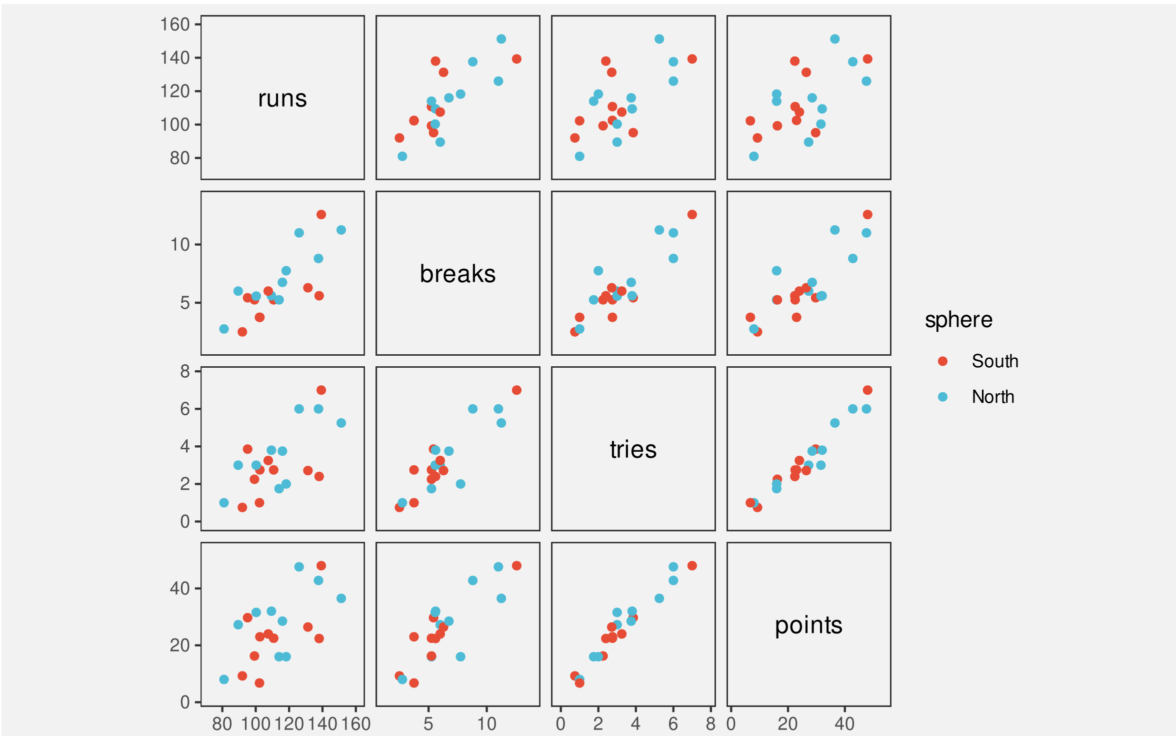


Figure 11.2: A scatter plot matrix of the quantitative performance measures for teams in the 2023 Rugby World Cup.

The emergent feature of position in space within each scatter plot allows us to decode the correlation between each pair of variables (Section 9.2) and we can also get a glimpse of multivariate correlations by comparing multiple scatterplots.

For example, looking down the scatter plots just above the diagonal of the matrix, we can see that: the number of runs is correlated loosely with the number of clean breaks; the number of clean breaks is more strongly correlated with the number of tries; and the number of tries is very strongly correlated with the number of points. This suggests a multivariate correlation between all four of these variables.

If we look down the first column of the matrix, we can see that: the number of runs is correlated loosely with the number of clean breaks, a little more loosely with the number of tries, and a little more loosely still with the number of points.

One negative, however, is that this decoding of multivariate relationships requires significant cognitive effort (Section 2.9). A scatter plot matrix is not particularly effective at allowing us to *effortlessly* decode multivariate relationships.

> A scatter plot matrix is effective for decoding individual data values from multiple variables at once, plus pairwise correlations. However, scatter plot matrices are less effective for decoding multivariate relationships.

11.2 Facetting

Figure 11.3 shows a line plot of the number of youth offenders per year in different Police Districts of New Zealand. This visualisation makes use of several effective encodings. For example, we encode year as horizontal position and we encode count as vertical position, although the line data symbols are **visual shapes** (Section 10.1), so the most effective decoding is to **data summaries**, like the overall downward trend for all districts.

The encoding of police district in Figure 11.3 is less effective. The district is encoded as **colour**, but because there are 12 different districts, we are exceeding the capacity of colour (Section 3.6). It is not easy to identify all of the different districts from the colours of the lines (not helped by the overplotting of the lines).

A better encoding for Police District would be **position**, which has a much higher capacity, but, similar to Figure 11.1, we have been forced to use colour because position has already been taken by year and the number of offenders.

Figure 11.4 shows a solution—a **facetted plot**—that, like a scatter plot matrix, reuses position to encode additional variables.[147] In Figure 11.4, each district is encoded as a **position in space** and we draw a separate plot for each district. This makes it very easy to decode the individual districts, much easier than it is with Figure 11.3.

The plot for each district also has an enclosing border, which helps to visually differentiate between districts (Section 2.7) and we have ordered the panels so that districts are near to other districts with a similar starting count of offenders (Section 5.1).

One downside to Figure 11.4 is that decoding and comparison of the number of offenders between districts and/or years is more difficult because the vertical and horizontal positions are **unaligned** for some comparisons (Section 4.2). In this case, we have mitigated that problem by drawing a grey line for all districts in each plot. Similar to grid lines, this makes it easier to compare the line for each district to other districts (Section 5.4).

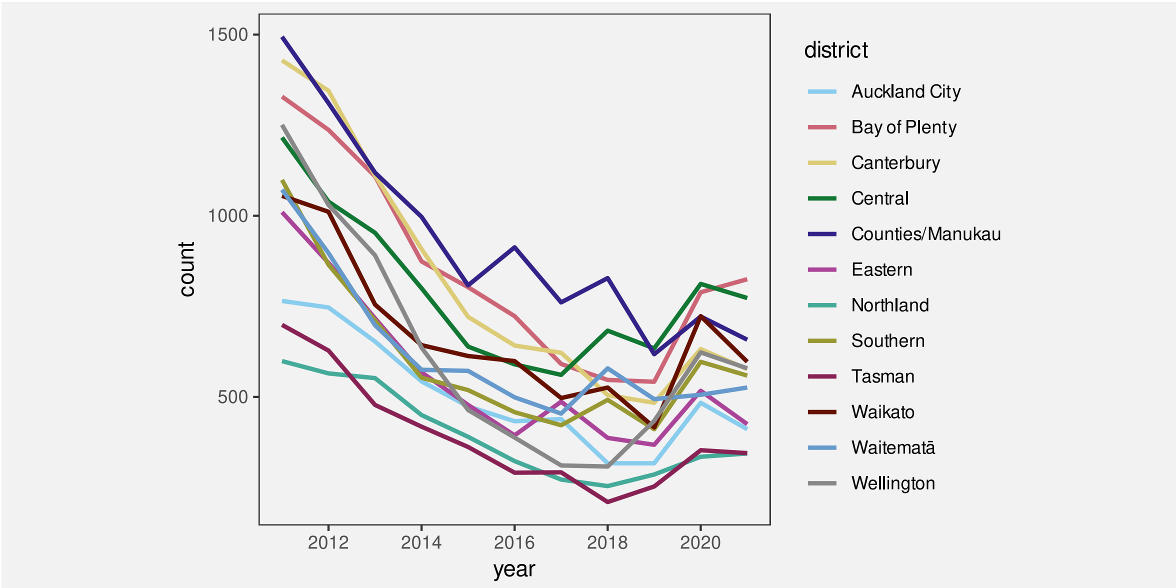


Figure 11.3: A data visualisation of the number of offenders per year in different Police Districts of New Zealand, from 2011 to 2021.

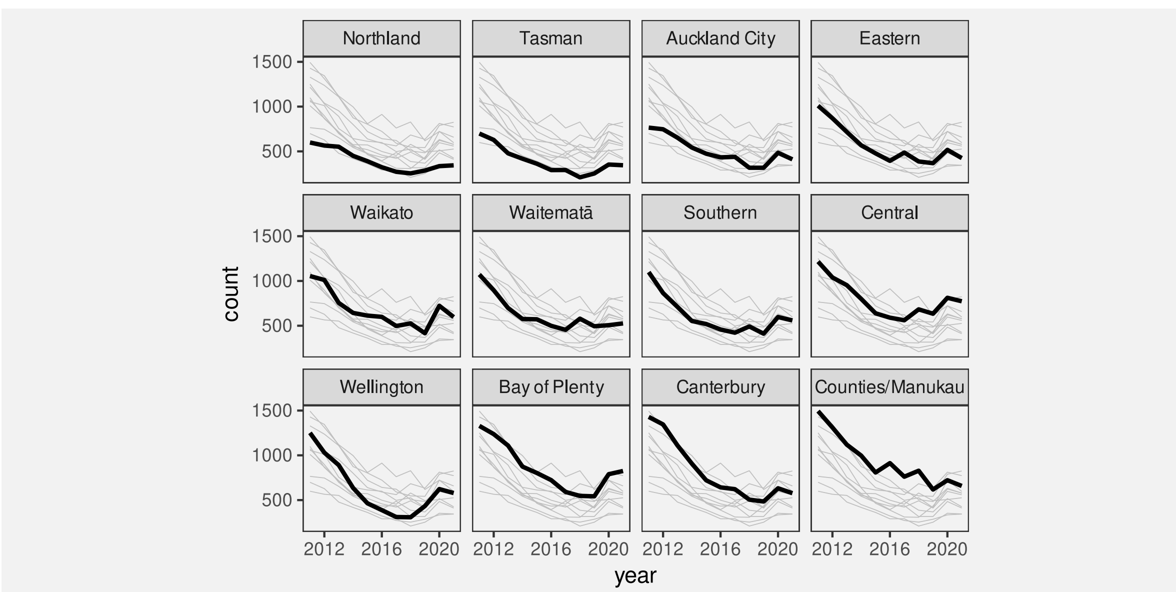


Figure 11.4: A data visualisation of the number of offenders per year in different Police Districts of New Zealand, from 2011 to 2021.

11.3 Case study: Bar plots

Figure 11.5 shows a stacked bar plot of the youth crime rate in New Zealand over time, broken down by age group. The data symbols are bars. The crime rate is encoded as the (vertical) **length** of the bars and the year is encoded as the horizontal **position** of the bars. These encodings are supplemented by an additional encoding of age as the vertical **position** of the bars. In effect, the final vertical position of each bar is a complicated encoding of crime rate plus age group.

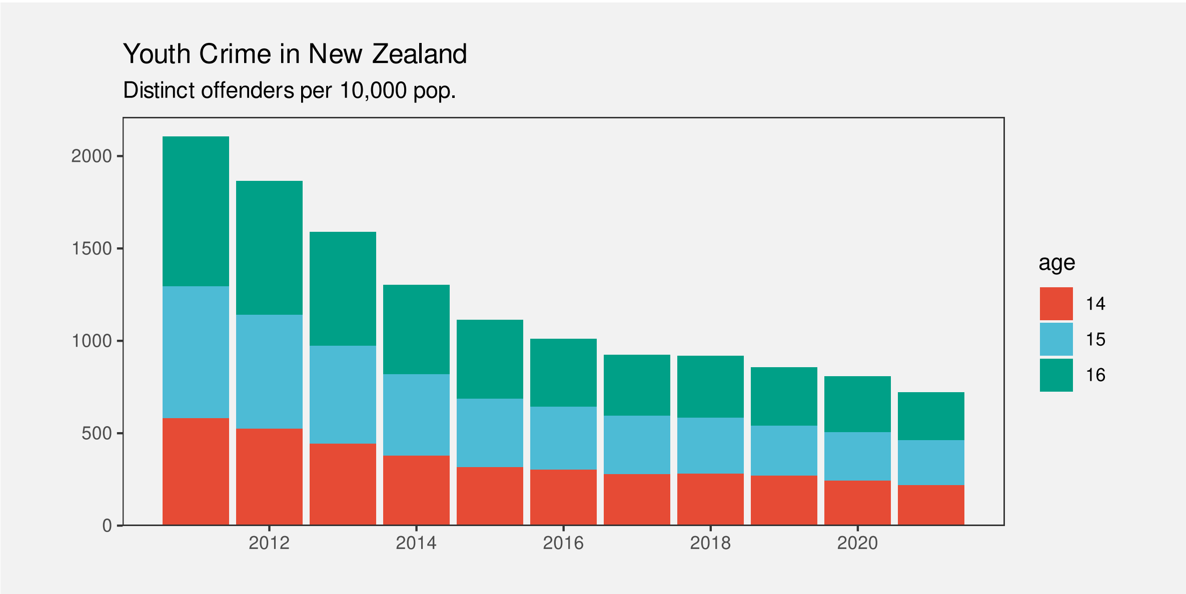


Figure 11.5: A bar plot of the number of offenders in each age group over time.

An alternative is shown in Figure 11.6, where we have the same data, but in the form of a side-by-side bar plot. This simple-seeming plot demonstrates another example of reusing position to encode multiple data values.

The crime rate is again encoded as the lengths of the bars and the year is encoded as the horizontal position of the bars. These encodings are supplemented by an additional encoding of age as the horizontal position of the bars.

In effect, the side-by-side bar plot is like a facetted plot, where each year is its own bar plot, with age group encoded as horizontal position and crime rate encoded as length. For example, Figure 11.7 shows the same data in a plot with one facet per year.

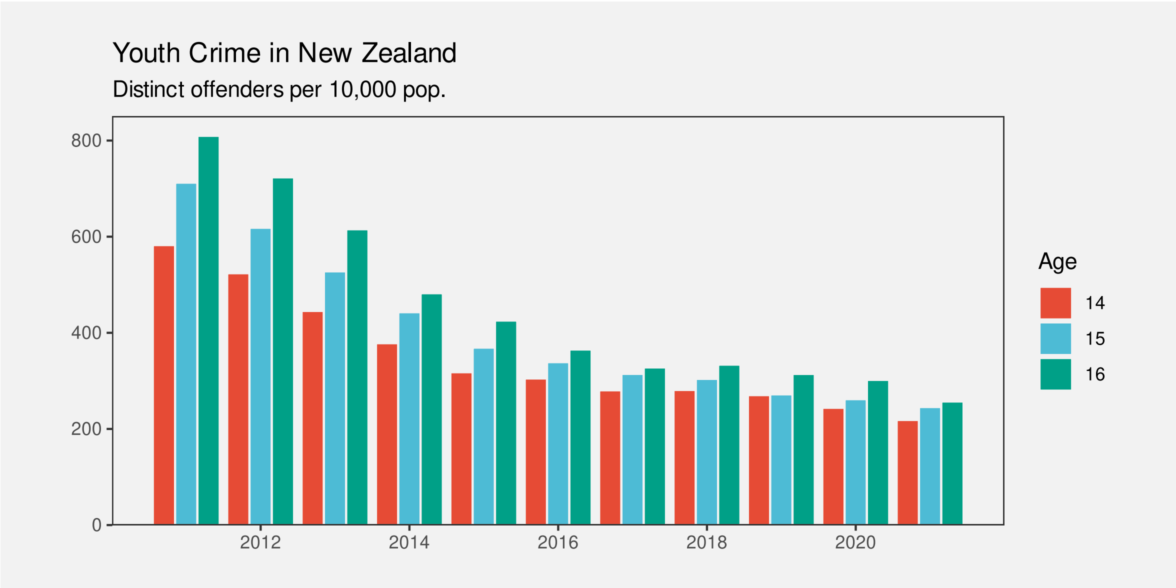


Figure 11.6: A bar plot of the number of offenders in each age group over time.

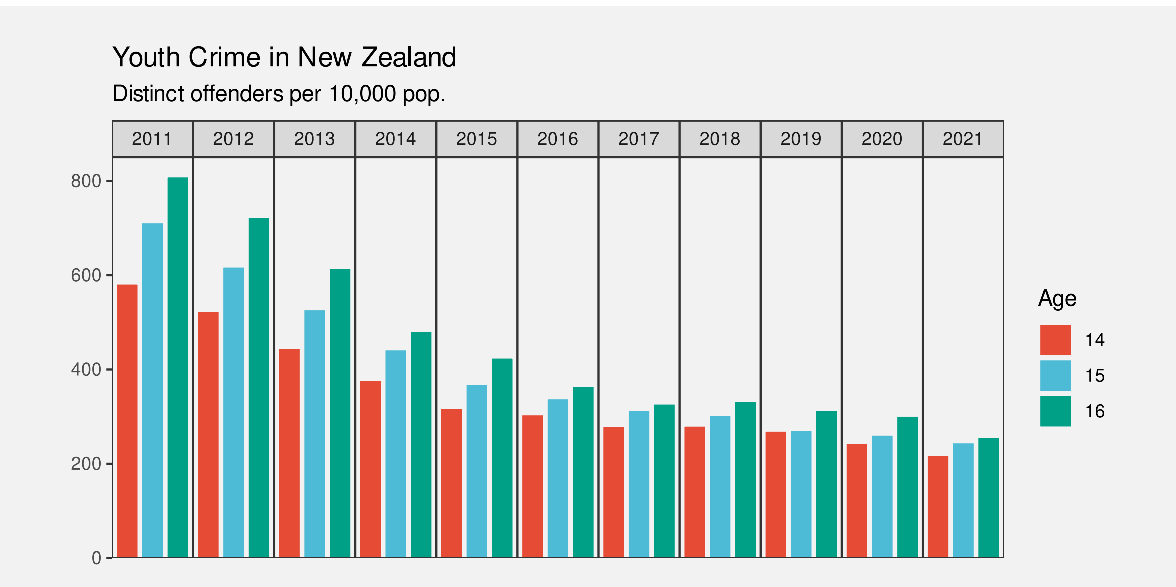


Figure 11.7: A bar plot of the number of offenders in each age group over time.

11.4 Non-cartesian encodings

Another way to encode more variables using **position** is to abandon the standard cartesian coordinate system (Section 4.5). For example, Figure 11.8 shows a 3D scatter plot of three of the performance measures for teams in the 2023 Rugby World Cup (Table 9.1).

Figure 11.8 takes advantage of the visual system's ability to interpret a 2D image as a 3D scene (Section 2.10). For example, we can see that the spheres in Figure 11.8 appear to lie roughly along a line in space and this decodes to a multivariate correlation between the three variables.[148] It is also possible to see that there are four teams somewhat separate in 3D space (not just in a 2D plane, as we saw in Figure 9.4).

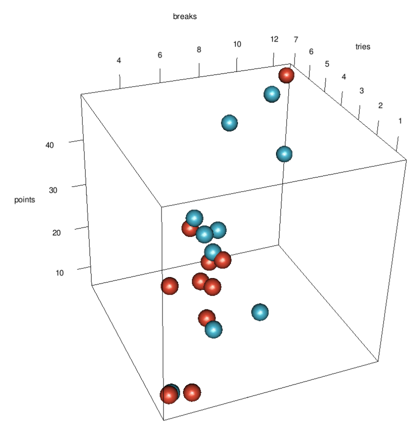


Figure 11.8: A data visualisation of three RWC variables

The value of a 3D scatter plot lies in this ability to decode multivariate **data summaries** from 3D **visual shapes**. Decoding individual data values is extremely difficult from a 3D scatter plot because data values are encoded as positions along dimensions that are not perpendicular (Section 4.5), may not be linear (Section 4.7), and are not aligned (Section 4.2). However, that should not be the goal of such a data visualisation; a 3D scatter plot is effective for decoding data summaries rather than raw data values.

Figure 11.9 shows another example of a 3D plot. This is a 3D version of the heatmap from Figure 1.2, which shows the rate of offending for different age groups across different years. The crime rate is encoded both as colour and as the height of a 3D bar.

Again, the decoding of positions to individual data values is not very accurate with this sort of 3D plot, although because there is a regular grid of bars, and there are not too many bars, we do not require much accuracy to decode year and age.

The decoding of the bar heights is also inaccurate, again because of non-linear and unaligned scales, but also because many bars are partially obscured.[149] This means that we cannot accurately decode the ratio of the highest bar to the heights of other bars; we are limited to ordinal comparisons at best. However, data summaries can be decoded from the overall visual shapes. For example, the general downward trend for all ages over time is much clearer in Figure 11.9 than it was in Figure 1.2. The implicit visual congruence with a descending stairway (Section 7.1) strongly supports the decoding of decreasing crime rates over time.

```
Warning in ggplot2::guide_colourbar(legend.ticks = ggplot2::element_blank(), : Arguments in
x Problematic argument:
* legend.ticks = ggplot2::element_blank()
i Did you misspell an argument name?
```

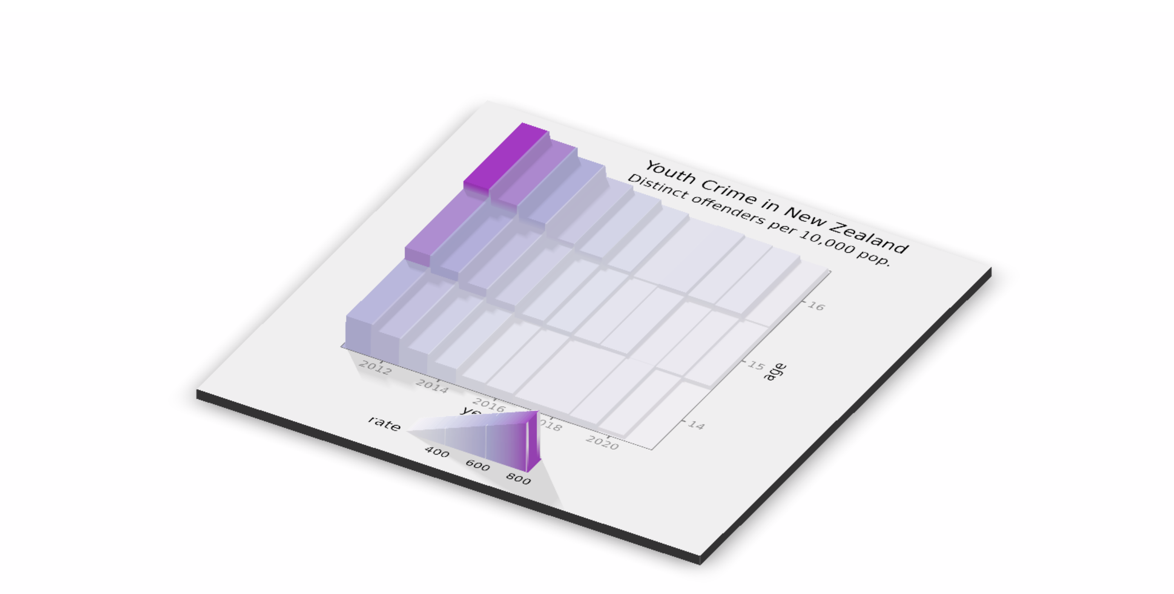


Figure 11.9: A data visualisation of youth crime by age

> A 3D plot is not effective for decoding raw data values, but it may be effective because the data symbols generate visual shapes in space, which can decode to three-dimensional data summaries, such as clusters, trends, and multivariate correlations.

A significant limitation of 3D plots is that they can only encode three variables using position. Although it is possible to produce views of more than three dimensions on a 2D page or screen, the implicit decoding of visual shapes is limited to three dimensions.[150]

11.5 Case study: Parallel coordinates plots

Figure 11.10 shows a parallel coordinates plot of the data from Figure 11.1. This is another example of a non-cartesian coordinate system. In Figure 11.10, all of the quantitative variables have been encoded as **position**, but those positions are along a series of vertical parallel axes, rather than just the normal two perpendicular axes.[151]

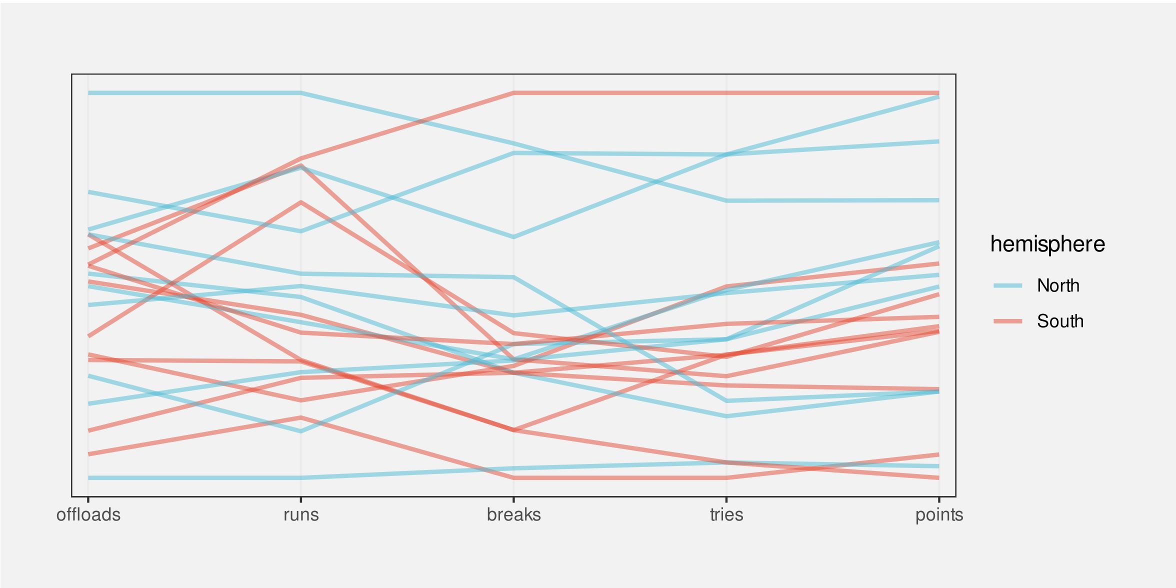


Figure 11.10: A parallel coordinates plot of the performance measures for teams in the 2023 Rugby World Cup. The data values from five quantitative variables are encoded as positions along five vertical, parallel axes. For each team, the data symbol is a line connecting the positions on the axes.

The parallel coordinates plot in Figure 11.10 encodes data summaries rather than raw data values (Chapter 8). All data values are standardised to a range between 0 and 1.[152] Table 11.1 shows some of the values that are actually encoded to visual features in Figure 11.10. Each variable is encoded as a different horizontal **position** and the values within each variable are encoded as vertical **position**. In this way, all quantitative data values are encoded as **position**.

In addition, the hemisphere is encoded as **colour** and a separate line data symbol is drawn for each country. This means that a visual shape is drawn for each country (Chapter 10).

Table 11.1: The data summaries that are encoded to visual features in Figure 11.10.

country	sphere	variable	value
Namibia	South	runs	0.16
Romania	North	runs	0.00
Chile	South	runs	0.30
Samoa	South	runs	0.31
Australia	South	runs	0.42
Georgia	North	runs	0.47

On one hand, a parallel coordinates plot appears to makes sense because it encodes multiple data values to **position**, which is an accurate visual feature for quantitative data. However, this advantage is immediately squandered because the values that are encoded are not the raw data values (Figure 8.2), so a parallel coordinates plot is not effective for decoding raw data values from the positions of the lines. Furthermore, the data symbols are lines, so each data symbol encodes multiple data values (Section 10.1), which makes it difficult to decode even individual data summary values from the lines.

The effectiveness of a parallel coordinates plot comes from the visual shapes of the line data symbols, which allow us to decode data summaries. For example, in Figure 11.10, we are able to identify a group of four lines at the top of the plot that are separate from the other lines, plus a group of three lines at the bottom. This grouping across at least three variables (clean breaks, tries, and points) is easier to see compared to Figure 11.1. We can also more clearly see that there are two teams that have similar numbers of runs to the top four, but then are mixed in with the other teams on the other variables. Furthermore, the fact that many of the lines are reasonably horizontal rather than criss-crossing each other suggests that there is an overall correlation between all four quantitative variables, especially the last three variables, and not just amongst the top four teams.

Compared to Figure 11.2, it is easier to decode multivariate correlations from Figure 11.10 and it is easier to decode multivariate clusters and outliers. A parallel coordinates plot can also cope with a larger number of variables; we just add another vertical axis rather than having to add an entire new row and column to a scatter plot matrix.

> A parallel coordinates plot can be effective at identifying high-dimensional groups or correlations because it encodes multiple data values from different variables to a single visual shape.

One major issue with a parallel coordinates plot is that the visual shapes created by the lines are extremely dependent on the order of the variables along the x-axis. For example, Figure 11.11 shows exactly the same data as Figure 11.10, but with a different ordering of the variables. It is now much harder to identify a group of four teams at the top and a group of three teams at the bottom. There is also now no clear decoding of parallel lines between variables, so no sense of correlation between variables (all neighbouring pairs of variables are less correlated).

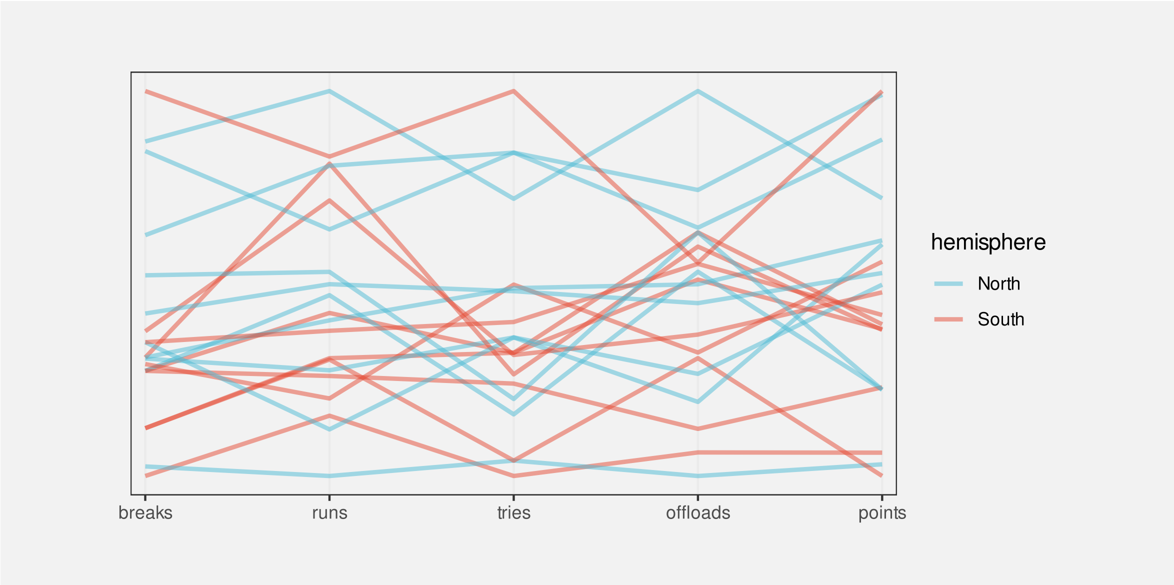


Figure 11.11: A data visualisation of the performance measures for teams in the 2023 Rugby World Cup. The order of the axes matters.

We can also have a large influence on the shapes in a parallel coordinates plot by inverting the direction of the axes. For example, Figure 11.12 shows the same data as Figure 11.10 and with the same ordering of the variables, but with the axes for line breaks and points inverted so that higher values are at the bottom and lower values are at the top. In other words, the positive relationships between runs and line breaks, between line breaks and tries, and between tries and points are now negative relationships (a higher number of tries correlates with a lower number of negative points).

The arrangement in Figure 11.12 shows that a negative correlation between two adjacent variables shows up as many crossings of lines at a very tight location (rather than lots of mostly parallel lines with few crossings for positive correlation). This demonstrates that it can be easier to identify a negative relationship than a positive one with a parallel coordinates plot. For example, the tight crossing of lines between runs and (negative) line breaks and between (negative) line breaks and tries stands out more in this plot than the roughly parallel lines for the same pairs of variables in Figure 11.10.[153] On the downside, it is now much harder to identify clusters. For example, it is more difficult to identify the group of four teams that all have high numbers of line breaks, tries, and points.

Parallel coordinates plots are often used on much larger data sets, in which case overplotting becomes a serious problem (Section 4.4). The use of colour for different groups and the use of

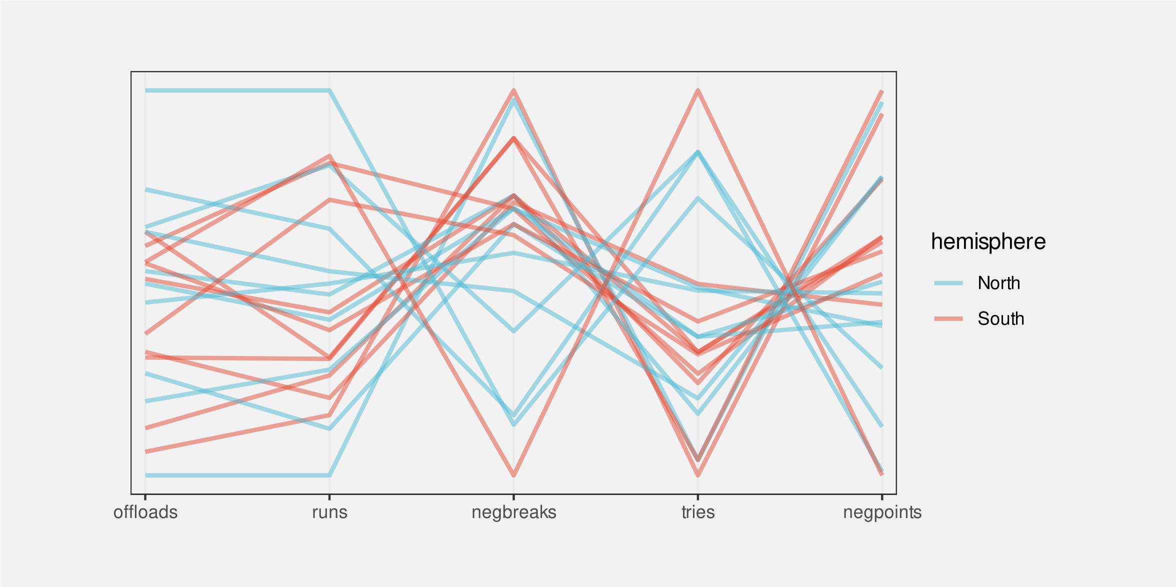


Figure 11.12: A data visualisation of the performance measures for teams in the 2023 Rugby World Cup. The direction of the axes matters.

semitransparent colours for the lines can be helpful.[154]

11.6 Case study: Profile plots

One difficulty with Figure 11.10 is that many of the lines overlap each other, which makes it difficult to decode the separate visual shapes. Figure 11.13 shows an alternative approach to visualising the same data, called a profile plot.[155] Similar to a parallel coordinates plot, a visual shape is drawn for each country, but each shape is drawn at its own position in space and the shape a filled polygon, effectively the lines in Figure 11.10 with the area below them filled in.

Like the parallel coordinates plot, this sort of data visualisation can be effective because of the overall shapes. For example, we can see a group of four shapes that are larger than the others. We can also see that Argentina and Fiji appear very similar, which was not easily apparent in previous plots.

Figure 11.13 uses the same data summaries as Figure 11.10 (Table 11.1), and similar encodings: each variable is encoded as a different horizontal **position** and the values within each variable are encoded as vertical **position** and hemisphere is encoded to **colour**. The data symbol is also similar: multiple data values are encoded to a single data symbol, in this case a polygon.

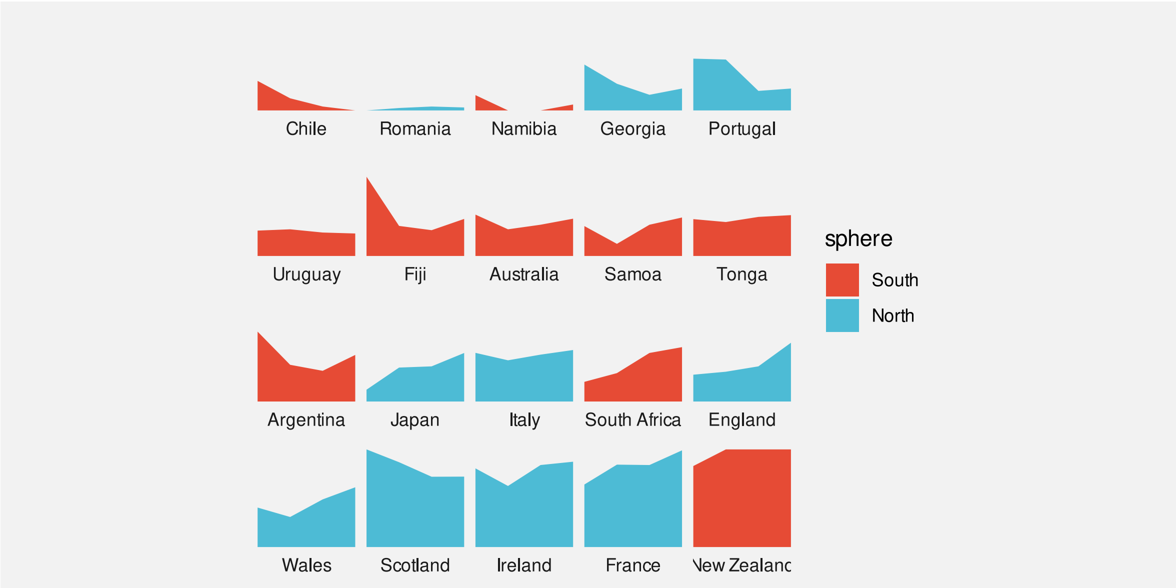


Figure 11.13: A data visualisation of the performance measures for teams in the 2023 Rugby World Cup.

The difference in Figure 11.13 is that, rather than having all of the data symbols on the same plot, each data symbol is drawn in its own **position in space**. This makes it easy to view each distinct data symbol.

The cluster of four larger shapes in Figure 11.13 is also easier to see because the countries have been ordered by the number of points scored (the right-hand height of the shape). As we have seen before, ordering data symbols can make a big difference (Section 5.1).

A major weakness with a profile plot like Figure 11.13 is that it does not scale well to large data sets. For example, adding another country to Figure 11.13 would take up additional space, but no more space would be necessary for a parallel coordinates plot like Figure 11.10 because we would just add another line.

11.7 Dangers of multiple encodings

Figure 11.1 showed that the naive approach of just using a separate visual feature for each data variable leads to complex data visualisations that are difficult to decode.

We have explored some alternative data visualisations of multiple variables that provide better decoding of individual data values, like facetting and scatter plot matrices, but they do not provide effortless decoding of multivariate data summaries, like correlation and clusters.

The alternatives that provide better decoding of multivariate data summaries, like 3D plots and parallel coordinate plots, rely on visual shapes that can be decoded to data summaries, but sacrifice the ability to accurately decode individual data values. One danger is that there is still a temptation to decode individual data values, which can result in erroneous or misleading conclusions.

For example, in Figure 11.8, we may be tempted to decode and compare the number of clean breaks for different teams based on the positions of the data points and the labels on the axes, but that is unlikely to result in an accurate decoding, even for ordinal comparisons.

It is perhaps safer to make it clear that only visual shapes should be decoded from the 3D scatter plot by removing the axis scales altogether, as in Figure 11.14.

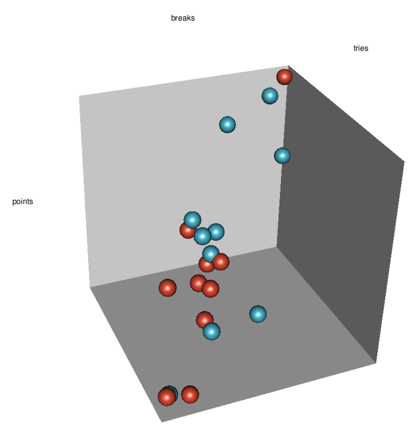


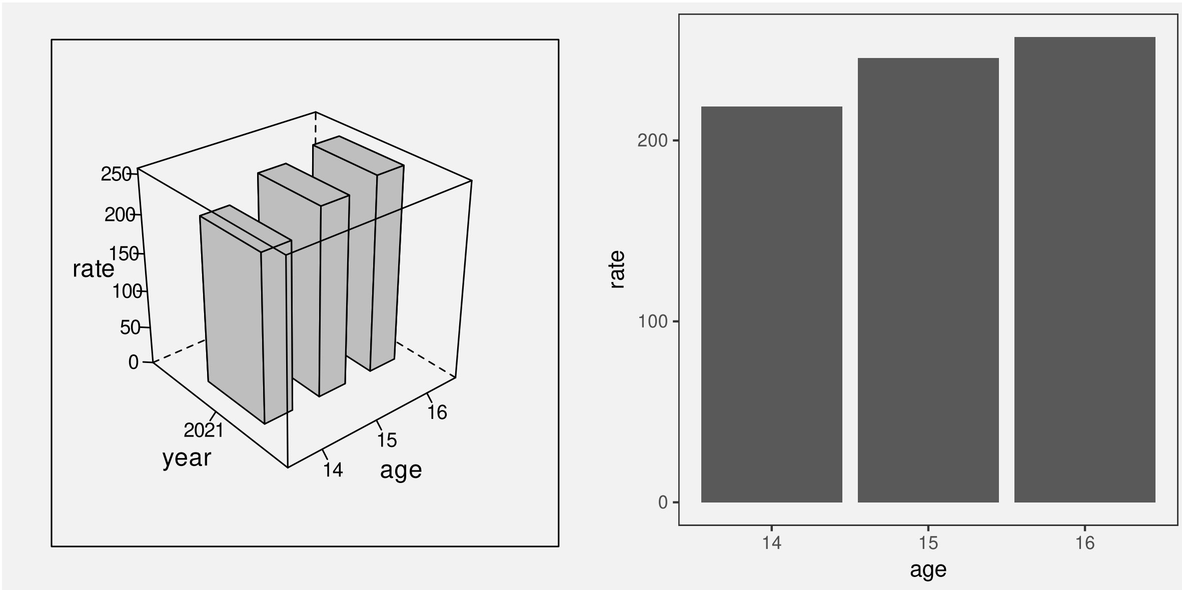
Figure 11.14: A data visualisation of three RWC variables

Given the problems with decoding individual data values, the classic objection to 3D plots is justified: if only two dimensions are required, then nothing is gained by adding a third dimension. There are really no arguments in favour of Figure 11.15a over Figure 11.15b.

11.8 Summary

When we visualise multiple variables at once, it is not effective to encode each different variable to a different visual feature because that usually forces us to make use of a visual feature that is either inappropriate or inaccurate.

An alternative is to reuse the same visual feature for multiple different variables. In particular, we can reuse **position** for more than just two variables, for example, to create a scatter plot matrix or a facetted plot. This at least allows us to accurately decode individual variables.



(a) 3D bar plot

(b) 2D bar plot

Figure 11.15: A data visualisation of youth crime by age for 2021

Another alternative is to use non-cartesian coordinates, for example 3D plots or parallel coordinates plots. These generate visual shapes that allow us to decode multivariate data summaries, like multivariate clusters and multivariate correlations. However, in order to gain multivariate data summaries, we typically have to sacrifice the ability to accurately decode individual data values.

No matter which approach we use, there is still a limit to how much information can effectively be displayed at once within a static data visualisation. This is one area where dynamic and interactive graphics can be useful because the viewer is able to rapidly switch between multiple views of the data.

Notes

146 The reuse of position is handled in a variety of ways within data visualisation frameworks. For example, the Grammar of Graphics (Wilkinson 2005) allows for *collision modifiers* to solve overlapping and separately allows for *facets*. GoFish (J. P. A. A. Satyanarayan 2026) introduces the idea of *geometric operators*, such as stacking, to the visualisation grammar.

In this book, we will just talk about encoding more than two variables to position.

[147] This method of producing a sub-plot for each level of a qualitative variable (or for each combination of levels for multiple qualitative variables) is called *facetting* in the Grammar of Graphics (Wilkinson 2005), but it is also known as *multipanel conditioning* plots or *trellis plots* (Becker, Cleveland, and Shyu 1996).

[148] This visual shape is more apparent if we are able to animate the image. In the online version of this book it should be possible to click and drag the mouse to rotate the image in space.

[149] The problem of elements being obscured is a significant one. The angle of view for Figure 11.9 and the placement of the plot elements was carefully chosen. As with the 3D scatter plot, this problem can be removed by allowing the 3D scene to be rotated interactively.

[150] A 3D plot is just a *projection* of three dimensions into 2D. More than three dimensions can also be projected to 2D and this can result in recognisable visual shapes in 2D or 3D, but decoding a meaningful data summary from those shapes becomes more difficult. This is another case where dynamic and interactive data visualisation can be very helpful (Cook et al. 1995).

[151] At a deeper level, a parallel coordinates plot represents a single *point* in n-dimensional space with a *line* through n vertices on the parallel coordinates plot and several points that lie on the same line in n-dimensional space translate to several lines that meet at the same point on a parallel coordinate plot (or a set of lines that are parallel).

Wegman (1990) provides a discussion from the point of view of statistical analysis.

Inselberg (2009) provides an extremely thorough explanation, derivation, and series of demonstrations, from a more abstract computational geometry perspective.
The early parts of Chapter 10 provide some accessible comments relevant to data visualisation.

[152] It is possible to use unscaled axes on a parallel coordinates plot, though that becomes untenable if the range of the values on different variables varies widely. There are also other ways to perform the scaling, for example, standardisation (in the sense of subtracting the mean and dividing by the standard deviation).

[153] There are many variations on parallel coordinates plots that attempt to mitigate some of the visual artifacts that can occur. For example, Pomerenke et al. (2019) suggest a way to reduce the higher visual impact of steeper, intersecting lines in a parallel coordinates plot.

[154] Parallel coordinates plots are an example of a data visualisation that benefits a lot from interactivity. The ability to select subsets of the lines can help with overplotting and the ability to rapidly experiment with different orderings and directions of the axes can help with determining the most effective view of the data.

[155] The idea of drawing lots of small repeated plots, with a consistent scale and style is referred to by Tufte (1983) as *small multiples*. This corresponds strongly with the idea of facetting (Section 11.2).

12 Visual Objects

Figure 12.1 shows a scatter plot of the number of clean breaks and the number of tries for countries at the 2023 Rugby World Cup (Table 9.1). This data visualisation is similar to Figure 9.4, but it adds an extra encoding. Country names are encoded as the **pattern** of the data symbols. This is a reasonable encoding, despite the large number of different countries, because each different shape only appears once. We are able to distinguish all countries from each other and, thanks to the legend, we can identify individual countries.

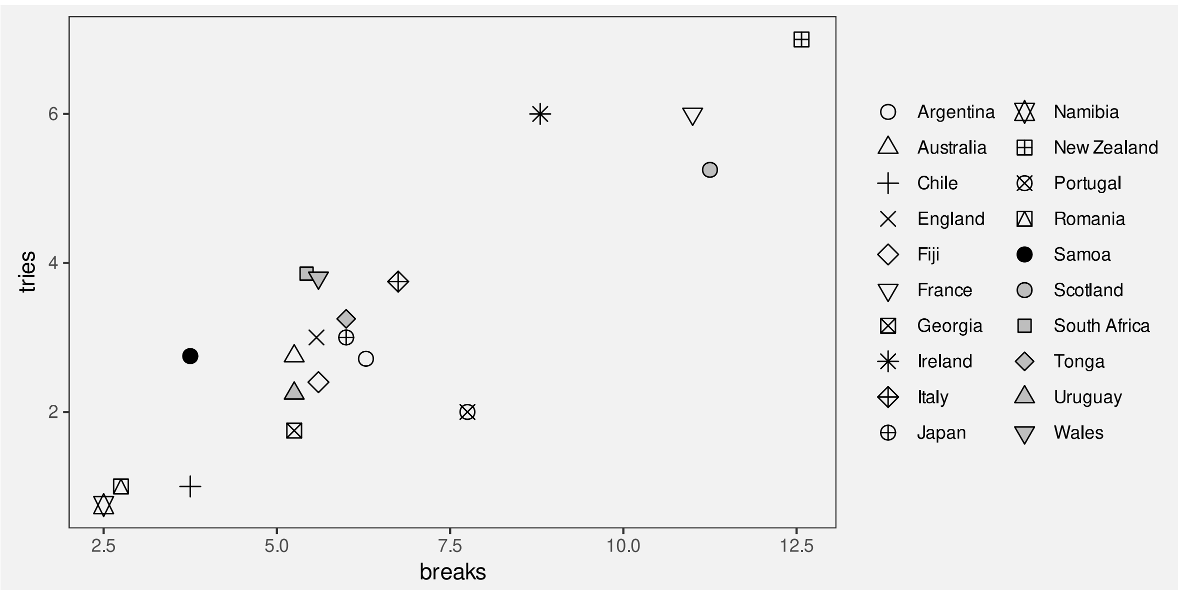


Figure 12.1: A scatter plot with a different shape per country

Figure 12.2 shows another scatter plot of the number of clean breaks and the number of tries for countries at the 2023 Rugby World Cup, but this time with flags as the data symbols. In one sense, this is just a variation on Figure 12.1 because the country names are encoded as the **pattern** of a data symbol. We can decode from the pattern of each data symbol to the relevant country.

The difference between the data symbols in Figure 12.1 and the data symbols on Figure 12.2 is that the flags are not only more complicated data symbols, but they are familiar symbols. We can decode that the flags are different, but we can also decode country names from the

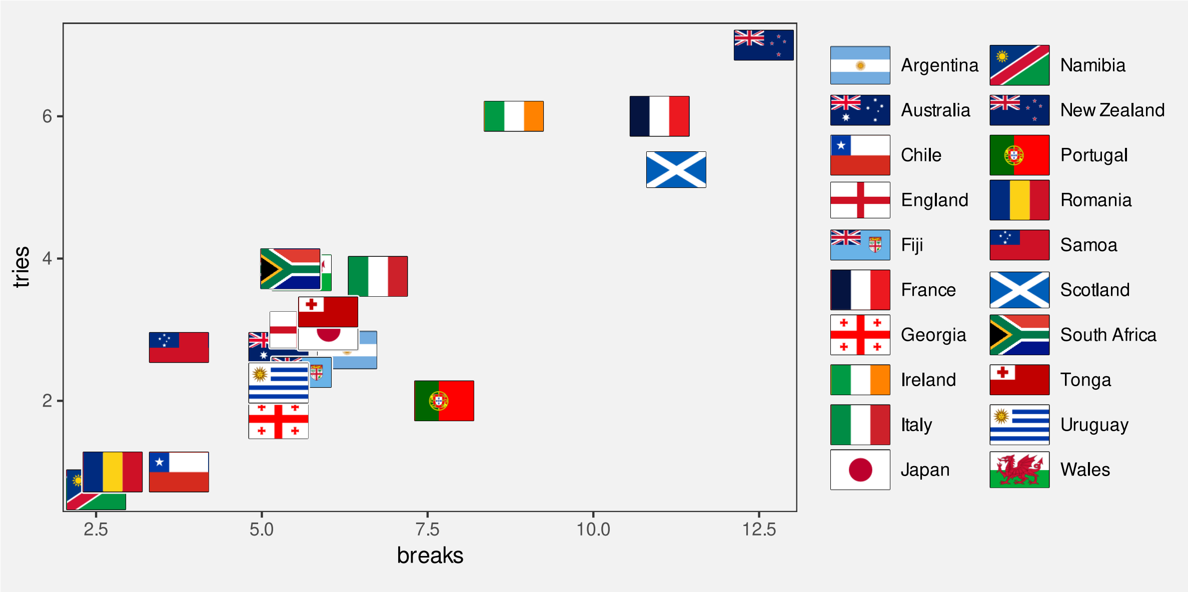


Figure 12.2: A scatter plot with flags as data symbols

flags. In the terms of Section 2.2, the flags are **visual objects** because they represent visual information that is combined with prior knowledge at a higher level of visual processing.

Rather than encoding the country name as the **pattern** of a simple data symbol (Figure 12.1), in Figure 12.2 we have encoded each country name as a different **visual object** (Figure 12.3).

Figure 12.3: We can encode data values using data symbols that are complex visual objects.

12.1 Learned decodings

We **encode** data values as visual features so that the viewers of a data visualisation can **decode** the data values from the visual features (Figure 3.6).

In Section 7.1, we saw that some encodings are effective because there are **implicit decodings**. For example, a longer bar in a bar plot naturally corresponds to a larger quantity. Basic visual features, like length, are effective thanks to *pre-existing* decodings that are built-in features of our visual system (Figure 7.2).

Part of the power of **visual objects** also comes from pre-existing decodings. However, these decodings are *learned* rather than being built-in. In some cases, we are able to **decode** values from a visual feature based on prior experience and learning—via a **learned decoding**—rather than due to an implicit encoding (Figure 12.4).[156]

Figure 12.4: Some visual objects have learned decodings, which means that we can decode values from the visual objects without having explicitly encoded values as visual features. This is similar to the idea of implicit encoding (Figure 7.2).

For example, Figure 12.5 shows three coloured circles. Without any explicit encoding, we can decode the red circle as "stop", the orange circle as "slow", and the green circle as "go". This is not a decoding that we are born with; it is a decoding that we learn through experience.

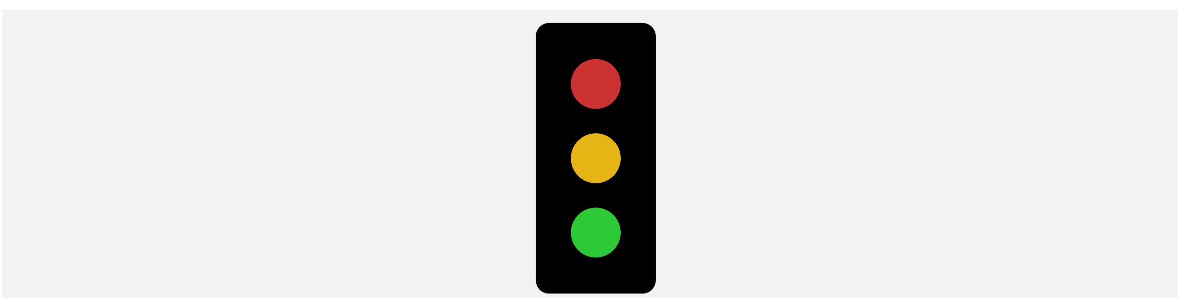


Figure 12.5: An example of visual symbols that have pre-existing, learned decodings.

The flags in Figure 12.2 provide examples of learned decodings. We associate each flag with a particular country because we have seen these flags before and we have learned the decoding from flag to country (Figure 12.4).

In terms of visual congruence (Chapter 7), a learned decoding strengthens the effectiveness of the explicit encoding because the learned decoding leads back to the same data values that we explicitly encoded (Figure 12.6). This is similar to having a congruent *implicit* decoding (Figure 7.3).

Figure 12.7 demonstrates this point by repeating the scatter plot, but removing the legend. We are still able to identify not only that the data symbols represent different countries, but

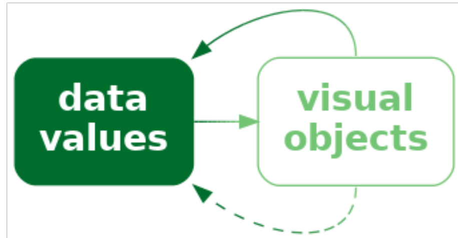


Figure 12.6: If we explicitly encode data values as visual objects in a way that is consistent with a learned decoding, we can more effectively decode data values from the visual objects.

we can identify (at least some of) the country names as well because of the learned decoding of the flags (Figure 12.8).

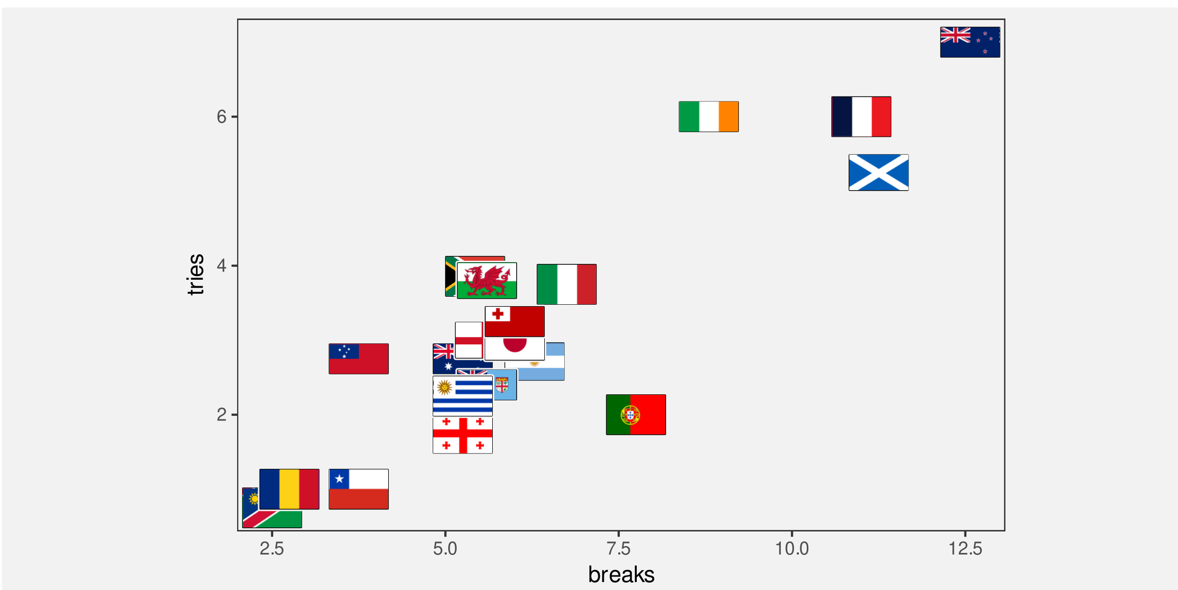


Figure 12.7: A scatter plot with flags as data symbols

Just to make the point completely clear, Figure 12.9 shows Figure 12.1 without a legend. Although we can still perceive that the data symbols are different from each other, there is no learned decoding for the shape of the data symbols in Figure 12.9, so they do not, by themselves decode to the country names. We can clearly see that Figure 12.7 contains more information than Figure 12.9.

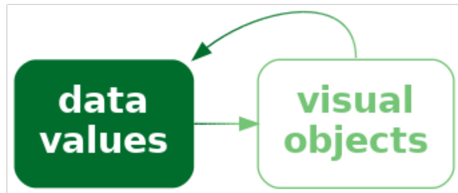


Figure 12.8: Some visual objects have learned decodings, which means that if we encode data values as the visual objects, decoding will be automatic.

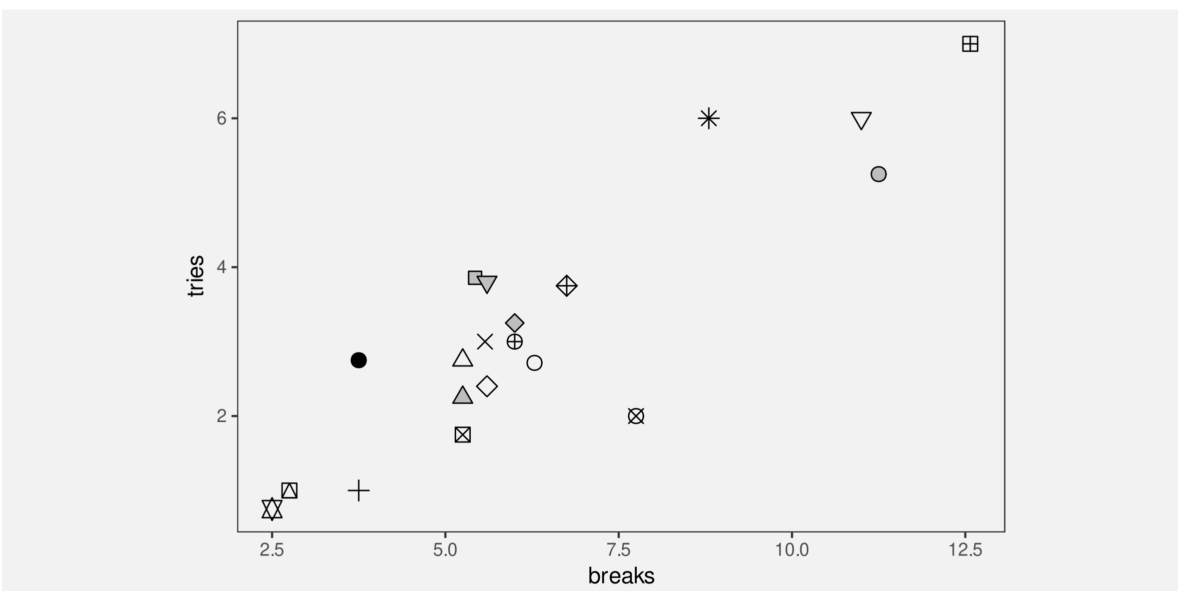


Figure 12.9: A scatter plot with a different data symbol pattern per country

> A plot that uses visual objects as data symbols can be effective because the visual objects can convey information based on learned decodings.

Figure 12.7 also demonstrates a weakness of **visual objects**. The flags work well for identifying the country names *only if* we are familiar with all of the different flags. For example, it may only be Australians and New Zealanders who can reliably identify the difference between their two flags.

> A plot that uses visual objects as data symbols may *not* be effective if the learned decodings have not been learned by the viewer.

Another significant problem is that the data symbols in Figure 12.7 are relatively complex. One of the reasons why data visualisations that use simple data symbols are effective, like bars in a bar plot or data points in a scatter plot, is because changes in the data are encoded as simple changes to visual features in the data symbols, like a change in the length of a bar (Section 2.5). The flags in Figure 12.7 contain many different visual features, like different colours, within the data symbol itself *and* differ from each other on many visual features at once, like differences in colours, patterns, and so on.

> A plot that uses visual objects as data symbols may *not* be effective if the data symbols contain many different visual features that all change at once.

12.2 Case study: Chernoff faces

Figure 12.10 visualises the 2023 Rugby World Cup data (Table 9.1) using Chernoff faces.[157] The data symbol in this visualisation is a little unusual; each country is represented by a combination of circles, curves, and lines that together produce a **visual object** that resembles a face (Figure 12.11). The encodings in this visualisation are also a little unusual. The number of points scored is encoded as the **curvature** of the smile, the number of tries is encoded as the **angle** of the eye brows, and the number of clean breaks is encoded as the horizontal **position** of the eyes.

The encodings in a Chernoff face are *not* very accurate (see Section 3.5; **curvature** is ranked *below* colour in terms of accuracy),[158] so Cheroff faces are *not* very effective for decoding raw data values.

However, the familiarity of Chernoff faces means that we are easily able to decode common combinations of the facial features. For example, we can see several sad, worried faces at the top-left of Figure 12.10 and several happy, perhaps even smug, faces at the bottom-right. We can decode clusters of countries that share similar combinations of performance measures. In

Figure 12.10: A Chernoff faces plot of the ...

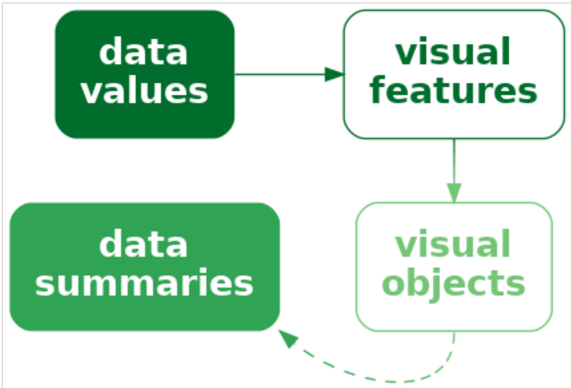


Figure 12.11: We can encode data values as the visual features of a complex data symbols that produces a complex visual objects.

other words, we can decode multivariate correlations and multi-dimensional groupings in the data.

In this case, the happier faces have been arranged to correspond to the more successful teams and the teams have been ordered from least points scored to most points scored (Section 5.1). The resulting proximity helps with perceiving groups of similar faces.

> A Chernoff face is effective at conveying multi-dimensional relationships because the data symbol is a familiar visual object from which we can easily decode simple human emotions.

On the downside, Chernoff faces suffer from the same problem as profile plots (Figure 11.13): they are only effective for a small amount of data. Furthermore, the faces that result is highly dependent on which variables are encoded as which features. This is similar to the problem that we saw with selecting the order and direction of axes in parallel coordinate plots (Section 11.5).

12.3 Visual features of visual objects

Figure 12.12 shows a variation of Figure 12.10. In this plot, the hemisphere of each country is encoded as the **colour** of the flag (large or small). Figure 12.12 shows that visual objects, although they are complex data symbols, can still have basic visual features, like **colour**.

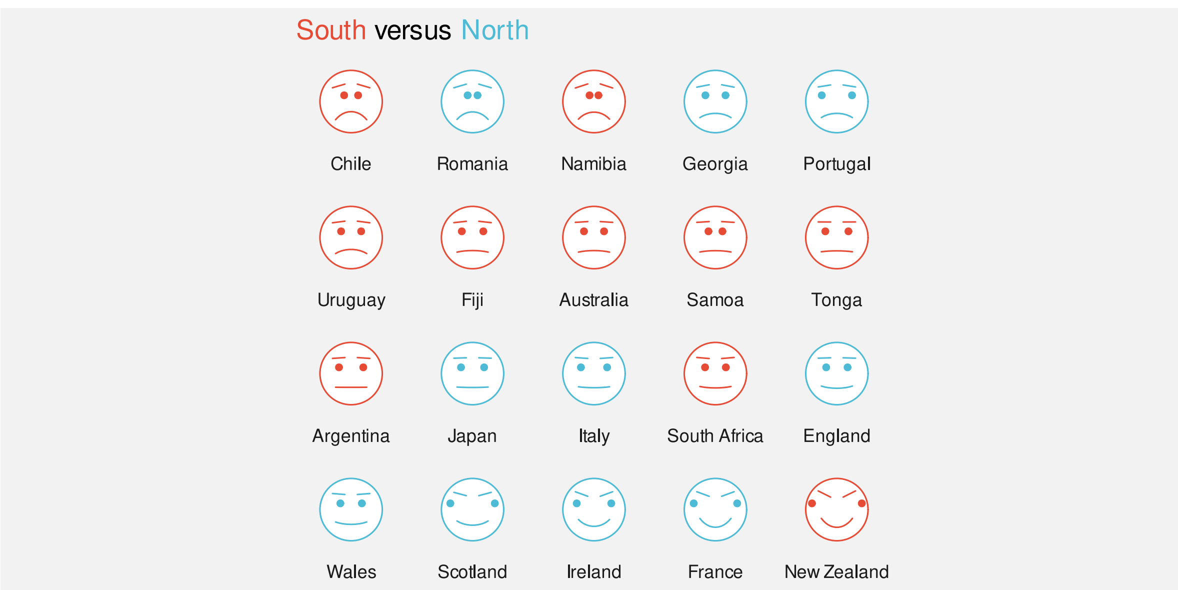


Figure 12.12: A Chernoff faces plot of the ...

This means that it is still possible to encode data values as basic visual features of visual objects, just as we can with much simpler data symbols, like data points. We saw in Section 10.4 that we can do this with other complex data symbols, visual shapes, as well.

Figure 12.13 shows another example, this time with a variation of Figure 12.7. In Figure 12.13, the hemisphere of each country is encoded as the **area** of the flag (large versus small). We can decode the hemisphere for each country just from this basic visual feature, separately from decoding the country name from the visual object.

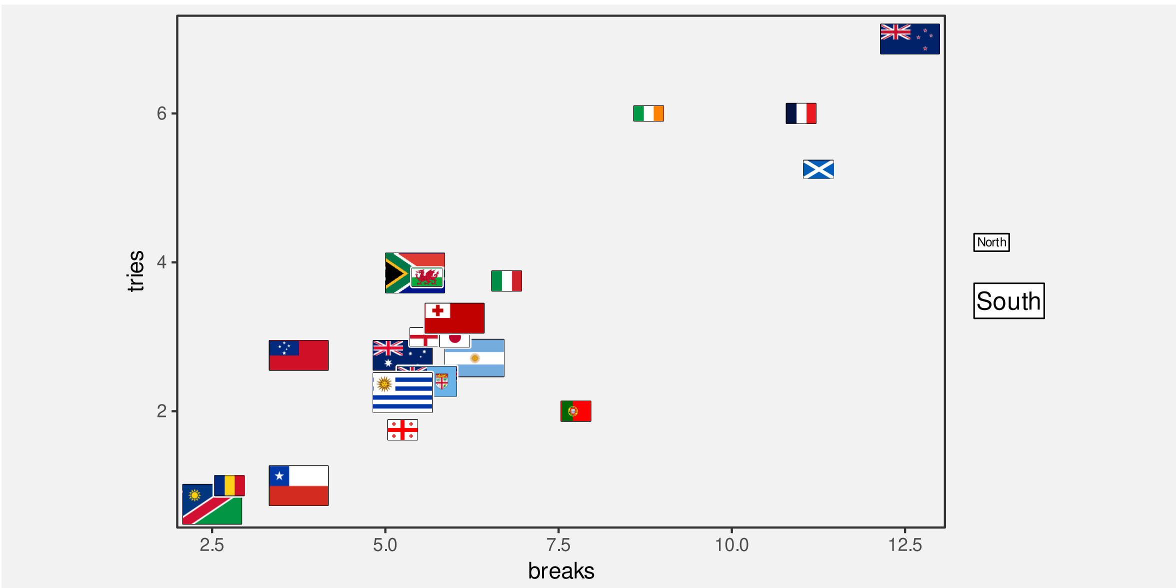


Figure 12.13: A scatter plot with flags as data symbols

12.4 Decoding visual objects

If data values have been encoded as a single consistent feature of a visual object, such as the colour or area of a visual object (Section 12.3), then decoding proceeds as for simpler data symbols like bars.

In Figure 12.2, we saw an example of encoding individual data values to visual objects: each country name is encoded as a separate flag. In this case, it is possible to decode an individual data value from each visual object. Assessing the accuracy of the decoding of quantitative data values from visual objects is difficult because there are very many possible visual objects. [159] On the other hand, the decoding of qualitative data values from visual objects, thanks to the very large range of possible objects, clearly has a very large capacity (Section 3.6). For example, there is no confusion between the large number of different flags in Figure 12.7 (apart from those silly Australian and New Zealand flags).

Figure 12.10 showed a different situation, where multiple data values are encoded within a single visual object. The result in this case is similar to when we encode multiple data values within a visual shape (Section 10.6). We are less able to decode individual data values, but we gain the ability to decode multivariate data summaries, such as multivariate clusters and multivariate correlations.

In some cases, it is also possible to obtain **visual summaries** of visual objects, similar to what is possible for much simpler data symbols, like data points in a scatter plot (Section 8.2). For example, in Figure 12.10, we can decode a trend of emotions from the rows of Chernoff faces, from more sad at the top to more happy at the bottom.[160]

12.5 Case study: Choropleth maps

Figure 12.14 shows a choropleth map of the crime rate for 2021 in each police district of New Zealand.[161] The crime rate data values for this map are shown in Table 8.3.

The data symbol in Figure 12.14 is a polygon representing the geographic boundary of each police district. This implicitly entails an encoding of a police district as the **position** of a polygon. Crime rate is encoded as the fill colour of each polygon.

The use of polygons as data symbols is effective because the polygons form familiar shapes with familiar spatial arrangements (familiar to a New Zealand audience at least). The polygons are **visual objects** that have a learned decoding to districts within New Zealand. The spatial arrangements are also a form of visual **congruence**; the arrangements mirror the real-world spatial relationships between the regions (Chapter 7).

> A choropleth map is effective because it encodes geographic regions to familiar visual objects that resemble the real world regions.

A weakness of Figure 12.14 is that it encodes the **quantitative** crime rate as colour (chroma and/or luminance). This means that we can only decode ordinal comparisons between regions (Section 6.2). For example, we can see that the highest crime rate occurs in the Bay of Plenty (the darkest region), but it is difficult to decode how much higher that crime rate is compared to the next highest crime rate.

> A choropleth map is *only* effective for making ordinal comparisons between geographic regions because it encodes quantitative values as colour.

Figure 12.15 shows a bar plot of the same data, which demonstrates the advantages and disadvantages of the choropleth map in Figure 12.14. In Figure 12.15, it is much easier to compare the crime rate values because crime rates are encoded as bar **length**. However, it is much harder to associate the different regions with their spatial arrangements because all we

Figure 12.14: A choropleth map of the crime rate for each police district for 2021.

have are the region names. Each police district is encoded as the vertical position of a bar, with no correspondence to the spatial arrangement of the regions in the real world.

Another potential weakness with choropleth maps is the fact that regions differ in terms of **area**. We know from Section 7.3 that larger data symbols are associated with larger data values. The larger polygons accurately reflect the fact that some police districts are geographically larger than others, but there is potential for larger polygons to be erroneously decoded as larger crime rates.

Two alternative approaches are shown in Figure 12.16. These both retain the region shapes of the map, but encode crime rates to the **area** of circles or the **length** of bars. These data symbols have a spatial arrangement that corresponds to the real-world arrangement of police districts and they provide a better decoding of crime rates from data symbols compared to the colours in the choropleth map (although area is still not the most accurate visual feature and the lengths are unaligned; Section 3.5).

12.6 Chart junk and clutter

Figure 12.17 shows a version of Figure 12.14 with two cartoon villians added in the top-left and bottom-right corners.[162] These sorts of annotations are often, derisively, referred to as **chart junk**, with the suggestion that they add no value and make it harder to extract the important information.[163]

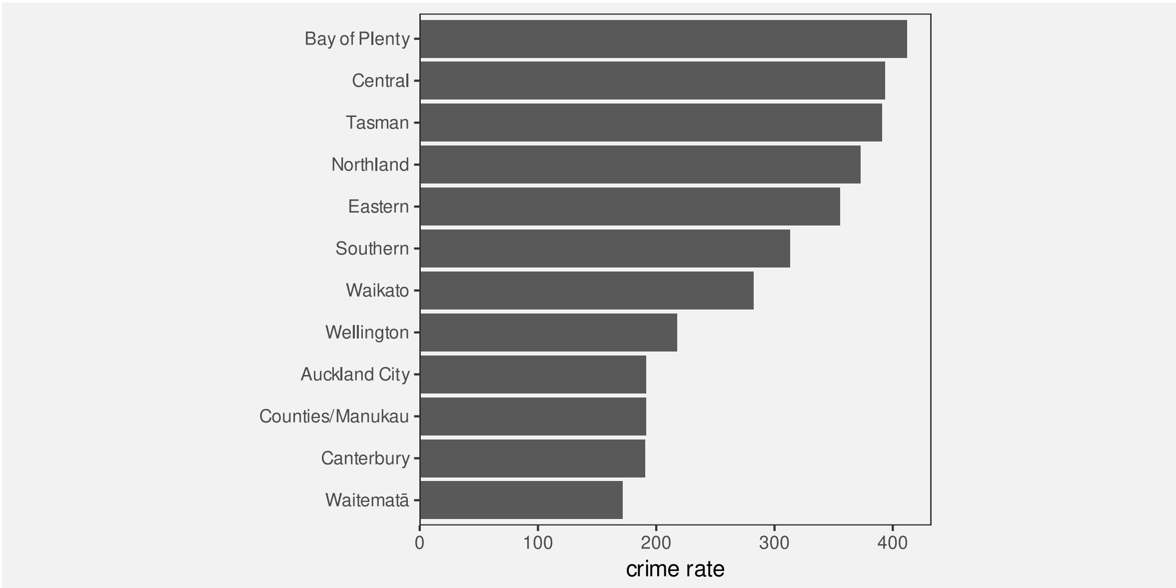
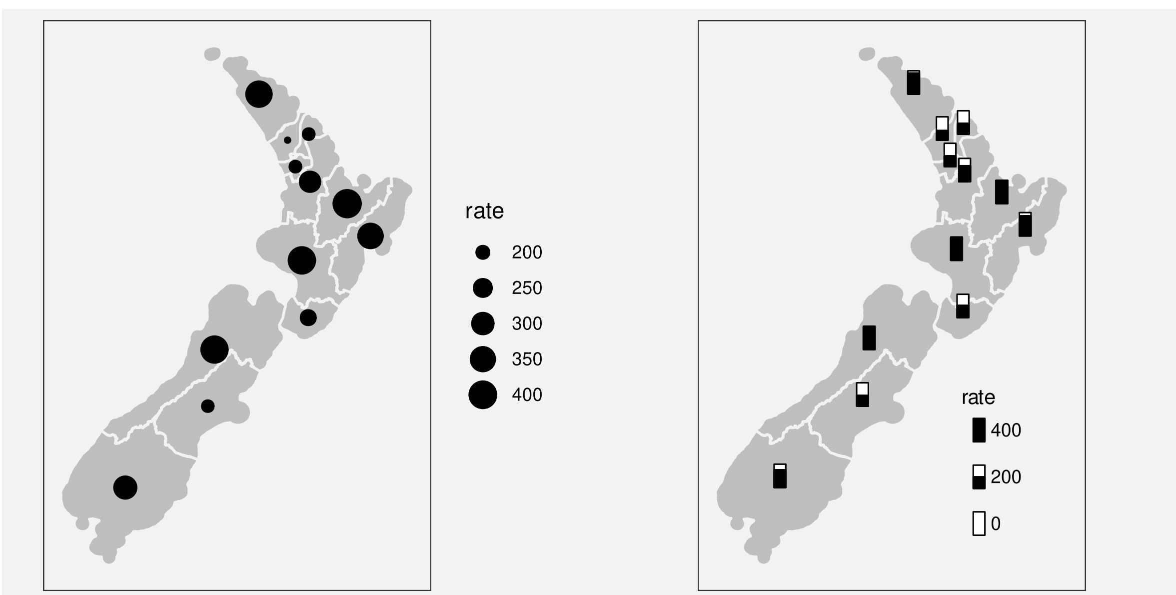


Figure 12.15: A bar plot of the data from Figure 12.14



(a) Dots as data symbols.

(b) Bars as data symbols.

Figure 12.16: Alternative map-based representations of the crime rate in each police district for 2021.

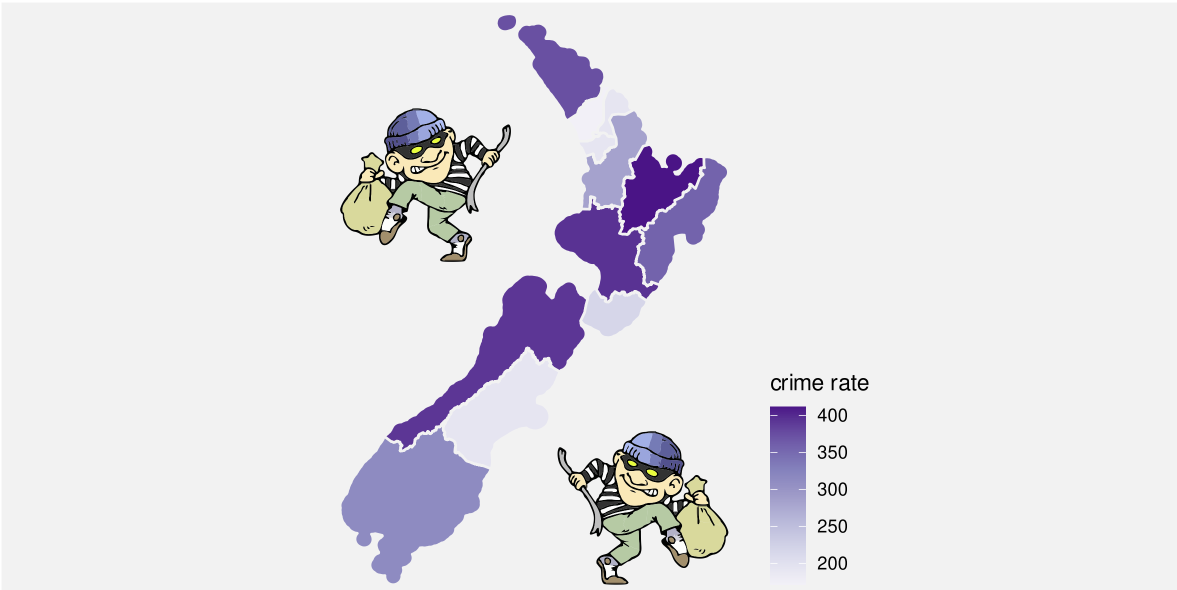


Figure 12.17: A choropleth map of the crime rate for each police district in 2021 (like Figure 12.14), with cartoon villian images added.

The cartoon villians are another example of **visual objects**. They (crudely) convey the concepts of crime and criminals. However, they are *not* **data symbols** because there is no encoding from the raw data to the cartoon villians. Their position is unrelated to the data, their colour is unrelated to the data, and their size is unrelated to the data. No information has been encoded in the visual objects, so no useful information can be decoded (Figure 12.18). The best that can be said is that they relate to the **metadata**, the general topic of the data visualisation.

Figure 12.18: Chart junk is a visual object that is *not* the result of encoding data values. It cannot provide any decoding to data values.

In Section 2.5 we learned that we should keep things simple and only present visual changes that correspond to changes in the data. In Section 2.6 we learned that our attention is drawn to the larger and brighter items within an image. The cartoon villians are relatively large, complex, and colourful and bear no connection to the raw data, so there is a good chance that they will distract from the visual features that do convey data values.

> Chart junk is *not* effective because it is impossible to decode data values from chart junk (no data values have been encoded) and because chart junk distracts from the visual elements that do represent data values.

However, it is important to point out that not all **visual objects** are chart junk. For example, the data symbols in Figure 12.17, the map regions, are also **visual objects**, but they differ from the cartoon villians because they are related to the data values.

Furthermore, **visual objects** that have no connection to the raw data may still have some positive effects. For example, the cartoon villians are not devoid of information; they convey the concept of crime more dramatically than the simple text label "crime rate". The more complex and interesting images may attract attention to the data visualisation as a whole and there is some evidence that such annotations improve the memorability of a data visualisation.[164]

> Visual objects that are not directly related to the data can still be effective because they can have a greater emotional impact and they can be more memorable.

Overall, visual objects that are not related to data values are *not* effective for decoding data values, so any other benefits have to be weighed against that cost.

12.7 Case study: Icons

Figure 12.19 shows a data visualisation of the number of pet dogs and pet cats globally.[165] The data symbols in this plot are images, or **icons**, of a cat and a dog, with the number of pets encoded as the height of each icon.[166] Icons are an example of **visual objects**; they are effective for decoding categories because they visually resemble physical objects, in this case, cats and dogs.

Unfortunately, because the cat and dog images are not simple shapes, it is harder to accurately decode the heights of the images, so it is harder to decode the number of cats and dogs. Even worse, we are likely to decode the the size, or **area**, of the images. The dog image is wider than the cat image so the number of dogs is visually over-emphasised.

Figure 12.20 shows two alternative uses of icons that attempt to resolve some of the problems in Figure 12.19. In Figure 12.20a, the images are distorted so that they have the same width and so that only the height of the images changes. While the fixed width helps with decoding the number of pets from the heights of the icons, this sort of distortion can remove the learned decoding of the icons if they become unrecognisable.

Figure 12.20b uses a bar to represent the number of pets and just has a small version of the icon at the base of each bar. We know that the bar is effective for decoding the number of pets and the small icons are effective for decoding the categories of cat and dog.

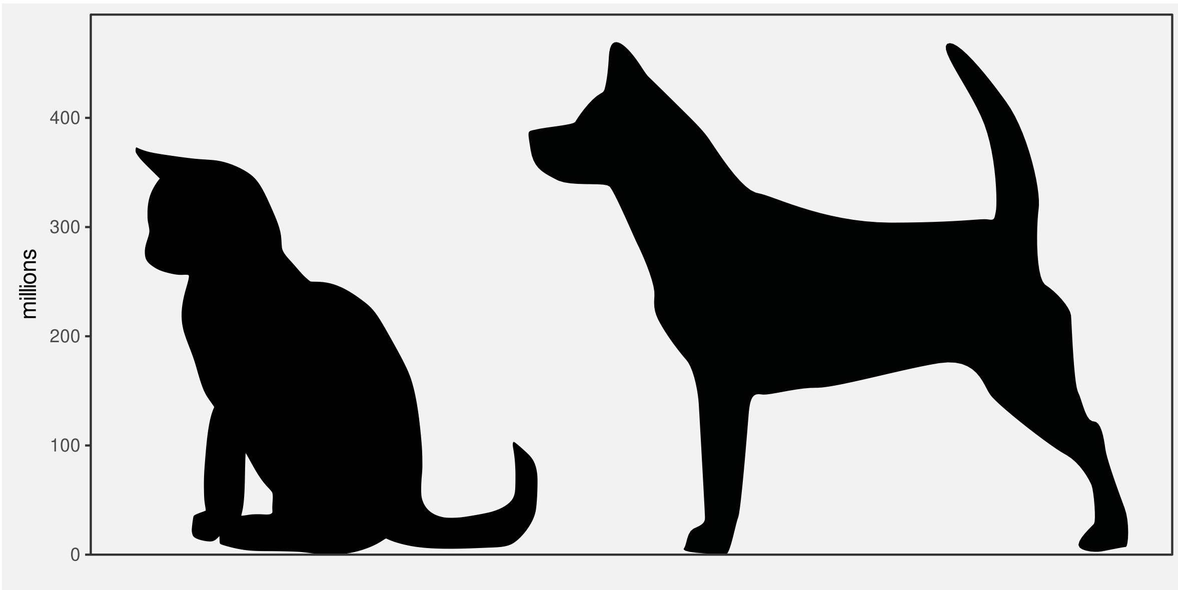
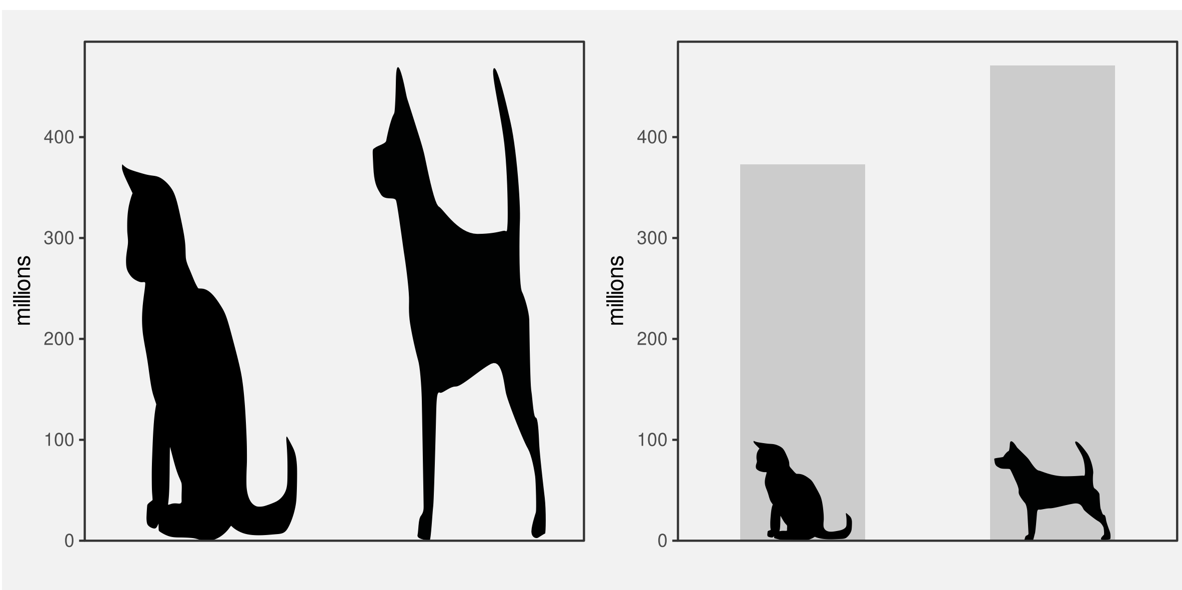


Figure 12.19: A data visualisation of the number of pet cats and pet dogs worldwide.

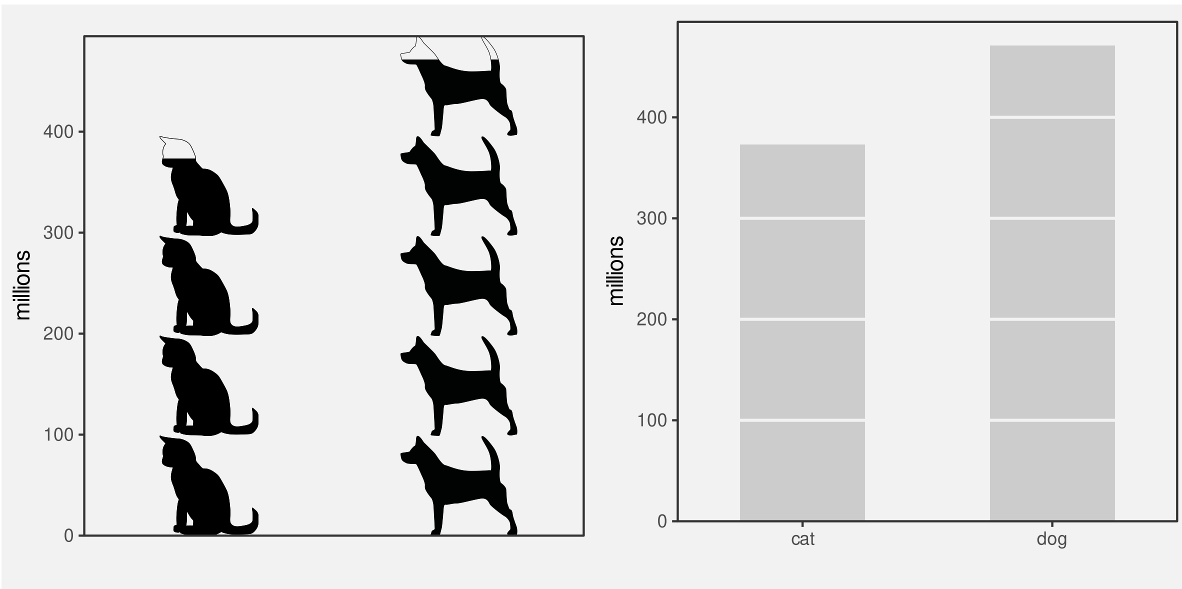


(a) distorted icons

(b) icon labels

Figure 12.20: Two alternative uses of icons.

Figure 12.21 shows two more alternatives to Figure 12.19. Figure 12.21a uses repetitions of the icons to represent the number of pets, still within a bar. In this case, each icon represents 100 million pets. The ability to count individual icons can improve the decoding of the overall data totals.[167] Figure 12.21b is a standard bar plot, but with each bar broken into chunks. We need the labels “cat” and “dog” to replace the icons, but it is possible to count the bar chunks in a similar way to counting the icons in Figure 12.21a, as well as decode the overall bar height, which we know is effective for decoding the number of pets.



(a) Icons are repeated to represent the number of pets.

(b) A basic bar plot, but with each bar broken into chunks.

Figure 12.21: Two more alternative data visualisations of the pet data.

12.8 Dangers of learned decodings

Figure 12.22 provides a demonstration of the Stroop Effect.[168] The task is to name the colour of each word. This task is difficult to do because there are two decodings involved and the two decodings are inconsistent. Each set of letters has a learned decoding based on the semantics of the letters—the word that the letters spell out—but there is also a learned decoding based on the actual **colour** of the letters—the colour names that we learn as children.

One issue with relying on learned decodings is that we may learn more than one association for the same visual feature. For example, the colour red is commonly associated with hot (versus blue for cold), but also with stop (versus green for go), plus also passion, or even

Figure 12.22: The Stroop effect. Try to name the *colour* of each word.

anger.[169] Furthermore, many associations are specific to a particular culture. For example, red is associated with luck and prosperity in Chinese culture, among other things.[170]

Relying on a learned decoding can cause confusion if there are multiple possible decodings. It may not be clear which data value should be decoded from the visual feature (Figure 12.23).

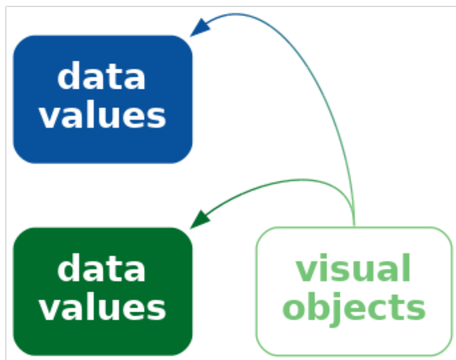


Figure 12.23: A visual object may have more than one learned decoding, which means that we may decode more than one value from the visual object.

The problem is much worse if the learned decoding(s) conflict with an explicit encoding of data values. For example, in Figure 12.2, if we use the incorrect flag for a country—we could swap the Australian and New Zealand flags—we not only introduce confusion, but a misleading decoding (Figure 12.24). This is similar to the problem of dissonant encodings (Section 7.5 and Figure 7.11).

Suppose that we made a similar change to Figure 12.1, swapping the open triangle for Australia and the square-with-a-plus symbol for New Zealand. That would not cause any problem because there is no learned decoding to conflict with. We only get the problem if we swap flags in Figure 12.2 because of the conflict with learned decodings.

> Employing a learned decoding incorrectly can at best weaken and at worst destroy the effectiveness of a data visualisation.

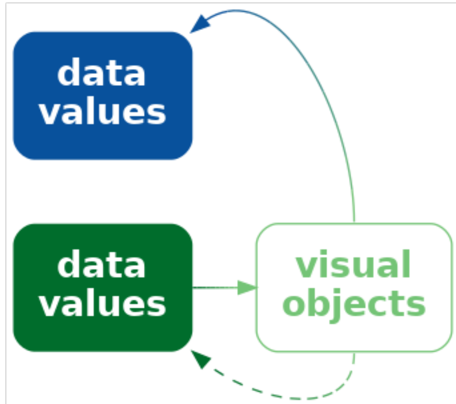


Figure 12.24: An explicit encoding may conflict with a learned decoding. This will lead to confusion and/or incorrect decoding of the visual object.

12.9 Dangers of visual objects

One issue with more complex data visualisations is that their effectiveness depends on a larger amount of learning and experience. If the viewer does not have the knowledge required to decode a visual object, then the data visualisation will fail (Figure 12.25). We saw an example of this in Figure 12.7, where the familiarity of different country flags will vary with nationality and age, for example.

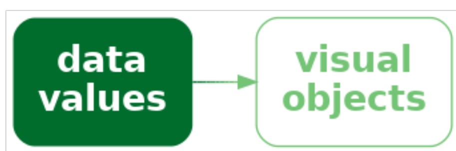


Figure 12.25: If we encode data values using data symbols that are complex visual objects, but there is no learned decoding, the encoding will be ineffective because there is no decoding back to the data value.

An extreme version of this problem is the use of not just complex or unusual data symbols, but entirely novel data visualisations.[171]

For example, Figure 12.26 shows an Andrews curves plot.[172] This data visualisation encodes each row of data to a line data symbol, similar to a parallel coordinates plot. However, the line for an Andrews curve is a more complex data summary and one that has no simple interpretation. In other words, it is not possible to decode even the individual data summaries from the data symbols on an Andrews curve. The x-axis and y-axis on an Andrews curve are of no real value, which is why they have not been labelled on Figure 12.26.

The value of an Andrews curve lies in the visual shapes that are produced by each line. If two rows of data are similar, then they will produce an Andrews curve with a similar visual shape. This means that we can decode clusters from lines that appear similar on an Andrews curve plot.

For example, there are four curves in Figure 12.26 that form a much higher peak than the other curves. The horizontal position of the peak and the vertical height of the peak is of no value, but the fact that the four lines all have a similar peak suggests that there are four countries that are similar to each other and different from all other teams on at least one performance measure.

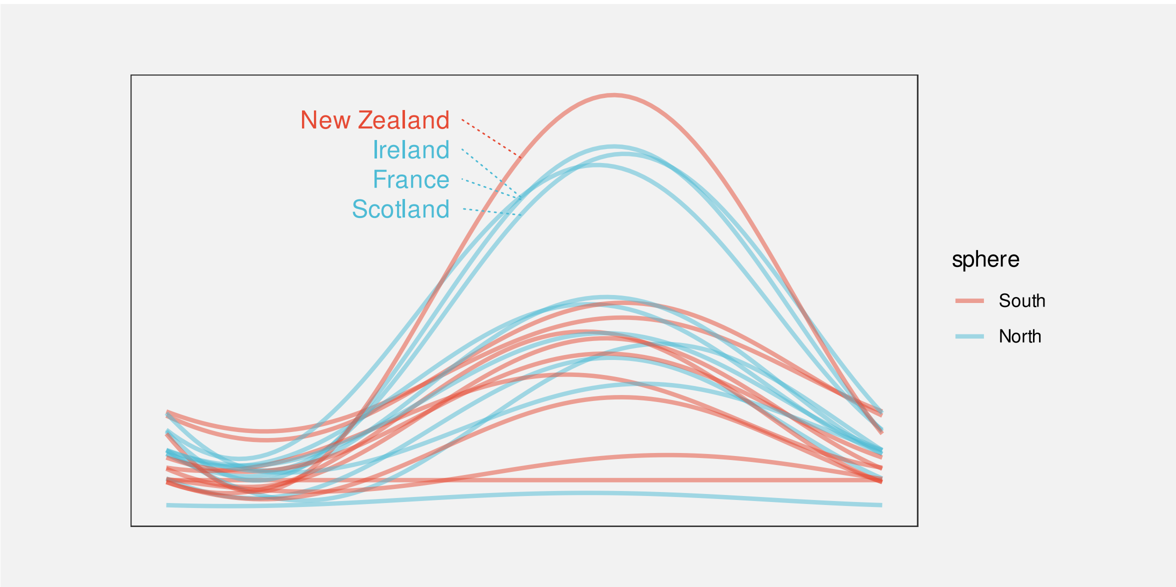


Figure 12.26: An Andrews plot of the performance measures for teams in the 2023 Rugby World Cup.

An andrews curve plot like Figure 12.26 is an example of a data visualisation that is difficult to decode without having learned how to decode the plot. Even if the decoding of similar lines as clusters of data rows is fairly intuitive, ignoring the position and height of peaks and valleys is not.

Parallel coordinates plots (Section 11.5) are another example of a data visualisation that requires learning in order to decode properly. In particular, decoding a tight crossing of lines as a negative correlation between variables is not necessarily an intuitive decoding. Ternary plots (Section 4.6) is yet another example of this sort. In this case, it is necessary to learn how to decode positions relative to the oblique axes.

12.10 Summary

Visual objects are relatively complex data symbols that have a learned decoding.

Encoding data values as visual objects is effective because the learned decoding provides a pre-existing decoding from the data symbol to data values. This can support an explicit encoding and can remove the need for axes and legends to explain the decoding.

However, visual objects may be distracting if they are not related to any encoding of data values and the increased complexity can make decoding more difficult.

It is also important not to create a conflict between an explicit encoding and a learned decoding. That will result in confusion or incorrect decoding of the data symbols.

Notes

[156] Schloss et al. (2019) refer to *inferred mappings*, in particular from colours to quantities, though this relates more to *implicit decodings* (Section 7.1) than learned decodings.

Anderson and Robinson (2022) explore the affective, or emotional, response to colours. It is unclear whether the decoding of colours to emotions is implicit or learned or some combination.

[157] Chernoff faces were invented by Herman Chernoff as a way to visualise high-dimensional data (Chernoff 1973).

[158] William S. Cleveland and McGill (1984) ranked curvature below area, but above colour in terms of accuracy for decoding quantitative data values (Figure 3.12).

[159] Vanderplas, Cook, and Hofmann (2020) allude to the difficulty of experimentally testing "rich, complex graphics that may require domain expertise in addition to the ability to read information from a visual display."

[160] Whitney and Yamanashi Leib (2018) point to a study that demonstrates the ability to summarise crowds of faces for "average emotional expression and gender."

[161] Choropleth maps are a very old form of data visualisation. There are examples from more than 200 years ago! (Dupin and Collon 1826)

[162] Image by rawpixel.com. Free for Personal and Business use.

[163] There have been quite passionate arguments against the use of visual embellishments in data visualisation. Tufte's exhortation to minimise the *data-ink ratio* (the proportion of ink that represents data values) is the prime example (Tufte 1983, chap. 4).

Stephen Few has also had quite a bit to say on the matter (e.g., Few 2011).

Ajani et al. (2022) summarise the general mood: "The visualization practitioner community prescribes two popular guidelines for creating clear and efficient visualizations: declutter and focus."

[164] The arguments in favour of visual embellishments mostly relate to assessing the effectiveness of data visualisation by measures *other than* accuracy and efficiency.

For example, Borgo et al. (2012) found evidence that *visual embellishments* improved memorability and a survey by Rahman et al. (2025) concluded that *annotations* could lead to improved comprehension, memorability, and recall.

Even arch-enemies of chart junk make some concessions: "Graphical embellishments can support the effectiveness of a data visualization in three potential ways: 1) by engaging the interest of the reader (i.e., getting them to read the content), 2) by drawing the reader's attention to particular items that merit emphasis, and 3) by making the message more memorable. Embellishments only enhance effectiveness, however, if they refrain from undermining the message by significantly distracting from it or misrepresenting it" (Few 2011).

There is also some evidence that visual objects may perform better on standard measures under some circumstances. For example, Haroz, Kosara, and Franconeri (2015) found evidence that encoding data values as pictographs rather than simple shapes produced *better* accuracy when cognitive load was higher (when the task was recall of a plot *prior* to the last observed plot).

[165] These counts were obtained from Statista.

[166] We use the term *icon* to describe a small recognisable image. These are also also known as *pictograms* (Haroz, Kosara, and Franconeri 2015; Brewer and Campbell 1998). Borgo et al. (2013) make further fine distinctions between *icons*, *indices*, *symbols*, *pictograms*, and *ideograms*.

[167] Haroz, Kosara, and Franconeri (2015) refer to the ability to rapidly count small numbers of items (*subitizing*) and recommend breaking bars into (a small number of) countable chunks to improve the decoding of overall height (as in Figure 12.21b). This is similar to the use of grid lines, but appears to rely on a different visual effect.

[168] The effect was demonstrated by Stroop (1935). Interestingly, there was little effect on response time for reading words that named colours that were coloured with conflicting colours, but a much larger effect for naming colours of words that had conflicting colour names.

[169] Brockmann (1991) gives the following examples of ambiguous colour association (attributed to Thorell and Smith (1990)):

green for process control engineers means "nominal or safe," for financial managers it means "profitable," and for health care workers it means "infected."

Arcia et al. (2016) report examples of viewers interpreting icons too literally. For example, a picture of an apple as "apple" rather than the intended general category of "fruit".

[170] Huang (2011) describes multiple associations in Chinese culture with the colour red.

[171] This relates to Kosslyn's *Principle of Appropriate Knowledge* (Kosslyn 2006): "Readers can interpret a display only if it builds on appropriate information that they have already stored in memory."

Pinker (1990) talks about the necessity of having a *graph schema* in order to be able to decode information from a graph.

Shah and Hoeffner (2002a) provide multiple examples of young students misinterpreting the message from line plots, which suggests that we need to *learn* how to make effective use of even basic data visualisation formats.

[172] Andrews curves were invented as a way to visualise high-dimensional data by David F. Andrews (Andrews 1972). The curve is a sum of sine and cosine waves with coefficients based on the data values.

13 Text

Figure 13.1 is a version of Figure 1.1 with one significant modification: all of the text has been removed. Although we can see that some values, or rather three sets of values, haved decreased, we cannot tell what the values are or by how much they have decreased. If there was ever any doubt, Figure 13.1 makes it clear that text is doing a lot of important work in a data visualisation.

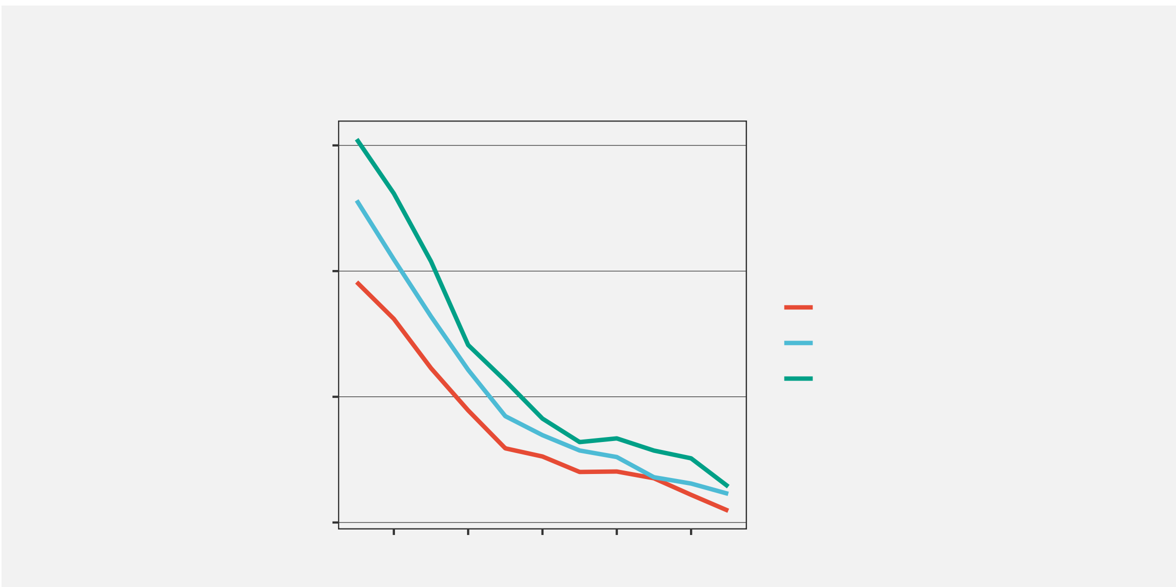
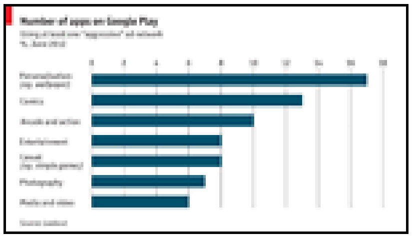


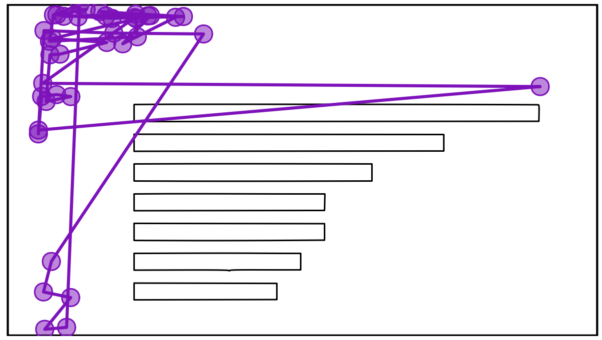
Figure 13.1: A line plot with all text removed. Without text, it is impossible to know what data values are represented in this plot.

Eye-tracking studies also show that people spend a lot of time looking at the text within a data visualisation. For example, in Figure 13.2, there are more fixations in the regions that contain text than anywhere else. There is also evidence that text elements are more memorable than the other elements of a data visualisation.[173]

This chapter looks at the role that text plays in making data visualisations work.[174]



(a) A thumbnail of the original data visualisation.



(b) Fixations and saccades from eye tracking data.

Figure 13.2: Eye tracking data from the MASSVIS project that shows fixations (circles) and saccades (lines between circles) for one person viewing an image for 10 seconds.

13.1 Visual features of text

One role that text serves within a data visualisation is as a standard data symbol. For example, Figure 13.3 is a version of the bar plot Figure 3.1 with white text added to each bar to show the exact number of offenders in each ethnic group. In Figure 13.3, there are bar data symbols and there are text data symbols (the white text).

Just as data values are encoded to the visual features of the bar data symbols in Figure 13.3, data values are also encoded to the visual features of the white text data symbols. For example, the ethnic groups are encoded as the vertical **position** of the bars *and* as the vertical **position** of the white text. The data values for the number of offenders in each ethnic group are encoded as the **length** of the bars *and* as the horizontal **position** of (the right edge of) the white text.

There is also a very important encoding of data values to the **pattern** visual feature of the white text. The data values for the number of offenders in each ethnic group are encoded as the **pattern** of the white text—the letters and digits that make up the text. Each white text data symbol has a different pattern because it is composed of a different set of digits.

Just as we can decode data values from the visual features of the bars in Figure 13.3, we can decode data values from the visual features of the text. For example, we can see just from the vertical positions of the white text, without reading the text values, that each text symbol corresponds to a different ethnic group. We can also see, just from the horizontal position of the text, without reading the text values, that some counts are (much) larger than others.[175] We can also see, just from the text pattern, without reading the text values, that the counts are different. Every white text label has a different pattern.

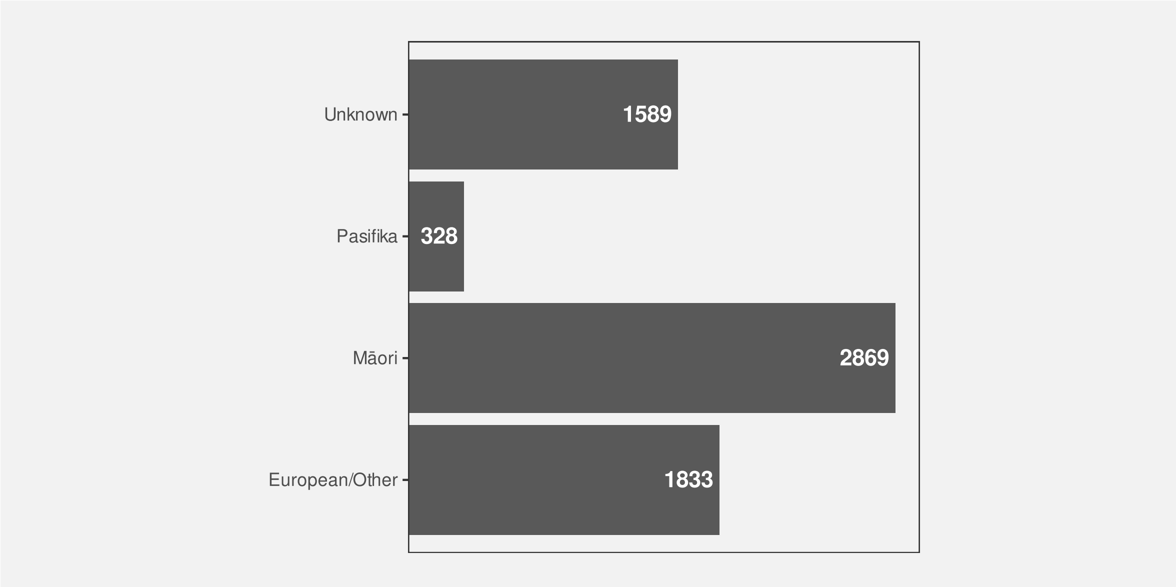


Figure 13.3: A bar plot of the total number of offenders for different ethnic groups. The data symbols in this plot are the horizontal bars *and* the text values at the end of each bar.

> Encoding data values as the position, colour, and pattern of text data symbols is effective for the same reasons that these visual features are effective for other types of data symbols.

However, the unique power of text comes from **decoding** the **pattern** of the text. Text is the ultimate example of a **learned encoding** (Section 12.1). We all learn to read text (and numbers), which means that, when we are presented with a text pattern, there is a pre-existing decoding to the semantic content of the text (Figure 13.4). For example, in Figure 13.3, we are able to decode the data value two thousand, eight hundred and sixty-nine (the count for Māori) from the text "2869".

> Encoding data values as the pattern of text is uniquely effective thanks to the learned decoding of information from specific combinations of digits and letters.

Figure 13.5 shows another example of text being used as data symbols. This scatter plot includes both data point data symbols, one for each country in the 2023 Rugby World Cup, and text data symbols, the names of the countries that finished the tournament in the top four places.

We can again see several encodings of data values as the basic visual features of text data

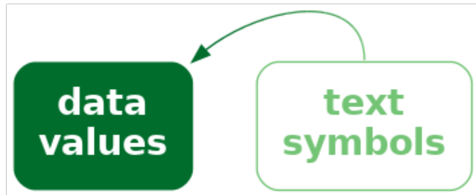


Figure 13.4: Some visual objects have learned decodings, which means that we can decode values from the visual objects without having explicitly encoded values as visual features. This is similar to the idea of implicit encoding (Figure 7.2).

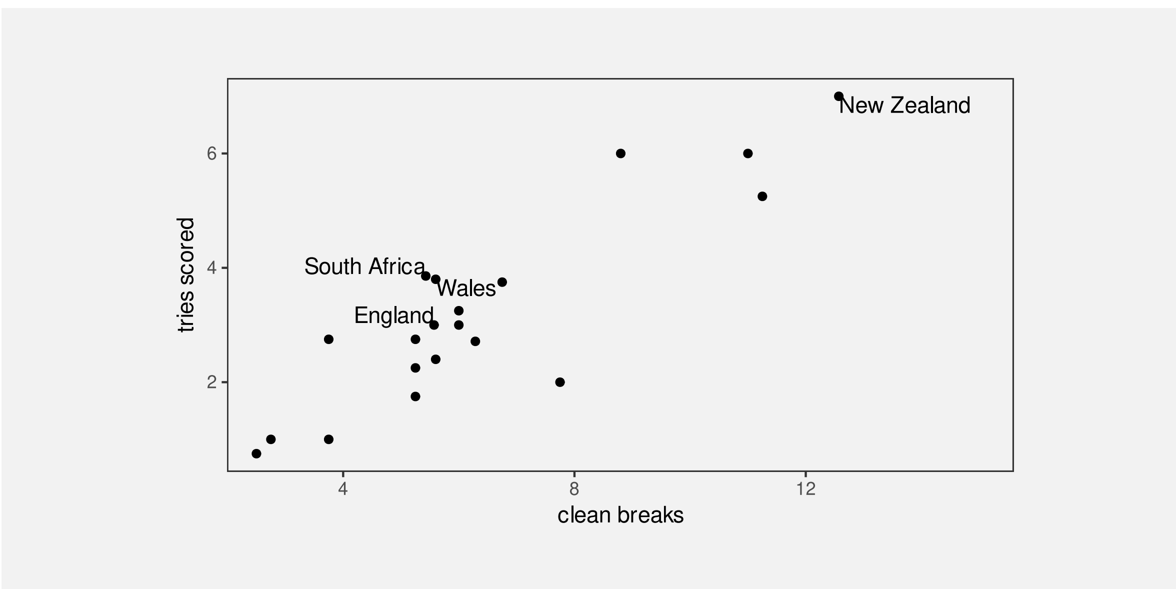


Figure 13.5: A scatter plot of the number of times a team breaks through the opposition defence and the number of tries that a team scores (both are per-game averages) for teams at the 2023 Rugby World Cup.

symbols. The horizontal positions of the country names encode the number of clean breaks for each country, just like the data points. The vertical positions of the country names encode the number of tries scored by each country, just like the data points. Again, there is also an encoding that just involves the text: the country name is encoded as the pattern of the text (the letters used for each text symbol).

There are simple decodings from the positions of the text symbols to data values. For example, we can decode the number of tries scored by each of the top-four teams from the vertical positions of the text and see that New Zealand scored more tries (on average) than the other three teams.

However, the most important decoding is from the patterns of the text symbols. These learned decodings allow us to identify each top-four country by name.

13.2 Text-specific visual features

In addition to the standard visual features like position and colour, there are some visual features that are specific to text data symbols. For example, Figure 13.6 shows a variation on Figure 13.5 where the hemisphere for each of the top-four teams is encoded as the weight and style of the text symbol: teams from the southern hemisphere are bold and italic. Another examples of a text-specific visual feature is the font family, for example, Arial versus Times Roman.[176]

We see uses of basic visual features and text-specific visual features in almost any piece of writing. For example, section headings are typically larger and bolder than the main text in a report. The Executive Summary in this book makes use of colour and angle to create visual connections between associated terms within a bullet-point list. These are examples of encoding qualitative groups, such as headings versus normal text, using just the visual features of text.

13.3 Decoding text

As we have seen already, data values can be encoded as the visual features of text, like position and colour, and data values can be decoded from the visual features of text just like for any other data symbol.

There are text-specific visual features, like font weight, font style, and font family and these are most useful for encoding and decoding qualitative information. Font family is a nominal feature, like hue, and it has a similarly limited capacity. Font weight and font style (or slope) in theory allows for the encoding of numerical values, but similar to area and angle, the relatively low accuracy means that we can only really decode ordinal data values.[177]

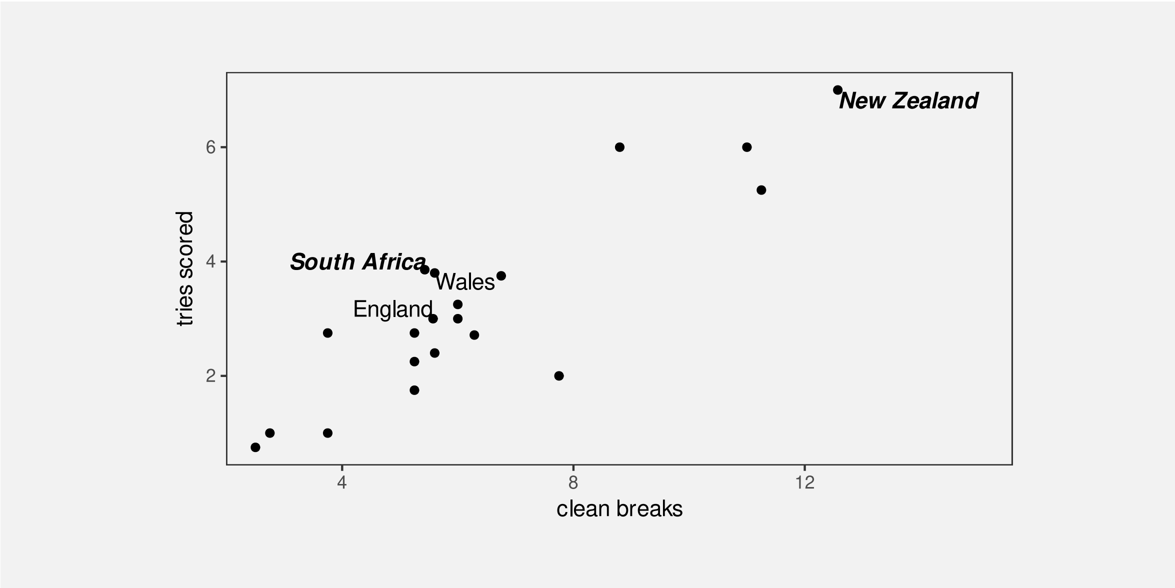


Figure 13.6: A scatter plot of the number of times a team breaks through the opposition defence and the number of tries that a team scores (both are per-game averages) for teams at the 2023 Rugby World Cup.

However, the most important decoding of text symbols is the learned decoding of text patterns—reading numbers and words from text symbols. How effective is this decoding?

In terms of the criteria that we discussed in Chapter 3, the value of a text pattern is that it allows us to decode data values very **accurately**. Text patterns are also very flexible because they are capable of encoding both **quantitative** and **qualitative** data values. Finally, text patterns have a very large **capacity**; we are able to distinguish between many different words and numbers.

> Encoding data values as the pattern of text is very effective because text patterns can be used to represent any sort of data and text patterns allow a very accurate decoding of data values.

The pattern of text is also almost unique in permitting decoding of *absolute* data values. In Section 4.1, we saw that we can decode positions very accurately, but only *relative* to other positions. It is not possible to decode an exact numeric value from a position. However, it is possible to decode an exact numeric value from a text pattern. Visual objects can also sometimes allow decoding of absolute data values (Section 12.1), but to a much more limited extent than text (Section 12.9). We will talk more about this important decoding of text in Section 13.7.

> We can decode absolute data values from a text pattern because there is a pre-existing, learned decoding of words and numbers.

Compared to much simpler data symbols like the bars in a bar plot, text data symbols are much more complex shapes. In terms of the visual processing model from Section 2.2, the text data symbols are similar to **visual objects**. This means that text can convey more sophisticated information, at the expense of more cognitive effort, and with a limit on how many shapes we can process simultaneously (Figure 2.3).[178]

> We can decode complex information from text patterns.

On the other hand, one weakness of text data symbols is that it requires more cognitive effort to compar data values from text symbols than it does to compare data values from simple visual features. For example, in Figure 3.1, it is quick and easy to approximate the ratio of Pasifika to Māori or European/Other to Māori from the lengths of the bars. Those comparisons are slower and take more conscious effort if we try to use the white text data symbols in Figure 13.3.

> It is harder to compare data values that are encoded as text patterns because it requires more effort to decode data values and to perform mental calculations.

In addition, our visual system cannot generate **visual summaries** of large amounts of text (Section 8.2). This is why a table of numbers, like Table 1.1, is ineffective for perceiving groups of values or trends in values.[179]

> Large numbers of text data symbols are *not* effective because the visual system cannot decode visual summaries from the text.

13.4 Case study: Word clouds

Figure 13.7 shows a word cloud data visualisation based on the text of the Wikipedia page on the Rugby World Cup.[180] This visualisation shows the frequency with which different words appear on the Wikipedia page. The data values that are being visualised are shown in Table 13.1.

This data visualisation is unusual because it consists entirely of text **data symbols**. The encodings in Figure 13.7 include encoding each word as the **pattern** of a text data symbol and encoding frequency as the **size** of a text data symbol. There is no need for a guide for the encoding from words to text pattern because there is a learned decoding (Figure 12.4); we can decode each text data symbol into a word data value because that decoding has already been learned.

Figure 13.7: A word cloud of the frequency of words in the Wikipedia page for the Rugby World Cup.

There is no guide for the encoding from word frequency to text size, so we cannot decode exact word frequencies. Table 13.1 shows the six most frequent words. However, the visualisation is still effective at conveying the *relative* frequencies. The larger words occurred more frequently. For example, we can see the names of the teams that were most successful at the tournament, like New Zealand and South Africa, are much larger than less successful teams.

Table 13.1: A table of the number of times different words occur in the Wikipedia page for the Rugby World Cup (for words that appear more than once). There are 287 different words, but only the first 6 are shown here.

word	freq
rugby	72
world	66
tournament	56
cup	54
new	41
zealand	38

The encoding of frequencies as the **size** of the words does not provide the best **accuracy** for decoding frequencies (Section 3.5) and this made worse by the fact that longer words become even larger. This means that we cannot identify the exact frequency of any one word, we cannot identify the quantitative difference in frequency between pairs of words, and we cannot easily perceive the distribution of frequencies across words. A word cloud is only effective for approximate ordinal comparisons of word frequencies (Chapter 3).

Frequency is also encoded to **position in space**, with more frequent words placed towards the centre of the word cloud. This is an example of redundant encoding; more frequent words are both larger *and* more central (Section 9.8).

> A word cloud is effective for conveying words because it encodes words to text data symbols. It is also partially effective for conveying relative frequencies of words because it encodes word frequency to both size and position in space.

Another benefit of encoding frequency as position in space is the efficient use of space. For example, Figure 13.8 shows the same data presented as a bar plot with the frequency of each word encoded to a separate bar. Although the accuracy is improved for comparing frequencies, the bar plot does not have as much room for the words themselves.

One weakness of the word cloud is that the **angle** of words changes with no corresponding change in the data values. This is an example of a **dissonant** encoding, which can cause confusion (Section 7.5). The viewer will notice the differences in angle and attempt to decode that difference to a difference in data values that does not exist.

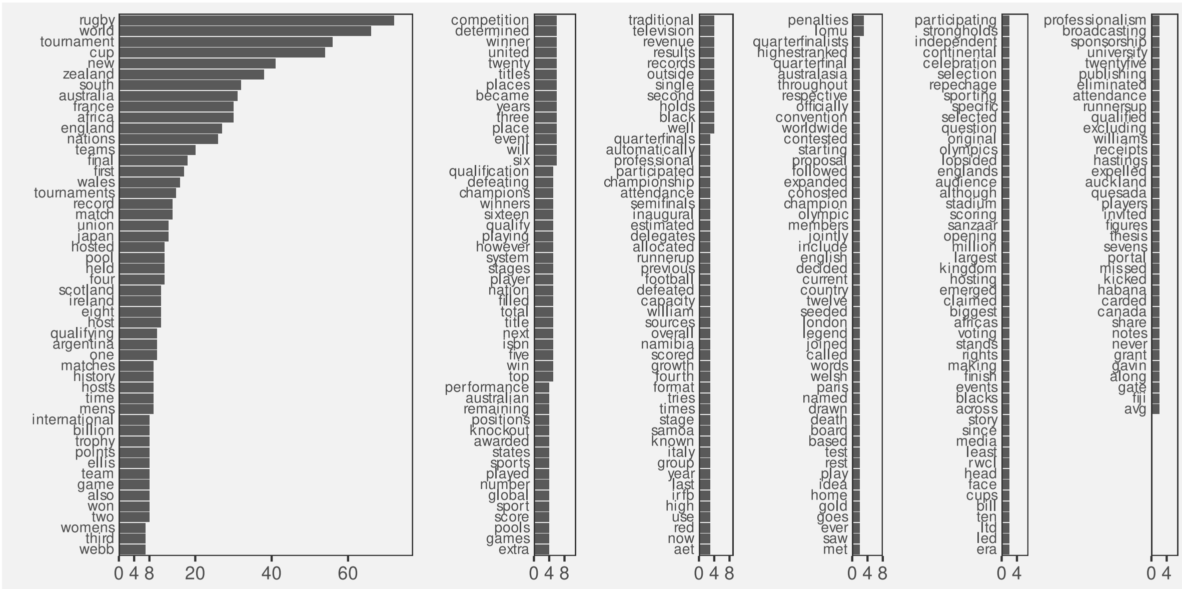


Figure 13.8: A bar plot of the frequency of different words in the Wikipedia page for the Rugby World Cup.

> The different angles of text in a word cloud is not effective because it does not encode any differences in the data.

13.5 Case study: Tables

Table 13.2 shows a subset of the 2023 Rugby World Cup data set (Table 9.1), just including the top eight nations in the tournament. We established in Section 13.3 that visualising data values in a table provides an excellent basis for accurately decoding individual data values, but imposes a larger cognitive burden for comparing values, and is a very poor for extracting visual summaries, such as trends.

For example, it is very difficult to determine from Table 13.2 whether there is any relationship between the values in the four numeric columns.

Table 13.2: Performance measures for the top 8 teams at the 2023 Rugby World Cup. Each measure is a per-game average because some teams played more games than others.

country	sphere	runs	breaks	tries	points
South Africa	South	95.14	5.429	3.857	29.71
England	North	100.3	5.571	3	31.57

Table 13.2: Performance measures for the top 8 teams at the 2023 Rugby World Cup. Each measure is a per-game average because some teams played more games than others.

country	sphere	runs	breaks	tries	points
Fiji	South	138	5.6	2.4	22.4
Wales	North	109.4	5.6	3.8	32
Argentina	South	131.3	6.286	2.714	26.43
Ireland	North	137.6	8.8	6	42.8
France	North	126	11	6	47.6
New Zealand	South	139.3	12.57	7	48

However, there are standard guidelines for formatting tables of values that can improve the readability of a table.[181] Table 13.3 shows the same data as Table 13.2, but with all of the numeric columns right-aligned, a consistent number of decimal places in each column, fewer decimal places than the data actually possess, and the rows have been ordered from the highest number of points per game to the lowest.

It is easier to see some structure in Table 13.2. For example, the number of tries appears to decrease as well as the number of points. We can also see some evidence of the number of breaks decreasing as well, though it is weaker.

There are some familiar explanations for the improvement in the table display. The reduction in detail, with fewer decimal places, means that there is less complexity in the visual field. The consistent number of decimal places means that less is changing, so differences are easier to detect (Section 2.5). Also, the consistent baseline provided by right-alignment means that the text **length** of the numbers (the number of digits in each number) provides a crude encoding of the size of the number, particularly in the column showing the number of clean breaks. However, it is worth noting that some of the **accuracy** of the text has been sacrificed in order to make some of these gains.

Table 13.3: Performance measures for the top 8 teams at the 2023 Rugby World Cup. This is a version of Table 13.2 with some standard table design guidelines applied.

country	sphere	runs	breaks	tries	points
New Zealand	South	139	12.6	7.0	48.0
France	North	126	11.0	6.0	47.6
Ireland	North	138	8.8	6.0	42.8
Wales	North	109	5.6	3.8	32.0
England	North	100	5.6	3.0	31.6
South Africa	South	95	5.4	3.9	29.7
Argentina	South	131	6.3	2.7	26.4
Fiji	South	138	5.6	2.4	22.4

For comparison, Figure 13.9 shows a scatter plot of the data from Table 13.2. The data symbol is a circle, with the number of tries encoded to horizontal **position**, the number of points encoded to vertical **position**, the number of clean breaks encoded to **size**, and the number of runs encoded to **colour** (shade). The relationships between tries and points and between breaks and both tries and points are still easier to perceive in this plot than in Table 13.3, but Table 13.3, even with its rounding, still provides better accuracy, particularly for breaks and runs.

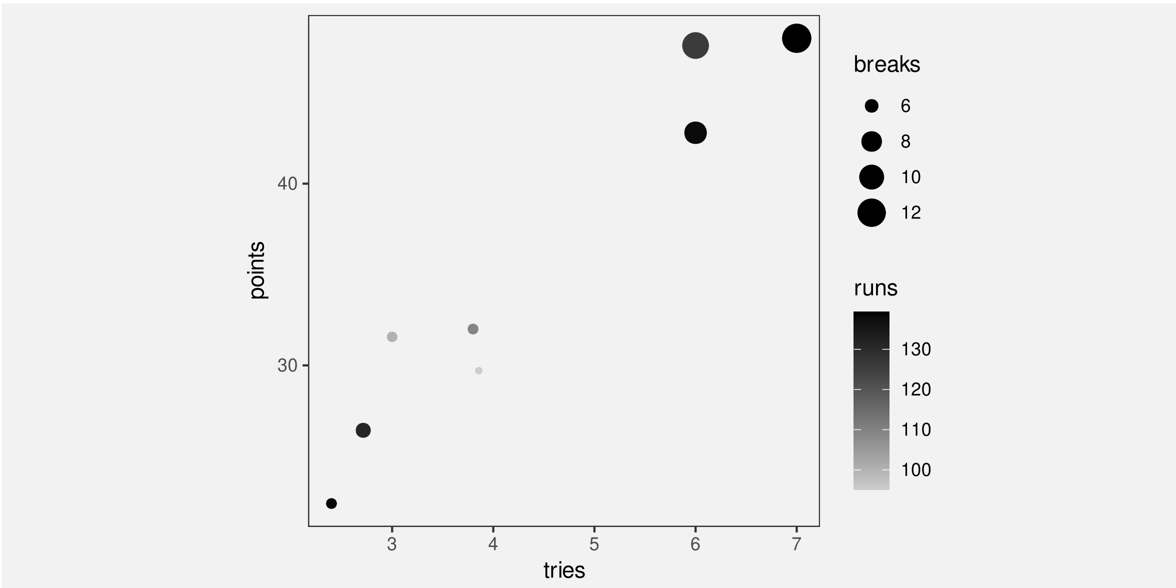


Figure 13.9: A scatter plot of the performance measures for the top 8 teams at the 2023 Rugby World Cup.

13.6 Case study: Stem-and-leaf plots

Figure Figure 13.10 shows a histogram of the number of tries scored by teams in the 2023 Rugby World Cup. The data symbols in Figure 13.10 are bars. The raw data values are the average number of tries per game for each team, but the lengths of the bars encode **data summaries**: the number of teams with an average number of tries within each range: 0 to 1, 1 to 2, and so on.

Because the borders of the bars are not drawn, we get an emergent shape from the combined bars (similar to a density plot; Section 10.3).

This histogram allows us to see features of the data summaries, such as a suggestion that there is a small subset of teams that score a higher number of tries than the other teams (a bimodal distribution; though in this case we have a very small amount of data). However, there is no way to decode the raw data values from this histogram.

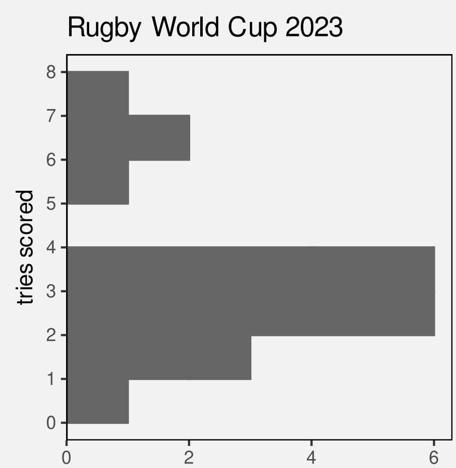


Figure 13.10: A histogram of the number of tries scored. The bars are horizontal for easy comparison with the stem-and-leaf plot.

Figure Figure 13.11 shows a stem-and-leaf diagram of the same data (with tries scored rounded to one decimal place).[182] The data symbol in this case is a text digit. Every team is encoded as a digit from 0 to 7, which represents the first decimal place for the average number of tries scored by each team. For example, New Zealand scored 7.0 tries per game on average and that is encoded as a `0` beside the `7` on the y-axis. The number of tries is also encoded as a combination of vertical position (the whole number of tries) and horizontal position (teams with the same whole number of tries are stacked horizontally, but also ordered by the first decimal place value).

The number of teams within each range (0 to 1, 1 to 2, etc) is encoded by the number of digits. Another way to look at it is that the data symbol is a piece of text consisting of several digits and the number of teams is encoded as the **length** of the piece of text.

As with the histogram in Figure 13.10, there are emergent shapes with two distinct humps visible, one taller than the other. However, in Figure 13.11 we are *also* able to decode the raw data values as well. Furthermore, because the data symbols are text digits, we are able to decode individual data values very accurately.

In summary, Figure 13.11 combines standard decodings (**length**) with **learned decodings** (text pattern) to provide access to both high-level features of data summaries *and* raw data values.

Figure 13.11: A stem-and-leaf plot of the number of tries scored.

Figure 13.12 shows a variation on Figure 13.11 with the hemisphere encoded as the colours of the digits. This provides another demonstration that we can use simple visual features like colour to encode data values with text data symbols.

Figure 13.13 shows another variation on Figure 13.11, this time with the hemisphere encoded as the **weight** of the digits. This demonstrates the idea that there are text-specific visual features that can be used to encode data values (Section 13.2).

Stem-and-leaf plots might be seen as an archaic form of data visualisation, but we can see that they are effective in many ways.[183]

Unfortunately, despite all of their advantages, stem-and-leaf plots have a limited usefulness because they can only accommodate very small data sets. There is a similarity between a stem-and-leaf plot and the stacked icon plot in Figure 12.21a. However, whereas an icon can easily be used to represent some multiple of objects, that option is not available for text digits.

13.7 Encoding encodings

Figure 13.14 shows a dot plot of the total number of points scored by Tier One nations at Rugby World Cups. This data visualisation contains no text data symbols. The only data symbols, in the traditional sense, are the coloured circles.

However, there are text elements within Figure 13.14, on both axes and in the legend. Most of the data visualisations that we have seen so far have contained axes and legends, but we have

Figure 13.12: A stem-and-leaf plot of the number of tries scored with hemisphere encoded as colour.

Figure 13.13: A stem-and-leaf plot of the number of tries scored with hemisphere encoded as font weight.

largely ignored them and concentrated just on the data symbols within a data visualisation. In this section, we look at the work that axes and legends, particularly the text elements of axes and legends, are doing within a data visualisation.

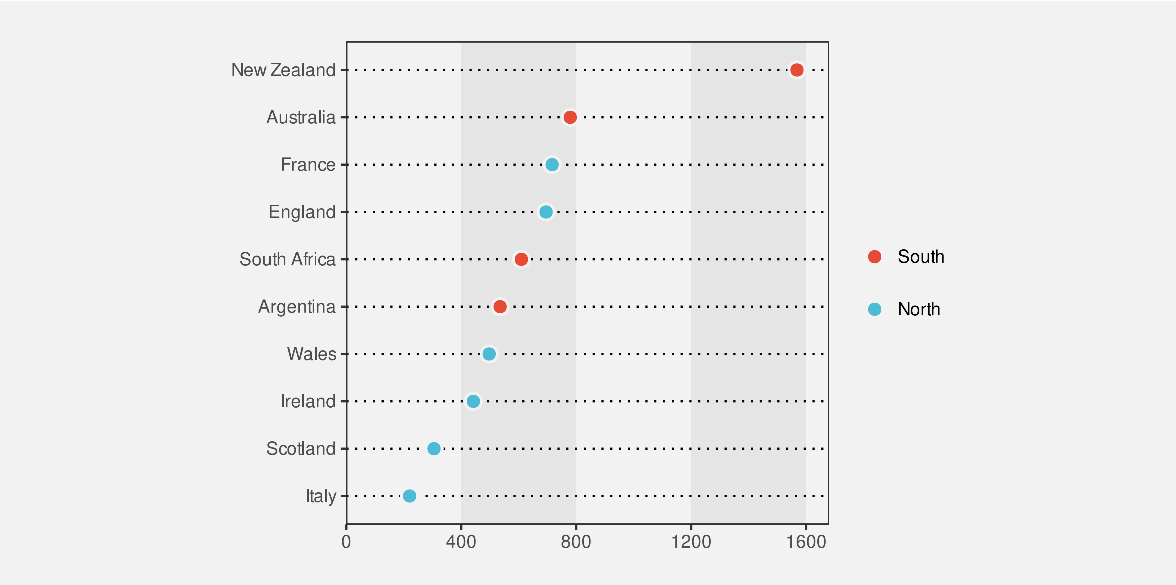


Figure 13.14: A dot plot of the total number of points scored by Tier One nations at Rugby World Cup matches.

There are very familiar encodings of data values to data symbols in Figure 13.14. The number of points is encoded as the horizontal position of the circles, The different teams are encoded as the vertical position of the circles, and the hemisphere of each team is encoded as the colour of the circles.

Although we have seen these encodings many times, we have not previously addressed the fact that, in addition to producing the location and colour of the circles, all of those encodings are represented within the data visualisation as either an axis or a legend. For example, the encoding of the number of points as horizontal position is represented in Figure 13.14 by the x-axis. We will now look more closely at those axes and legends.[184]

The x-axis in Figure 13.14 consists of five text labels, typically referred to as tick labels, plus five small line segments, typically referred to as tick marks. For example, there is a tick label "400" with a corresponding tick mark.

A key insight is that, although the tick labels and the tick marks are not usually considered to be data symbols, and the values that they represent, in this case zero to sixteen hundred in steps of four hundred, are not usually considered to be data values, both the tick labels and the tick marks are just **encodings** of information. For example, the tick label "400" on the x-axis is an encoding of the number four hundred: the horizontal position of the text encodes the number four hundred and the text pattern encodes the number four hundred. Similarly, the number four hundred is encoded as the horizontal position of the corresponding tick mark.

In other words, the x-axis in Figure 13.14 is an **encoding** of the information that the number of points is **encoded** as horizontal position.

Furthermore, because the tick labels and tick marks are encodings of information, it is possible to **decode** information from the tick marks and tick labels. In particular, as we discussed in Section 13.3, it is possible to decode the number 400 very accurately from the tick label "400". The proximity of the tick mark to the tick label means that we group the tick label, and the decoded value four hundred, with the tick mark. (Section 2.7).

> Text on axes and legends are similar to data symbols because we use them to encode and decode information.

The tick label "400" on the x-axis is even more important because it is the *only* way to decode the number 400 from Figure 13.14 (Section 13.3). The tick label is what establishes an absolute value of four hundred as a horizontal position. The horizontal position of the circle data symbols are decoded relative to that absolute position (Section 4.1). For example, we can decode that Ireland has scored a little over four hundred points in Rugby World Cup games based on the position of the circle for Ireland *relative* to the position of the tick label "400". There is no way to decode the number four hundred from the horizontal position of the circle for Ireland *on its own*.

> Text on axes and legends are essential because they are often the only way to decode absolute data values from a data visualisation.

Similarly, although with a little more work, we can establish that New Zealand has scored approximately eight hundred more points than Australia, by comparing the horizontal positions of the relevant circles with the tick labels that decode to sixteen hundred and eight hundred (and then performing the required mental arithmetic).

We can use similar reasoning to establish the role of the tick labels on the y-axis. In this case, the values that are being encoded are actually data values—the names of the teams. The tick labels might not usually be called data symbols, but they are encodings of the team names. Furthermore, the tick labels are the *only* way to decode the actual team names. All that we can decode from the vertical position of the circles is that there are different teams. It is the fact that each tick labels shares the same vertical position as a circle that allows us to decode the team names from the circles, *via* the tick labels.[185]

The legend in Figure 13.14 also works in a similar way. The legend consists of circles and text labels, arranged in rows so that each circle forms a visual group with one of the text labels. The text labels encode the hemisphere of the teams and they are the *only* way to decode the hemisphere values. The colours of the circles, and the proximity of the circles to the text labels, allow us to decode a hemisphere from a colour and the fact that the same colour encoding is used for the circle data symbols in the plot allow us to decode a hemisphere from a data symbol.

Although the circles in the legend are not usually considered data symbols, they do look a lot like the data symbols in the plot. This makes it even clearer that axes and legends are just another example of encoding and decoding information.

Another point to note about the text on axes and legends is that there tends to be a small number of tick labels and legend labels, whereas the number of data symbols can be quite large. This makes sense from what we know about how well decoding works from text (Section 13.3). The purpose of tick labels and legend labels is to allow the decoding of absolute values with great accuracy. This is the situation where text excels.

A further point is that tick labels tend to be regularly spaced and they tend to encode "nice" numbers, for example, multiples of ten. As we saw earlier, if we want to decode the difference between tick labels, in addition to decoding individual tick labels, we are also required to perform mental to obtain the difference between the decoded values. The regular sequence of "nice" values makes that mental arithmetic much easier.

13.8 Case study: Direct Labelling

Figure 13.15 shows a line plot of the number of offenders for different ethnic groups from 2011 to 2021. The ethnic groups are encoded as the colours of the lines, but instead of a legend to explain this encoding, we have drawn labels directly on the plot, a technique sometimes described as *direct labelling*.

As we saw in Section 13.7, the decoding of a legend requires several steps of indirection: from text label in the legend to coloured data symbol in the legend to coloured data symbol in the plot. The direct labelling in Figure 13.15 reduces the number of steps. The text label can be decoded to give the ethnic group, then the proximity of the label to a line, plus the common colour encoding, visually groups the line with the label (Section 2.7).

This technique reduces the amount of cognition required and also reduces the memory load because information has to be rembered between each step. Furthermore, the proximity of labels to data symbols reduces the number of fixations required to perform the decoding (Section 2.3), which again reduces the memory load.[186]

13.9 Encoding information

Figure 13.16 shows a very simple bar plot of the youth crime rate in New Zealand for 2021, broken down by ethnic group. There are no text data symbols in this plot, but in addition to the text on the axes, there is text in the overall plot title and there is text in the x-axis title.

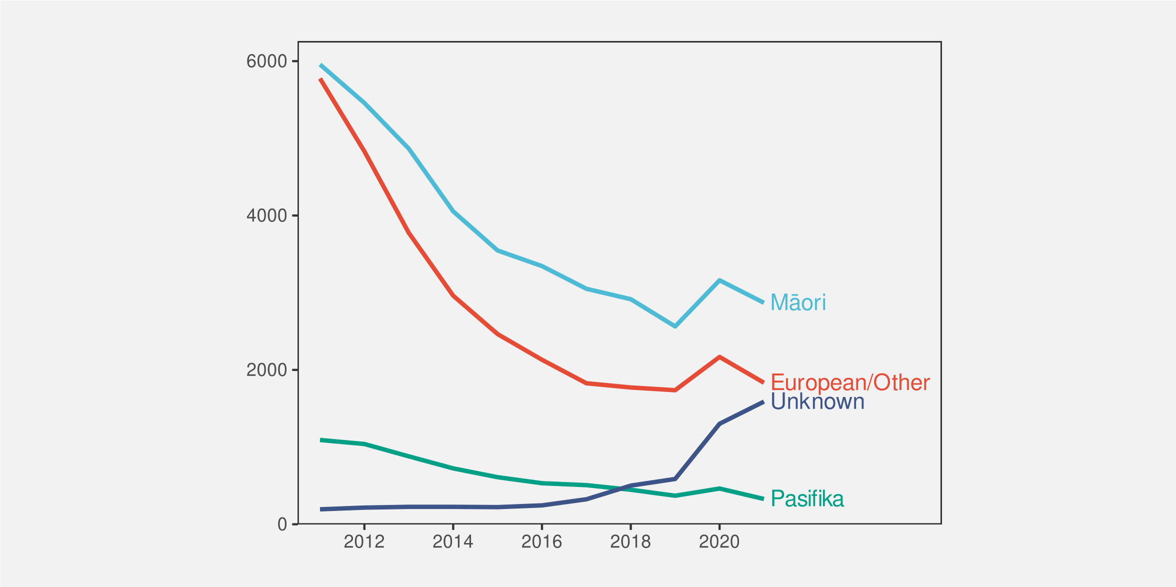


Figure 13.15: A line plot with …

Many of the data visualisations that we have seen previously have contained titles like these, but, as with the axes and legends, we have largely ignored their role. This section looks at the important work that titles and other larger pieces of text are performing within a data visualisation.

The first important point is that, although we are no longer encoding data values as data symbols, there is still an encoding happening. For example, the x-axis title encodes information about the data values that are being encoded as horizontal position, in this case the crime rate. The x-axis title does not encode any particular crime rate data value, but it does encoded a description of what those data values mean.

In other words, the x-axis title encodes **metadata** rather than raw data values (Figure 13.17).

The overall plot title is similar, though more general again. This is not an encoding of specific data values, or a description of the data values, but a description of the overall topic of the data visualisation. A title may also summarise the overall message that should be drawn from the data visualisation.[187]

The titles on a data visualisation contain information like descriptions of variables and units of measurement and references to data sources. These are all important pieces of information, but they are more complex, more abstract, and far fewer in number than data values. Fortunately, text is very capable of expressing complex information and works very well in small amounts, so it is ideal for this role.

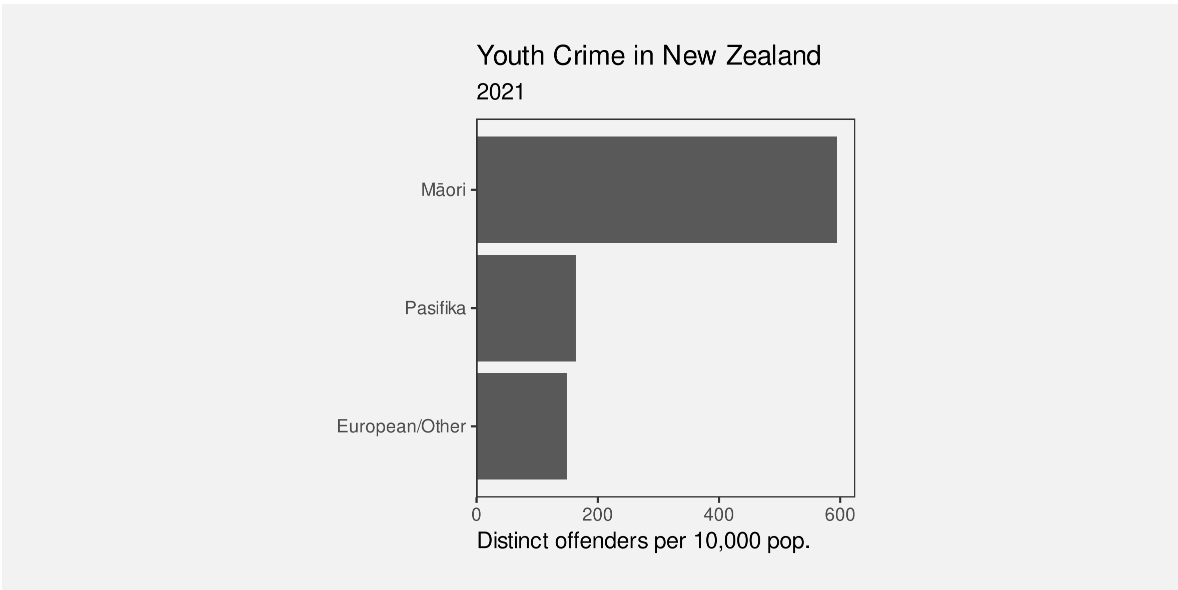


Figure 13.16: A bar plot of the crime rate for different ethnic groups.

Figure 13.17: Titles can be viewed as an encoding of metadata as text. We can encode complex information using text and we can decode text with extremely high fidelity.

To emphasise the unique ability of text to encode complex information, Figure 13.18 shows a variation on Figure 13.16 that uses small icons to express the plot title.[188] Icons like text, have learned decodings, so that we can decode "youth", "crime", and "New Zealand" from the icons in the title. However, the decodings of icons are far less precise, for example, we could also decode "child" or just "person" from the first icon and we could decode "prison" or "police" from the second icon. Furthermore, it is much harder to compose pictograms in the way that text can be composed into phrases and sentences.

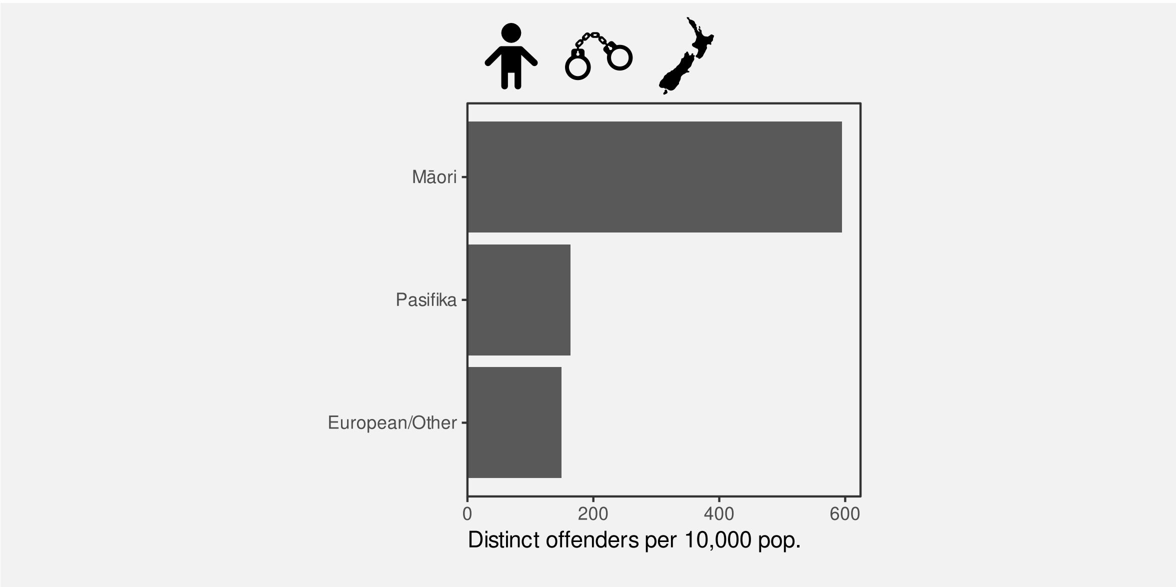


Figure 13.18: A bar plot of the crime rate for different ethnic groups.

> Text is the only way to encode the type of complex and abstract information that is required for titles and axis labels.

Another use of text in a data visualisation is to label specific points of interest.[189] For example, Figure 13.19 shows a variation of Figure 9.4, with text added to help explain the result of the Rugby World Cup final to disillusioned New Zealanders. This demonstrates further the value of text as the only way to encode complex and nuanced information.[190]

Figure 13.19 also demonstrates that text is an effective way of directing attention (Section 2.6). Text labels can explicitly describe important areas of the data visualisation or even important information to extract from the data visualisation.

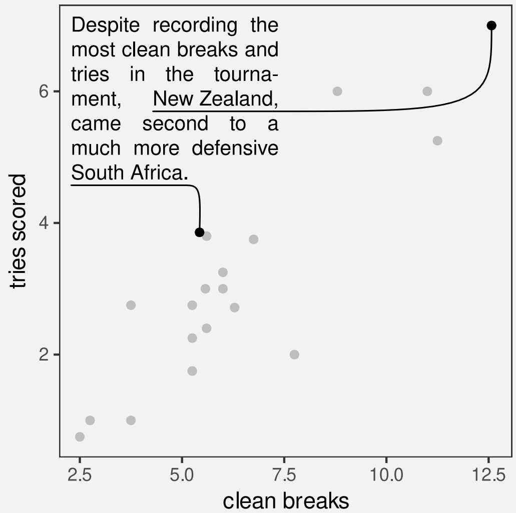


Figure 13.19: The number of times a team breaks through the opposition defence and the number of tries that a team scores (both are per-game averages) for teams in the 2023 Rugby World Cup.

13.10 Dangers of text

Figure 13.20 shows a data visualisation of the number of gun deaths in Florida between 1990 and 2012, covering the enactment of Florida's "stand your ground" law that loosened the constraints on an individual's rights to use deadly force.[191] The title of the plot points out that gun deaths *decreased* after the law was enacted, which is perhaps surprising given that use of deadly force was less constrained after the introduction of the law.

However, if we look closely at the plot itself, we can see that the y-axis scale is inverted. A downward trend is actually an *increase* in gun deaths, so the data shows that gun deaths actually increased after the law was enacted.

The point of this example is to demonstrate that the text within a data visualisation is not only given a lot of attention, but it can have a dominant effect on the message that the viewer takes away.[192]

Figure 13.20 is an example of a deliberately misleading data visualisation, which we obviously want to avoid. However, the broader point is that we need to be at least as careful with our encoding of information in text labels as we are with our encoding of data values to visual features. We have a responsibility to make sure that there is a consistent message from both data symbols and text labels.

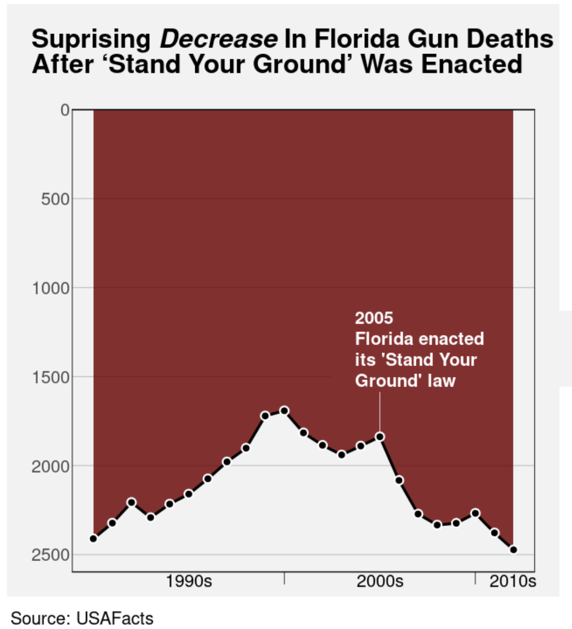


Figure 13.20: The title influences our perception; the message from the data symbols is actually quite different.

Figure 13.21 shows that we are also responsible at a more fundamental level for making sure that text labels are legible and easy to understand.[193] Figure 13.21 shows data on the fortunes of New Zealand political parties since the 2023 election. The title is not misleading in this case, but it is not easy to understand. The main data values that are encoded in the bars of Figure 13.21 are the difference between the percentage of votes that each party gained at the election and the percentage of respondents who would vote for each party according to the most recent 30-day rolling average of poll results since the election. It is not clear what the best title for this plot would be, but it is also clear that the current title is not the best.

In addition, the text labels at the end of each bar in Figure 13.21 are complicated, quite small and congested, and they are not regularly formatted. For example, the numbers in the labels on the bars to the right do not line up, so they are not easy to read and compare (Section 13.5).

Although text is extremely flexible and expressive, it is also complex and it is quite easy to generate a garbled or confusing piece of text.[194] Just because it is *possible* to encode complex information as text labels does not mean that it will always be easy to decode complex information from a text label.

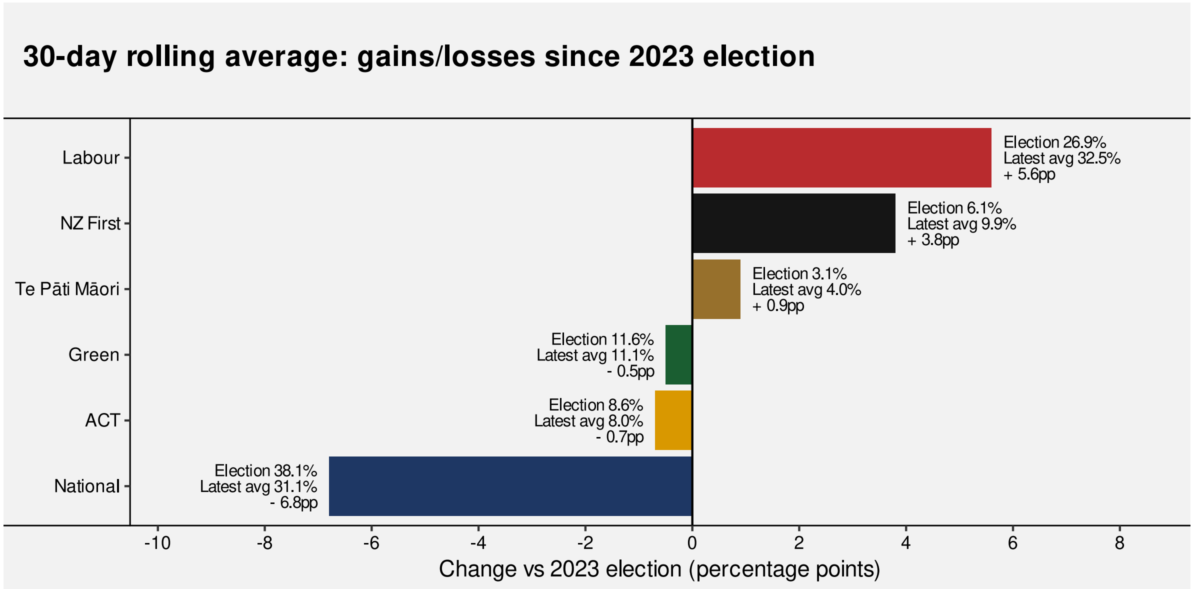


Figure 13.21: The changing fortunes of New Zealand political parties since the 2023 election.

13.11 Case study: Linear pairwise plots

Figure 13.22 shows a correlation matrix of seven performance measures for teams in the 2023 Rugby World Cup (Table 9.1). We can see that many measures have quite a strong positive correlation (dark red circles), but one measure (the number of tackles per game) is negatively correlated with all other measures (though often only very weakly).

When the number of variables becomes large, a correlation matrix becomes an inefficient use of space.[195] An alternative is to produce a *linear* plot of pairwise correlations, as shown in Figure 13.23.

The data symbols in Figure 13.23 are circular data points. The correlation between a pair of measures is encoded as the horizontal position of the circles and the vertical position encodes for different pairs of measures.

The different pairs of measures are also encoded as the pattern of the tick labels on the y-axis. This encoding of the tick labels is an essential encoding because we cannot decode the pairs of measures from the vertical position of the circles alone.

However, the encoding of the pairs of measures is an example of a poor text encoding because it is difficult to decode the variable names from the text pattern. Each tick label consists of two measure names separated by a colon. That makes a piece of text that is not a recognisable word, or a recognisable expression, so it is more difficult for our learned decoding to extract the information about the measures.

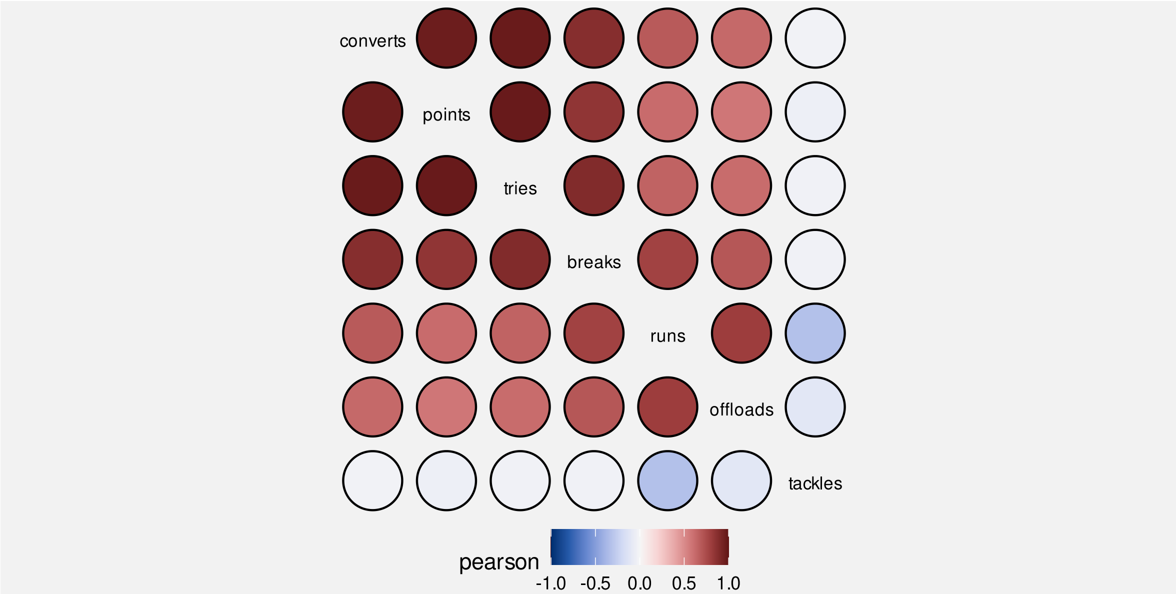


Figure 13.22: Correlations between pairs of measures of performance for teams in the 2023 Rugby World Cup.

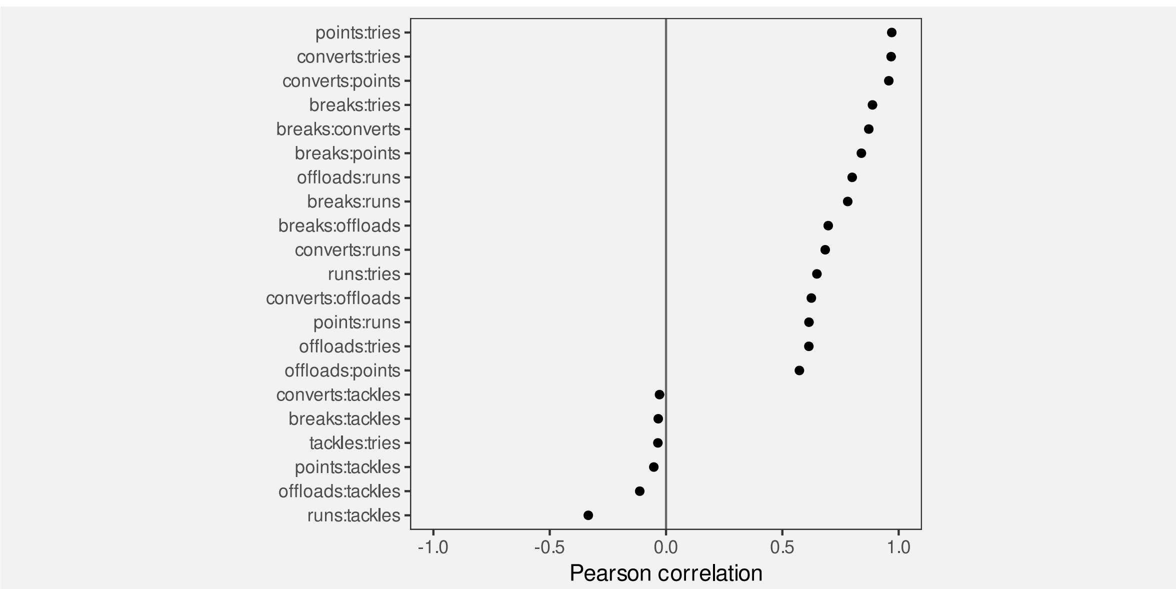


Figure 13.23: Correlations between pairs of measures of performance for teams in the 2023 Rugby World Cup.

Figure 13.24 shows improved tick labels. The formatting in these labels makes it easier to decode individual measure names for each pair of measures.

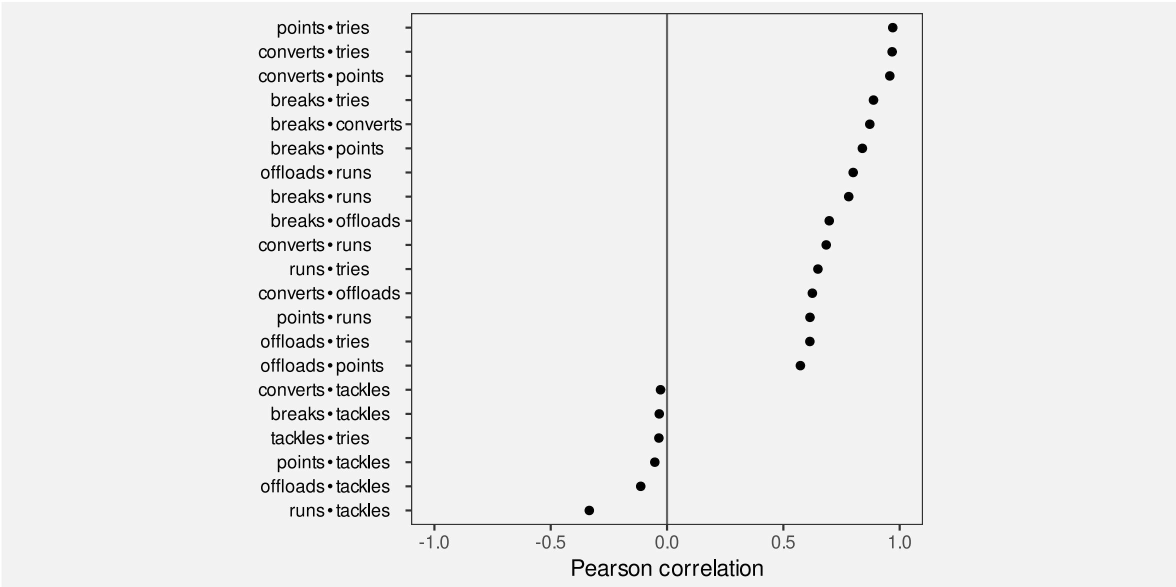


Figure 13.24: Correlations between pairs of measures of performance for teams in the 2023 Rugby World Cup.

13.12 Case study: Alt text

A data visualisation as a conversion of information into a visual representation (Section 1.2). We saw in Section 6.7 that a data visualisation that makes use of a colour encoding can take into account red-green colour blindness, but what about a low-vision or blind audience?

The standard approach to provide support for making visual content accessible is to provide some form of **alt text**, which consists of a purely text-based description of the image. A text description of a data visualisation is accessible because it can be recognised and read out by a screen reader.

Ideally, any image will be accompanied by alt text and accessibility is even mandated in many public service organisations.[196] The alt text for a data visualisation will ideally describe what variables are being displayed and how they are encoded (what sort of plot is it), data summaries like trends, clusters, outliers, and correlations, and include the main message of the visualisation, including causal explanations and wider implications.[197]

In terms of the framework of this book, alt text is a purely text-based encoding. However, more important than that is the fact that alt text is *not* a visual decoding. The alt text for a data visualisation will be decoded by listening to a screen reader rather than by any visual pathway. The early processing of auditory signals occurs in a different part of the brain than visual input.

On the other hand, because text is language, the higher-level processing of auditory alt text involves the same learned decodings as visual text. This means that alt text is capable of communicating both simple and complex information, but only in relatively small amounts. Unsurprisingly, alt text is not effective for communicating large amounts of individual data values.

13.13 Summary

Data values can be encoded as the visual features of text, such as position and colour, just like for other data symbols.

The pattern of text—the characters used in text—is particularly important because we have a learned decoding of text. We can read data values from text.

We can use text to encode any type of data, we can represent a very large number of different categories, and we can decode individual data values from text extremely accurately.

On the other hand, decoding a large number of data values from text is very slow and we cannot generate visual summaries from text.

The text elements within axes and legends are essential components of a data visualisation because they provide the only precise decoding of exact data values. This is what allows the decoding of the main data symbols within a data visualisation.

Titles and annotations are very important components of a data visualisation because they are the only way to encode complex and higher-level information, such as metadata and the overall subject matter of a data visualisation.

Text is very valuable in a data visualisation for accurately encoding a small number of data values, or to express complex information. But text is not appropriate for representing a large number of raw data values.

Notes

[173] And how well a data visualisation can be recalled is another way to measure its effectiveness: "a memorable visualization is often also an effective one." Borkin et al. (2016).

Only a thumbnail version of the original image is shown in Figure 13.2 in order to comply with terms of use of the MASSVIS data set.

[174] Brath (2020) provides a very thorough treatment of the use of text in data visualisation and is the source of many ideas in this chapter.

Murrell (2025) provides an more in-depth version of much of this chapter.

[175] This is similar to the way that data values can be encoded and decoded using the visual features of visual shapes and visual objects (Section 10.4 and Section 12.3).

[176] Brath and Banissi (2016) provides a much longer list of text-specific visual features.

[177] These assessments, and more, can also be found in Figure 14 of Brath and Banissi (2016), though the information there is largely speculative and by analogy, rather than being based on experimental evidence.

[178] Although text initially arrives as visual input and therefore follows the visual processing pathways discussed in Section 2.2, there are also separate verbal processing pathways in the brain that are involved in the learned decoding of text as language (Ware 2021, 315–16)

[179] Ware (2021) neatly summarises when to use text and when not to use text:

"When a large number of data points must be represented in a visualization, use symbols instead of words or pictorial icons." (Guideline G8.18)

"Use words directly on the chart where the number of symbolic objects in each category is relatively few and where space is available." (Guideline G8.19)

Why is a picture is worth a thousand words? Because 1000 words is 1000 fixations, identifying 1000 visual objects, and then interpreting their combined meaning.

[180] The Rugby World Cup page of Wikipedia (2024).

[181] Examples of guidelines for tables include Ehrenberg (1977), (Wainer 1992), and Schwabish (2020).

[182] The invention of stem-and-leaf plots is attributed to Tukey (1977).

[183] "If we are going to make a mark, it may as well be a meaningful one. The simplest - and most useful - meaningful mark is a digit" (Tufte 1983).

[184] Another name for a *legend* is a *key*. Wilkinson (2005) collects all axes and legends within the single concept of a *guide*.

[185] In the case of a dot plot, like Figure 13.14, the dotted lines are also a big help in associating each tick label with a circle. The dotted lines are an example of using *connection* to create a visual group (Section 2.7).

[186] Shah and Hoeffner (2002b) include the following advice for legends: "reduce the difficulty viewers face in keeping track of graphic referents because of the demands imposed on working memory."

Carpenter and Shah (1998) give evidence from eye tracking that there is a memory load from switching between legend and plot: "viewers must continuously reexamine the labels to refresh their memory."

[187] Suggestions for what to include in a plot title (and exhortations that we must always include a title!) are a popular inclusion in data visualisation guidelines. For example, from William S. Cleveland (1985b):

1. Describe everything that is graphed.
2. Draw attention to the important features of the data.
3. Describe the conclusions that are drawn from the data on the graph.

[188] Child image by Creative Stall from Noun Project (CC BY 3.0)

Handcuffs image by Uswa KDT from Noun Project (CC BY 3.0)

New Zealand image by Sharon Faria from Noun Project (CC BY 3.0)

[189] These sorts of text labels are often referred to as *annotations*.

[190] We have described data visualisations mostly in terms of encoding *data values* because that is what is encoded in data symbols. We have mostly only considered the encoding and decoding of very simple forms of information: data values or data summaries. In this chapter, the addition of text vastly expands the range of information that we can encode and decode. Text is very powerful because it can convey not just data values, but information, knowledge, understanding, and wisdom (Ackoff 1989).

[191] Figure 13.20 is based on a Business Insider data visualisation, using data from USA Facts.

[192] Kong, Liu, and Karahalios (2019) found that titles that are inconsistent with the data symbols bias the decoded information, even more so when recalling the message of the data visualisation later on.

Kim, Setlur, and Agrawala (2021) found that captions (another form of text label) that are consistent with the message in the data symbols make the decoding of the message more effective.

[193] This plot is based on a plot in the November 15-21 edition of the New Zealand Listener.

[194] The text of this book no doubt provides multiple examples in support of this claim.

[195] The plots in this case study, and the idea of *linear* pairwise plots, come from version 1.0.1 of the {bullseye} package for R (Chinwan and Hurley 2025).

[196] For example, the New Zealand Government Web Accessibility Standard ("Web Accessibility Standard 1.2" 2025) mandates conformance with the W3C Web Content Accessibility Guidelines (World Wide Web Consortium (W3C) 2025).

[197] There are many guidelines to assist with generating useful alt text, from the very practical (World Wide Web Consortium (W3C) Web Accessibility Initiative (WAI) 2025) to the more theoretical (A. L. A. A. Satyanarayan 2022).

14 Encoding Structure

Figure 14.1 shows a line plot of the rate of youth crime in New Zealand for different ethnic groups and a pie chart showing the proportions of different types of crime.[198] There is a problem with this figure because the line plot and the pie chart share the same colour palette even though they are visualising different variables (ethnicity and crime type). For example, the ethnic group European/Other is encoded as the orange colour of a line *and* the crime type Theft is also encoded as an orange colour, in this case the colour of a wedge.

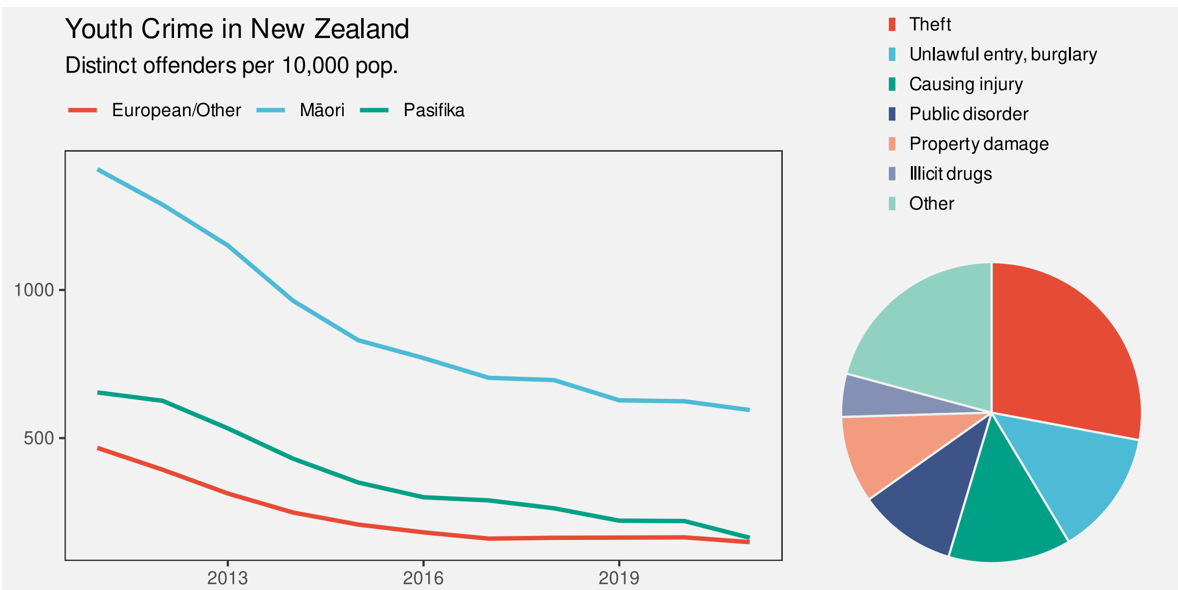


Figure 14.1

The conflicting encodings in Figure 14.1 create a confusing data visualisation because the same visual feature decodes to multiple data values (Figure 14.2). There is nothing wrong with the encodings in the line plot by itself or with the encodings in the pie chart by itself, but the overall combination of elements within Figure 14.1 is ineffective.

The problem in Figure 14.1 involves the coordination, or lack thereof, between multiple elements of a data visualisation. In this chapter we look at the arrangement and coordination of the elements of a data visualisation.[199]

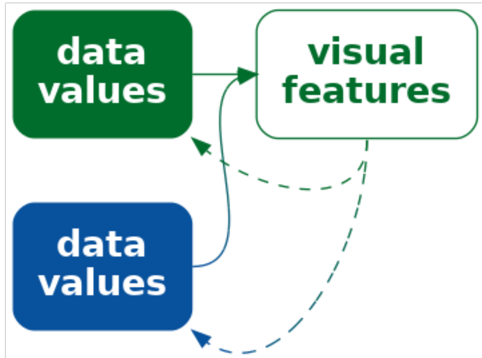


Figure 14.2: If we encode more than one data value to the same visual feature, it is unclear which is the correct decoding.

14.1 Encoding organisation

Figure 14.3 shows a scatter plot of the number of clean breaks and the number of tries for countries at the 2023 Rugby World Cup (Table 9.1). This data visualisation is similar to Figure 9.4, but with an additional encoding of the hemisphere for each country as the colour of the data points.

The legend in Figure 14.3 encodes the colour encoding: which colour is used to encode each hemisphere. In Section 13.7, we described how the proximity of the labels “South” and “North” to the coloured points in the legend allow us to associate the colours with the different hemispheres. The different values of hemisphere are encoded as the vertical positions of the points and the text labels; both the orange point and the label “South” have the same vertical position.

What about the horizontal position of the points and the text in the legend? These do not encode hemisphere values because they are the same for both North and South. What about the position of the legend title, “hemisphere”?

All of the elements of the legend are positioned together and to the right of the main plot. Their proximity to each other (and separation from the main plot) mean that we see the legend as a separate group (Section 2.7). This creates a visual organisation for the plot, which helps the viewer to navigate and comprehend the different elements of the plot.[200] This is analogous to reading a body of text, which is split into paragraphs and sections with headings to help the reader navigate the text.

One way to express this is that the **position** of the elements of the legend **encode** the **organisation** of the overall data visualisation (Figure 14.4). Our visual system **decodes** the legend as a separate element based on the proximity of the text and points in the legend.

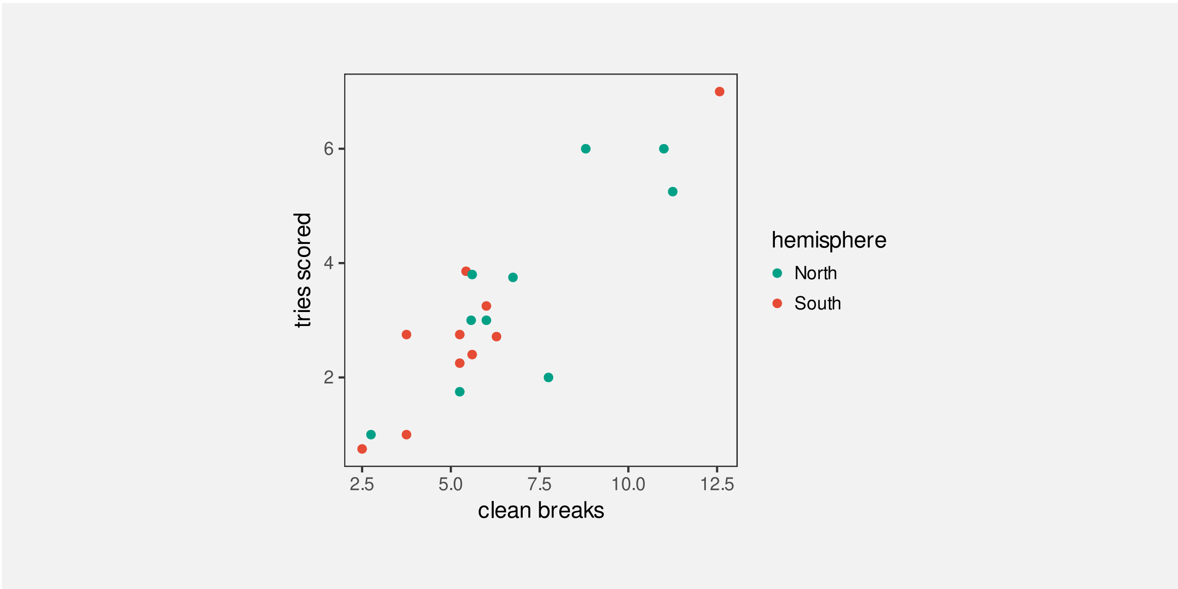


Figure 14.3: A scatter plot of the number of times a team breaks through the opposition defence and the number of tries that a team scores (both are per-game averages) for teams at the 2023 Rugby World Cup.

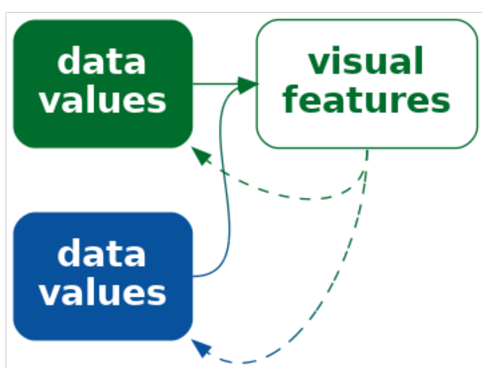


Figure 14.4: We can encode the structure of a data visualisation as position so that elements that are part of the same structure are close to each other.

> One way in which a data visualisation can be effective is if it provides information about how to read the plot as well as information about the data itself.

Another way to say it is that each element of the data visualisation in Figure 14.3 belongs to one of two groups: the main plot or the legend. There is a **qualitative** value—plot or legend—associated with each element of the data visualisation. These qualitative values are encoded as the **position** of the elements within the data visualisation: the plot is on the left and the legend is on the right.

Although we are no longer talking about encoding data values as data symbols, we are still talking about encoding. We are just encoding and decoding a different sort of information—the organisation of a data visualisation.

Figure 14.5 shows a variation of Figure 14.3, with exactly the same plot elements, but in a different arrangement. There is greater separation of the legend from the main plot. The legend has also been moved so that its top is aligned with the top of the main plot. Similarly, the axis titles have been moved so that they align with the top and right sides of the plot, and the y-axis label has been rotated to horizontal. More subtly, the right edge of the y-axis label is aligned with the right edge of the y-axis tick labels and the points in the legend are aligned with the left edge of the legend title.

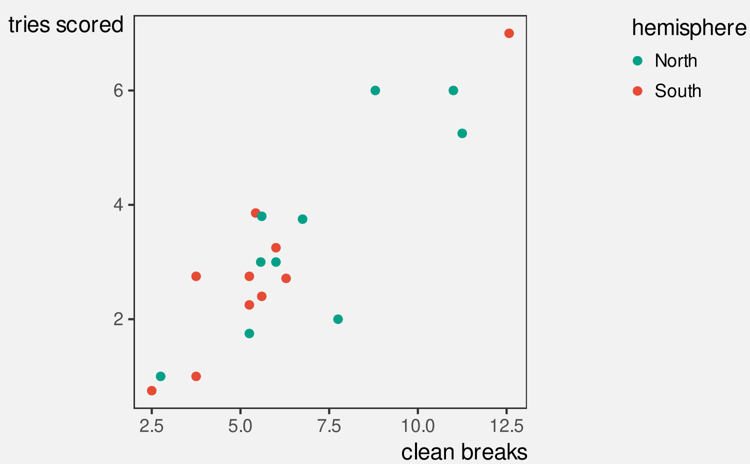


Figure 14.5: A scatter plot of the number of times a team breaks through the opposition defence and the number of tries that a team scores (both are per-game averages) for teams at the 2023 Rugby World Cup.

All of these changes are aimed at making the data visualisation more orderly (Section 2.8).[201] The legend is more obviously a separate element from the main plot and the axis titles more obviously connect to the main plot. A more orderly arrangement of the visual elements will provide fewer distractions and lead to a more effective data visualisation because the important differences in the data will be more readily perceived (Section 2.5).

The strong alignment and regular patterns within facetted plots like Figure 11.4 are another effective example of keeping everything else the same in order to champion the changes in the data. Conversely, another weakness of placing data symbols on maps, like in Figure 12.16b, is that the lack of regular positioning and alignment makes it harder to attend to the important differences between data symbols.

14.2 Consistent encodings

Figure 14.6 shows a variation on Figure 14.3 with a modification to the legend. In this plot, the hemisphere of a country is encoded using orange and green within the main plot, but then hemisphere is encoded using light blue and brown in the legend. This is clearly an ineffective data visualisation. It is possible to decode that there are two groups of data symbols in the plot, but it is impossible to decode which of those two groups is North and which is South.[^inconsisten-dissonant] The point of Figure 14.6 is merely to highlight the critical importance of maintaining consistent encodings between different elements of a plot.[202]

Figure 14.7 demonstrates that we can harness consistent encodings to actually improve upon the original Figure 14.3. In Figure 14.7, the encoding of hemisphere as the colour of points is not only consistent between the main plot and the legend, but it has been extended to the colour of the legend text as well. The visual grouping of elements that relate to the same hemisphere is now stronger both within the legend and across the entire data visualisation (Section 2.7).

Another consistency in Figure 14.7 (and the original Figure 14.3) is the use of the same data symbol in both the main plot and the legend. Circular data symbols are used in both cases. That consistency, like the use of consistent colours, might again appear obvious, but it is not always true. For example, the data symbols in Figure 13.22 are circles, but the legend is a rectangular colour gradient.

Figure 14.8 shows a variation of Figure 13.22 with a legend that not only shares the same colour encoding, but also shares the same data symbol. It is much easier to compare the colour of one of the circular data symbols with the colours of the circles in the legend because they are all circles.

Figures Figure 6.16 and Figure 6.17 are other examples of this consistent encoding between legends and the main plot. The legends in those cases are drawn so that they mirror the *positions* of the bars in the main plot as well as the colours of the bars in the main plot.

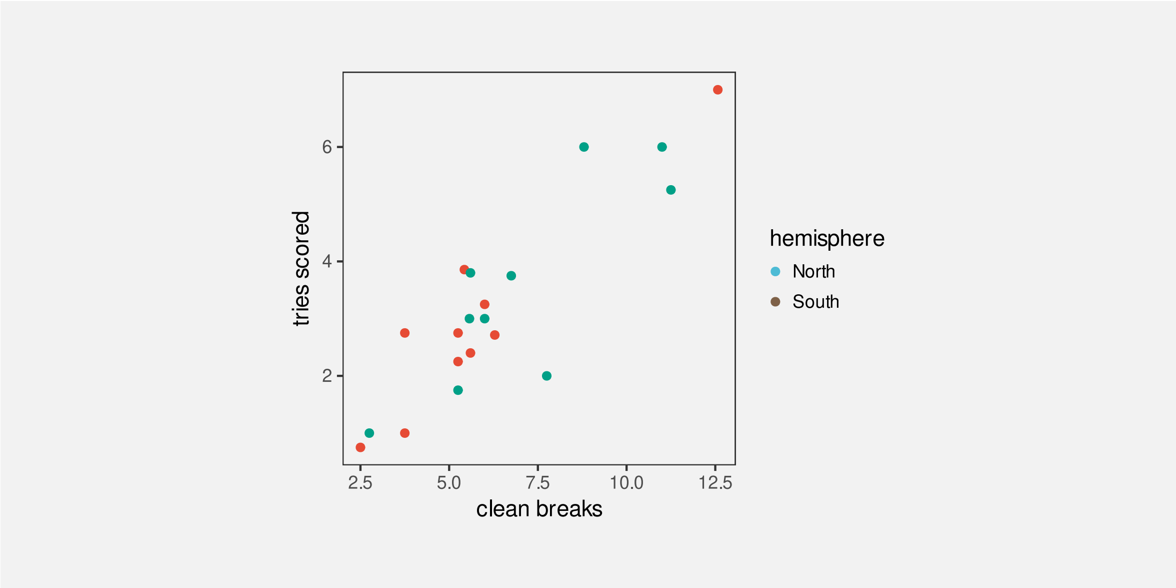


Figure 14.6: A scatter plot of the number of times a team breaks through the opposition defence and the number of tries that a team scores (both are per-game averages) for teams at the 2023 Rugby World Cup.

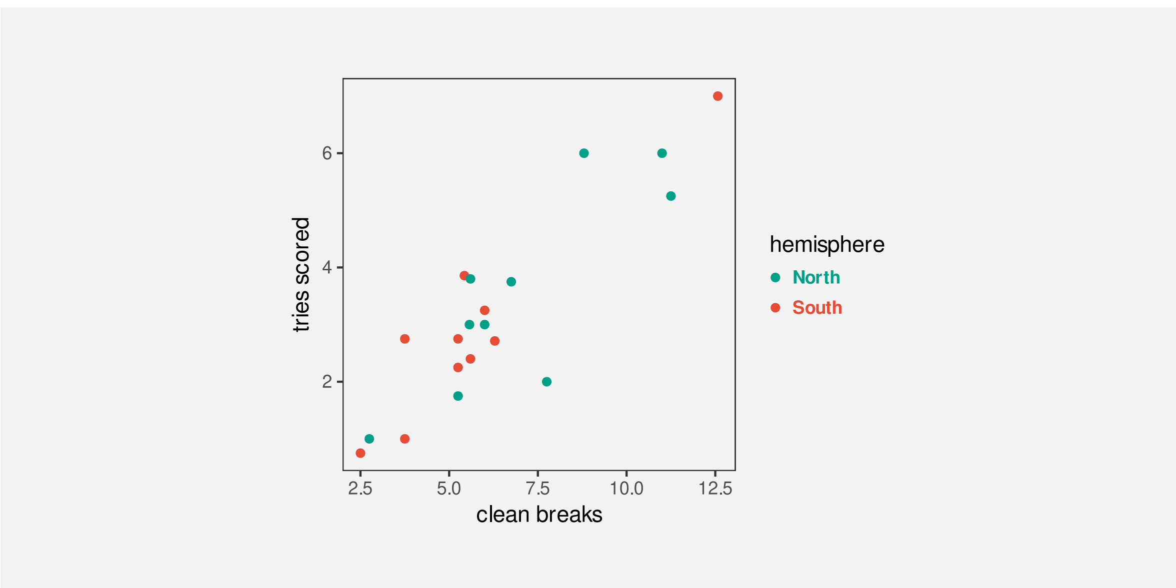


Figure 14.7: A scatter plot of the number of times a team breaks through the opposition defence and the number of tries that a team scores (both are per-game averages) for teams at the 2023 Rugby World Cup.

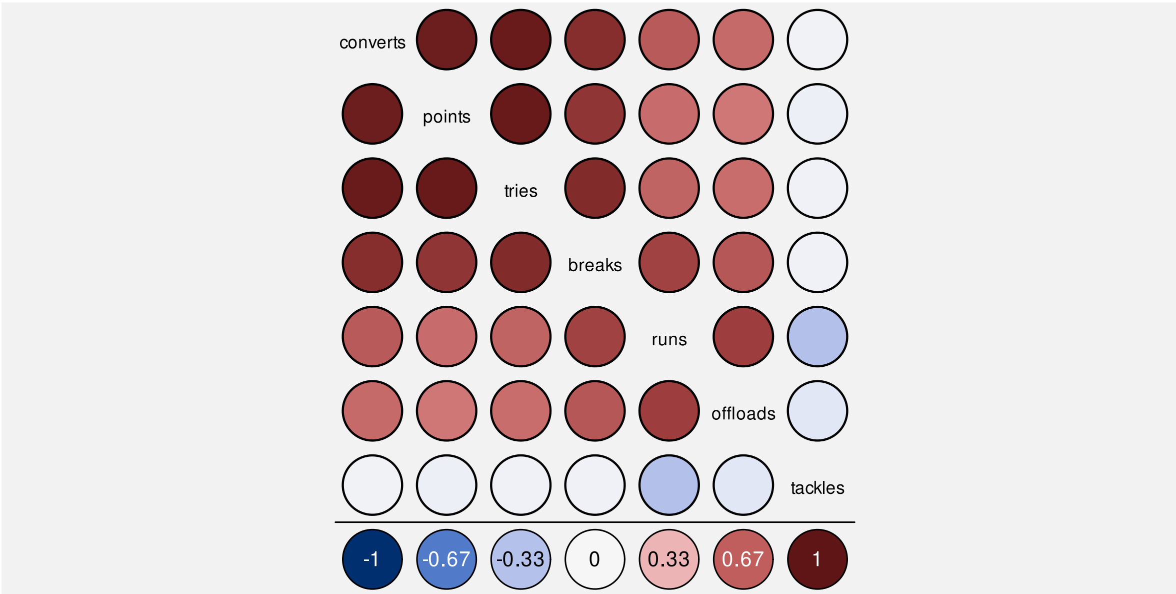


Figure 14.8: Correlations between pairs of measures of performance for teams in the 2023 Rugby World Cup.

14.3 Dangers of inconsistent encodings

Figure 14.9 shows a bar plot of the impact of a proposed tax bill during the second Trump administration.[203] Each bar represents the predicted change in annual income for a particular income bracket. Lower incomes are to the left and higher incomes are to the right. There are labels to describe each income bracket and there are labels that describe each change in income. Where the change in income is large, the label for change in income is written within the bar (at the top of the bar), but for smaller changes in income, the bar is too small to fit the label, so the label is placed just above the bar. For the two leftmost bars, the change in income is actually negative and the change in income label is placed *below* the bar. The labels that describe the income brackets are mostly below the bars, but for the two leftmost bars the labels for the income brackets are placed above the bars.

In summary, there are two labels for each bar, one below the bar and one above (or at the top of) the bar. This consistent positioning of the labels encourages the viewer to decode two groups of labels: one group above the bars and one group below the bars. However, this will mislead us into comparing, for example, “-1K” (a change in income) for the leftmost bar with “4.3M” (an income bracket) for the rightmost bar. The correct comparison is between “-1K”, which is the change in for the lowest income bracket, and “389.3K”, which is the change in income for the highest income bracket.[204]

Just as it makes no sense to change the encoding of data values within a data visualisation (Figure 14.6) or across multiple data visualisations within the same document (Figure 14.1),

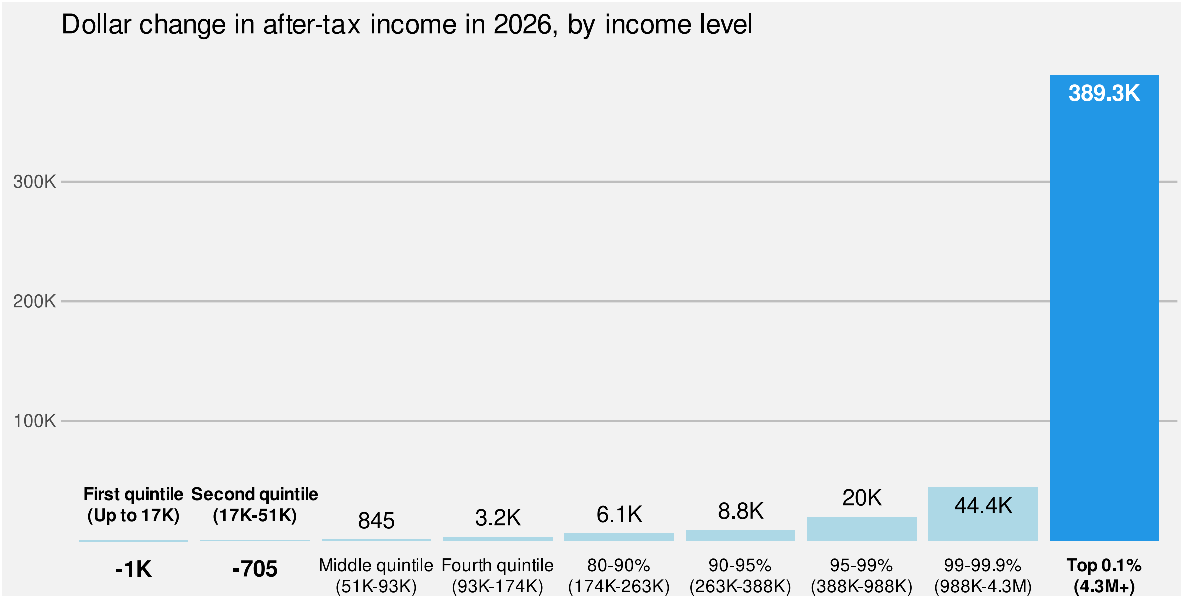


Figure 14.9: A bar plot of the distributional effects of Trump's "Big, Beautiful Bill".

it makes no sense to change the encoding of *structure* within a data visualisation. In this case, all labels above the bars should belong to the same group of labels (changes in income) and all labels below the bars should belong to the same group of labels (income brackets).

14.4 Encoding importance

Figure 14.10 shows a variation on Figure 14.7 with many elements of the plot drawn in light grey. The effect of this change is to create two visual groups, one lighter and less saturated than the other. This emphasises the main data symbols in the main plot and in the legend, and de-emphasises the axes and labels. Our attention is drawn to the darker and more saturated elements first (Section 2.6), with the other elements in the background, which we can focus on if we need the additional information.[205]

By comparison, in Figure 14.7 there is a similar visual impact from all elements of the plot, so it is less clear where to look first. All of the elements of the data visualisation compete for our attention. Figure 14.11 emphasises this point by making all axis and legend text black and adding black grid lines. The data symbols are now lost amongst the non-data elements of the plot.[206]

The use of colour in Figure 14.10 is another example of encoding (qualitative) non-data values as visual features. In this case, we are encoding the **importance** of different visual elements as visual features (Figure 14.12). We decode the differences in colour to a more important group of elements and a less important group of elements.

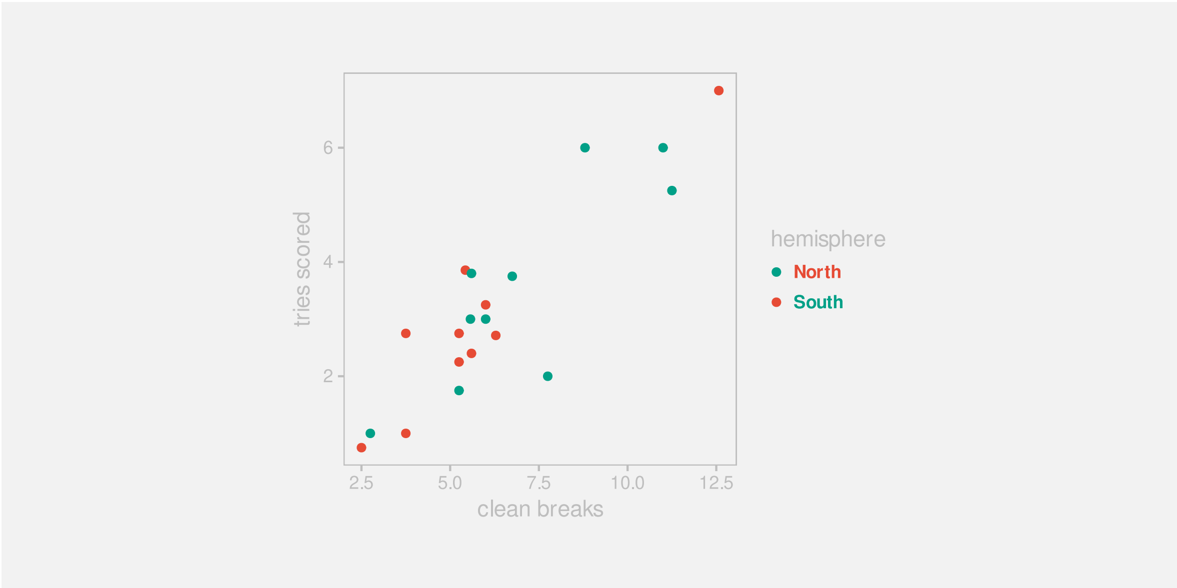


Figure 14.10: A scatter plot of the number of times a team breaks through the opposition defence and the number of tries that a team scores (both are per-game averages) for teams at the 2023 Rugby World Cup.

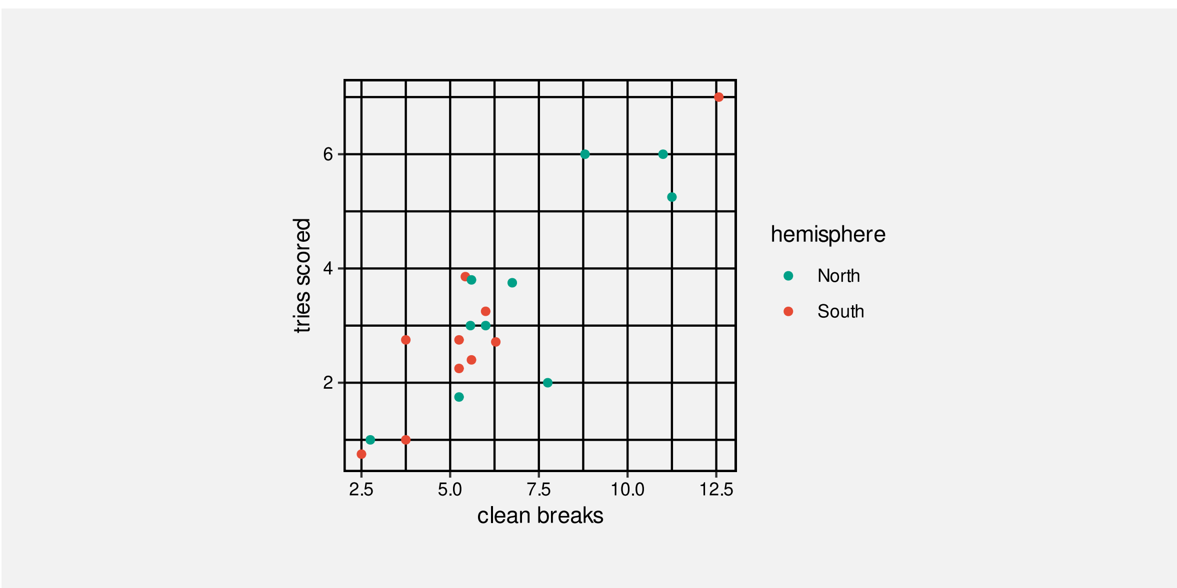


Figure 14.11: A scatter plot of the number of times a team breaks through the opposition defence and the number of tries that a team scores (both are per-game averages) for teams at the 2023 Rugby World Cup.

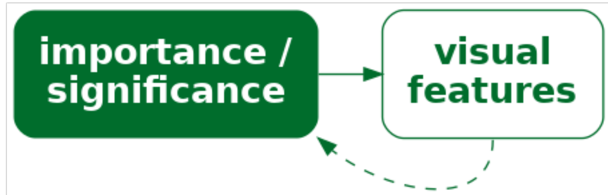


Figure 14.12: We can encode the importance of elements within a data visualisation as visual features so that elements that are more important are more visible.

Figure 14.13 shows a similar application of importance. In this case, we are highlighting two data points (New Zealand and South Africa). The distinct colour and larger size of the two points encodes the importance of those points and draws attention to those points, with everything else as background context.

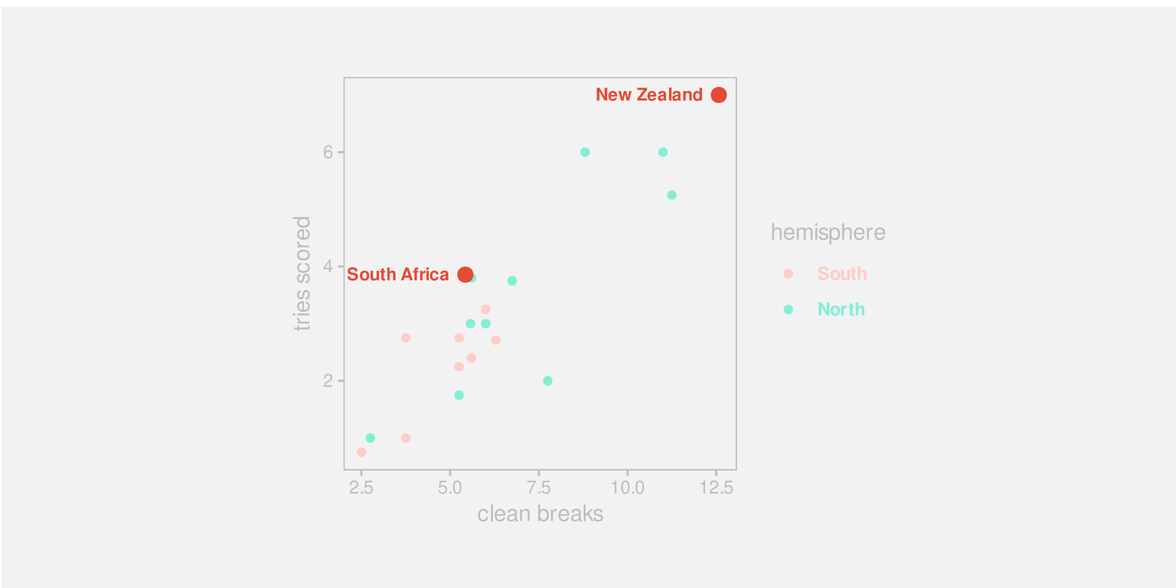


Figure 14.13: A scatter plot of the number of times a team breaks through the opposition defence and the number of tries that a team scores (both are per-game averages) for teams at the 2023 Rugby World Cup.

14.5 Dangers of encoding importance

When we encode importance, we are relying on basic properties of the visual system to draw attention to specific elements of a plot (Section 2.6). As we saw in Section 2.10, those basic properties of the visual system can also work against us. Our attention will be drawn to large and bright elements of a data visualisation automatically.

For example, in Figure 14.14, our eye is drawn to the tiger's eye, which distracts us from the more important heights of the bars. There are clear overlaps here with the ideas of clutter and chart junk (Section 12.6).

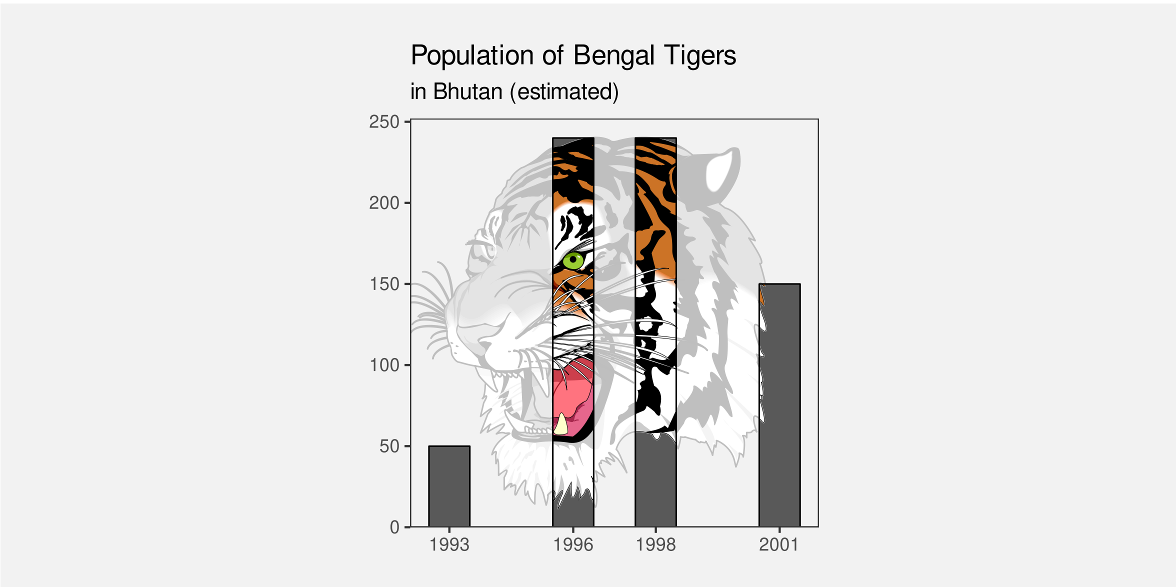


Figure 14.14: A bar plot of the estimated population of Bengal Tigers in Bhutan. The background image in this plot distracts attention from the most important data symbols.

14.6 Summary

Just as we can encode data values to visual features, in order to communicate information about the data, we can encode non-data information to visual features, in order to communicate important aspects of the structure of a data visualisation.

The structure or organisation of a data visualisation, or the importance of specific elements, can be effectively encoded as the position or colour of different elements of the data visualisation.

Communicating structure and importance helps the viewer to navigate within the data visualusation, which makes it easier to decode information from the data symbols within a data visualisation.

Notes

[198] This data visualisation was based on figures from the Youth Justice Indicators Report from 2021.

[199] We might describe the topic of this chapter as the overall *design* of a data visualisation. Much of what we discuss will have a basis in the properties of the visual system, but much will also echo the basic design principles espoused in Robin Willams' CRAP design principles (Williams 2014).

[200] Cabouat et al. (2025) report that "if a learner found a visualization more readable, they felt it required less mental effort to parse relevant information from it for learning."

[201] *Alignment* is also one of the CRAP design principles (Williams 2014).

[202] A consistent encoding aligns with the CRAP design principle of *repetition* (Williams 2014).

[203] This data visualisation is based on a plot from the Popular Information web site by Judd Legum. The data are from the Penn Wharton Budget Model.

[204] This is precisely the mistake that was made when this data visualisation was discussed on The Majority Report (~6:00).

[205] The idea of making a clear visual distinction between different elements of a data visualisation mirrors the CRAP design principle of *contrast*.

[206] This also relates to Tukey's *data-ink ratio* (Tufte 1983).

15 Review

In this book, we have focused on the different ways that information can be encoded to create a data visualisaiton and how well information can be decoded from a data visualisation.

There are two main ways that we can apply this knowledge: we can use it to evaluate the effectiveness of an existing data visualisation; or we can use it to design a new data visualisation. Either way, the first questions we should ask are:

- What information do we want to decode from the data visualisation? *and*
- Is that information present in the data (either as raw data values or data summaries)?

We can only hope to decode information from a data visualisation if we have the appropriate data.

Assuming that we have the relevant data, the next questions are:

- What are the data symbols?
- What is being encoded (raw data values and/or data summaries)? *and*
- What encodings are being used? *and*
- How many encodings are there?

For each data value or summary and its encoding, we must then ask:

- Can we decode quantitative data values and is the decoding accurate? *or*
- Can we decode qualitative data values and is there enough capacity?
- Is the encoding congruent? *or*
- Is the encoding dissonant?
- Can we decode raw data values? *or*
- Can we decode data summaries?

- Do encodings interact to create emergent features?
- Does each data value or summary have its own data symbol? *or*
- Are multiple data values or summaries encoded as a single data symbol?
- Do the encodings combine to create recognisable visual objects?

Looking beyond the data symbols, we should also identify all text elements and consider the overall design:

- Can we decode absolute data values or data summaries from the text?
- Can we decode the overall context from the text?
- Can we decode the main message of the data visualisation from the text?
- Can we decode the structure of the data visualisation?
- Are the encodings consistent?
- Is our attention drawn to the most important elements?

Overall, the goal of a data visualisation is to take advantage of the visual system to rapidly and effortlessly communicate data values and data summaries:

- What are you asking the visual system to do? *and*
- Is the visual system good at that task?
- Do all visual differences correspond to differences in the data? *and*
- Are all differences in the data represented by visual differences?

15.1 Case study: A letter-value plot

Figure 15.1 shows a letter-value plot of taxi time for over 450,000 flights within the United States, broken down by airline.[207] The letter-value plot is an example of a novel data visualisation, so this provides a good opportunity to analyse whether this data visualisation represents an effective encoding of information.[208]

The purpose of a letter-value plot is to visualise the distribution of a set of data values. The information that we want to decode is about the distribution of the data values: the average value, the spread of the values, the skewness, and so on.

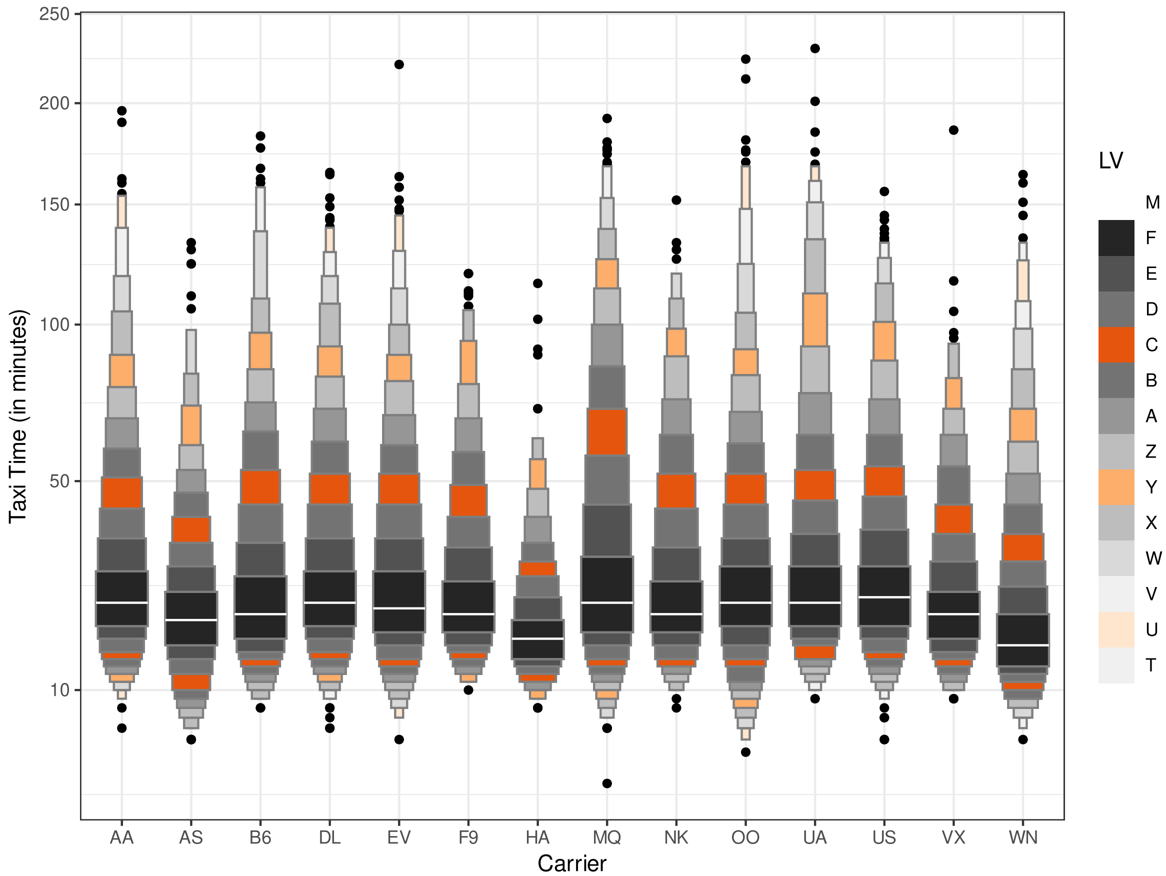


Figure 15.1: A letter-value plot of taxi times for over 450,000 flights within the United States.

The main motivation for the letter-value plot design is that a normal boxplot produces far too many outliers for cases like this that involve such a large amount of data, as demonstrated in Figure 15.2. Furthermore, alternatives to the boxplot, like violin plots, that fix the problem of too many outliers, are reliant on tuning parameters, like kernels and bandwidths (Section 10.3).

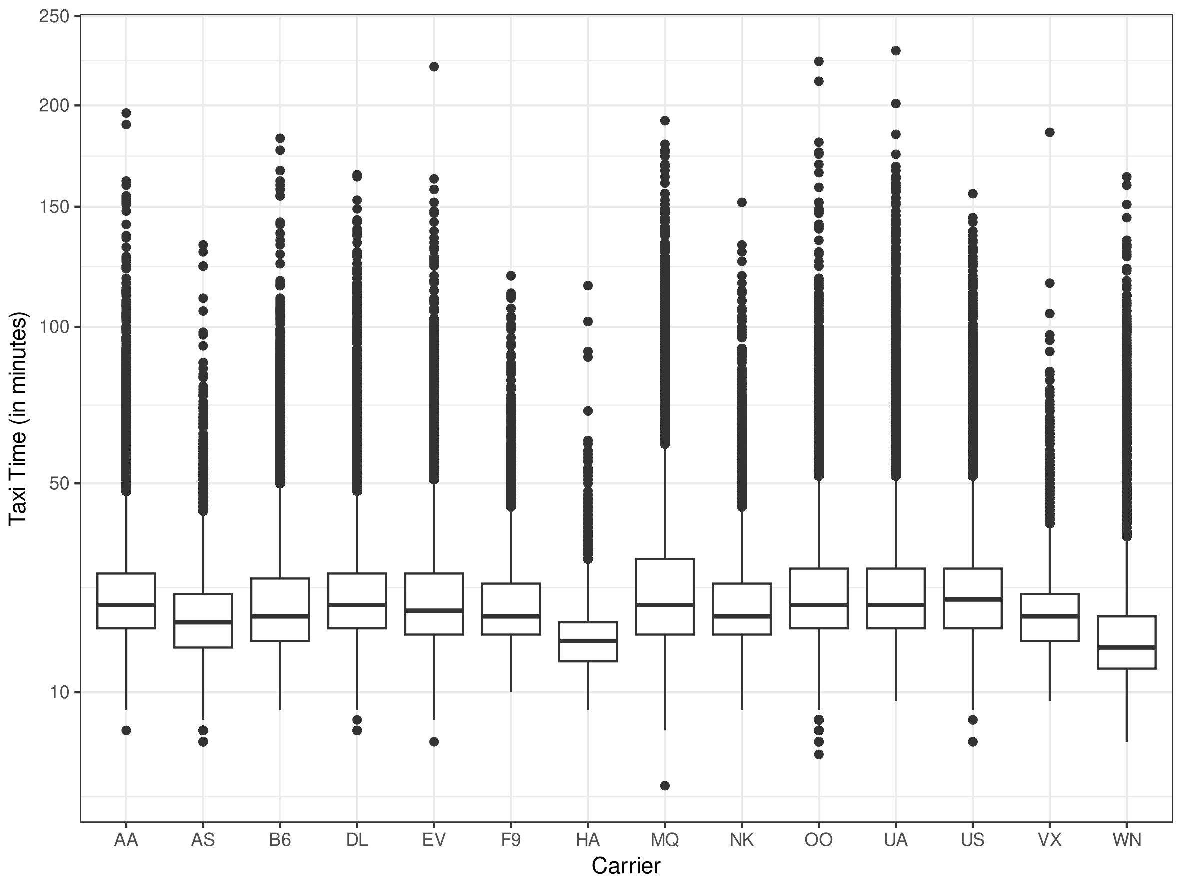


Figure 15.2: A box plot of taxi times for over 450,000 flights within the United States.

The letter-value plot aims to fix the problem of outliers, while at the same time, like the original boxplot, remain free of tuning parameters. This is achieved by displaying as many letter values as are supported by the amount of data, with only individual data values beyond those limits shown as individual data points.

15.1.1 Letter values

The well-known boxplot is based on the median and the upper and lower quartiles of a data set, but these data summaries are just the first two levels of a more general concept of *letter* *values*.[209]

Letter values are order statistics, where the ith letter value approximates the quantile corresponding to a tail area that contains 2^{-i} of the observations in the data set. The first letter value is the median, which splits the data into halves (2^{-1}), the second letter values are the upper and lower quartiles, which leave a quarter of the data in each tail (2^{-2}), and so on. The amount of data between letter value i and letter value $(i + 1)$ is $2^{-(i+1)}$. For example, the amount of data between the upper quartile (letter value 2) and letter value 3 is an eighth (2^{-3}).

The letter values are named M for median, F for fourths, E for eighths, then, following the alphabet in reverse, D for sixteenths, C for thirty-seconds, down to A for one-hundred and twenth-eighths, then starting again at the end of the alphabet, in reverse alphabetic order, Z for two-hundred and fifty-sixths, Y for five-hundred and twelfths, and so on.

15.1.2 Tearing down

We will use the framework that we have developed in this book to analyse the data visualisation in Figure 15.1. We will break Figure 15.1 down into the encodings that are being used and evaluate the effectiveness of those encodings.

Data symbols There are two types of data symbol in a letter-value plot: complex data symbols consisting of multiple rectangles, which we will call *letter-value* data symbols, to represent the distribution of the data values; and very simple data points, which represent the outliers (if any).

Encoding visual features In Figure 15.1, there is a separate letter-value symbol for each airline, and the airlines (Carrier), which are qualitative data values, are encoded as the horizontal **position** of the letter-value symbols. The encoding of airline as the position of data symbols is very effective; we can easily differentiate between the airlines.

The taxi times, are encoded as the letter-value data symbols within the plot. However, the letter-value plot encodes **data summaries**—letter values—rather than raw data values (Chapter 8).[210] Consequently, all that we can expect to decode from the plot is information about the letter values, such as the location of specific letter values, plus possibly data summaries of letter values, such as the spread of letter values.

The number of letter values that are displayed (before resorting to showing individual points for outliers) is based on the amount of data values, so there are fewer rectangles for airlines with fewer tax times. For example, airline HA has fewer taxi times than airline AA, so ten letter values are displayed for airline HA, but thirteen letter values are displayed for airline AA.[211]

Each letter value is encoded as the vertical **position** of the top (or bottom) of a rectangle within the letter-value symbol. The median (letter value 1) is marked as a single white line rather than a rectangle. This is effective for decoding individual letter values for an airline. For example, we can see that one of the letter values for the AA airline is slightly greater than 50.

The *depth* of each letter value—how many data values lie beyond that letter value—is encoded as the width (**length**) of a rectangle within the letter-value symbol. For example, the depth of the median is half the total number of data values. This is also an encoding of a data summary. However, the actual encoding is only based on the *order* of the depth of the letter values, with regular decrements from widest at the median (letter value 1) to narrowest at the final letter value (e.g., letter value 13 for airline AA). In effect, we are encoding a summary of a data summary. Using **length** for the encoding is effective, but we can only decode ordinal information about the depth of the letter values because that is all that we have encoded. For example, the width of the rectangle that encodes the depth of the fifth letter value (the bright orange rectangle) for the AA airline is just the fourth-widest rectangle. In particular, that rectangle is *not* proportional to the actual depth of the fifth letter value. There is approximately one thirty-second of the data values beyond the fifth letter value, so this rectangle would have to be sixteen times narrower than the median for the widths to be proportional to the depth of the letter values.

The *data density*—the number of data values between adjacent letter values—is encoded as the **colour** of a rectangle. The **luminance** of the rectangle varies from darkest for the highest density (approximately one quarter of the data values lie between the median and the upper quartile) to lightest for lowest density (approximately one ten-thousandth of the data values between the most extreme letter values). In addition, the **hue/chroma** of a rectangle encodes a categorisation of the letter values (yet another data summary): every fourth letter value is orange, while all the other letter values are grey.

The luminance encoding of data density only allows us to decode ordinal information about the data density between letter values. Like the widths of the rectangles, the shades of the rectangles steadily decrease upwards from the median and downwards from the median. We can also identify rectangles that represent the same letter value for different airlines by finding rectangles with the same shade in the letter-value symbols for two airlines. This is less effective due to the relatively limited capacity of luminance encoding (Section 6.2). For example, the different shades of grey are much more difficult to identify and compare between different airlines than, for example, the different horizontal locations for different airlines.

The chroma encoding of every fourth letter value is very effective for decoding between those letter values and all of the others. This is essentially an encoding of two groups (orange versus grey). It is also very effective for identifying the same letter value across different airlines. The difference in chroma between the oranges and the greys makes the oranges easy to identify and there are only three different luminances for the orange rectangles, so they are more easily differentiated. For example, it is easy to identify the rectangle for the fifth letter value for all airlines; these are the brightest orange rectangles.

The outliers in a letter-value plot are encoded differently. The individual taxi times for outliers are encoded as the vertical position of data points, with the airline encoded as the horizontal position, as for the letter-value symbols. These allow for very simple and effective decodings. It is even possible to rapidly decode the number of outliers for each airline thanks to the small number of data symbols involved (Section 8.2).

Table 15.1 summarises the encodings that are involved in a letter-value plot and what we can decode from them. There are several effective encodings for both quantitative and qualitative data values, but the encoding of data density and the encoding of letter-value depth both lose some information. Those are quantitative data summaries that have only ordinal encodings. One overall point is that this is quite a large number of encodings for a single data visualisation. This means that the encodings in Figure 15.1 produce a relatively complex data visualisation. There are many visual features changing, which makes each individual encoding in Figure 15.1 more difficult to decode (Section 2.5). Compare this with Figure 15.2 which, for all its faults, is a much simpler data visualisation because it involves fewer changes in visual features.

In defence of Figure 15.1, the encoding and decoding of individual data values and data summaries that we have described, while potentially useful, are not the primary aim of the data visualisation. Any weaknesses of these encodings are less important than how well we can decode the *distribution* of the data values, which we will come to later.

Table 15.1: The encodings involved in a letter-value plot.

Value	Raw/Summary	Type	Encoding	Decoding
airline	raw data value	qualitative	position (horiz)	airline
taxi time outlier	raw data value	quantitative	position (vert)	taxi time
taxi letter value	summary of taxi time	quantitative	position (vert)	taxi letter value
letter value depth	summary of letter value	qualitative	length (width)	letter value depth order
taxi density	summary of taxi time	quantitative	colour (luminance)	taxi density order
fourth letter value	summary of letter value	qualitative	colour (chroma)	fourth letter value

Combining encodings There is one obvious interaction between the encodings in Figure 15.1. The width of the rectangles (letter-value depth order) and the height of the rectangles (difference between adjacent letter values) together produce the emergent feature of **area** for the rectangles.

Unfortunately, this area does not correspond to any data value or summary of the data. Letter value depth order multiplied by difference between letter values does not give a meaningful value. This makes Figure 15.1 confusing because the rectangles areas do not decode to anything useful (Section 9.7).[212]

Furthermore, the **implicit decoding** of the areas of the rectangles in the letter-value data symbols suggests that, for example, the (darkest) rectangle just above the mean is about five times the size of the highest (lightest) rectangle for the "AA" airline. However, the proportion of the data that lies between the median and the next letter value is actually over sixteen thousand times more than the proportion of the data that lies between the top two letter values. In other words, the area of the rectangles is actually *misleading*.

There is also a **redundant encoding** in Figure 15.1 (Section 9.8). Because the order of a letter value is related to the data density order, both the luminance of each rectangle in the letter-value symbol and the width of each rectangle decode to essentially the same information. For example, the bright orange rectangles, which correspond to the data density of the fifth letter value, all have exactly the same width. This has the benefit of providing more than one way to decode information. However, given that we have already identified that there are a large number of encodings in Figure 15.1, this may be a luxury that we can ill afford.

Non-linear scales An extra complication with the encoding of vertical position in Figure 15.1 is that the vertical scale is non-linear (Section 4.7); it is actually the *square root* of the letter values that are encoded as the vertical position of the tops (or bottoms) of the rectangles in the letter-value data symbols. This makes the decoding of letter values and the comparison of letter values much more difficult because the same physical difference does not decode to the same difference between letter values. For example, for the AA airline, although most of the letter values appear to be similar distances apart, in terms of taxi data values, the letter values are in reality further apart for more extreme letter values.

We have already established that the areas of the rectangles in the letter-value data symbols provide no useful information and are potentially misleading. The non-linear scale only makes that potentially misleading information even more misleading.

Visual shapes The letter-value data symbols encode multiple letter values within a single data symbol, which produces a **visual shape** (Chapter 10). We have a data symbol that is widest at the median of the data values and narrows towards either extreme.

This allows us to decode useful data summaries about the distribution of the taxi times. For example, the fact that all of the letter-value data symbols narrow more rapidly below the mean than they do above the mean suggests a skewed distribution of taxi times for all airlines (or at least a skewed distribution of the square root of taxi times).

It is also possible to identify airlines that have either similar or different distributions based on similar or different visual shapes. For example, the letter-value data symbol for the HA airline is clearly shorter than the other data symbols, which tells us that the range of taxi times is smaller for the HA airline.

These are examples of effective decodings of data summaries from the visual shapes that are produced by the letter-value data symbols. This effectiveness is important because decoding information about the distribution of taxi times is the main purpose of a letter-value plot.

One caveat is that the changes in the widths of the visual shapes are potentially misleading because they do not narrow anywhere near as fast as the density of the taxi times. In other words, the shape of the letter-value data symbols does not provide an accurate decoding of quantitative features of the distribution of taxi times (actual rate of change), only broad qualitative features (faster or slower change). For example, we can decode *whether* the distribution is symmetric or skewed, but we cannot decode *how* skewed the distribution is.

Encoding encodings We have already discussed that the y-axis scale is non-linear. The square root of taxi times, rather then the raw taxi times, are encoded as the vertical position of the tops (or bottoms) of rectangles. This creates the usual problems when it comes to encoding this non-linear scale as tick marks and text labels. For example, the square-root taxi time of 10 is encoded as a position about two-thirds of the way up the y-axis, which accurately reflects the fact that 10 is approximately two-thirds of the range of square-root taxi times. However, this value of 10 is also encoded as the text “100”.

On one hand, this is a valuable encoding of the value 10 because it saves us the mental effort of having to reverse the square root transformation. If we had encoded this position as the text “10”, then we would be required to perform the mental calculation to decode the raw taxi time of 100.

On the other hand, encoding the square root taxi times as text that corresponds to the original taxi times causes problems because we cannot linearly interpolate between the text values. We can correctly decode the precise taxi time 100 from the text “100”, and 50 from “50”, but it is very difficult to decode any taxi times in between.

The x-axis is more straightforward because it reflects a simpler encoding. Each airline is encoded as the horizontal position of a letter-value data symbol. On the x-axis, we encode each airline as the horizontal position of a tick mark and a text label. A vertical grid line connecting the letter-value data symbols and the tick marks helps to form a strong visual grouping of data symbols with their respective labels (Section 2.7).

However, the encoding of airlines to text is missing a learned encoding (Section 13.3), at least for infrequent travellers internal to the United States of America. For example, not all viewers will immediately decode the text “MQ” to the name “Envoy Air”. In effect, the x-axis tick labels only involve an encoding of airline as the **pattern** of the text. We can only tell that there are different airlines because they use different combinations of letters.

The legend on the right of Figure 15.1 provides information about the colour encoding of the data density between letter values. The data density is encoded as the luminance of a stack of squares, with every fourth square having a different chroma (orange). In addition, the letter values are encoded as individual text characters alongside the squares.

One problem is that there is no learned decoding for the letter values. For example, a letter value of "Z" is not immediately decoded as a meaningful data value. Not all viewers will immediately decode "Z" to a tail area of 2^{-8}. As with the x-axis, the encoding only provides us with a different text pattern for each letter value. We can identify that the letter values are different from each other, but little more.

In particular, there is no way to decode the absolute data density from the legend. The luminance of the squares only gives an indication of the data density order thanks to the implicit decoding of darker to more (darker squares are higher density; Section 6.2)

Another minor problem is that the letter values are placed alongside each rectangle, but they really correspond to the boundaries between the rectangles in the letter-value data symbols. For example, the true position of letter value "F" is at the top (or bottom) of the black rectangle in each letter-value data symbol.

Finally, there is no explanation of the width of the rectangles in the letter-value data symbols. There is no legend that explains how the depth order of the letter values is encoded as the width of the letter-value data symbols.

Encoding structure Another problem with the legend in Figure 15.1 is that the squares within the legend are not consistent with the rectangles in the letter-value data symbols (Section 14.2). Specifically, the rectangles in the letter-value data symbols have a grey border, but the squares in the legend have no border. This also makes it harder to match the colours in the legend with the colours in letter-value data symbols because the surrounding colours are not the same (Section 6.3).

A final small inconsistency in the legend is that the luminance of the squares in the legend decrease from top to bottom, while the luminance of the rectangles in the letter-value data symbols decreases from the middle both up and down, with the clearest decrease upwards, from middle to top.

There are also several horizontal grid lines in Figure 15.1. While these can help with decoding individual letter values that lie exactly on a grid line, thanks to the non-linear y-scale, decoding letter values relative to the grid lines is restricted to ordinal comparisons. The large number of grid lines also adds to the overall visual complexity of Figure 15.1 that we discussed earlier.

15.1.3 Building up

Having used the framework in this book to identify a number of problems with the encodings in Figure 15.1, we now attempt to use the information to make improvements to Figure 15.1.

Figure 15.3 shows a data visualisation that is based on the same taxi time data as Figure 15.1, but uses a different set of encodings. One solution to the complexity of Figure 15.1 would be to avoid multiple encodings by producing several, simpler data visualisations for different purposes. For example, one data visualisation that just shows the distribution shapes and another that shows the location of specific letter-value taxi times. However, Figure 15.3 still attempts to fit all of the information from Figure 15.1 into a single data visualisation.

Figure 15.3 does not aim to be an optimal solution, but rather a demonstration that we can use our framework to suggest changes based on deliberate reasoning and that we can provide justifications for those changes.

Data symbols One issue that we identified with Figure 15.1 was the high complexity of the letter-value data symbols, which arises from having multiple encodings within a single data symbol. One option is to have more data symbols, each with fewer encodings.

There are *three* types of data symbols in Figure 15.3. As in Figure 15.1, there are data points to represent outliers and there is a letter-value data symbol to represent the distribution of taxi times. In addition, there are light-coloured rectangles that represent the individual letter values.

Encoding visual features A number of encodings are the same as before. For example, the airline and taxi times for outliers are encoded as the position of data points, though the axes have been swapped, so that taxi times are now horizontal positions and airlines are vertical positions.

Also as before, instead of raw taxi times, letter-value data summaries are encoded (other than for outliers). However, the encoding of letter values is different. Letter values are encoded as the position of coloured rectangles so that the edges of the rectangles decode to the individual letter values and the width of the rectangles decodes to differences between letter values. There are also rectangles that span the range of the outliers. These encodings are effective for decoding quantitative letter values from the edges and widths of the rectangles.

In addition, letter values (or the spaces between) are encoded as the colour of the rectangles. Pairs of consecutive letter values are encoded to different hues from the Okabe-Ito colour palette (Section 6.10), with light grey reserved for the outlier rectangles. Each hue is repeated twice to reduce the amount of change, while still allowing easy decoding. Although this encoding loses the quantitative information of the letter values, the qualitative encoding is effective for identifying the same letter values across different airlines.

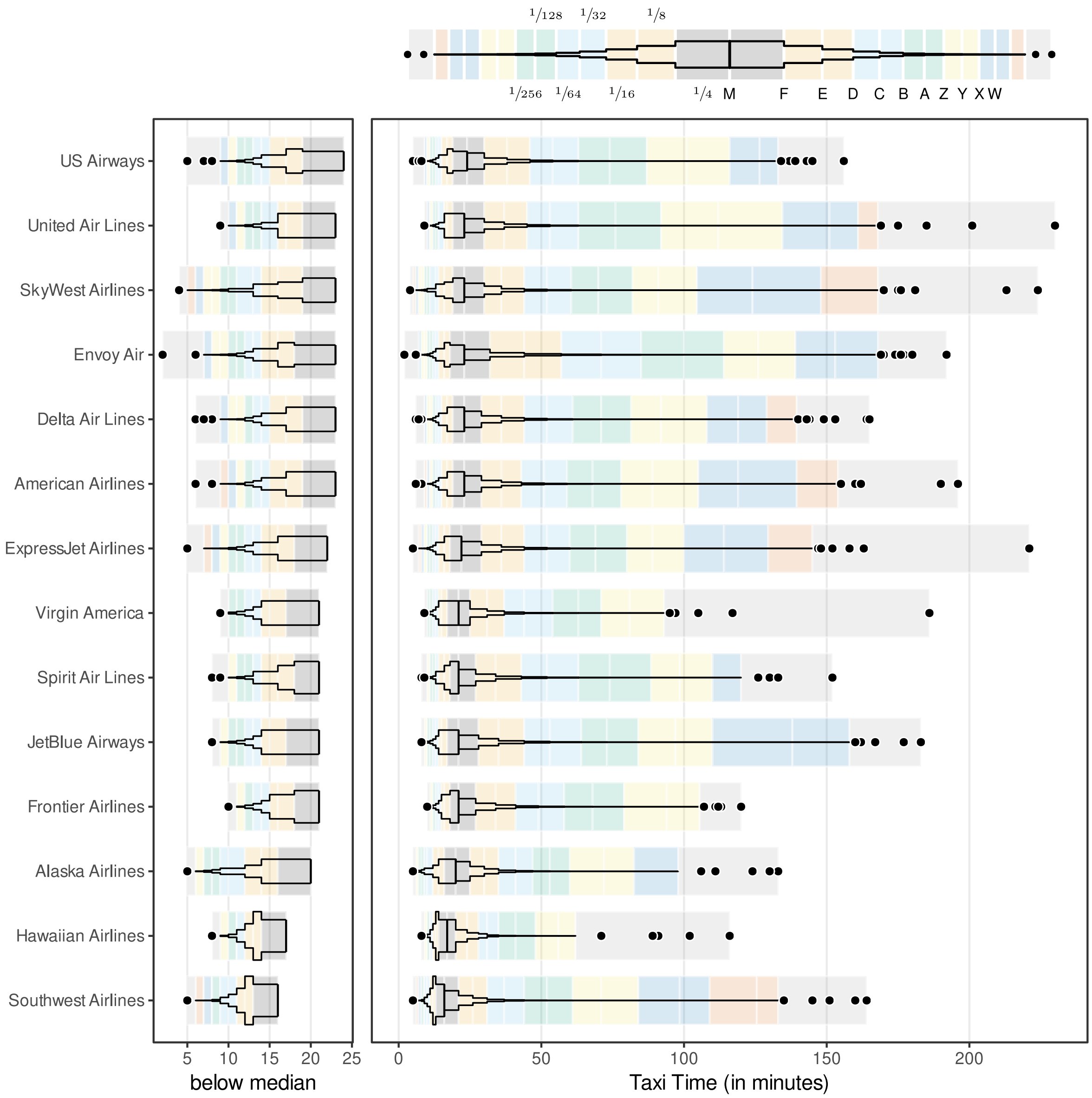


Figure 15.3: An improved letter-value plot?

The combined encodings of letter values mean that it is easy to decode and compare individual letter values. For example, we can see from Figure 15.3 that letter value E for Southwest Airlines is approximately 25 minutes and we can see that Envoy Air has the highest letter value B.

Visual shapes The letter values are also encoded as the horizontal positions of a black polygon for each airline. The data density of the taxi times is encoded as the vertical positions of the polygon so that the total area of each letter-value data symbol is the same (i.e., the polygon reflects the density distribution of the taxi times). This means that the narrowness of the polygon properly encodes the amount of data between different letter values. In other words, the implicit decoding of the area of the polygons is not misleading.

The polygon for each airline is of little use for decoding individual letter values, but it produces a visual shape that allows us to decode data summaries of the distribution of taxi times for different airlines (and the coloured rectangles provide a way to decode individual letter values). For example, we can see from Figure 15.3 that all airlines are right-skewed, but Hawaiian Airlines and Southwest Airlines are more skewed than the other airlines.

Detail about the spread of values within the long tails (longer taxi times) can be decoded from the coloured rectangles. For example, Envoy Air has a fatter tail (more longer taxi times) than other airlines because, for each letter value, the Envoy Air rectangle is further to the right than for other airlines (light blues are further right, greens are further right, yellows are further right, etc).

Non-linear scales Another feature of Figure 15.1 that created some problems was the use of a non-linear scale. In Figure 15.3, the encoding of taxi values and letter values to horizontal position is linear. This makes it possible to decode individual letter value and outlier taxi times and compare them between airlines. For example, we can see that the longest taxi time for Southwest Airlines is approximately 50 minutes longer than the longest taxi time for Hawaiian Airlines.

Because the linear scale makes it even more difficult to see the distribution below the median, an extra panel has been added that shows the distribution below the median with a higher-resolution scale. For example, we can see that Hawaiian Airlines and Southwest Airlines have a much more truncated distribution below the median than all other airlines.

Encoding encodings We identified several issues with the labelling of axes and the legend in Figure 15.1 and these have been improved in Figure 15.3.

The airline has been encoded as the full airline names in the text labels on the y-axis. These text labels are full words with learned decodings. This makes it possible not only to differentiate between airlines, but also to decode the actual airline name for each letter-value data symbol. For example, we can see that the bottom airline is Southwest Airlines rather than just WN.

The airlines are also encoded as the vertical position of letter-value symbols, which now run horizontally rather than vertically (just a swapping of axes from Figure 15.1). This is to allow space for the airline labels.

The legend in Figure 15.3 has changed because there are different data symbols and encodings involved, but it is also enhanced in several ways: the legend has a polygon similar to the letter-value data symbol that is used for each airline (but based on a standard normal distribution); the legend has coloured rectangles corresponding to the rectangles in the main plot (but with widths based on a standard normal distribution); the letter values are placed at the correct positions on the borders of the bars; and examples of the data density between letter values is encoded as vulgar-fraction text labels. These changes make the decoding of letter-value names, letter-value positions, and data density all more effective.

As an additional improvement, the airlines have been ordered by median. This simplifies the data visualisation a little because there is a simple monotonic trend in the median lines, rather than a series of increases and decreases. It also makes comparisons easier between more similar airlines because they are now closer together (Section 5.1).

Overall, the most important decoding of the distribution of taxi times is improved because the visual shape of the polygon provides this decoding while not being misleading. This is supplemented by the coloured rectangles, which show more detail about the tails of the distribution. In addition, thanks to linear scales and improved axes and legends, the decoding and comparison of individual taxi times is also more effective. Figure 15.3 is not perfect, but it demonstrates that the framework that we have developed allows us not only to deconstruct and identify issues with an existing data visualisation, it also allows us to think of ways that we can make improvements.

Notes

[207] This plot reproduces Figure 2 from Hofmann, Wickham, and Kafadar (2017), which was produced with the R package {lvplot} (Wickham and Hofmann 2025).

[208] In order to incorporate as many points as possible, this analysis just focuses on the default {lvplot} settings that are used in Figure 15.1 and even sinks to the level of critiquing the specific data set (and non-linear scales) that are used in this specific plot. Letter-value plots and the {lvplot} package are capable of a much broader range of results than the specific instance that we focus on in this section.

[209] The invention of letter values is attributed to Tukey (1977).

[210] Letter values are an interesting data summary in the sense that each letter value can correspond to a raw data value. In other words, letter values are a subset of the original raw data values, chosen to represent boundaries beyond which a certain proportion of the raw data values lie.

This is not true for all letter values because a letter value may fall between two raw data values.

[211] The calculation of the number of letter values to display is based on the uncertainty in the estimate of the letter values (assuming Gaussian data). We stop once a 95% confidence interval for a letter value overlaps with the neighbouring letter value.

This calculation is relevant only for the default letter-value plot. Variations of the letter-value plot design allow for different calculations of the number of letter values to display.

[212] This point is only relevant to the default letter-value plot. A variation of the letter-value plot design involves a different encoding of width so that the area of the rectangle reflects the proportion of the data between the letter values.

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